MCQs

Life Sciences | Biotechnology

Practice Book for

GRE

CSIR-UGC-NET

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ICMR

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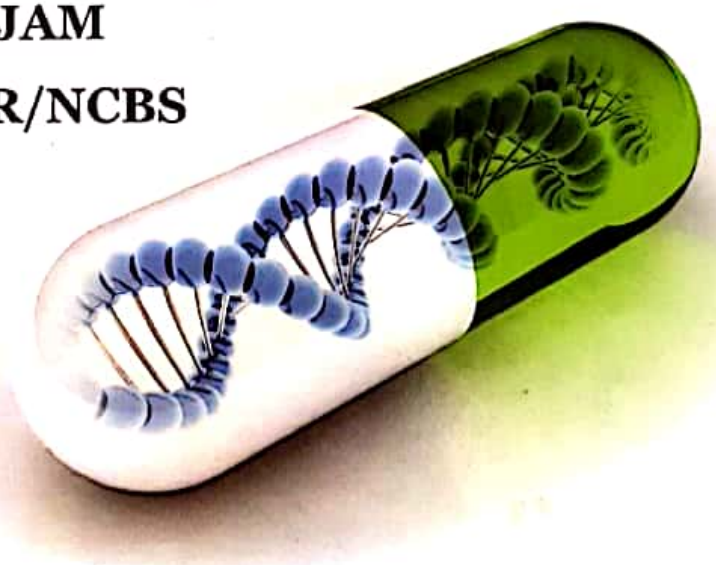

Pranav Kumar | Satyendra Singh

Fourth editio

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MCQs

Life Sciences | Biotechnology

Fourth edition

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Preface

MCQs, Life Sciences/Biotechnology, an exclusive fundamentals search based package of multiple choice questions, is basically designed for students to help them revise, consolidate and polish their knowledge and skills. This book is an attempt to provide concise yet comprehensive coverage of a true concept and application based factual and synthetic questions to develop in-depth understanding of the subject and to assess and review knowledge. This book contains more than 2500 factual and synthetic questions accompanied with their answers. Questions contained in the book have been designed to be similar in pattern used in CSIR-UGC-NET, GRE, DBT-JRF, ICMR, GATE, JNU (Ph.D), TIFR/NCBS (Ph.D), and IISc.

It also contains previous years questions from above mentioned examinations along with their answers to measure the level of understanding and their applications. This book is divided into Biochemistry, Cell Biology, Molecular Biology and Genetics, Microbiology, Physiology and Evolutionary Biology and contains two types of questions: Multiple Choice Questions and Fill in the blanks; which are designed to enable the student to identify pattern of examinations easily. For the sake of better self-assessment as well as thorough and comprehensive understanding of the subject, we have kept it topic/chapter-wise.

We therefore, recommend that only after a thorough and patient study of the reference books, one may undertake the sample questions from the respective chapters.

Pranav Kumar

Satyendra Singh

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Chapter 01

Amino acids and proteins

Amino acids and Peptides

1. Which of the following statements about standard amino acids are *correct*?

   P. Lysine has butylammonium side chain.

   Q. Average mass of an amino acid residue is ~110Da.

   R. Universal genetic code specifies only the 22 standard amino acids.

   S. Proline has a secondary amino group.

   a. P and Q
   b. Q and S
   c. P, Q, R and S
   d. only Q and R

2. Which of the following statement is *not* true?

   a. The pI is the pH value at which a protein has overall charge of +1.
   b. At a pH value equal to pI, a protein will not move in the electric field.
   c. An acidic protein will have a pI less than 7.
   d. A basic protein will have a pI greater than 7.

3. Which of the following statements is *not* true about the amino acids?

   a. Only 22 amino acids are used in protein synthesis.
   b. All amino acids exist in two stereoisomeric forms.
   c. Amino acids have an N-terminus, C-terminus, and R groups.
   d. There are more than 300 different kinds of amino acids present in the cell.

4. Choose the *incorrect* statement about standard amino acids.

   a. Aspartate has the smallest pI value.
   b. Arginine has the largest pI value.
   c. pK_a value of side chain is maximum for aspartate.
   d. Side chain of isoleucine has maximum hydropathy value.

5. Which of the following is *not* true about alanine?

   a. At pH 1, the overall charge is +1.
   b. At pH 1, it will move towards cathode.
   c. An equimolar mixture of d and *l* alanine does not rotate the plane polarized light.
   d. It contains branched side chain.

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6. Which of the following amino acid if added to an acidic solution will have buffer action on solution at the physiological pH?
   a. Alanine
   b. Glycine
   c. Histidine
   d. Arginine

7. What is the net charge (+, 0, −) of the amino acids glutamine and arginine at pH 10.76?
   a. − and 0
   b. − and −
   c. + and −
   d. + and 0

8. Which of the following is an imino acid?
   a. Histidine
   b. Glycine
   c. Cysteine
   d. Proline **[IISc]**

9. Which pair of amino acids will have the highest absorbance at 280 nm? (Assume equimolar concentrations)
   a. Thr and His
   b. Phe and Pro
   c. Trp and Tyr
   d. Phe and His **[GATE]**

10. Which of the following amino acid has a nonpolar and aliphatic R group?
   a. Tyrosine
   b. Histidine
   c. Glutamate
   d. Leucine

11. Which amino acid residue is most likely to be found in the interior of a water soluble globular protein?
   a. Ser
   b. Arg
   c. Val
   d. Asp **[GATE]**

12. Non-polar amino acid residues are found mostly
   a. in the core of proteins
   b. on the surface of proteins
   c. on alpha helices
   d. in non specific region **[IISc]**

13. The mean molecular weight of an amino acid in a typical globular protein is
   a. 70
   b. 110
   c. 150
   d. 90 **[IISc]**

14. If the molecular mass of an amino acid is 150 daltons, the molecular mass of a tripeptide will be
   a. 450
   b. 486
   c. 504
   d. 414 **[JNU]**

15. The closest estimate for the number of amino acid residues in a protein with a molecular mass of 85 kDa is
   a. 710
   b. 7100
   c. 8500
   d. 85 **[IISc]**

16. If one arginine has a molecular mass of 174 Daltons, then what would be the molecular mass (in Daltons) of a circular polymer of 38 arginines?
   a. 6612
   b. 5928
   c. 5946
   d. 7278

17. The molecular formula for glycine is $C_2H_5O_2N$. What would be the molecular formula for a linear oligomer made by linking ten glycine molecules together by condensation synthesis?
   a. $C_{20}H_{50}O_{20}N_{10}$
   b. $C_{20}H_{32}O_{11}N_{10}$
   c. $C_{20}H_{40}O_{10}N_{10}$
   d. $C_{20}H_{68}O_{29}N_{10}$

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18. A protein X was fused with GFP in a vector and expressed in *E. coli*. The length of the protein X is 1000 amino acid residues and the molecular weight of GFP is 27 kilodaltons. What is the total approximate molecular mass of the fusion protein in daltons?

a. 137000
b. 138000
c. 83000
d. 270000

19. An amino acid has three ionizable groups with pKa's of 2.0, 10.5 and 3.8. What is the pI of this amino acid?

a. 6.25
b. 9.05
c. 5.43
d. 2.9 **[IISc]**

20. At the isoelectric point of an enzyme, which is 5, it was observed that there are 10 positively charged groups and 10 negatively charged groups. When the enzyme was titrated with alkali starting from pH 5 to give pH 7, 3 positively charged groups were deprotonated. The net charge on the protein at pH 7 would be

a. +2
b. −3
c. −7
d. −10 **[JNU]**

21. Imagine you are generating a site-directed mutant of a protein in which a given residue is replaced by another residue. Which one of the following substitutions will result in the highest isoelectric point of the mutant compared to the wild type? (X→Y means residue X in the wild type is replaced by residue Y in the mutant).

a. Asp → Arg
b. Arg → Asp
c. Tyr → Lys
d. Gly → Arg

22. Which of the following statements is *false*?

a. Amino acids tend to be least soluble in water at their isoelectric point.
b. Sickle cell Hb and normal Hb have same value of negative charge.
c. Amino acids are made visible on the chromatogram by treatment with ninhydrin.
d. The net charge on an amino acid is a function of the pH of the solution.

23. Which one of the following peptides lacks a chiral center at the first position, is devoid of a hydrogen in the backbone amide at the second position and the side-chain of the third residue has a pK_a close to the physiological pH?

a. Gly-Pro-Tyr
b. Pro-Gly-Tyr
c. Pro-Gly-His
d. Gly-Pro-His **[IISc]**

24. What is the value of net charge in a peptide, GWYQR, at pH 10.5?

a. −1
b. 0
c. −2
d. +1

25. From the following sets, pick the one that contains exclusively hydrophobic amino acids

a. Asp, Glu, Lys, Arg, Ser
b. Arg, Phe, Tyr, Trp, Asn
c. Ala, Ile, Leu, Phe, Val
d. Ala, Arg, Phe, Leu, Tyr **[IISc]**

26. Which of the sequence below would be most likely termed homologous to the following?

Leu-Arg-Cys-Lys-Gln-Gly-Ser?

a. Ser-Gly-Gln-Lys-Cys-Arg-Leu
b. Ile-Arg-Cys-Arg-Gln-Gly-Thr
c. Val-Lys-Ala-Arg-Asp-His-Thr
d. Leu-Cys-Arg-Gln-Lys-Ser-Gly

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27. Which of the following peptides can be easily detected by absorbance at 280 nm?

P. Leu – Tyr – Met – Ala – Glu
Q. Ser – Thr – Trp – Val – Ile – Leu
R. Asp – Ala – Glu – Gln – Ser – Asp – Lys
S. Thr – Tyr – Trp – Val – Ile

a. P, Q and S
b. P and S
c. Q and R
d. Q and S

[GATE]

28. Of the groups of amino acids indicated below, which one group of amino acids can be phosphorylated?

a. Thr, Tyr, Asn
b. Ser, Thr, Tyr
c. Ser, Phe, Asn
d. Ser, Thr, Phe

[IISc]

29. In cell-free lysates of *E. coli*, radioactive isotope, [γ^{32}P] ATP can be used to label the following

a. closed circular duplex DNA.
b. protein containing serine, threonine or tyrosine.
c. protein devoid of serine, threonine and tyrosine.
d. peptidoglycan.

[IISc]

30. Choose the mismatch pair

Amino acid	*Group present in side chain*
a. Tyrosine	phenolic group
b. Arginine	guanidino group
c. Histidine	imidazole group
d. Cysteine	imino group

31. Presence of specific functional group in an amino acid gives a specific colour reaction and used as a test for that amino acid. Choose the *incorrect* combinations

Name of test	*Amino acid*
a. Hopkins cole	Tryptophan
b. Pauly	Histidine
c. Nitroprusside	Cysteine
d. Millon	Arginine

32. The high solubility of amino acids in water is due to

a. presence of side chain
b. dipolar ion structure
c. unipolarity
d. the hydrophilic nature of the amino group

33. Amino acids found in proteins that are formed by post-translational modification of one of the common amino acids (derived amino acids) include which of the following?

a. Isoleucine
b. γ-carboxyglutamate
c. threonine
d. 4-hydroxyproline

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34. Choose the *incorrect* statement.

a. At isoelectric point, protein has minimum solubility.

b. Among standard amino acids, tryptophan has highest molecular weight.

c. Selenocysteine is derived from cysteine.

d. Histidine contains three ionizable groups.

35. Three properties (1, 2, 3) of an amino acid are given. Mark one amino acid which has all the three properties

1. contains an imino group
2. with ninhydrin formed a yellow derivative
3. is a non essential amino acid

a. Glutamic acid

b. Proline

c. Lysine

d. Histidine

36. Which of the following substitutions, introduced by site-directed mutagenesis, in a protein, is expected to bring about the maximum change in the isoelectric point in the mutant protein compared to the wild-type protein?

a. Asparagine to serine

b. Lysine to arginine

c. Arginine to glutamate

d. Glutamate to aspartate

[IISc]

37. Which statement about amino acids is *correct*?

a. They always have at least one amino group and at least one carboxyl group.

b. In the formation of proteins, it is a condensation reaction that links the amino group of one amino acid to the variable side chain of the adjacent amino acid.

c. The variable side chains of all of the amino acids are highly reactive and carry a charge at neutral pH.

d. The peptide bond that links amino acids together in a protein is a type of ionic bond, which explains why proteins are unstable at high temperatures.

38. The peptide bond has a *backbone* of atoms in which of the following sequences?

a. C - N - N - C

b. C - C - C - N

c. C - C - N - C

d. N - C - C - C

39. The peptide bond is

P. partially double bonded

Q. polar

R. planar

S. mostly *trans* in proteins

a. P and Q

b. R and S

c. P and S

d. P, Q, R and S

40. A Ramachandran plot describes, for a particular amino acid, the sterically permitted angles for

P. rotation about the C_α-C_β bond

Q. rotation about the C_α-C bond

R. rotation about the N-C_α bond

S. rotation about the C_α-H bond

a. Q and R

b. P and Q

c. R and S

d. Q and S

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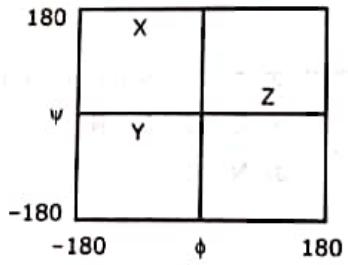
41. The property of resonance as applied to protein structure is responsible for
    a. the prevention of rotation about the alpha carbon.
    b. the partial double bond character of the side chain-alpha carbon bond.
    c. the planar nature of the peptide bond.
    d. the ability of cochlear cells to detect sound waves by mechanotransduction.

42. The peptide bond in proteins is
    a. planar, but rotates to three preferred dihedral angles.
    b. nonpolar, but rotates to three preferred dihedral angles.
    c. nonpolar, and fixed in a trans conformation.
    d. planar, and usually found in a trans configuration. [JNU]

43. Which one of the following bonds in proteins has a partial double bond character?
    a. Cα – C
    b. Cα – N
    c. C – N
    d. C – O [IISc]

44. On a Ramachandran plot the entries for hemoglobin would be clustered around
    a. all four corners
    b. the extended chain conformation
    c. the left-handed α-helix conformation
    d. the right-handed α-helix conformation

45. In the following Ramachandran diagram, which types of secondary structure do the regions marked X, Y and Z represent?



a.	X : right handed α-helix	Y : left handed α-helix	Z : β-sheet
b.	X : left handed α-helix	Y : right handed α-helix	Z : β-sheet
c.	X : β-sheet	Y : right handed α-helix	Z : left handed α-helix
d.	X : β-sheet	Y : left handed α-helix	Z : right handed α-helix [TIFR]

46. Which of the following statements related to the Ramachandran plot is *not* true?
    a. Almost all peptide bonds in proteins are cis.
    b. The restriction of ϕ and ψ angles limits structures accessible to unfolded form of proteins.
    c. A configuration that has $\phi = 90°$ and $\Psi = -90°$ is disallowed.
    d. Three-quarters of possible (ϕ, ψ) combination are excluded by their steric clashes. [JNU]

47. What should be the net charge at pH = 1 on the tetrapeptide : Glu—Gly—Ala—Lys?
    a. +1
    b. +2
    c. -1
    d. zero

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48. Aspartame, the synthetic sweetener is
   a. a dipeptide
   b. also secreted by the posterior pituitary gland in response to thirst
   c. a glucose derivative
   d. not digested (i.e. hydrolyzed) by humans

49. For the pentapeptide, Glu-Met-Arg-Thr-Gly
   a. number of charged groups at pH 7 is 4.
   b. the net charge at pH 1 is +1.
   c. the sequence using one letter symbols is E-T-R-M-G.
   d. the net charge at pH 1 is −2.

Protein structure

1. If there are 20 kinds of amino acids and they can be assembled in any order, then how many different proteins of 100 amino acids long can theoretically be produced?
   a. 20×100
   b. 20^{100}
   c. 100^{20}
   d. $20 \times 20 \times 100$

2. How many different pentapeptides are there that contain one residue each of Gly, Asp, Tyr, Cys and Leu?
   a. 625
   b. 120
   c. 32
   d. none

3. In five-residue long peptide each amino acid can adopt three conformations. The total number of conformations that the peptide can adopt is
   a. 15
   b. 243
   c. 8
   d. 125 **[IISc]**

4. If one needs to use the natural 20 amino acid residues, with no restrictions on the number of times a given residue type can be used, what is the total number of possible tripeptide sequences?
   a. 20×3
   b. $20 \times 19 \times 18$
   c. 4000
   d. 8000 **[IISc]**

5. Enormous diversity of protein molecules is due to
   a. sequence of amino acids
   b. R-group of amino acids
   c. peptide bonds
   d. amino group of amino acids

6. An alpha helix represents
   a. primary structure of a protein
   b. secondary structure of a protein
   c. tertiary structure of a protein
   d. aggregation of proteins **[JNU]**

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7. The shape of the polypeptide backbone making α-helix in proteins is
   a. like the English letter "α"
   b. right handed spiral
   c. left handed spiral
   d. like the Greek letter "α" **[IISc]**

8. Hydrogen bonds in α-helices are
   a. more numerous than van der Waals interactions.
   b. not present at phenylalanine residues.
   c. analogous to the steps in a spiral staircase.
   d. roughly parallel to the helix axis.

9. An α-helix is made up of 78 amino acid residues. What will be the value of its axial length and also length when polypeptide chain is fully extended respectively?
   a. 117 Å, 117Å
   b. 281 Å, 117 Å
   c. 117 Å, 281 Å
   d. 281 Å, 281 Å

10. Consider three polypeptides of 15 residues each. They adopt distinct conformations corresponding to right-handed α-helical structure, left-handed α-helical structure and single strand of a β-sheet structure. Considering the shortest distance between the first and the last residues as the length of the structure, which one of the following statements is *true*?
    a. Right-handed α-helical structure is longest.
    b. Left-handed α-helical structure is longest.
    c. β-strand structure is longest.
    d. All the three polypeptides have identical length. **[IISc]**

11. Which one of the following statements about protein secondary structure is *correct*?
    a. An α-helix is primarily stabilized by ionic interactions between the side chains of the amino acids.
    b. β-sheets exist only in antiparallel form.
    c. β turns often contain proline.
    d. An α-helix can be composed of more than one polypeptide chain. **[GATE]**

12. The formation of an alpha helix results in a dipole along the helix, with a positive N-terminus and a negative C-terminus. What is the most likely explanation for this phenomenon?
    a. The helix dipole is the sum of each peptide bond dipole moment.
    b. The N-terminus of a protein contains a positively charged amine group, while the C-terminus contains a negatively charged carboxl group.
    c. The peptide carbonyl oxygens that hydrogen bond with amide hydrogens *point* toward the C-terminus.
    d. Side chain residues in alpha helices extend outward, perpendicular to the axis of the helix.

13. The mirror image of right handed α-helix with all L-amino acids will appear as
    a. left-handed α-helix with L-amino acids.
    b. left-handed α-helix with D-amino acids.
    c. right-handed α-helix with L-amino acids.
    d. right-handed α-helix with D-amino acids. **[IISc]**

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14. A twenty residue peptide composed of the sequence $(L - Ala)_{20}$ forms a right handed alpha helix in methanol. If every alternate L - Ala residue is replaced with D - Ala, the peptide will probably
    a. form a right handed alpha helix
    b. form a left handed alpha helix
    c. form ten residues of right handed helix followed by ten residues of a left handed helix
    d. not form a helix of any kind

[IISc]

15. Which of the following has a quaternary structure?
    a. α-chymotrypsin
    b. Hemoglobin
    c. Insulin
    d. Myoglobin

[GATE]

16. For a protein to have a quaternary structure, it must
    a. have four amino acids
    b. consist of two or more polypeptide subunits
    c. consist of four polypeptide subunits
    d. have at least four disulfide bridges

17. The forces that maintain the three-dimensional structure of a protein is mainly
    a. non-covalent
    b. covalent
    c. coordinate
    d. covalent and non-covalent

[JNU]

18. The following type of interaction is mainly responsible for aggregation of proteins in dilute solutions
    a. hydrogen bonds
    b. hydrophobic interactions
    c. disulfide bonds
    d. peptide bonds

[GATE]

19. Protein structures are confirmed by weak bonds, the example of which is
    a. hydrophobic
    b. disulphide
    c. peptide
    d. all of the above

20. A favorable charge-charge interaction between R group in an α-helix is expected to occur when the interacting side chains are separated by
    a. one-two residues
    b. three-four residues
    c. five-six residues
    d. seven-eight residues

[IISc]

21. Which one of the following is an example of a non-covalent interaction in proteins?
    a. Salt bridge
    b. Disulfide bridge
    c. Peptide bond
    d. Phosphodiester bond

[IISc]

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22. What does a protein lose when it denatures?
    a. Its primary structure
    b. Its peptide bonds
    c. Its sequence of amino acids
    d. Its three dimensional shape

23. Which one of the following is the least important factor as far as protein folding is concerned?
    a. Hydrophobic effect
    b. Hydrogen bonding
    c. Electrostatic interactions
    d. Quaternary association **[IISc]**

24. In the α-helix,
    P. the hydrogen bonds are roughly parallel to the axis of the helix.
    Q. the R groups on the amino acid residues are found on the outside of the helix spiral.
    R. the hydrogen bonds are roughly perpendicular to the axis of the helix.
    S. the R groups on the amino acid residues are stacked within the interior of the helix.
    a. P and Q
    b. P and S
    c. R and S
    d. Q and R

25. In how many ways can eight cysteine residues in a protein form disulphide bonds?
    a. 8
    b. 16
    c. 24
    d. 28 **[IISc]**

26. When you boil an egg, you convert the albumin into a white solid mass. In chemical terms you would say that
    a. the protein was dehydrated by heat
    b. the protein was denatured by heat
    c. the protein was cross-linked by heat
    d. the protein was degraded by heat **[JNU]**

27. Thermal denaturation of proteins involves
    a. conformational change in the protein
    b. random cleavage of the peptide bonds
    c. covalent modification of certain amino acids
    d. increase in its isoelectric point **[JNU]**

28. DNA polymerase contains a lysine residue that is important for binding to DNA. Mutations were found that converted this lysine to either glutamate, glycine, valine or arginine. Which mutations would be predicted to be the most and least harmful to the ability of the enzyme to bind DNA?

	Most	*Least*
a.	Glycine	Arginine
b.	Arginine	Glycine
c.	Glutamate	Valine
d.	Glutamate	Arginine

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29. Which statement about proteins is *correct*?
   a. Protein quaternary structure is determined solely by the primary amino acid sequence.
   b. Protein sequences that span a lipid bilayer membrane contain a number of charged amino acids.
   c. Examples of protein secondary structure include an alpha helix or a beta pleated sheet.
   d. Protein tertiary structure is the result of the interaction of two or more independent polypeptides.

30. Which of the following statement about protein structure is *correct*?
   a. The extended β-configuration is not found in globular proteins.
   b. The stability of the α-helix is mainly due to hydrophobic interactions.
   c. Globular proteins tend to fold into configurations that keep hydrophobic side chains in the interior of the molecule.
   d. The primary structure of a peptide does not influence the formation of the native three-dimensional configuration.

31. Water has a high dielectric constant value of 80. Because of its presence in biological systems compared with low dielectric solvents like hydrocarbons, it should
   a. strengthen electrostatic interactions but weaken hydrophobic interactions.
   b. weaken electrostatic interactions but strengthen hydrophobic interactions.
   c. strengthen both electrostatic as well as hydrophobic interactions.
   d. weaken both electrostatic as well as hydrophobic interactions.

Globular and Fibrous proteins

1. Hemoglobin is
   P. a monomeric protein
   Q. an oligomeric protein
   R. a chromoprotein
   S. an allosteric protein
   a. P and Q
   b. Q and S
   c. Q and R
   d. Q, R and S

2. The quaternary structure of human hemoglobin is best described as a
   a. dimer of two myoglobin dimers
   b. tetramer of identical subunits
   c. tetramer of four different subunits
   d. tetramer of two different subunits

3. When Glu at the sixth position in the β-chain of human hemoglobin is replaced by Val,
   a. secondary structures are significantly altered
   b. tertiary structure is significantly altered
   c. quaternary structure is significantly altered
   d. no significant structural change takes place **[IISc]**

4. Difference between the R- and T-states of hemoglobin is
   a. dimeric and tetrameric structure
   b. oxy and deoxy states
   c. binding to ferrous and ferric ions
   d. folded and unfolded states **[IISc]**

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5. Binding of oxygen to hemoglobin occurs when Fe is in the following oxidation state [IISc]

a. +1
b. +2
c. +3
d. +4

6. The advantage of hemoglobin having a high histidine content is

a. histidine binds to oxygen
b. histidine carries oxygen to the tissues
c. histidine imparts buffering capacity to hemoglobin
d. 'R' group of histidine has low pK_a [GATE]

7. Which of the following statement is *false*?

a. Homozygotes for HbS results in an elevated BPG concentration in their erythrocytes.
b. Hill constant, n can never be larger than the number of ligand binding sites on the protein.
c. Peptide groups, with few exceptions, assume the trans configuration.
d. None of the above.

8. Which of the following statement is/are *true*?

P. Cooperativity coefficient is a parameter in Hill's plot.
Q. Bohr's effect is observed in myoglobin.
R. Myoglobin is an allosterically-regulated protein.
S. At low pH oxygen binds to hemoglobin weaker than at high pH.

a. P only
b. R only
c. P and S
d. Q and R

9. Which of the following statements are *true* about myoglobin?

P. It is a monomeric protein.
Q. It contains four protoheme IX prosthetic groups.
R. The molecule has eight major helical segments.
S. Oxygen binding exhibits cooperative kinetics.

a. P, Q and R
b. P and Q
c. P and R
d. P, Q, R and S

10. The residue histidine can act as a ligand for the heme iron in proteins such as myoglobin because

a. histidine side chain has pK_a close to pH 5.0.
b. one of the nitrogens has a lone pair of electrons available for coordination.
c. the aromatic ring of the histidine can act favorably with the heme iron.
d. five membered rings have a favourable dipolar interactions with ferrous or ferric irons. [IISc]

11. Which of the following statements is *incorrect*?

a. Mb consists of a single polypeptide chain in addition to a heme prosthetic group.
b. The red colour associated with the oxygenated forms of Mb and Hb is due to heme.
c. Hb has a higher affinity for oxygen compared to Mb as seen from the oxygen binding curves.
d. The α and β chains of the Hb tetramer are slightly shorter than the single chain of Mb. [GATE]

12. Which of the following statements about oxygen binding and release by hemoglobin is *correct*?

a. On binding oxygen, the iron of the heme prosthetic group is oxidized to the ferric state.
b. Lowering the pH accelerates the release of oxygen from oxyhemoglobin.

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c. A high concentration of 2,3-BPG in the erythrocyte enhances the binding of oxygen by hemoglobin.
d. Oxygen binding by any one of the four heme groups occurs independently of the other three.

13. The hemoglobin consists of two alpha and two beta chains each of which has a heme group with an iron atom at the centre. The iron in the heme binds one O_2 molecule. Therefore one molecule of hemoglobin can bind four oxygen molecules.

Indicate which of the given statement is *not* correct?

a. Changes in the pH of the blood affects the binding of oxygen by hemoglobin.
b. Carbon dioxide from respiring cells diffuse into the blood and lowers the pH of the blood which enables the hemoglobin to release O_2.
c. Most of the CO_2 produced by the cells are transported in the blood in the form of bicarbonate ions.
d. The CO_2 can bind to the iron of the heme which is free. **[JNU]**

14. In sickle cell anaemia, the basis of the malfunction of the hemoglobin molecule is

a. substitution of a single amino acid
b. incorrect secondary structure
c. faulty binding of the heme groups
d. reduced affinity for oxygen

15. Which of the following describes a major effect of sickle cell anaemia?

a. Absence of biphosphoglycerate binding of hemoglobin.
b. Substitution of two proximal histidines.
c. Decreased solubility of the deoxy form of hemoglobin.
d. A P_{50} value for hemoglobin is similar to that of myoglobin.

16. Which one of the following pairs refers to paralogous proteins?

a. Rat myoglobin and rat hemoglobin.
b. Rat myoglobin and mouse myoglobin.
c. Rat myoglobin and mouse hemoglobin.
d. Rat myoglobin and rat cytochrome. **[IISc]**

17. An example of a structural protein which is also an enzyme is

a. actin
b. myosin
c. trypsin
d. troponin **[IISc]**

18. Most abundant protein in the human body is

a. hemoglobin
b. keratin
c. collagen
d. immunoglobulin

19. An example of a trimeric protein is

a. lysozyme
b. hemoglobin
c. keratin
d. collagen **[IISc]**

20. Collagen is best described as

a. an α-helical structural protein
b. a coiled-coil found in hair
c. a cross-linked globular protein
d. a triple-helical fibrous protein

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21. Collagen is a fibrous protein consisting of three peptide chains intertwined in a triple helix. What repetitive motif is responsible for collagens ability to form the triple helix?
    a. Ala-X-Y
    b. Cys-X-Y
    c. Gly-X-Y
    d. Pro-X-Y

22. Which one of the following is unlikely to be water-soluble protein?
    a. Ribonuclease
    b. Myoglobin
    c. Collagen
    d. Lysozyme

23. Which of the following is *not* correct about the collagen triple-helix?
    a. is rich in glycine
    b. is rich in proline
    c. is rich in hydroxyproline
    d. is an α helix

[IISc]

24. Choose the *incorrect* statement with regard to collagen molecule.
    a. It is formed from three α-chains.
    b. Glycine accounts for one-third of amino acids found within collagens.
    c. It contains less hydroxyproline than elastin.
    d. Ascorbate is essential for hydroxylation of proline.

25. Which of the following statements are *true* about the structure of collagen?
    P. Fundamental unit of collagen is tropocollagen.
    Q. H-bond between three α-chains holds the helix together.
    R. Rich in proline and glycine.
    S. Three α-chains wind around one another in a characteristic left handed triple helix.
    a. P, Q and R
    b. P and Q
    c. P and R
    d. P, Q, R and S

26. Which of the following cofactors derived from a vitamin is involved in the formation of hydroxyproline during collagen synthesis?
    a. Pyridoxal phosphate
    b. Biotin
    c. Thiamine pyrophosphate
    d. Ascorbic acid

27. Collagen occurs in different forms, which are usually classified on the basis of the
    a. type of carbohydrate present
    b. cysteine content
    c. hydroxyproline and hydroxylysine content
    d. types of peptide chains present

28. Choose the mispair

Fibrous Protein	*Most abundant amino acids*
a. Silk fibrion	Gly, Ala, Ser
b. Collagen	Phe, Ser, Pro
c. Elastin	Gly, Ala, Val
d. α-Keratin	Ser, Glu, Cys

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29. Which of the following is *not* a feature of α-keratin?
    a. Presence of –S–S– cross-link between adjacent α-helices.
    b. Presence of intrachain H-bonding.
    c. Contains predominant amino acid residues like Phe, Cys, Met.
    d. Presence of intrachain –S–S– cross-link.

30. Desmosine is
    P. a derivative of four lysine residues
    Q. a derivative of four histidine residues
    R. is found in the blood clotting protein thrombin
    S. is found in fibrous protein elastin
    Choose the right combination
    a. P and R
    b. P and S
    c. Q and R
    d. Q and S

31. Disulfide bonds do not form in the cytosol but keratin intermediate filaments in the skin are cross-linked by disulfide bonds. This can be explained by the following
    a. intermediate filaments are transported to the cell exterior via the secretory pathway that allows the formation of disulfide bonds.
    b. Keratinocytes do not have a strongly reducing cytosol thereby allowing the formation of disulfide bonds in the cytosol.
    c. Keratinocytes lack glucose-6-phosphate dehydrogenase so that their cytosol is less reducing.
    d. Disulfide bonds in keratin are formed after cells have died and therefore cannot maintain the reducing environment in the absence of metabolic activity and cellular contents are oxidized. **[JNU]**

Protein techniques
(Electrophoresis, Chromatography, Spectroscopy and Centrifugation)

1. The isoelectric point of alanine is 6.0. If alanine is dissolved in a buffer of pH 3.0 and subjected to electrophoresis
    a. it will not migrate to either anode or cathode
    b. it will migrate to the cathode
    c. some will migrate to the anode and some to the cathode
    d. it will migrate to the anode

2. The rate of migration of a protein in an SDS-polyacrylamide gel is not influenced by
    a. size of the protein
    b. charge of the protein
    c. pore size of the gel
    d. strength of the electric field **[IISc]**

3. Which of the following statement is *false*?
    a. β-mercaptoethanol acts as reducing agent and used to break disulfide bonds.
    b. Bromophenol blue used as tracking dye during electrophoresis.
    c. Coomassie brilliant blue used for staining proteins.
    d. Sodium dodecyl sulphate is a cationic detergent.

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4. SDS is used in polyacrylamide gel electrophoresis of a mixture of proteins for their efficient separation on the gel. SDS in this experiment is used to
   a. solubilize the protein.
   b. stabilize the protein.
   c. decrease the surface tension of buffer.
   d. have uniform charge density on the proteins.

5. Sodium dodecyl sulphate upon addition into water can form spherical micelles. In this micelle the hydrocarbon tail would be
   a. cleaved on micelle formation.
   b. in the interior of the micelle.
   c. towards the surface of the micelle.
   d. none of the above. [JNU]

6. If the pH becomes lower than the isoelectric point of a protein, then how will the protein respond in an electrophoretic system? It will
   a. become denatured.
   b. migrate to the negative pole.
   c. migrate to the positive pole.
   d. remain stationary and unchanged.

7. Which of the following electrophoretic procedures does *not* depend on the charge of the protein?
   a. Moving boundary electrophoresis.
   b. Zone electrophoresis.
   c. Polyacrylamide gel electrophoresis in sodium dodecyl sulphate.
   d. Isoelectric focusing.

8. In two-dimensional electrophoresis, proteins are separated
   a. first by their charge and then by their mass.
   b. first by their mass and then by their charge.
   c. only on the basis of their charge.
   d. only on the basis of their mass.

9. Two-dimensional (2-D) gel electrophoresis performed under denaturing conditions can be used to separate proteins according to which of the following characteristics?

First Dimension	Second Dimension
a. subunit molecular weight	density
b. density	charge
c. amino acid composition	charge
d. isoelectric point (pI)	subunit molecular weight

10. A mixture of amino acids consisting of glycine, lysine, arginine, histidine, aspartic acid and glutamic acid was placed in the centre of a paper strip, moistened with buffer of pH 6 and electric current applied. The migration of amino acids was as follows.
    a. Glycine, lysine and histidine moved towards the anode. Aspartic acid and glutamic acid moved towards the cathode while arginine remained near the origin.
    b. Aspartic acid and glutamic acid remained near the origin and lysine, histidine and glycine moved towards the anode while arginine moved towards the cathode.

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c. Glycine remained near the origin. Lysine, arginine and histidine moved towards the cathode while aspartic acid and glutamic acid moved towards the anode.

d. All amino acids remained near the origin. **[GATE]**

11. SDS and NaCl are used in the purification of genomic DNA from peripheral blood samples. Their functions in the process are respectively.

a. salting out of DNA – denaturation of protein.

b. salting out of protein – denaturation of DNA.

c. denaturation of protein – salting out of DNA.

d. denaturation of protein – salting out of proteins.

12. The relative viscosity of a protein solution is affected by

a. temperature, shape and molecular weight.

b. temperature and molecular weight.

c. molecular weight.

d. temperature and shape. **[IISc]**

13. Proteins may be separated according to size by

a. reverse phase chromatography

b. ion exchange chromatography

c. molecular exclusion chromatography

d. isoelectric focusing

14. The elution of a protein from size exclusion chromatography depends on

a. the size of the protein alone.

b. the size and shape of the protein.

c. the size and surface charge distribution of the protein.

d. the shape and surface charge distribution of the protein. **[IISc]**

15. A solution of a mixture of proteins is loaded onto a gel filtration column that has a size exclusion limit of 75,000 Da. The proteins have the following molecular mass in Daltons-A:11237; B:66000; C:42000; D:90000. The proteins will elute from the column in the following order

a. A, B, C, D

b. D, B, C, A

c. A, C, B, D

d. D, A, B, C **[IISc]**

16. A mixture of lysine, aspartic acid, tyrosine and alanine is passed through an anion exchange column at pH 8.0. What is the likely order of elution of amino acids when eluted with a buffer of reducing pH gradient?

a. Tyr, Lys, Asp, Ala

b. Lys, Ala, Asp, Tyr

c. Asp, Tyr, Ala, Lys

d. Lys, Tyr, Ala, Asp **[IISc]**

17. Anion exchangers preferentially bind proteins with

a. acidic isoelectric points

b. basic isoelectric points

c. no charge

d. abundant hydrophobic groups **[IISc]**

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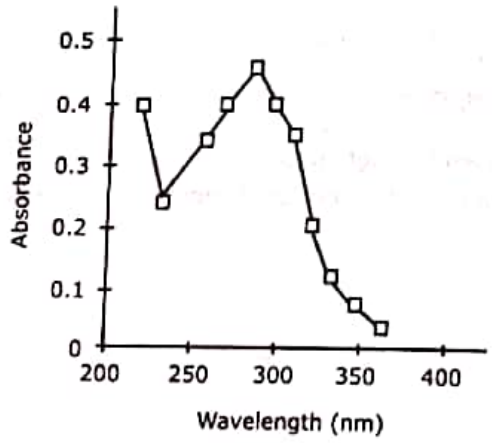
18. Separation of one protein from other proteins on the basis of molecular size can be achieved by
    P. polyacrylamide gel electrophoresis.
    Q. dialysis.
    R. gel filtration (molecular exclusion chromatography).
    S. ion-exchange chromatography.
    a. P, Q and R
    b. P and Q
    c. P and R
    d. P, Q, R and S

19. A protein was purified to homogeneity. Determination of the molecular weight by molecular exclusion chromatography yields 60 kDa. Chromatography in the presence of 6 M urea yields a 30 kDa species. When the chromatography is repeated in the presence of 6 M urea and 10 mM β-mercaptoethanol, a single molecular species of 15 kDa results. The protein is a
    a. homodimer of 30 kDa
    b. homodimer of 15 kDa
    c. homotetramer of 30 kDa
    d. homotetramer of 15 kDa

20. Electrophoresis of a purified protein named X in the presence of sodium dodecyl sulfate and 2-mercaptoethanol, shows a single band of 45 kDa. In gel filtration column chromatography, protein X elutes between alcohol dehydrogenase (160 kDa) and β-amylase (190 kDa). How many identical subunits is protein X composed of?
    a. One
    b. Two
    c. Three
    d. Four
    [GATE]

21. Molecular weight of an unknown protein can be found out by
    a. electrophoresis
    b. ion-exchange chromatography
    c. affinity chromatography
    d. none of the above
    [GATE]

22. The spectrum shown below is of a sample dissolved in buffer, pH 7.0 in a 1 cm pathlength cuvette.



The sample is most likely to be
    a. a solution of DNA
    b. a solution of NADH
    c. a solution of 2,4-dinitrophenyl phosphate
    d. a solution of protein

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23. Although protein concentration can be estimated by measuring the absorbance of a protein solution at 280 nm, this technique does *not* yield an accurate measure of absolute protein concentration. Why?
    a. Protein solutions refract light optimally at 280 nm, thus interfering with the measurement.
    b. Proteins scatter light optimally at this wavelength, thus interfering with the concentration measurement.
    c. The aromatic amino acid content varies from protein to protein.
    d. Cysteine residues absorb light optimally at this wavelength.

24. According to Beer-Lambert's law, what will be the absorbance when $I = I_0/10$?
    a. 0
    b. 1
    c. 0.1
    d. 100

    [IISc]

25. A solution of tryptophan has an absorbance at 280 nm of 0.54 in a 0.5 cm pathlength cuvette. Given the absorbance coefficient (ε) for tryptophan is 5.4×10^3 L mol^{-1} cm^{-1}, the concentration of the solution is
    a. 0.2 mM
    b. 20 mM
    c. 1×10^{-3} M
    d. 0.1 mM

    [GATE]

26. The molar absorption coefficient (extinction coefficient) of NADH at 340 nm is 6,220 L/mole/cm, whereas NAD does not absorb at 340 nm. What absorbance will be observed when light at 340 nm passes through a 1 cm cuvette containing 10 µM NADH and 10 µM NAD?
    a. 0.031
    b. 0.062
    c. 0.124
    d. 0.310

    [IISc]

27. Changes in protein conformation can be detected by
    a. ultraviolet absorption spectroscopy
    b. fluorescence emission
    c. circular dichroism
    d. all of the above

    [GATE]

28. Using circular dichroism one can obtain
    a. relative proportions of α-helices, β-sheets and loops in proteins
    b. three-dimensional structure of a protein
    c. amino acid sequence of a protein
    d. quaternary structure of a protein

    [IISc]

29. The secondary structure of a protein cannot be determined by one of the following techniques
    a. NMR spectroscopy
    b. X-ray crystallography
    c. fluorescence spectroscopy
    d. circular dichroism

    [IISc]

30. What is the most common approach for the determination of precise 3-D structure of globular proteins?
    a. Circular dichroism
    b. Mass spectrometry
    c. Infrared spectroscopy
    d. X-ray diffraction

    [IISc]

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31. The void volume of a gel filtration column, V_0 = 30 mL. A monomeric protein with a known molecular weight of 25 kDa elutes at a volume of 45 mL. The protein that you are trying to purify elutes at a volume of 35 mL. Which of the following is a valid conclusion?

a. your protein is repelled by the gel filtration material
b. your protein Mr=25 kDa
c. your protein Mr>25 kDa
d. your protein Mr<25 kDa **[MKU]**

32. A homogeneous protein of native molecular weight 100,000 gave a single band of molecular weight 50,000 on SDS-PAGE in presence of β-mercaptoethanol. N-terminal analysis gave two amino acids alanine and leucine in equal proportions. Hence the

a. protein is a homodimer.
b. protein is contaminated with another protein.
c. protein has two polypeptides linked by disulfide bridges.
d. none of the above.

33. A mixture of four proteins of pIs 11, 7, 5, and 3 are loaded on DEAE anion-exchange column equilibrated with low ionic strength buffer of pH 8. Which of the four proteins would be expected to be retained on the column?

a. Protein with pI 11 but not the others.
b. Proteins with pIs 11 and 7 but not 5 and 3.
c. Proteins with pIs 7, 5 and 3.
d. Protein with pI 7 but not the others. **[GATE]**

Protein sequencing

1. Choose the mismatch

a. phenyl isothiocyanate – Hydrolysis of peptide bonds on the carboxyl side of lysine and arginine residues
b. dansyl chloride – Identification of the amino-terminal residue of a peptide
c. chymotrypsin – Hydrolysis of peptide bonds on the carboxyl side of aromatic residues
d. CNBr – Cleavage of peptide bonds on the carboxyl side of methionines

2. Which of the following statements are *correct* about Edman reaction?

P. The peptide does not have to be hydrolyzed for recovery of the reacted N-terminal amino acid.
Q. The derivation of the N-terminal amino acid is identified by its fluorescent properties.
R. The reaction can be carried out repeatedly on the same sample of the peptide for sequencing.
S. The reaction can be used in reverse for the synthesis of peptides.

a. P, Q and R
b. P and Q
c. P and R
d. P, Q, R and S

3. A protein treated with fluorodinitrobenzene (FDNB) and subjected to acid hydrolysis yielded two different amino acids with their α-amino groups linked to DNB. A reasonable explanation for this result is that

a. the protein contained more than one free amino group.
b. the protein contained more than one N-terminal amino acid.
c. the protein contained two basic amino acid residues.
d. the protein contained amide groups.

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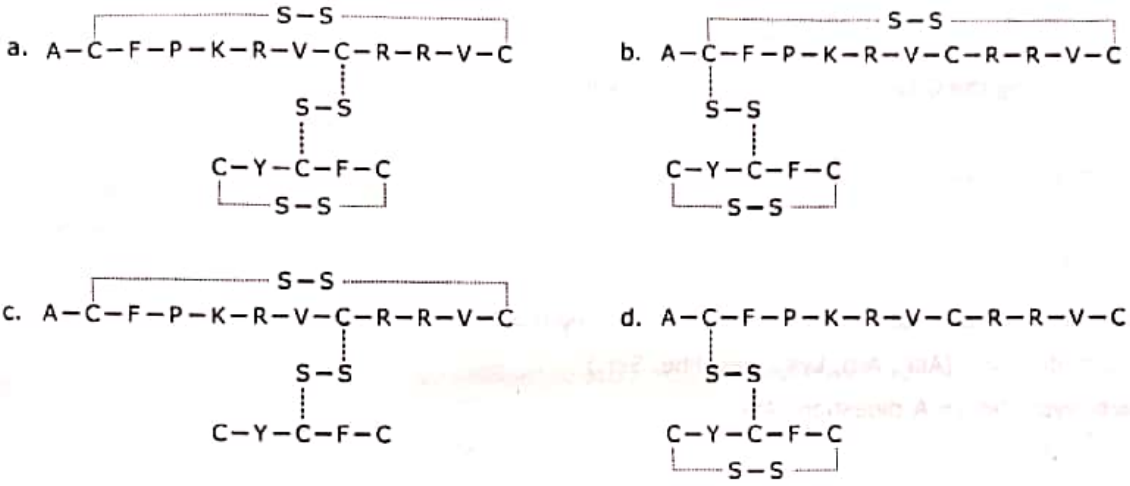
4. N-terminal amino acids are usually determined by Sanger's method using
   a. ninhydrin reagent
   b. Fluoro-2,4, dinitrobenzene/dansyl chloride
   c. hydrazine
   d. concentrated nitric acid **[GATE]**

5. A polypeptide is reduced by β-mercaptoethanol to yield two peptide fragments with the following sequences:
   Fragment 1: A-C-F-P-K-R-V-C-R-R-V-C
   Fragment 2: C-Y-C-F-C
   The non-reduced polypeptide is digested with thermolysine and yields the following fragments:
   (A, C, C, V); (F, P, K, R); (R, R, C, C, V, Y, C, C, F).
   Indicate the positions of disulfide bridges in the polypeptide.



6. Chymotrypsin in the small intestine hydrolyzes peptide linkage containing
   a. phenylalanine
   b. alanine
   c. methionine
   d. valine

7. Histones have very high percentage of arginine and lysine residues (15-30%). For this class of proteins which of the following reagents would be suitable choice for generating peptides in the determination of the amino acid sequence of the protein
   a. cyanogen bromide
   b. thermolysin
   c. trypsin
   d. N-bromosuccinamide **[GATE]**

8. Cleavage of the following peptide with chymotrypsin
   Gly – Arg – Ala – Ser – Phe – Gly – Asn – Lys – Trp – Glu – Val results in
   a. 2 fragments
   b. 3 fragments
   c. 4 fragments
   d. no cleavage **[GATE]**

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9. The peptide, Ala-Arg-Gln-Met-Thr-Trp-Lys-Val, upon treatment with cyanogen bromide produces
   a. Ala - Arg - Gln - Met + Thr - Trp - Lys - Val
   b. Ala - Arg - Gln - Met - Thr - Trp + Lys - Val
   c. Ala - Arg + Gln - Met - Thr - Trp - Lys - Val
   d. Ala - Arg - Gln + Met - Thr - Trp - Lys - Val **[IISc]**

10. A group of scientists have identified the following peptide having anticancerous properties, Leucine - Phenylalanine - Asparate - Glycine - Lysine - Histidine - Glutamate – Alanine. Further work has indicated that its anticancerous activity can be increased by hydrolysing the 'Leucine-Phyenylalanine' dipeptide from the protein. Choose the most appropriate approach to achieve this objective
   a. by trypsin hydrolysis of the protein.
   b. by chymotrypsin hydrolysis of the protein.
   c. by using antibody against the 'Leucine-Phenylalanine' dipeptide.
   d. by chemical hydrolysis of the peptide bond.

11. For determining the C-terminal amino acid of a polypeptide chain, the reagent that would be useful is
   a. trypsin
   b. carboxypeptidase
   c. phenyl isothiocyanate
   d. 1 (N) HCl **[GATE]**

12. What will be the polypeptide sequence that generated the following results:
   1. Acid hydrolysis: (Ala_2, Arg, Lys_2, Met, Phe, Ser_2)
   2. Carboxypeptidase A digestion: Ala
   3. Trypsin digestion: (Ala, Arg), (Lys, Phe, Ser), (Lys), (Ala, Met, Ser)
   4. CNBr treatment: (Ala, Arg, Lys_2, Met, Phe, Ser), (Ala, Ser)
   5. Thermolysine digestion: (Ala), (Ala, Arg, Ser), (Lys_2, Met, Phe, Ser)

   a. Ala-Lys-Ser-Arg-Lys-Phe-Met-Ser-Ala
   b. Ala-Arg-Ser-Phe-Lys-Lys-Met-Ser
   c. Ser-Arg-Ala-Phe-Lys-Ser-Met-Lys-Ala
   d. Ala-Arg-Ser-Phe-Lys-Lys-Met-Ser-Ala

Amino acid metabolism

1. Choose the *incorrect* pair

Compound	*Precursor amino acid(s)*
a. Porphyrin	Serine
b. Creatine	Glycine, Arginine, Methionine
c. Glutathione	Glycine, Glutamate, Cysteine
d. Spermine	Methionine, Ornithine

2. What is *not* true about histamine?
   a. It is produced on decarboxylation of histidine.
   b. It causes allergy.
   c. It lowers blood pressure.
   d. Diphenyl hydramine hydrochloride activates histamine.

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3. Epinephrine is formed from norepinephrine by
   a. decarboxylation
   b. oxidative deamination
   c. O-methylation
   d. N-methylation

4. Which of the following amino acids is the precursor for the thyroid hormones T_3 and T_4?
   a. tryptophan
   b. lysine
   c. tyrosine
   d. histidine

5. Which of the following reactions in the biosynthesis of porphyrins is the rate-limiting step?
   a. δ-aminolevulinate synthetase
   b. porphyrinogen synthetase
   c. uroporphyrinogen synthetase
   d. decarboxylation of uroporphyrinogen III

6. All of the following statements regarding the degradation of porphyrins are correct *except*
   a. the formation of biliverdin is the first step in the degradation of heme.
   b. reduction of biliverdin yields bilirubin.
   c. bilirubin is conjugated to glucuronic acid in the liver.
   d. free (unconjugated) bilirubin may be reduced to urobilinogen in the liver.

7. Creatine is formed from which of the following amino acids?
   a. Glycine, lysine and methionine.
   b. Glycine, arginine and S-adenosylmethionine.
   c. Glycine, ornithine and S-adenosylmethionine.
   d. Glycine and S-adenosylmethionine.

8. Amino acid decarboxylation is *not* involved in the synthesis of
   a. serotonin from tryptophan
   b. histamine from histidine
   c. GABA from glutamate
   d. glucuronic acid

9. Which of the following is *false* about porphyrin?
   a. It contains 4 linked pyrrole rings.
   b. All porphyrins are derived from common precursor, δ-aminolevulinic acid.
   c. It is synthesized through succinate-glycine pathway.
   d. All nitrogens of porphyrin in heme is derived from valine.

10. The rate-limiting step in the biosynthetic pathway for catecholamines is
    a. the hydroxylation of tyrosine
    b. the hydroxylation of phenylalanine to tyrosine
    c. the formation of dopamine
    d. the reduction of biopterin

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11. The two important biochemical reactions of nitrogen metabolism are shown below:

$$NO_2^- + 8H^+ + 6e^- \xrightarrow{\text{Enzyme}} NH_4^+ + 2H_2O$$

$$\text{Glutamate} + NH_3 + ATP + Mg^{2+} \xrightarrow{\text{Enzyme}} \text{Glutamine} + ADP + P_i$$

Which one of the following pairs of enzymes is *correct* for the above reactions respectively?

a. Nitrite reductase and glutamate dehydrogenase.
b. Nitrate reductase and glutamine synthetase.
c. Nitrite reductase and glutamine synthetase.
d. Nitrite reductase and glutamate synthase. [GATE]

12. Which of the following statement is *false*?

a. Nitrogen fixation by the nitrogenase complex requires eight electrons.
b. Conversion of nitrogen to ammonia ($N_2 + 3H_2 \rightarrow 2NH_3$) is an endergonic process.
c. Certain free living aerobic bacteria are also capable of nitrogen fixation.
d. The nitrogen fixing nitrogenase complex is oxygen-labile.

13. Nitrogenase reductase of nitrogenase complex

P. contains [4Fe–4S] complex
Q. is $\alpha_2\beta_2$ tetramer
R. is a dimer of identical subunits
S. contains MoFe-cofactor

a. P and R
b. Q and S
c. P and Q
d. R and S

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Chapter 02

Nucleic acids

Nucleosides and Nucleotides

1. On hydrolysis a nucleoside would not yield
   a. purine
   b. pyrimidine
   c. pentose sugar
   d. phosphoric acid

2. The chemical name of thymine is
   a. 2-oxy-4-aminopyrimidine
   b. 2,4-dioxy-5-methylpyrimidine
   c. 2,4-dioxypyrimidine
   d. none of the above **[JNU]**

3. The most abundant intracellular free nucleotide is
   a. ATP
   b. FAD^+
   c. NAD^+
   d. $NADP^+$ **[JNU]**

4. N-Glycosyl linkage joins 1^{st} carbon of pentose sugar with
   a. N-9 of pyrimidine
   b. N-9 of purine
   c. N-3 of pyrimidine
   d. N-3 of purine

5. High energy bond present in ATP is
   P. phosphoester bond
   Q. ether bond
   R. phosphoanhydride bond
   S. phosphodiester bond
   a. only P
   b. only R
   c. both P and Q
   d. both P and S

6. Deoxyribose in a nucleotide is a
   a. primary alcohol
   b. secondary alcohol
   c. tertiary alcohol
   d. phenol **[IISc]**

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7. Which statement about nucleotides is *correct*? Nucleotides
   a. such as ATP are used in the in vivo synthesis of nucleic acids.
   b. are composed of only a pentose sugar and a nitrogenous purine or pyrimidine base.
   c. such as GTP replace ATP in the synthesis of RNA molecules.
   d. contain a deoxyribose sugar if they are components of RNA, and a ribose sugar if they are components of DNA.

8. Correct statements regarding nucleotide structure include which of the following?
   a. Removal of the 5'-phosphate group from guanylic acid (GMP) yields guanine.
   b. Removal of the 5'-phosphate group from adenylic acid (AMP) yields adenosine.
   c. Removal of the ribose 5-phosphate from thymidylic acid (TMP) yields uracil.
   d. Removal of the ribose 5-phosphate from cytidylic acid (CMP) yields cytidine.

9. The purine nucleotides act as the components of
   a. FAD^+
   b. NAD^+
   c. $NADP^+$
   d. all of the above

[JNU]

Nucleic acids

1. A radioisotope used to label proteins differentially from nucleic acids is
   a. ^{32}P
   b. ^{14}C
   c. ^{35}S
   d. ^{15}N

2. Sugar–phosphate backbone in nucleic acid
   P. hydrophobic in nature
   Q. hydrophilic in nature
   R. negatively charged
   S. show polarity
   a. P and Q
   b. Q and S
   c. Q, R and S
   d. P, R and S

3. The information in a genetic nucleic acid resides in the
   a. number of nucleotides
   b. kinds of nucleotides
   c. sequence of nucleotides
   d. all of the above

4. One of the strands of a double-stranded, 10 kbp DNA duplex has the following numbers of base residues: adenine(a)–3800, thymine(t)–2600. The base composition of the whole double-stranded molecule will be
   a. a–5200, t–5200, g–4800, c–4800
   b. a–7600, t–7600, g–2400, c–2400V
   c. a–6400, t–3600, g–6400, c–3600
   d. a–6400, t–6400, g–3600, c–3600

5. What is the number of hydrogen bonds in a B–DNA of 1000 base–pairs with base adenine occurring 75% of the times in one strand and 10% in the other strand? [JNU]
   a. 2150
   b. 2820
   c. 2250
   d. 2750

[IISc]

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6. Which of the following statement is *true*?
   a. The ratio of G to A in double-stranded DNA is 1:1.
   b. The ratio of T in double-stranded DNA to U in single-stranded RNA is 1:2.
   c. The ratio of G to C in single-stranded RNA is 1:1.
   d. The ratio of G to C in double-stranded DNA is 1:1.

7. For a double - stranded DNA molecule, which of the following statements is *wrong*?
   a. A/T ratio is constant
   b. G is always equal to C
   c. A+T is always equal to G+C
   d. (A+G)/(C+T) ratio is constant

8. Chargaff's rules state that
   a. in RNA, A=U, and in DNA, A=T
   b. G=C in both RNA and DNA
   c. (A+T)/(G+C) is always 1
   d. A+G/T+C= 1

9. A double stranded DNA has 30% Thymine. The percentage of cytosine is
   a. 30%
   b. 20%
   c. 70%
   d. 15% **[GATE]**

10. If RNA in TMV contains 28% cytosine then what percentage of the bases in TMV is adenine?
   a. 30%
   b. 20%
   c. 60%
   d. none

11. In a given nucleic acid G + A is not equal to C + T content. This indicates that the sample is
   a. AT rich
   b. GC rich
   c. ssDNA
   d. dsDNA

12. The G+C content of bacteriophage 13 double stranded DNA is 68%. What would you expect the G+C content of its mRNA?
   a. about 68%
   b. about 34%
   c. about 32%
   d. about 86% **[GATE]**

13. Which is the correct statement of the following pertaining to the mass of bases present in a double stranded DNA with 50% GC content?
   a. A = T
   b. C > G
   c. A > T
   d. T > A **[GATE]**

14. If a sample of DNA is found to have the base composition (mole ratios) of adenine, 40; thymine, 22; guanine, 19; and cytosine, 19, what conclusion can be drawn?
   a. The DNA is a circular duplex.
   b. The DNA is a linear duplex.
   c. The DNA is single-stranded.
   d. The DNA has highly repetitive sequences.

15. A nucleic acid sample has 20% A residues and 20% G. The proportion of T residues is
   a. 20%
   b. 30%
   c. 50%
   d. Information incomplete **[JNU]**

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16. If the DNA of a species has the mole fraction of G + C = 0.36, calculate the mole fraction of A.
    a. 0.64
    b. 1.28
    c. 0.36
    d. 0.32

17. If the ratio (A+G)/(T+C) in one strand of DNA is 0.7, what is the same ratio in the complementary strand?
    a. 0.7
    b. 1.43
    c. 0.35
    d. none of the above

18. What is the approximate molecular weight of duplex DNA required to code for glyceraldehyde phosphate dehydrogenase (MW 40,000)?
    a. About 70,0000
    b. About 94,000
    c. About 45,000
    d. About 21,300

19. DNA base stacking is diagnosed by
    a. hyperchromacity
    b. pH change
    c. electrophoretic mobility
    d. none of the above

20. The hyperchromic effect refers to
    a. a change in the optical rotatory dispersion (ORD) of a DNA solution upon heating.
    b. a maximum rate of denaturation versus temperature for duplex DNA.
    c. an increase in the absorbance of light at 260 nm upon denaturation of DNA.
    d. an increase in the absorbance of light at 260 nm when DNA-RNA hybrids are annealed.

21. Human DNA contains 31% thymine. 5% of cytosines are modified to 5-methylcytosine after replication. What percent of human DNA is 5-methylcytosine?
    a. 0.475%
    b. 0.95%
    c. 1.55%
    d. 1.9%

22. The melting temperature (T_m) of DNA is
    a. directly proportional to the length of the DNA chain
    b. directly proportional to the GC content
    c. directly proportional to the AT content
    d. not related to the base composition

23. Melting of DNA results in
    a. decrease in optical density
    b. increase in optical density
    c. no change in optical density
    d. none of the above

24. One strand of double-stranded DNA is mutated, changing all cytosines to uracils. After one round of replication of the mutated DNA strand, the melting temperature of the resulting DNA will
    a. be higher
    b. be lower
    c. remain the same
    d. insufficient data

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25. Which of the following reagents and conditions will denature dsDNA?

P. Heat

Q. Urea

R. Extreme pH

S. Ethanol

a. P and Q
b. P, Q and R
c. P and R
d. P, Q, R and S

26. The melting temperature, T_m, of a DNA duplex is defined as the temperature at which half the molecules have dissociated into single strands. T_m will be maximal at

a. low ionic strength and high DNA concentration.
b. high ionic strength and high DNA concentration.
c. high ionic strength and low DNA concentration.
d. low ionic strength and low DNA concentration. **[IISc]**

27. Which of the following factors favour strand separation in double stranded DNA melting?

a. Hydrogen bonding between bases.
b. Repulsion between phosphate groups.
c. van der Waals interaction between bases.
d. High content of G + C. **[IISc]**

28. If a solution of double stranded DNA is heated above its melting temperature, its absorbance will

a. decrease
b. increase
c. remain unchanged
d. initially increase and then decrease **[GATE]**

29. The T_m of a DNA molecule, when measured in 0.15 M NaCl was 60°C. What will be its T_m, compared to 60°C, if measured in the presence of 0.15 M NaCl and 0.5 M Urea?

a. lower
b. higher
c. no change
d. zero **[IISc]**

30. The base sequences of three deoxyoligonucleotides, P, Q and R, that adopt double helical B-DNA conformation are given below;

P – AAGCG

Q – CCTGA

R – GACTA

Identify the most accurate statement (in principle) from the following. Thermodynamically,

a. P is the most stable
b. Q is the most stable
c. R is the most stable
d. P and Q are equally stable **[IISc]**

31. GC-rich DNA has higher stability than AT-rich DNA because

a. fewer Na^+ ions bind to AT bp
b. GC base pairs have three H-bonds
c. GC base pairs wind the double helix more tightly
d. stacking interactions are stronger in GC bp

32. The double-stranded DNA molecule loses its viscosity upon

a. denaturation
b. filtration
c. sedimentation
d. concentration **[JNU]**

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DNA

1. Choose the *incorrect* statement
   a. DNA acts as genetic material
   b. RNA acts as genetic material
   c. RNA acts as genetic material when DNA is not present
   d. RNA may act as genetic material even when DNA is present

2. Which of the following findings first gave a strong indication that DNA was the genetic material?
   a. RNA synthesis depends on a DNA template.
   b. DNA is in the nuclei of all eukaryotic cells.
   c. viral nucleic acid can be DNA.
   d. transforming activity is due to DNA.

3. DNA is the genetic material can be evidenced by the fact that
   a. chromosomes are made up of DNA
   b. DNA is not present in cytoplasm
   c. transformation and transduction in bacteria are caused by DNA only
   d. DNA is concentrated in nucleus [GATE]

4. DNA
   a. is more susceptible than RNA to degradation at high pH.
   b. has catalytic activity.
   c. can hybridize with other DNA molecules but not with RNA.
   d. has fewer hydroxyl group than RNA.

5. Which of the following statements is *true* of duplex DNA?
   a. The planes of the bases lie parallel to the helix axis.
   b. The chains have a backbone of linked glycosides.
   c. Unless the DNA is circular, the 3'-OH groups of each chain are at opposite ends of the molecule.
   d. The duplex structure is stabilized only by hydrogen bonding between bases.

6. Helical structure of DNA is determined by
   a. electron diffraction measurement
   b. neutron diffraction measurement
   c. X-ray diffraction measurement
   d. diffraction of visible light [JNU]

7. A non-supercoiled B-DNA molecule is composed of 4,800 base pairs. How many helical turns are present?
   a. 10
   b. 380
   c. 480
   d. impossible to determine

8. Which of the following statements about the structure of B-DNA is *incorrect*?
   a. Within the double helix, there are 10 bases per turn of the helix.
   b. Separation of the two strands of the double helix requires untwisting of the helix.
   c. Molar amount of adenine plus thymine is equal to the molar amount of guanine plus cytosine.
   d. The double helix contains antiparallel chains. [GATE]

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9. The major groove of DNA is lined by
   a. N3 of purine and N1 of pyrimidine
   b. N7 of purine and O2 of pyrimidine
   c. N7 of purine and C4 of pyrimidine
   d. none of the above

[GATE]

10. For which of the following property does the A-form double helix DNA not differ from that of B-form double helix?
   a. appearance of the major and minor grooves
   b. thickness of the helix
   c. tilt of the bases
   d. polarity of the two complementary strands

[GATE]

11. Feulgen staining of DNA is based on the reaction of Schiff's reagent with
   a. free aldehyde group of pentose sugar
   b. purine at N-7 position
   c. deoxyribose at C-2 position
   d. pyrimidine at N-1 position

12. The Z-DNA helix
   a. has fewer base pairs per turn than the B-DNA.
   b. is favoured by an alternating GC sequence.
   c. tends to be found at the 3' end of genes.
   d. is the most common conformation of DNA.

[GATE]

13. Which of the following favours Z-DNA conformation?
   a. 2'-OH steric hindrance
   b. anti-G and C2'-endo sugar
   c. syn-G and C3'-endo sugar
   d. G4-DNA quartet

14. Z-DNA is a third type of DNA helix with structure
   a. CGATGC
   b. ATGCCG
   c. GCTAAT
   d. CGCGCG

[GATE]

15. Which of the following is *not* true about DNA?
   a. A-DNA is usually in a left handed helix structure.
   b. B-DNA is the most commonly found structure *in vivo*.
   c. Z-DNA is found in purine-pyrimidine repeat sequences.
   d. RNA-DNA hybrids are in A-DNA shape.

16. A double-stranded DNA plasmid harbors a sequence which has a propensity to undergo a B form to Z form transition. Which one of the following conditions will facilitate this transition?
   a. Intercalation of ethidium bromide into a covalently closed circular plasmid.
   b. Introduction of negative supercoiling in the plasmid by topoisomerases.
   c. Introduction of positive supercoils in the plasmid by reverse gyrase.
   d. Introduction of a nick in the plasmid.

[IISc]

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17. The stability of DNA to alkaline hydrolysis relative to RNA, is due to
    a. the greater stability of ribose over deoxyribose
    b. DNA having β-glycosidic bonds and RNA having α-glycosidic bonds
    c. DNA being more insoluble in water than RNA
    d. the absence of a 2'-hydroxyl to assist in the hydrolysis [JNU]

18. Which of the following statements are *correct*?
    P. In Hoogsteen base pairing N7 position of adenine pairs with N3 position of thymine.
    Q. RNA structures are always C3'-endo.
    R. Intrastrand H-bonds responsible for 2- and 3-D structure of tRNA.
    S. Hydrophobic base stacking interaction is responsible for stability of dsDNA.
    a. P and Q
    b. P, Q and R
    c. P and R
    d. P, Q, R and S

19. Find out the *wrong* statement.
    a. Super helical density is the ratio of change in linking number and linking number in relaxed state.
    b. Circular DNA from bacteria are usually negatively supercoiled having value of superhelical density about 0.06.
    c. Linking number of right handed dsDNA is considered as negative.
    d. For relaxed DNA, linking number and twisting number value remain same.

20. If the DNA content of a single human sperm is $\sim 10^{-12}$ g, then the total DNA content of a mature human red blood cell would be
    a. 10^{-12} g
    b. 10^{-6} g
    c. 2×10^{-12} g
    d. none of the above

21. What is the number of hydrogen bonds in a B-DNA of 1000 base pairs with the proportion of bases C, A, and T in one of the strands corresponding to 60%, 30% and 10% respectively?
    a. 2600
    b. 2400
    c. 2200
    d. 2000 [IISc]

22. There are two single stranded DNA of 10 nucleotides each. What is the probability that they form a double stranded DNA with all the 10 base pairs in Watson-Crick pairing?
    a. $(1/4)^{10}$
    b. $(1/10)^4$
    c. $(1/4)^5$
    d. $(1/5)^4$ [IISc]

RNA

1. An aqueous solution of a phosphorous containing biopolymer shows UV absorption spectra. Optical density at 260 nm increases much when the solution is made alkaline and heated but not much if the solution is simply heated. The biopolymer can be
    a. deoxyribonucleic acid
    b. ribonucleic acid
    c. phospholipid
    d. phosphoprotein

2. RNA is very much susceptible to hydrolysis in alkali because
    a. it contains uracil residues in its structure.
    b. its 2'-OH group participate in intramolecular cleavage of phosphodiester backbone.
    c. cleavage occurs in the glycosylic bonds of purine bases.
    d. cleavage occurs in the glycosylic bonds of pyrimidine bases. [GATE]

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3. Which one of the following choice best completes the following sentence? tRNA
   a. contains the information necessary for the synthesis of specific protein.
   b. must exist in 20 different forms, one for each amino acid.
   c. is the largest of the RNA species.
   d. has little or no secondary structure.

4. Which of the following is *false* statement?
   a. tRNA binds an amino acid covalently.
   b. tRNA has many modified bases.
   c. tRNA is present only in eukaryotic cells.
   d. tRNA carries anticodon.

5. Which of the following most correctly describes mammalian mRNAs?
   a. They are usually transcribed from both DNA strands.
   b. They are normally double-stranded.
   c. They have an overall negative charge at neutral pH.
   d. Their ratio of ribose to purine bases equals 1.

6. Which of the following is *not* a characteristic of t-RNA?
   a. It contains a codon
   b. It contains an anticodon
   c. It can become attached covalently to an amino acid
   d. It interacts with mRNA during transcription **[GATE]**

7. The following sequence best arranges tRNA, rRNA and DNA in order of increasing molecular weights
   a. tRNA, DNA, rRNA
   b. tRNA, rRNA, DNA
   c. rRNA, tRNA, DNA
   d. rRNA, DNA, tRNA **[GATE]**

8. The molecular weight of an mRNA that codes for a protein of molecular weight 75,000 is closest to
   a. 6000
   b. 60000
   c. 600000
   d. 600 **[IISc]**

9. Identify the *wrong* statement amongst the following.
   a. Nucleotide bases in tRNA are extensively modified after transcription.
   b. One particular amino acid may have multiple tRNA species acting as its adaptors.
   c. One particular tRNA is always specific for one particular amino acid.
   d. The size of a tRNA generally varies from 600 bases to 1000 bases. **[JNU]**

10. RNA can be specifically radiolabeled by feeding the cell with radioactive
    a. adenosine
    b. guanosine
    c. thymine
    d. uridine **[JNU]**

11. Tobacco mosaic virus has RNA rather than DNA as its genetic material. In a hypothetical situation where RNA from a tobacco mosaic virus is mixed with proteins from a related DNA virus, the result could be a hybrid virus. If that virus were to infect a cell and reproduce, what would the resulting "offspring" viruses be like?
    a. Tobacco mosaic virus.
    b. The related DNA virus.
    c. A hybrid: tobacco mosaic virus RNA and protein from the DNA virus.
    d. A hybrid: tobacco mosaic virus protein and nucleic acid from the DNA virus. **[MKU]**

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Electrophoresis and Sequencing

1. In agarose gel electrophoresis
   a. DNA migrates towards the negative electrode.
   b. supercoiled plasmids migrate slower than their nicked counterparts.
   c. larger molecules migrate faster than smaller molecules.
   d. ethidium bromide can be used to visualize the DNA.

2. The enzymatic method of DNA sequencing
   a. uses RNA as template.
   b. uses dideoxyribonucleoside triphosphates, in which the deoxyribose 3'-OH is missing.
   c. uses dideoxyribonucleoside triphosphates, in which the deoxyribose 3'-OH is present.
   d. uses different chemical treatment to cleave DNA preferentially at A, T, C or G. [JNU]

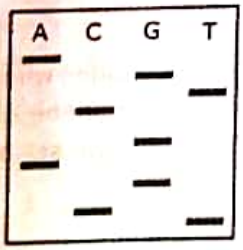
3. DNA sequencing by Sanger's method involves the use of
   a. ribonucleotide
   b. 3'-deoxyribonucleotide
   c. 2',3'-dideoxyribonucleotide
   d. fluorodinitrobenzene [GATE]

4. A DNA sequencing reaction was performed with the fragment 5'-XXXXGCGATCGYYYY-3' as the template, dideoxy GTP, all the four dNTPs, and the required primers and enzyme. XXXX and YYYY in the given DNA fragment represent primer binding sites. The set of fragments obtained during the reaction will be (the primers are not shown in the amplified fragments)
   a. 5'-CGATCGC-3' only
   b. 5'-CG-3', 5'-CGCTAG-3', 5'-CGCTAGC-3'
   c. 5'-CG-3', 5'-CGATCG-3', 5'-CGATCGC-3'
   d. 5'-G-3', 5'-GCG-3', 5'-GCGATCG-3'

5. In Sanger's method of DNA sequencing, the growing DNA chains are terminated because
   a. DNA polymerase is not very processive
   b. a radioactive nucleotide is incorporated
   c. the substrates become limiting
   d. a phosphodiester bond cannot be made [JNU]

6. If the sequence reading from the bottom to the top of a dideoxy sequencing gel is TAACGT, the sequence of the template is
   a. 5'-TGCAAT-3'
   b. 3'-TGCAAT-5'
   c. 5'-ACGTTA-3'
   d. 3'-ACGTTA-5' [JNU]

7. Shown below is the autoradiogram of an electrophoresis gel obtained during the sequencing of a single stranded DNA by Sanger's method. The base sequence of the DNA is



   a. 3'-AGTCGAGCT-5'
   b. 5'-TCAGCTCGA-3'
   c. 3'-TCAGCTCGA-5'
   d. 5'-AGTCGAGCT-3' [GATE]

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Nucleotide metabolism

1. How many carbons of the purine ring are contributed by the folate one-carbon pool during purine biosynthesis?
   a. 0
   b. 1
   c. 2
   d. 3

2. The atoms of the pyrimidine ring are derived from
   a. glutamine
   b. carbamoyl phosphate and aspartate
   c. glutamine and N^{10} formyl tetrahydrofolate
   d. aspartate and N^{10} formyl tetrahydrofolate

3. The committed step in the de novo synthesis of purine nucleotides is the formation of
   a. ribose 5-phosphate from ribose 1-phosphate
   b. 5'-phosphoribosyl-1-pyrophosphate (PRPP) from ribose 5-phosphate
   c. 5'-phosphoribosylamine from PRPP
   d. 5'-phosphoribosylglycinamide from 5'-phosphoribosylamine

4. Deoxyribonucleotides are formed by reduction of
   a. ribonucleosides
   b. ribonucleoside monophosphates
   c. ribonucleoside diphosphates
   d. ribonucleoside triphosphates

5. The end product of purine catabolism in normal humans is
   a. urea
   b. uric acid
   c. creatinine
   d. xanthine

6. The cellular concentration of 5'-phosphoribosyl-1-pyrophosphate (PRPP) is controlled by modulators of PRPP synthetase activity, including
   P. inorganic phosphate
   Q. nucleoside diphosphates
   R. 2,3-diphosphoglycerate
   S. nucleoside triphosphates
   a. P and Q
   b. P, Q and R
   c. P and R
   d. P, Q, R and S

7. Mammalian kinases are able to convert which of the following nucleosides to nucleotides?
   P. Adenosine
   Q. Inosine
   R. Cytidine
   S. Guanine
   a. P and Q
   b. P, Q and R
   c. P and R
   d. P, Q, R and S

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8. Which of the following statement is *not* true about Lesch–Nyhan syndrome?
   a. Occurs due to lack of enzyme HGPRTase.
   b. Occurs due to accumulation of urate.
   c. Occurs due to lack of enzyme sphingomylenase.
   d. Excessive amount of purine production occurs.

9. Which of the following contributes nitrogen atoms to both purine and pyrimidine rings?
   a. Aspartate
   b. Carbamoyl phosphate
   c. Carbon dioxide
   d. Glutamine

10. Thymidylate synthetase is an enzyme involved in the synthesis of TMP from
   a. dUMP
   b. UMP
   c. thymidine
   d. TDP

[IISc]

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Chapter 03

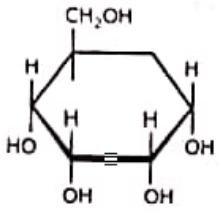
Carbohydrates and Lipids

Carbohydrates

Monosaccharides and Disaccharides

1. Which of the following statement best characterizes glucose?
   a. It usually exists in the furanose form.
   b. It is a ketose.
   c. Carbon 2 is the anomeric carbon atom.
   d. It forms part of the disaccharide sucrose.

2. Which three of the following molecules will rotate plane-polarized light?

P. 

Q. $H_3N^+-CH_2-COO^-$

R. $CH_2OH-CHOH-CHOH-CHO$

S. $CH_2OH-CHOH-CHO$

a. P, Q and S
b. P, Q and R
c. P, R and S
d. Q, R and S

[IISc]

3. Which of the following statements are *correct*?
   P. Anomers differ in configuration at the glycosidic carbon.
   Q. Epimers differ in configuration at any carbon other than glycosidic carbon.
   R. In aldose, C1 is the anomeric carbon.
   S. Epimers differ in chemical properties.
   a. P and Q
   b. R and S
   c. P, Q and R
   d. P, Q, R and S

4. Glucose and mannose are epimers. This means that
   a. they are mirror images of each other
   b. one is an aldose, the other a ketose

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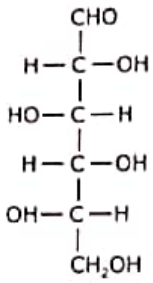
c. they rotate plane polarized light in opposite directions
d. they differ only in the configuration of one carbon atom [JNU]

5. Glucose and galactose are two isomeric monosaccharide known as
a. anomers
b. epimers
c. enantiomers
d. conformers [IISc]

6. Molecular structure of glucose is given below. Identify the *incorrect* statement.



a. It is L-form of glucose
b. It is an aldose
c. It contains four chiral centers
d. It gives 20 different stereoisomer

7. Choose the mismatch
a. D-glucose and D-fructose : anomer
b. D-glucose and D-mannose : epimer
c. α-D-glucose and β-D-glucose : anomer
d. D-glucose and L-glucose : enantiomer

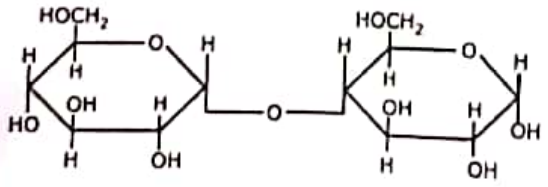
8. Glucose gives the silver mirror test with ammonical $AgNO_3$ because it contains functional group
a. ketonic
b. amide
c. ester
d. aldehyde

9. What is *not* true about monosaccharides? They are
a. colourless
b. soluble in water
c. sweet in taste
d. soluble in non-polar solvent

10. Cyclic form of pentose or hexose sugars are formed as a result of covalent bond between carbonyl group and hydroxyl group. It may be
a. hemiacetal
b. hemiketal
c. furanose or pyranose
d. all of the above

11. Ring structure of glucose is due to formation of hemiacetal and ring formation between
a. C_1 and C_5
b. C_1 and C_4
c. C_1 and C_3
d. C_2 and C_4

12. The molecular structure of a disaccharide is given below. Identify the compound.



a. Lactose
b. Maltose
c. Sucrose
d. Cellobiose [GATE]

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13. Which of the following statements about disaccharides are *correct*?

P. Trehalose and sucrose are non-reducing sugar.

Q. Cellobiose contains β (1 → 4) glycosidic bond.

R. Lactose is galactose β (1 → 4) glucose.

S. Sucrose is dextrorotatory.

a. P and Q
b. Q and R
c. P and S
d. P, Q, R and S

14. Which of the following is *correct* statement?

a. All monosaccharides in their hemiacetal and hemiketal form are reducing sugar.
b. Sucrose is a reducing sugar.
c. All disaccharides formed from head to tail condensation are non-reducing sugar.
d. Reducing sugar donot exhibit mutarotation.

15. Which of the following statement about sucrose is *false*?

a. No free aldehyde or ketone group in sucrose.
b. Contain an α-1,2 glycosidic bond.
c. Oxygen bridge between glucose and fructose is between the anomeric carbons.
d. Sucrose can reduce Fehling's solution.

16. Choose the mismatch

a. amylose — contains α-1,4 glycosidic bond.
b. D-glucose-1-phosphate — phosphoric acid ester of glucose.
c. glycosides — forms when sugar reacts with acid.
d. acetal — forms when hemiacetal reacts with alcohol.

17. Which of the following statement is *correct*?

P. Change in the optical rotation of sucrose upon hydrolysis is known as inversion.

Q. Fructose is a ketose sugar having a furan ring.

R. A monosaccharide with 'n' chiral center will form 2^n number of stereoisomer.

S. The most abundant organic molecule synthesized by living system is cellulose.

a. P and Q
b. P, Q and R
c. Q, R and S
d. P, Q, R and S

18. The enzyme invertase (sucrose) is allowed to act on a solution of sucrose until the optical rotation of the solution becomes zero. What fraction of the sucrose has been hydrolyzed?

a. 0.84
b. 0.63
c. 0
d. 1

19. Choose the *correct* statements.

P. All monosaccharides are reducing sugars.

Q. All disaccharides having head to tail condensation are reducing sugar.

R. Ascorbic acid is an aldonic acid.

S. Sucrose is dextrorotatory.

a. P and Q
b. Q and R
c. P, Q and R
d. P, Q, R and S

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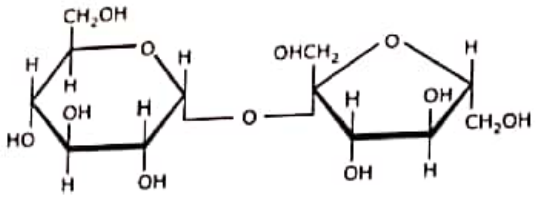
20. The sugar whose structure is shown below



a. is a substrate for β-galactosidase.
b. contains a β 1,2 linkage.
c. contains two hexose sugars.
d. contains one hexose and one pentose sugar. [IISc]

Polysaccharides

1. O-glycosidic bond in a polysaccharide forms between
   a. anomeric carbon and alkoxy oxygen
   b. anomeric oxygen and alkoxy carbon
   c. anomeric hydrogen and alkoxy carbon
   d. anomeric hydrogen and alkoxy oxygen

2. The sugar residues of amylose are
   a. in α-1,4 linkages
   b. in β-1,4 linkages
   c. galactose units only
   d. fructose units only

3. Which of the following statement is *incorrect*?
   a. Amylose is a water soluble fraction of starch.
   b. Amylopectin is a water insoluble fraction of starch.
   c. Amylopectin fraction has both α-1,4 and α-1,6 glycosidic linkage.
   d. None of the above.

4. Choose the *incorrect* statement(s).
   P. Cellulose is a linear homopolymer of D-glucose units.
   Q. Chitin is a polymer of N-acetyl-D-glucosamines with β (1 → 4) glycosidic bond.
   R. Agarose is a chain of alternating D-galactose and 3,6-anhydro-galactose.
   S. Dextran is a β (1 → 6) linked glucan.
   a. P and S
   b. only R
   c. P and Q
   d. only S

5. Which of the following statements are *true*?
   P. Salivary α-amylase is an endoamylase.
   Q. β-amylase is an exoamylase.
   R. α-amylase and β-amylase cleave α (1 → 4) glycosidic bond.
   S. β-amylase is the most prevalent form of amylase in animals.
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

6. Cellulose is indigestible by humans because we lack the enzyme that hydrolyzes
   a. α-1,4 glycosidic bonds
   b. α-1,6 glycosidic bonds
   c. β-1,4 glycosidic bonds
   d. long-chain polysaccharides

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7. Choose the *incorrect* statement(s).
   1. Aldose oxidizes to give aldonic acid.
   2. Reduction of glucose gives sorbitol.
   3. Glycogen is highly branched form of amylopectin.
   4. Glycogen is a major structural polysaccharide.
   5. Most abundant monosaccharide in nature is D-glucose.
   6. All monosaccharides are optically active except dihydroxy acetone.

   a. only 1
   b. both 1 and 2
   c. only 4
   d. 3, 4 and 5

8. Glycogen
   P. is highly branched form of amylopectin
   Q. contains α (1 $\rightarrow$ 4) and α (1 $\rightarrow$ 6) glycosidic bond
   R. is linear unbranched homopolysaccharide of D-glucose
   S. skeletal muscle contains greatest amount of body glycogen in human

   a. P and Q
   b. R and S
   c. P, Q and S
   d. Q and S

9. *Correct* statements regarding the polysaccharides include which of the following?
   P. Amylose is a linear unbranched polymer of D-glucose with α-1, 4-glycosidic linkages.
   Q. Amylopectin is a branched polymer of D-glucose with α-1,4-glycosidic linkages and with α-1,6 branching points every 25 to 30 D-glucose residues.
   R. Cellulose is composed of linear chains of D-glucose units joined by α-1,4-glycosidic linkages.
   S. Dextran is a polymer of D-glucose present in yeast and bacteria.

   a. P and Q
   b. Q and R
   c. R and S
   d. P, Q and S

10. Which of the following is *not* true about glycosaminoglycan?
    a. Negatively charged heteropolysaccharide.
    b. Make complex with protein and forms proteoglycan.
    c. Composed of repeating disaccharide unit.
    d. Keratan sulfate contains acidic sugar.

11. Which of the following is/are *true* about glycosaminoglycans (GSGs)?
    P. GSGs are linear polymer of repeating disaccharides.
    Q. GSGs have negative charge.
    R. Heparin, a natural anticoagulant, is a GSG.
    S. Hyaluronic acid is a proteoglycan.

    a. only P
    b. only S
    c. P and S
    d. P, Q and R

12. Peptidoglycan
    P. is heteropolymer of NAG and NAM linked through β (1 $\rightarrow$ 4) glycosidic bond.
    Q. is covalently linked with tetrapeptide : L-Ala — D-Glu — L-Lys — D-Ala.
    R. is present in bacterial cell wall.
    S. is sensitive to lysozyme which cleaves bond between NAM and NAG.

    a. only P
    b. only S
    c. P and S
    d. P, Q, R and S

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Glycogenesis, Glycogenolysis and Gluconeogenesis

1. Which of the following are common features of both glycogen synthesis and glycogen breakdown?
   a. Both require UDP-glucose.
   b. Both involve glucose 1-phosphate.
   c. Both are driven in part by the hydrolysis of pyrophosphate.
   d. Both use the same enzyme for branching and debranching.

2. Free glucose is formed during glycogenolysis from
   a. glucose residues in 1,4 linkage to the main chain.
   b. glucose residues in 1,6 linkage to the main chain.
   c. glucose 1-phosphate hydrolysis.
   d. glucose 1,6-diphosphate hydrolysis.

3. Glycogen synthase is characterized by all of the following statements *except*
   a. the enzyme exists in active and inactive forms.
   b. uridine diphosphate glucose is a substrate.
   c. it is activated by phosphorylation.
   d. it requires a primer strand of glycogen.

4. Starting from a glucose residue in glycogen, how many net ATP molecules will be formed in the glycolysis of the residues to pyruvate?
   a. One
   b. Two
   c. Three
   d. Four

5. In terms of energy yield, phosphorolysis is preferable to hydrolysis in the breakdown of glycogen or starch because
   a. most phosphorylases have lower K_m values than do the corresponding phosphatases.
   b. glucose 1-phosphate yield more ATP than does free glucose when subsequently catabolized to pyruvate.
   c. the products of hydrolysis cannot be metabolized by the glycolytic pathway.
   d. the abundance of inorganic phosphate in the cell ensures that the reaction will function in the degradative and not the synthetic direction.

6. The functions of gluconeogenesis are described by all of the following statements *except*
   a. it maintains blood sugar levels during fasting.
   b. it is useful during strenuous exercise.
   c. it plays a role in countering metabolic acidosis.
   d. it allows the use of acetyl coenzyme A for glucose production.

7. Which of the following compounds is a positive allosteric regulator of the enzyme pyruvate carboxylase?
   a. Adenosine triphosphate
   b. Acetyl coenzyme A
   c. Biotin
   d. Phosphoenolpyruvate

8. Which of the following prosthetic group is required by pyruvate carboxylase, an enzyme involved in gluconeogenetic pathway?
   a. Pyridoxal phosphate
   b. Biotin
   c. Tetrahydrofolate
   d. TPP

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9. Choose the *incorrect* statement.
   a. Gluconeogenesis occurs in liver and to a smaller extent in kidney in animals.
   b. Amino acids leucine and lysine does not participate in gluconeogenesis.
   c. During glycogenolysis terminal glucose residue removes from non-reducing end.
   d. Glycogen phosphorylase, an enzyme involved in glycogenolysis requires biotin as cofactor.

10. Suppose a glycogen molecule with 8000 glucose residues. If branches occur every eight residues, how many reducing ends does the molecule have?
   a. 1
   b. 500
   c. 8000
   d. 499

11. Glycogen phosphorylase exists in two forms in skeletal muscle. The active form phosphorylase *a* is generated from phosphorylase *b* by
   a. reversible dimerization of phosphorylase *b* triggered by calcium ion.
   b. proteolytic cleavage of a decapeptide from the N-terminus of phosphorylase *b*.
   c. protonation of the active-site histidine residue by a decrease in intracellular pH.
   d. ATP-dependent phosphorylation of a specific serine residue on each subunit. **[JNU]**

Lipids

Fatty acids

1. Which of the following statement is *incorrect* about fatty acids?
   a. All non-essential fatty acids can be synthesized from acetyl-coA.
   b. Essential fatty acids are unsaturated.
   c. Most naturally occurring fatty acids contain even number of carbon atoms.
   d. Unsaturated fatty acid produce more energy than saturated one of the same size when oxidized.

2. Which of the following fatty acids (FAs) of similar size have higher melting point?
   a. FAs having a *cis* double bond.
   b. FAs having a *trans* double bond.
   c. Both have same melting point.
   d. Melting point does not depend upon *cis* and *trans* configuration.

3. Which of the following compound is *not* amphipathic?
   a. Phosphatidylcholine
   b. Cholesterol
   c. Oleic acid
   d. Succinate **[IISc]**

4. Which of the following substance represents an unsaturated fatty acid?
   a. Palmitate
   b. Stearate
   c. Choline
   d. Oleate **[MKU]**

5. Physical property of fatty acid depends on
   a. length of hydrocarbon chains
   b. degree of unsaturation
   c. branching
   d. all of the above

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6. Which of the following fatty acids has the lowest melting point?
   a. Fatty acids with sites of unsaturation with cis double bonds.
   b. Fatty acids with sites of unsaturation with *trans* double bonds.
   c. Fatty acids with *no* sites of unsaturation.
   d. Fatty acids with longer hydrophobic tails. [JNU]

7. A scientist wanted to genetically engineer a new type of corn plant that could withstand cold temperatures. He decided to try to change the composition of the plant's membranes to be expected to improve the cold tolerance of the plants? This can be done by
   a. increasing the length of the fatty acyl chains.
   b. eliminating all steroids.
   c. increasing the frequency of unsaturated fatty acyl chains.
   d. decreasing the frequency of unsaturated fatty acyl chains.

8. Which of the following glycerophospholipid is the major lipid of mitochondrial membrane?
   a. Phosphatidyl choline
   b. Lecithin
   c. Phosphatidyl inositol
   d. Cardiolipin

9. Most of the free fatty acids are transported in the blood
   a. inside the red blood cells
   b. as lipoproteins
   c. combined with glucose
   d. bound to albumin

10. Which of the following is *not* a biological role of fatty acids?
    a. Fatty acid is essential component of plasma membrane.
    b. Fatty acid are stored as triacylglycerols in body.
    c. Fatty acid acts as intracellular second messengers.
    d. Fatty acids mainly contain odd number of carbons.

11. Which fatty acid (having 15 carbons) will have highest melting point?
    a. Single double bond
    b. More than one double bond
    c. Only single bond
    d. None of them

Triacylglycerol, Phospholipid and Glycolipid

1. Some unsaturated fats used in cooking, such as olive oil, spoil rapidly upon exposure to air at room temperature because
   a. unsaturated fats are susceptible to oxidation by molecular oxygen.
   b. of low melting point of unsaturated fat.
   c. carboxyl group oxidized upon exposure to air.
   d. none.

2. Arrange a triacylglycerol, a diacylglycerol and a monoacylglycerol, all containing only palmitic acid, in order of increasing solubility in water
   a. diacylglycerol, monoacylglycerol, triacylglycerol.
   b. triacylglycerol, diacylglycerol, monoacylglycerol.
   c. monoacylglycerol, diacylglycerol, triacylglycerol.
   d. all have similar solubility.

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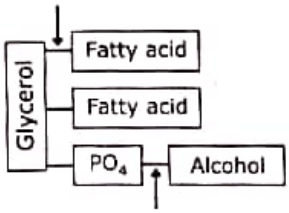
3. Choose the *wrong* statement(s).
   P. Unsaturated fatty acids (FAs) have low melting points.
   Q. Cell of cold blooded animal have a higher proportion of unsaturated FAs than warm blooded animals.
   R. Phosphoglycerides are amphipathic.
   S. Triacylglycerols are polar and hydrophilic lipid.
   T. Spoiling of fat/oil after long time is called rancidification.
   a. only P
   b. both P and Q
   c. only S
   d. R, S and T

4. Assuming they all had the same number of carbon atoms, which of the following has the maximum number of C—H bonds?
   a. An unsaturated fat
   b. A polyunsaturated fat
   c. A polysaccharide
   d. A saturated fat

5. Molecular structure of phosphoglycerides is given below. Name the correct bond depicted A and B



   a. A – ester bond B – ether bond
   b. A – ester bond B – amide bond
   c. A – ether bond B – amide bond
   d. A – ester bond B – phosphoester bond

6. Which of the following lipids have a net negative charge?
   a. Phosphatidylcholine
   b. Cholesterol
   c. Phosphatidylserine
   d. Phosphatidylethanolamine

7. Alkaline hydrolysis of a triglyceride is
   a. saponification
   b. esterification
   c. hydrogenation
   d. dehydration [JNU]

8. Which is *not* correct about saponification number?
   P. Smaller the saponification number, higher the MP of fat.
   Q. It is the number of miligram of KOH required to saponify one gram of fat.
   R. It measures the extent of unsaturation in fats.
   S. It measures the average molecular weight of fats.
   a. P, Q
   b. P, R
   c. P, Q, R
   d. P, Q, R, S

9. All of the following statements describe phosphoglycerides *except*
   a. they are both amphipathic and amphoteric.
   b. they arise from phosphatidic acid.
   c. they are found in cell membranes.
   d. they are a major store of metabolic energy.

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10. An essential building block of phosphatidic acid and phosphatidylcholine
    a. glycerol
    b. lysine
    c. cholesterol
    d. glucose [IISc]

11. Instead of ester link plasmalogens possess an ether link in position
    a. α-
    b. β-
    c. γ-
    d. none of the above

12. Which of the following does not belong to glycosphingolipids?
    a. Cerebrosides
    b. Gangliosides
    c. Globosides
    d. Sphingomyelin

13. The combination of an amino alcohol, fatty acid and sialic acid forms
    a. phospholipids
    b. sulpholipids
    c. glycolipids
    d. aminolipids

14. Gangliosides contain
    a. a ceramide structure
    b. glucose or galactose
    c. sialic acid
    d. all of the above [GATE]

15. A ganglioside must contain in its structure
    a. N-acetyl neuraminic acid (NANA), hexoses, sphingosine, long chain fatty acid.
    b. NANA, a hexose, a fatty acid, sphingosine, phosphorylcholine.
    c. NANA, sphingosine, ethanolamine.
    d. NANA, hexoses, fatty acid, glycerol. [GATE]

16. Gangliosides have which of the following characteristics?
    a. They contain sugars and sugar derivatives.
    b. They contain sialic acid (N-acetylneuraminic acid).
    c. They are derived from cerebrosides.
    d. All of the above.

17. The concentrations of sphingomyelins are increased in
    a. Gaucher's disease
    b. Fabry's disease
    c. Febrile disease
    d. Niemann-Pick disease

18. Tay-Sachs disease, caused due to accumulation of GM2 gangliosides, a complex sphingolipid is a/an
    a. autosomal dominant genetic disorder
    b. sex chromosomal recessive genetic disorder
    c. sex chromosomal dominant genetic disorder
    d. autosomal recessive genetic disorder

19. *Correct* statement(s) about inherited sphingolipid disorders include which of the following?
    a. Gaucher's disease is due to a deficiency in the degradation of glucocerebrosides.
    b. Tay-Sachs disease is due to a deficiency of lysosomal hexosaminidase A.
    c. Niemann-Pick disease is due to a deficiency of sphingomyelinase.
    d. all of the above.

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20. Which of the following statements are *correct*?
    1. Double bonds in unsaturated fatty acids are generally in the *cis* configuration.
    2. In polyunsaturated fatty acid, double bonds are almost never conjugated.
    3. Sphingoglycolipids are involved in biological recognition.
    4. Prostaglandins contain a five membered ring of carbon atoms.
    5. Prostaglandins regulate the synthesis of the intracellular messenger cAMP.

    a. 1 and 5
    b. 1, 2 and 3
    c. 3, 4 and 5
    d. 1, 2, 3 and 4

21. The prostaglandins are synthesized from
    a. arachidonic acid
    b. oleic acid
    c. linoleic acid
    d. linolenic acid

22. Which of the following is a *false* statement regarding Thromboxane A_2 (TXA_2)?
    a. Is derived from the membrane phospholipid.
    b. Its production is decreased by non-steroidal anti-inflammatory drugs.
    c. Causes platelet aggregation.
    d. Causes vasodialation.

Cholesterol and Lipoproteins

1. Which of the following statement about natural sterols is *incorrect*?
    a. Cholesterol is the most abundant sterol in animal tissue.
    b. All the carbon atoms of cholesterol are derived from acetyl-CoA.
    c. β-sitosterol is the most abundant plant sterol.
    d. Dietary β-sitosterol and cholesterol are absorbed to about the same extent in the intestine of normal human.

    [GATE]

2. Cholesterol is essential for normal membrane functions because it
    a. cannot be made by higher organisms, *e.g.* mammals.
    b. spans the thickness of the bilayer.
    c. keeps membranes fluid.
    d. catalyzes lipid flip-flop in the bilayer.

3. Cholesterol can act as a precursor for each of the following compounds *except*
    a. dihydroxycholecalciferol
    b. testosterone
    c. glycocholic acid
    d. cholecystokinin

4. Which one of the following processes is common to the synthesis of all steroid hormones?
    a. Conversion of pregnenolone to progesterone
    b. Cholesterol side chain cleavage
    c. Isomerization
    d. Dehydrogenation

    [IISc]

5. The UV rays of sunlight are responsible for the formation of the vitamin D in the skin is from
    a. cholesterol
    b. ergasterol
    c. cholecalciferol
    d. calciferol

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6. What are bile salts?
   a. Charged phospholipids
   b. Amphipathic cholesterol analogs with detergent properties
   c. Esterified cholesterols
   d. Hydrolyzed forms of triacylglycerols

7. Bile salts act as detergents in the duodenum because
   a. of their amphipathic structures
   b. they contain hydrophobic groups
   c. they carry an overall positive charge
   d. they are zwitterions at the pH of the duodenum

8. Which one of the following statement about lipoproteins is *true*?
   a. Molecular mass of lipoproteins is directly proportional to their density.
   b. The percent protein content in lipoproteins increases with molecular mass.
   c. Density of a lipoprotein decreases with increase in protein content.
   d. Molecular mass of lipoproteins is inversely proportional to their density. **[GATE]**

9. A patient has a genetic defect resulting in deficiency of lipoprotein lipase. After eating a meal containing a large amount of fat, one would expect to see a plasma elevation of
   a. chylomicrons
   b. VLDLs
   c. LDLs
   d. HDLs

10. Dietary triacylglycerols containing long-chain fatty acyl moieties are digested, absorbed from the small intenstine, and stored in adipose tissue cells. The following events occur at different times and sites during the process
    1. formation of chylomicrons
    2. hydrolysis by hormone-sensitive triacylglycerol lipase
    3. micellar formation with bile salts
    4. resynthesis from 2-acylglycerols
    5. hydrolysis by lipoprotein lipase
    6. transport in the circulation as free fatty acids bound to serum albumin

    The correct order of these events is given by which of the following sequences?
    a. 3, 1, 5, 6, 4, 2
    b. 3, 4, 1, 2, 6, 5
    c. 1, 4, 3, 5, 6, 2
    d. 3, 4, 1, 5, 2, 6

Fatty acid metabolism

1. Which of the following statement is *incorrect*?
   a. Fatty acid synthesis occurs in cytosol of animal cell.
   b. Fatty acid desaturation and elongation occurs in ER.
   c. In diabetes mellitus ketone bodies production increases and results in ketoacidosis.
   d. None of the above.

2. The acetyl groups required for cytoplasmic fatty acid synthesis appear in the cytoplasm as a result of the activity of
   a. citrate synthetase
   b. Isocitrate dehydrogenase
   c. citrate lyase
   d. thiolase

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3. The most important source of reducing equivalents for fatty acid synthesis in the liver is
   a. oxidation of glucuronic acid
   b. oxidation of acetyl coenzyme A
   c. the tricarboxylic acid cycle
   d. the pentose phosphate pathway

4. Which of the following vitamins is necessary as a coenzyme in the initial steps of fatty acid synthesis?
   a. Biotin
   b. Thiamine
   c. Vitamin D
   d. Riboflavin

5. Which of the following cofactor is essential for the activity of acetyl-CoA carboxylase?
   a. NAD^+
   b. Biotin
   c. TPP
   d. Vitamin B_6

6. All of the following statements about acetyl-CoA carboxylase are correct *except*
   a. it catalyzes the rate-limiting step of fatty acid synthesis.
   b. it requires biotin.
   c. it is inhibited by cAMP-mediated phosphorylation.
   d. it is activated by palmitoyl-CoA.

7. The removal of two-carbon units from a fatty acyl coenzyme A involves four sequential reactions. Which of the following best describes the reaction sequence?
   a. Oxidation, dehydration, oxidation, cleavage
   b. Reduction, dehydration, reduction, cleavage
   c. Dehydrogenation, hydration, dehydrogenation, cleavage
   d. Hydrogenation, dehydration, hydrogenation, cleavage

8. β-oxidation of phytanic acid, a fatty acid derived from chlorophyll and found in cow's milk, cannot take place without prior oxidation at the α-carbon. Which of the following statement best explains the requirement for α-oxidation?
   a. The chain length of phytanic acid is too long for the mitochondrial β-oxidation system to handle.
   b. The β-carbon in phytanic acid is blocked by a methyl group.
   c. Phytanic acid cannot be activated by coenzyme A prior to β-oxidation.
   d. The mitochondrial acyl carnitine transferase will not transport phytanic acid into the mitochondrial matrix.

9. Differences between the systems involved in the β-oxidation of fatty acids and in their biosynthesis include the
   P. intracellular location of the enzymes involved.
   Q. nature of the thiol acyl carriers.
   R. oxidation-reduction state of required pyridine nucleotide cofactors.
   S. participation of high-energy phosphate bonds.
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

10. Which of the following is common to both fatty acid synthesis and degradation?
    a. The oxidation/reduction reactions occur between the α and the β carbons of the fatty acid.
    b. The biochemical nature of the reductant/oxidant.
    c. The intracellular location of the metabolic pathways.
    d. The nature of the two carbon unit.

[IISc]

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11. If the 16-carbon saturated fatty acid palmitate is oxidized completely to carbon dioxide and water (via the β-oxidation pathway and the citric acid cycle), and all of the energy conserving products are used to drive ATP synthesis in the mitochondria, the net yield of ATP per molecule of palmitate is around

a. 3
b. 10
c. 25
d. 129

12. β-oxidation of a 16 carbon fatty acid and a 17 carbon fatty acid leads to formation of

a. (8 Acetyl CoA) and (8 Acetyl CoA + CO_2), respectively.
b. (8 Acetyl CoA) and (7 Acetyl CoA + 1 Propionyl CoA), respectively.
c. (5 Propionyl CoA + 1 CO_2) and (5 Propionyl CoA + 2 CO_2), respectively.
d. (5 Propionyl CoA + 1 CO_2) and (5 Propionyl CoA + 1 Acetyl CoA), respectively. [GATE]

13. Ketone bodies are formed due to

a. breakdown in the β-oxidation pathway which would produce an excess of acetyl-CoA.
b. inhibition of fatty acid activation and transport.
c. channelisation of oxaloacetate from the citric acid cycle to gluconeogenesis.
d. inhibition of cAMP production. [IISc]

Cholesterol metabolism

1. Which of the following compound provides the carbon atoms for cholesterol synthesis?

a. Pregnenolone
b. Acetyl-CoA
c. Phytanic acid
d. Succinic acid

2. Which of the following compounds is a precursor in the biosynthesis of both cholesterol and β-hydroxybutyrate (a ketone body)?

a. Farnesyl pyrophosphate
b. β-Hydroxy-β-methylglutaryl coenzyme A (HMG CoA)
c. Methylmalonyl CoA
d. Mevalonate

3. Which of the following step is *not* involved in cholesterol synthesis?

a. Formation of mevalonate from acetate.
b. Conversion of mevalonate into isopentenyl pyrophosphate.
c. Polymerization of six isoprene units into squalene.
d. Formation of malonyl CoA from acetyl-CoA.

4. Which of the following is *false* about cholesterol?

a. Cholesterol is the most abundant sterol found in human tissue.
b. Cholic acid is a precursor in the biosynthesis of cholesterol.
c. All the carbon atoms of cholesterol are derived from acetyl-CoA.
d. HMG CoA reductase is the main regulatory enzyme of cholesterol synthesis.

5. A common step involved in the biosynthesis of all steroid hormones is

a. conversion of cortisol to corticosterone.
b. cholesterol side chain cleavage.
c. aromatization.
d. dehydrogenation. [IISc]

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Chapter 04

Enzymes and Vitamins

Enzymes : General features

1. Which of the following statement is *true*? Enzyme catalysis of a chemical reaction
   a. increases the forward and reverse reaction rates.
   b. decreases ΔG° so that the reaction can proceed spontaneously.
   c. increases the energy of transition state.
   d. decreases the entropy of reaction. **[GATE]**

2. Enzymes increase the rate of reactions by
   a. increasing the free energy of activation.
   b. increasing the free-energy change of the reaction.
   c. changing the equilibrium constant of the reaction.
   d. decreasing the energy of activation.

3. The function of an enzyme is to
   a. cause chemical reaction that would not otherwise take place.
   b. change the rates of chemical reactions.
   c. control the equilibrium points of reactions.
   d. change the directions of reactions.

4. Binding energy released in enzyme–substrate complex formation causes the
   a. decrease in the free energy of product.
   b. decrease in the free energy of substrate.
   c. increase in the activation energy.
   d. decrease in the activation energy.

5. Which of the following is a *false* statement?
   a. Coenzymes may be derived from vitamins.
   b. Coenzymes may be prosthetic groups.
   c. Coenzymes increase the activation energy of the enzymatic reaction.
   d. All statements are correct.

6. An enzyme catalyzing the reaction Glycerate-3-phosphate $\rightleftharpoons$ Glycerate-2-phosphate would have an EC number with first digit
   a. 2
   b. 4
   c. 5
   d. none of the above **[JNU]**

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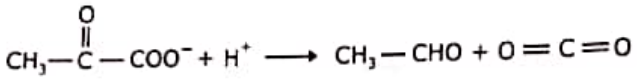
7. The enzymes where catalysis involves transfer of electrons are named as
   a. isomerases
   b. transferases
   c. oxidoreductases
   d. lyases [GATE]

8. An enzyme that catalyzes the conversion of an aldose sugar to a ketose sugar would be classified as one of the
   a. oxidoreductases
   b. transferases
   c. hydrolases
   d. isomerases

9. Which of the following enzyme transfers a phosphate group from ATP to another molecule?
   a. Phosphatase
   b. Phosphodiesterase
   c. Kinase
   d. Esterase

10. Which type of enzyme catalyzes the following reaction?



   a. lyase
   b. ligase
   c. isomerase
   d. oxidoreductase

11. Which one of the following coenzymes pairs correctly with the group transferred?
   a. CoA-electron
   b. $NADP^+$-phosphoryl
   c. thiamine pyrophosphate
   d. biotin-carbon dioxide [GATE]

12. Which of the following vitamins is a precursor of coenzyme that is required in enzymatic reactions involving transfer of acyl groups?
   a. Lipoic acid
   b. Biotin
   c. p-aminobenzoic acid
   d. Riboflavin [GATE]

13. Enzyme glutathione peroxidase, catalyzes destruction of H_2O_2, contains
   a. Zn
   b. Fe
   c. Se
   d. Mo

14. Lyases, a class of enzyme catalyses
   a. hydrolysis reaction
   b. group transfer reaction
   c. oxidation and reduction reaction
   d. addition of groups to double bond and vice versa

15. Which one of the following is a cofactor and not a coenzyme?
   a. Biotin
   b. Tetrahydrofolic acid
   c. Copper
   d. Methylcobalamin

16. Which of the following statement(s) is/are *correct* about coenzymes?
   P. Tetrahydrofolate transfer $-CH_3$, $-CH_2$, $-CHO$ and $-CHNH$ group
   Q. S-adenosylmethionine transfer $-CH_3$ group
   R. Biotin transfer CO_2 group
   S. TPP transfer acyl group
   a. P, Q and R
   b. P and R
   c. R and S
   d. P, Q, R and S

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17. Which of the following combinations is *correct*?
    a. Amylase – peptide bond
    b. Protease – phosphodiester bond
    c. Lipase – ester bond
    d. Nucleases – glycosidic bond

18. In an intact cell, the free energy change (ΔG) associated with an enzyme-catalyzed reaction is frequently different from the standard free energy change ($\Delta G°'$) of the same reaction because in the intact cell the **[IISc]**
    a. activation energy is different
    b. reaction is always near equilibrium
    c. enzyme may be regulated allosterically
    d. reactants are not at 1 M concentrations

19. Ribozymes are
    a. enzymes which use ribose as substrates
    b. enzymes working on DNA
    c. RNAs with enzyme's activity
    d. enzyme-RNA complexes

20. Abzymes are **[IISc]**
    a. enzymes that are highly specific like antibodies
    b. antibodies that have catalytic activities
    c. also referred to as zymogens
    d. enzymes that hydrolyze antibodies

21. Catalytic antibodies function as enzymes on the principle of **[IISc]**
    a. enzymatic conversion of antibodies
    b. stabilizing transition state analogue of substrates
    c. antigen-antibody affinity
    d. monoclonal antibodies with chemical capability

22. Lactic dehydrogenase (LDH) is essential for lactic acid formation. This is an example of
    a. chemienzyme
    b. abzyme
    c. apoenzyme
    d. isoenzyme

23. Isozymes can be characterized by
    a. the different chemical reactions that they catalyze.
    b. the differences in their elution profile from a size-exclusion column.
    c. the differences in their amino acid sequences.
    d. all of the above. **[IISc]**

24. Zymogen is
    a. the active form of an enzyme.
    b. the complex formed between an enzyme and its substrate.
    c. the inactive form of an enzyme activated by cleavage.
    d. the unfolded form of the enzyme. **[IISc]**

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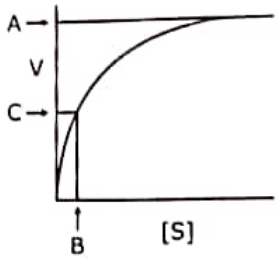
Enzyme kinetics

1. A typical saturation curve for an enzyme-catalyzed reaction is shown in the following figure. The parameters, which can be derived from this equation, is indicated by A, B and C. Tick the correct parameter that corresponds to B



a. K_m
b. V_{max}
c. K_i
d. K_d [JNU]

2. Conversion of substrate S present in large excess to product P by an enzyme increased in a linear fashion for about 10 minutes, after which it was seen to plateau. This is because the enzyme has
   a. reached its K_m
   b. reached its V_{max}
   c. been inactivated after 10 minutes of activity
   d. switched its substrate specificity [IISc]

3. The enzyme following Michaelis–Menten kinetics show a characteristic graph when substrate concentration is plotted against velocity, the nature of graph will be
   a. sigmoidal
   b. parabolic
   c. hyperbolic
   d. straight line

4. Michaelis–Menten equation states that
   a. rate of enzymatic reaction is independent of substrate concentration.
   b. rate of non-enzymatic reaction is proportional to substrate concentration.
   c. K_m is the enzyme–substrate complex association constant.
   d. enzyme–substrate complex formation is essential in enzymatic reaction.

5. The K_m of an enzyme is
   a. one half of the V_{max}.
   b. a dissociation constant.
   c. the substrate concentration that gives maximal velocity.
   d. the substrate concentration that gives half maximal velocity.

6. The Michaelis constant, K_m is [GATE]
   P. numerically equal to V_{max}.
   Q. dependent on enzyme concentration.
   R. independent of pH.
   S. numerically equal to $V_{max}/2$.
   a. P and Q
   b. P, Q and R
   c. P, Q, R and S
   d. none is correct

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7. Lower value of Michaelis constant shows
   a. greater affinity of the enzyme for the substrate
   b. less affinity of the enzyme for the substrate
   c. enzyme is allosteric
   d. question is incomplete

8. When substrate concentration is increased for a simple enzymatic reaction, the reaction order progresses from first order to zero order with respect to substrate concentration, because:
   a. enzyme-product complexes prevent further binding of more substrate.
   b. significant product formation results in inhibition of the reaction.
   c. all of the active sites of the enzyme are saturated with substrate at high substrate concentrations.
   d. substrate-substrate interactions at high concentrations become strong.

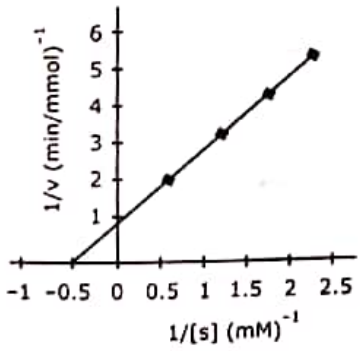
9. In a Michaelis-Menten enzyme mechanism, what substrate concentration (relative to K_m) is needed for the reaction rate to be 0.5 V_{max}.
   a. 1/9 K_m
   b. 1/3 K_m
   c. K_m
   d. 1/4 K_m

10. If the substrate concentration in an enzyme catalyzed reaction is equal to 1/2 K_m, the initial reaction velocity will be
   a. 0.25 V_{max}
   b. 0.33 V_{max}
   c. 0.50 V_{max}
   d. 0.67 V_{max}

11. At what [S] is the velocity (V_0) of an enzyme-catalyzed reaction is 25% of the V_{max}?
   a. 3/4 K_m
   b. 4 K_m
   c. 1/3 K_m
   d. 1/4 K_m **[IISc]**

12. A enzyme has Glu^{76} and Asp^{52} at active site. The pI for Glu is 5.6 and for Asp is 4.5. The enzyme function when Glu is in protonated form and Asp in deprotonated form. The pH where enzyme will show maximum activity will be
   a. 4.5
   b. 5.05
   c. 10.1
   d. 5.6

13. The graph shows a Lineweaver-Burk plot for an enzyme catalyzed reaction



Which of the following statement is *correct*?
   a. The V_{max} is 5 mmol/min and with competitive inhibition V_{max} remains unchanged.
   b. K_m is 2 mmol/min and with competitive inhibition both K_m and V_{max} decrease.
   c. K_m is 0.5 mM and with competitive inhibition V_{max} increase but K_m remains unchanged.
   d. K_m is 2.0 mM and with competitive inhibition K_m increase but V_{max} remains unchanged. **[GATE]**

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14. What is meant by the steady-state assumption that underlies the Michaelis-Menten relationship between substrate concentration and reaction velocity?

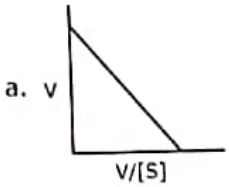
a. The reaction velocity is linearly related to substrate concentration.

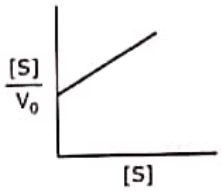
b. The reaction velocity is independent of substrate concentration.

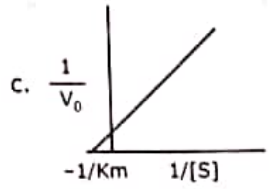
c. The rate of breakdown of the enzyme-substrate complex equals the rate of formation of the complex.

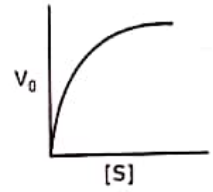
d. The rate of formation of product equals the rate of disappearance of substrate.

15. Which of the following is an Eadie-Hofstee plot showing enzyme kinetics?

a. 

b. 

c. 

d. 

16. What is the turnover number (i.e. the k_{cat} or enzymatic activity) for an enzyme catalyzed reaction which has a maximum velocity of 3.47 moles liter^{-1} min^{-1} when the enzyme concentration is 6.85 × 10^{-1} g liter^{-1}? Molecular weight of the enzyme is 12, 800 g mol^{-1}?

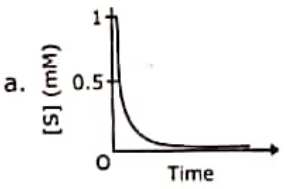
a. 6.48×10^{4} min^{-1}

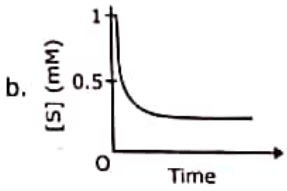
b. 5.258×10^{-3} min^{-1}

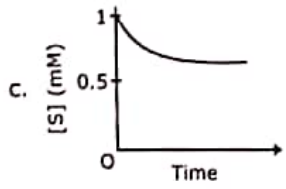
c. 1.579×10^{5} min^{-1}

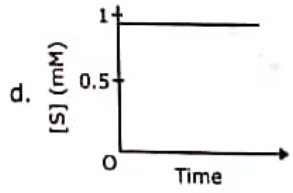
d. 1.27×10^{-5} min^{-1} [IISc]

17. For the enzyme catalyzed reactions S $\rightleftharpoons$ P, K_{eq} = 5, which of the following curves best represents the change in [S] that occurs with time when S is added to a solution containing the appropriate enzyme?

a. 

b. 

c. 

d. 

18. The effects of pH on enzyme-catalyzed reactions include which of the following?

P. The direction of the reaction may be influenced by the [H^+].

Q. The ionization state of dissociating groups on the enzyme may be modified.

R. The ionization state of the substrate may be modified.

S. The protein may be denatured at certain pH values.

a. P and Q

b. P and R

c. P, Q and R

d. P, Q, R and S

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Enzymatic inhibition

1. Which of the following inhibitor types is expected to change the K_m of an enzyme, but not the V_{max}?
   a. Competitive
   b. Non-competitive
   c. Uncompetitive
   d. All inhibitors [IISc]

2. A competitive inhibitor
   a. increases the K_m of an enzyme.
   b. decreases the K_m of an enzyme.
   c. increases both the V_{max} and the K_m of an enzyme.
   d. decreases the K_m but increases the V_{max} of an enzyme. [JNU]

3. Which molecule binds to the active site of an enzyme?
   a. Allosteric activator
   b. Allosteric inhibitor
   c. Noncompetitive inhibitor
   d. Competitive inhibitor

4. Sulpha drugs are effective antiblotic which inhibits the formation of folic acid by acting as a
   a. non-competitive inhibitor
   b. competitive inhibitor
   c. uncompetitive inhibitor
   d. irreversible inhibitor

5. The effects of a competitive inhibitor on the kinetics of an enzyme reaction include which of the following?
   P. The V_{max} is not changed.
   Q. Increased concentrations of substrate reverse the inhibition.
   R. The K_m is increased.
   S. The inhibitor binds to a site on the enzyme other than the catalytic site.
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

6. Drinking methanol can be fatal because the host alcohol dehydrogenase converts it to formaldehyde. One of the ways to detoxify methanol poisoning is to
   a. feed the patient with glucose
   b. give an intravenous dose of steroids to the patient
   c. make the patient drink ethanol
   d. make the patient drink lemon juice [IISc]

7. Which one of the following statement is *true* about non-competitive inhibition?
   a. K_m increases
   b. K_m decreases
   c. V_{max} increases
   d. V_{max} decreases [IISc]

8. In non-competitive type of enzymatic inhibition
   a. the V_{max} decreases and K_m remain unchanged
   b. the V_{max} remain unchanged and K_m increases
   c. V_{max} and K_m both decreases
   d. V_{max} decreases and K_m increases

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9. You think you may have found a new inhibitor (I) of the HIV protease, so you perform a few experiments to characterize its behavior. You find that if you add an excess of synthetic substrate to the protease in the presence of the inhibitor, enzyme actitivity increases but only reaches 50% of the V_{max} observed in the absence of the inhibitor. The affinity of the substrate for the enzyme is unaffected by the inhibitor. Furthermore, if you treat the enzyme with the inhibitor, then dialyze the protein, you find that full activity can be restored. What class of inhibitor is I?

a. Irreversible
b. Uncompetitive
c. Noncompetitive
d. Competitive

Regulatory enzymes

1. Which of the following regulatory actions involve a reversible covalent modification of an enzyme?
   a. Allosteric modulation
   b. Competitive inhibition
   c. Conversion of zymogen to active enzyme
   d. Phosphorylation of a serine hydroxyl on the enzyme

2. An allosteric modulator influences enzyme activity by
   a. competing for the catalytic site with the substrate.
   b. binding to a site on the enzyme molecule distinct from the catalytic site.
   c. changing the specificity of the enzyme for its substrate.
   d. none of the above.

3. Which of the following statements are *correct*?
   P. Allosteric enzymes don't obey Michaelis-Menten kinetics.
   Q. Some regulatory enzymes are modulated by reversible covalent modification.
   R. Allosteric enzymes undergo reversible covalent modification.
   S. Phosphorylation of regulatory enzymes is most common type of reversible covalent modification.
   a. P and Q
   b. P, Q and R
   c. P, Q and S
   d. P, Q, R and S

4. Which of the following statements are *true* about regulatory enzymes?
   P. Produce a sigmoidal curve
   Q. Produce hyperbolic curve
   R. Generally reflects cooperativity
   S. Always show allosteric property
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

5. Which of the following statement is *incorrect*?
   a. High ATP concentration inhibits the enzyme phosphofructokinase allostericallly.
   b. In end product inhibition, the end product of a metabolic pathway acts as an allosteric inhibitor on enzyme controlling the first step of the pathway.
   c. Temperature below freezing point causes inactivation of enzymes.
   d. In homotropic enzymes, for which substrate and modulator are identical, the active site and regulator site are the different.

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6. Which of the following processes is *not* an example of allosteric regulation?
   a. Regulation of phosphofructokinase activity by fructose 2,6-bisphosphate.
   b. Inactivtion of nitrogenase by ADP-ribosylation.
   c. Regulation of the lac operon by allolactose in *E. coli*.
   d. Catabolite repression by CAP in *E. coli*.

Enzymatic reactions

1. The reason for the presence of histidine at the active site of enzymes is
   a. it is the only polar residue with a five-membered ring at the side chain
   b. it can form salt bridges
   c. its pK_a is close to 7
   d. it can interact with the substrates better than any other residue **[IISc]**

2. Lysozyme has a pH optimum centered around pH 5.0. The active site of lysozyme contains a glutamic acid residue (pK_a = 5.5) and an aspartic acid residue (pK_a = 4.0). Which of the following statements is *correct* about the mechanism of lysozyme?
   a. The glutamic acid residue is in a more polar environment than the aspartic acid.
   b. During the entire catalytic mechanism, the aspartic acid residue remains unprotonated.
   c. During the mechanism, an oxyanion transition state forms.
   d. The glutamic acid residue acts as a general base catalyst.

3. A common feature of all serine proteases is
   a. autocatalytic activation of zymogen precursor.
   b. tight binding of pancreatic trypsin inhibitor.
   c. cleavage of protein on the carboxyl site of serine residues.
   d. presence of Ser-His-Asp catalytic triad at the active site.

4. Which of the following statement is *correct* about RNase A?
   a. It is an exonuclease.
   b. It cleaves phosphodiester bond between pyrimidine and purine nucleotide residues.
   c. Its hydrolysis product is a polynucleotide with a 3'-phosphate terminus.
   d. Its hydrolysis product is a polynucleotide with a 5'-phosphate terminus. **[GATE]**

5. Which of the following intermediate compound is involved when a peptide is hydrolyzed by chymotrypsin?
   a. An ester between the substrate's acyl carbon and the serine of the active site.
   b. A thioester between the substrate's acyl carbon and the cysteine of the active site.
   c. An amide between the substrate's acyl carbon and the lysine of the active site.
   d. An amide between the substrate's acyl carbon and the asparagines of the active site.

6. Which of the following is/are *true* when DIPF inhibits chymotrypsin activity?
   P. $[E]_T$ decrease
   Q. apparent V_{max} decrease
   R. K_m decrease
   S. V_{max} increase
   a. P only
   b. P and Q only
   c. R only
   d. S only

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7. RNase A cannot catalyze the hydrolysis of DNA because
   a. DNA is very stable
   b. DNA lacks 2'OH required for the formation of 2'3'-cyclic reaction intermediate
   c. DNA lacks 2'OH required for the formation of covalent linkage with Ser^{56}
   d. DNA is double stranded

8. Aspartic protease
   P. possesses two aspartic acid residues at the active site.
   Q. includes enzymes like pepsin, cathepsin and renin.
   R. involves formation of intermediate amide dihydrate.
   S. present in AIDS virus HIV-1.
   a. P and S
   b. Q, R and S
   c. P, Q, R and S
   d. P, Q, R but not S

Vitamins

1. Most vitamins are
   a. covalently linked to enzymes
   b. synthesized by humans
   c. precursors to coenzymes
   d. precursors to proteins [IISc]

2. Vitamin D is derived from which of the following precursors by the action of UV light?
   a. 7–dehydrocholesterol
   b. lanosterol
   c. glycocholate
   d. squalene epoxide [GATE]

3. Thiamine deficiency leads to
   a. scurvy
   b. beri-beri
   c. night blindness
   d. pellagra

4. All of the following vitamins act as antioxidant *except*
   a. vitamin D
   b. vitamin A
   c. vitamin E
   d. vitamin C

5. Which vitamin is necessary for coenzyme A synthesis?
   a. Ascorbic acid
   b. Pyridoxine
   c. Biotin
   d. Pantothenic acid

6. A deficiency of vitamin B_{12} causes which of the following disorders?
   a. Cheilosis
   b. Beriberi
   c. Pernicious anemia
   d. Scurvy

7. Which of the following vitamins would most likely become deficient in a person who develops a completely carnivorous lifestyle?
   a. Thiamine
   b. Niacin
   c. Cobalamin
   d. Vitamin C

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8. Biotin is involved in which of the following types of reactions?
   a. Hydroxylations
   b. Carboxylations
   c. Decarboxylations
   d. Dehydrations

9. Which of the following coenzymes does *not* contain adenine?
   a. Thiamine pyrophosphate
   b. S-adenosylmethionine
   c. Coenzyme A
   d. NADH

10. Which of the following answers complete the sentence correctly? During the reduction of FAD,
   a. a flavin group is transferred
   b. an equivalent of a hydride ion is transferred
   c. the isoalloxazine ring becomes charged
   d. two hydrogen atoms are added to the isoalloxazine ring

11. Which of the following is *not* a dietary antioxidant?
   a. Vitamin C
   b. Lipoic acid
   c. Vitamin K
   d. Beta-carotene

12. Which of the following cofactor is required for conversion of proline residues into hydroxyproline in procollagen?
   a. TPP
   b. Pantothenate
   c. Niacin
   d. Ascorbic acid

13. Which of the following coenzymes acts as an electron sink to promote catalysis?
   a. Biotin
   b. Lipoamide
   c. CoA
   d. Pyridoxal phosphate

14. The visual pigment retinal binds to the protein rhodopsin by the same type of linkage as that used by another coenzyme. What is this coenzyme?
   a. Thiamine pyrophosphate
   b. Tetrahydrofolate
   c. Pyridoxal phosphate
   d. S-adenosylmethionine

15. Pantothenic acid is a constituent of the coenzyme involved in which of the following?
   a. Decarboxylation
   b. Acetylation
   c. Dehydrogenation
   d. Reduction

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16. The coenzyme involved in oxidative decarboxylation is
    a. thiamine pyrophosphate
    b. biotin
    c. NAD^+
    d. pyridoxal phosphate

[IISc]

17. Which of the following vitamins does *not* act as a precursor for coenzymes?
    a. Biotin
    b. Thiamine
    c. Folic acid
    d. Ascorbic acid

[IISc]

18. Match the metal ions, vitamins and related compounds given in Column A with their specific functions in microbial metabolism listed in Column B. Choose the mismatch.

Column A	*Column B*
a. p-aminobenzoic acid	A. precursor of tetrahydrofolic acid, a coenzyme involved in transfer of one-carbon units
b. pantothenic acid	B. precursor of coenyme A and of the prosthetic group of acyl carrier proteins
c. vitamin B_{12}	C. involved in rearrangement reactions (e.g. glutamate mutase)
d. Cu^{2+}	D. present in carbonic anhydrase

[GATE]

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Chapter 05

Cell Biology

Eukaryotic cell : Structure and function

1. Which of the following criteria can be used to distinguish between prokaryotic and eukaryotic cells?
   a. Whether or not they have plasma membrane.
   b. Whether or not they have a nucleus (membrane-enclosed nucleoplasm).
   c. Comparison of the sequences of homologous proteins.
   d. Comparison of the signature sequences of 16S rRNA.

2. Which of the following statements is *not* correct?
   P. Cytoskeleton is present in eukaryotic cells and absent in prokaryotic cells.
   Q. Streaming, endocytosis and exocytosis occur only in eukaryotic cells.
   R. Circular DNA in cytosol occurs only in prokaryotic cells.
   S. First eukaryotic cell on earth born more than 3 billion years ago.
   a. P
   b. Q
   c. R
   d. S

3. Which one of the following arrangement of the metal ions Na^+, K^+, Mg^{2+} and Ca^{2+} in the order of decreasing concentrations is correct with respect to a quiescent mammalian cell?
   a. K^+, Na^+, Mg^{2+}, Ca^{2+}
   b. Na^+, K^+, Mg^{2+}, Ca^{2+}
   c. Mg^{2+}, K^+, Na^+, Ca^{2+}
   d. Ca^{2+}, Mg^{2+}, K^+, Na^+ **[IISc]**

4. Which of the following is *correct* statement? In animal
   a. concentration of sodium is high inside the cell than outside.
   b. concentration of calcium is high inside the cell than outside.
   c. concentration of potassium is high inside the cell than outside.
   d. concentration of magnesium is high inside the cell than outside.

5. You have homogenized plant tissue and would like to separate chloroplasts from nuclei. Which of the following methods would be most suitable?
   a. Polyacrylamide gel electrophoresis.
   b. Differential centrifugation using sucrose gradients.
   c. Equilibrium density gradient centrifugation on CsCl gradients.
   d. Gel filtration. **[JNU]**

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Plasma membrane : Structure and transport

1. In the fluid-mosaic model of the membrane
   a. the protein is arranged in layers.
   b. the lipid has no specific arrangement.
   c. the lipid is fluid and arranged in a bilayer with functional protein embedded in them.
   d. lipids and proteins are not arranged in any particular order. [IISc]

2. According to Robertson's unit membrane model of plasma membrane
   a. proteins on cytoplasmic and non-cytoplasmic sides are same.
   b. all proteins are transmembrane proteins.
   c. there is no space between lipid bilayers.
   d. none.

3. The primary thermodynamic factor that favours the formation of a lipid bilayer in aqueous surroundings is
   a. hydrogen bonds
   b. hydrophobic force
   c. ionic interactions
   d. covalent bonds [IISc]

4. In biological membrane, integral proteins and lipids interact mainly by
   a. covalent bond
   b. H-bond
   c. hydrophobic interactions
   d. van der Waals force

5. What property of biomembranes is responsible for their self-sealing nature?
   a. Hydrophilicity of the phospholipid head groups.
   b. Presence of protein in biomembranes.
   c. Presence of cholesterol in biomembranes.
   d. Hydrophobicity of the fatty acid side chains of phospholipids. [GATE]

6. Most abundant lipid in plasma membrane is
   a. cholesterol
   b. sterol
   c. glycolipid
   d. phospholipids

7. Which one of the following is *not* a membrane lipid?
   a. Cholesterol
   b. Phosphoglyceride
   c. Cerebroside
   d. Choline

8. Which of the following lipids is commonly found in biological membranes?
   a. Monoglycerides
   b. Diglycerides
   c. Triglycerides
   d. None of the above [IISc]

9. Lipids that are found in biological membranes
   a. are amphipathic
   b. are commonly referred to as triacylglycerols
   c. contain only unsaturated fatty acyl chains
   d. are normally covalently associated with proteins

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10. A poikilothermic organism living in the Arctic would have, compared to that organism living in the temperate climate zone, a plasma membrane richer in
    a. cholesterol
    b. long chain fatty acids
    c. protein
    d. unsaturated fatty acids

11. In plasma membrane, carbohydrate is present on the
    a. both layers of lipid
    b. only on cytoplasmic side of lipid bilayer
    c. only on non-cytoplasmic side of lipid bilayer
    d. none of the above

12. Which of the following statements is *true*?
    a. Membranes consist of only cholesterol and proteins.
    b. All membrane proteins are glycoproteins.
    c. Phospholipids and proteins are the main components of biological membranes.
    d. All membranes have the same lipid to protein ratio.

13. The following statement about biological membrane is *correct*
    a. they are symmetric bileaflet structure.
    b. their constituent lipid and protein moieties are held together by covalent interaction.
    c. they are rigid assemblies of protein, lipid and carbohydrates.
    d. their lipid moieties are amphipathic in nature.

14. A protein which spans the lipid bilayer
    a. cannot diffuse in the plane of the membrane.
    b. cannot have any attachment to cytoplasmic components.
    c. cannot have bound carbohydrate.
    d. usually has both hydrophobic and hydrophilic regions. [JNU]

15. Which of the following are *true* about proteins that span biological membranes?
    P. Proteins that span biological membranes often contain α-helices.
    Q. The amino acids would be hydrophobic in nature.
    R. An α-helix is especially suited to cross a membrane because all of the amide hydrogen atoms and carbonyl oxygen atoms of the peptide backbone take part in intrachain hydrogen bonds, thus stabilizing these polar atoms in a hydrophobic environment.
    a. P and Q
    b. P and R
    c. P, R and Q
    d. R and Q

16. Proteins that are embedded in or reach across a membrane are characterized by
    P. a stretch of at least 16 to 21 hydrophobic amino acids in their primary structure.
    Q. a high lysine, arginine and histidine content.
    R. a high glutamate and aspartate content.
    S. at least one α-helical or β-sheet domain.
    a. P and Q
    b. P and R
    c. P and S
    d. P, Q, R and S

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17. Which statement is *false*?
    a. Phospholipids, glycolipids, and sphingomyelin are all amphipathic lipids found in natural membranes.
    b. Sterols are found in membranes of animal cells but not in the membranes of plants cells.
    c. Many membrane components are amphipathic.
    d. Integral proteins in natural membranes may span the whole membrane or may be anchored to the membrane by lipid anchors.

18. Carbohydrates present on the plasma membrane
    a. have structural role
    b. form channel
    c. act as carrier
    d. help in molecular recognization

19. Liposome is
    a. closed, solvent filled vesicles which have a single bilayer of fatty acid chain.
    b. closed, solvent filled vesicles which have a single layer of fatty acid chain.
    c. a structure made up of protein.
    d. both a and c.

20. Which of the following is a function of plasma membrane?
    a. Structural barrier and cell communication.
    b. Metabolic activities and cell adhesion.
    c. Mass flow regulation, active transport, diffusion, endocytosis and exocytosis.
    d. All of the above

21. Which of the following statements about biological membranes is *not* true?
    a. They contain carbohydrates that are covalently bound to proteins and lipids.
    b. They are large sheet-like structures with closed boundaries.
    c. They are symmetric because of the symmetric nature of lipid bilayers.
    d. They contain specific proteins that mediate their distinctive functions. **[GATE]**

22. Generally accepted features of biological membrane include all of the following *except*
    a. asymmetric arrangement of lipids.
    b. rapid diffusion of inorganic ions across lipid bilayer.
    c. lateral diffusion of lipids.
    d. lateral diffusion of integral and peripheral proteins.

23. Fluorescence Recovery After Photobleaching (FRAP)
    P. is typically used to measure the rotational motion of membrane lipids and proteins.
    Q. is typically used to measure lateral diffusion of membrane lipids and proteins.
    R. involves fluorescent labeling of cell-surface molecules.
    S. has demonstrated that lipids diffuse freely over short distances, but generally not over longer distances, in fibroblast membranes.
    a. P and Q
    b. P, Q and R
    c. Q, R and S
    d. R and S

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24. A patch-clamp device is used to
    a. measure the strength of an electrochemical gradient.
    b. study the properties of individual neurotransmitters.
    c. infuse different kinds of ions into of an axon.
    d. study the properties of individual membrane channel.

25. The distribution of transmembrane protein in the plane of a cell membrane can best be visualized by which of the following
    a. thin-section electron microscopy
    b. freeze-fracture electron microscopy
    c. scanning electron microscopy
    d. SDS gel electrophoresis

Membrane transport

1. The order of the predicted rate of diffusion across a lipid bilayer (highest to lowest) is
    a. toluene, galactose, phenylalanine, chloride ion
    b. chloride ion, phenylalanine, galactose, toluene
    c. galactose, phenylalanine, chloride ion, toluene
    d. toluene, phenylalanine, galactose, chloride ion

[GATE]

2. The correct decreasing order of permeability through a lipid bilayer of the molecules/ions isoleucine, tyrosine, O_2 and Na^+ is
    a. O_2 > Na^+ > Isoleucine, > Tyrosine
    b. O_2 > Isoleucine > Tyrosine > Na^+
    c. Isoleucine > Tyrosine > O_2 > Na^+
    d. Isoleucine > Tyrosine > Na^+ > O_2

[GATE]

3. All membrane processes, such as pumping and channeling of molecules are carried out by
    a. lipid
    b. carbohydrate
    c. nucleic acid
    d. protein

4. Which of the following statements about membrane transport proteins is *incorrect*?
    a. Carrier proteins are similar to enzymes in that they show saturation.
    b. Carrier proteins can facilitate both active and passive transport.
    c. Channel proteins can facilitate both active and passive transport.
    d. The Na^+/Glucose transport protein carries out secondary active transport.

5. Movement of substances across cell membrane is controlled by the
    a. size of permeating particles
    b. permeability of membrane
    c. membrane proteins
    d. all of the above

6. A membrane transport protein is said to be a carrier protein if
    a. it forms an open pore through which a molecule can diffuse.
    b. an electrochemical gradient is necessary for transport to occur.
    c. it only allows transport down a concentration gradient.
    d. it binds to the molecule and changes shape during transport.

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7. Select the *false* statement.
   a. Facilitated diffusion is faster than simple diffusion.
   b. Facilitated diffusion exhibits saturation kinetics.
   c. Facilitated diffusion is not selective.
   d. Facilitated diffusion can be inhibited by agents known to denature proteins.

8. Valinomycin, a cyclic peptide
   P. acts as ionophore
   Q. facilitates the transport of K^+
   R. acts as a carrier for K^+ transport
   S. facilitates passive transport of cation
   a. P and Q
   b. Q and S
   c. Q, R and S
   d. P, Q, R and S

9. In simple diffusion process, molecules cross the plasma membrane
   a. against concentration gradient
   b. along concentration gradient
   c. do not depend on concentration
   d. with the help of energy

10. A passive membrane transport protein
   a. will require a direct source of energy for the transport to occur.
   b. can only transport a molecule down a gradient.
   c. involves a carrier type transport mechanism only.
   d. can move a molecule up a gradient if a membrane potential exists.

11. Diffusion across the plasma membrane is more rapid if a substance is
   a. a protein
   b. hydrophilic
   c. high in its oil : water partition coefficient
   d. large and globular in shape

12. A difference between simple diffusion and facilitated transport is that facilitated transport
   a. is concentration dependent
   b. occurs across plasma membrane
   c. requires membrane proteins
   d. utilizes a substance moving with its concentration gradient

13. Erythrocyte glucose transporter specifically transports glucose down its concentration gradient and exhibits hyperbolic saturation kinetics. This is an example of
   a. active mediated transport
   b. passive mediated transport
   c. non-mediated transport
   d. group translocation

14. Primary active transport
   P. requires carrier proteins
   Q. transport solutes against concentration gradient
   R. coupled to ATP hydrolysis
   S. driven by movement of a cotransported ion down its gradient
   a. P, Q and R
   b. P and S
   c. R and S
   d. P, Q, R and S

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15. Free fatty acids enter cell by
a. passive diffusion
b. primary active transport
c. cannot enter into the cell
d. secondary active transport

16. Which of the following is *not* related to facilitated diffusion?
a. Solutes are moved by a change in shape in the carrier protein.
b. The process is passive.
c. The process requires energy input.
d. Both a and c.

17. The concentration of an electrically neutral substance within a certain type of blood cell is much higher than it is in the surrounding blood plasma, yet the substance continues to move into the cell. The process by which this substance moves into the cell is called
a. osmosis
b. simple diffusion
c. facilitated diffusion
d. active transport

18. When an ion or solute is moved against a concentration gradient using energy, the process is called
a. diffusion
b. transport
c. active transport
d. regulated diffusion [IISc]

19. Which one of the following is a *correct* statement for Na-K ATPase?
a. It gives out 3 Na-ions and takes in 2 K-ions.
b. It gives out 2 Na-ions and takes in 3 K-ions.
c. It gives out 3 Ca-ions and takes in 2 K-ions.
d. It gives out 3 Na-ions and takes in 2 Ca-ions. [JNU]

20. Which of the following effects of the steroid digitalis is observed after treatment of congestive heart failure?
a. decrease in cytosolic sodium levels
b. inhibition of Na^+- K^+-ATPase
c. decrease in the force of heart muscle contraction
d. stimulation of the plasma membrane ion pump

21. Na^+ and K^+ pump
P. is a heterodimer composed of an α-subunit and a glycoprotein β-subunit
Q. contains α-subunit which is responsible for ATP hydrolysis and ion pumping
R. contains multipass transmembrane α-subunit
S. is palytoxin sensitive
a. P and Q
b. Q, R and S
c. P, Q, R and S
d. P, Q and S

22. Transport across a membrane is said to be coupled when
a. two molecules are transported across the membrane in the same direction.
b. membrane transport is coupled to an energy source, such as ATP hydrolysis.
c. transport of one ion down its gradient provides the energy to transport another molecule against its gradient.
d. both the concentration gradient and membrane potential determine the rate of transport across the membrane.

23. You wish to design a new drug which will act as an ionophore to deliver Ca^{2+} across the nerve cell membrane. This drug would most likely be
a. hydrophobic on the outside and hydrophilic on the inside
b. insoluble in lipids
c. soluble in proteins [JNU]
d. smaller than 0.001 nm in diameter

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24. Which of the following are *correct* about transport ATPase?

P. P, F, and V type transport ATPase transport ions only.

Q. Na^+/K^+ pump is an example of P type transport ATPase.

R. All members of the ABC superfamily contain two cytosolic ATP-binding domains.

S. F and V-class ATPase do not form phosphoprotein intermediates and transport only protons.

a. P and R
b. Q, R and S
c. P, Q, R and S
d. P, Q and R

25. The process by which a cell secretes macro-molecules by fusing a vesicle to the plasma membrane is called

a. endocytosis
b. exocytosis
c. pinocytosis
d. phagocytosis [IISc]

26. Receptor mediated endocytosis from plasma membrane requires which one of the following coat proteins?

a. Clathrin
b. SNARE
c. Arrestin
d. Glycophorin [IISc]

27. Choose the *incorrect* statement?

a. LDL receptor is a single transmembrane glycoprotein.
b. LDL contains a large protein called apoB100.
c. LDL contains cholesterol in the form of cholesterol ester.
d. Uptake of LDL occurs through COP I coated receptor mediated endocytosis.

Membrane potential

1. The value of membrane potential in a nerve cell during resting stage

a. negative inside the cell with respect to outside
b. positive inside the cell with respect to outside
c. both side negative
d. both side positive

2. Membrane potentials inside and outside of the membrane are caused by the unequal distribution of

a. ions
b. fatty acids
c. glucose
d. amino acids

3. The action potential results from

a. decrease in negative charge inside the nerve fibre.
b. increase in positive charge outside the nerve fibre.
c. opening of voltage gated sodium channels.
d. activation of the sodium-potassium pump. [IISc]

4. The hyperpolarization of the cell membrane associated with the

a. activation of voltage-gated K^+ channels.
b. activation of the Na^+ leaky channel.
c. activation of the Ca^{2+} voltage gated channel.
d. activation of voltage-gated Na^+ channels.

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5. The changes in the electrical potential of a neuron that constitute the action potential occur in the following order
   a. depolarization → resting potential → hyperpolarization → resting potential.
   b. resting potential → depolarization → hyperpolarization → resting potential.
   c. resting potential → hyperpolarization → depolarization → resting potential.
   d. resting potential → hyperpolarization → resting potential → depolarization.

6. Nicotinic acetylcholine receptors
   P. expressed in muscle cells is a ligand-gated channel.
   Q. present on postsynaptic membrane.
   R. contain α-subunits that binds to the acetylcholine.
   S. allow passage of both Na^+ and K^+ ions in hydrated form.
   a. P, Q, R
   b. P, Q, R, S
   c. P and R only
   d. Q only

7. Which of the following is *false* statement?
   a. Binding of acetylcholine with nicotinic acetylcholine receptor produces a net depolarization of ~15 mV.
   b. Neurotransmitter receptors fall into two classes: ligand gated ion channels and GPCR.
   c. An action potential at a presynaptic axon terminal opens voltage-gated Ca^{2+} channels.
   d. Acetylcholine is hydrolyzed to acetate and choline by acetylcholinesterase in postsynaptic cells.

8. Which of the following statements are *correct*?
   P. Axons with larger diameters conduct impulses at higher speeds than axons with smaller diameters do.
   Q. According to the all-or-none principle, if a stimulus is strong enough to generate an action potential, the impulse generated is of a constant size. A stronger stimulus does not generate a large action potential.
   R. A graded potential is a small deviation from the resting membrane potential that occurs because ligand-gated or mechanically gated channels open or close.
   S. The resting membrane potential is determined by two major factors– unequal distribution of ions in the extracellular fluid and cytosol and selective movement of ions along concentration.
   a. P and Q
   b. P and R
   c. P, R and S
   d. P, Q, R and S

9. Which of the following statements are *correct* about synapses?
   P. A chemical synapse produces one way information transfer from a presynaptic to a postsynaptic neuron.
   Q. An excitatory neurotransmitter is one that can depolarize the postsynaptic neuron's membrane, bringing the membrane potential closer to threshold. An inhibitory neurotransmitter hyperpolarizes the membrane of the postsynaptic neuron, moving it further from threshold.
   R. There are two major types of neurotransmitter receptors: ionotropic and metabotropic receptors.
   S. The postsynaptic neuron is an integrator. It receives excitatory and inhibitory signals, integrates them, and then responds accordingly.
   a. P and Q
   b. P and R
   c. P, R and S
   d. P, Q, R and S

10. Fibroblasts and most other non-neuronal cells exhibit it an inside-negative electric potential. However, when they are depolarized, fibroblasts do not produce an action potential even though the concentrations of Na^+ and K^+ inside and outside fibroblasts are identical to those associated with neurons. Fibroblasts do not generate an action potential because
    a. of absence of voltage gated ion-channel.
    b. there is no Na^+-K^+ pump.
    c. their membranes have large number of *porin*.
    d. their voltage gated Na^+ ion channels work at very high rate.

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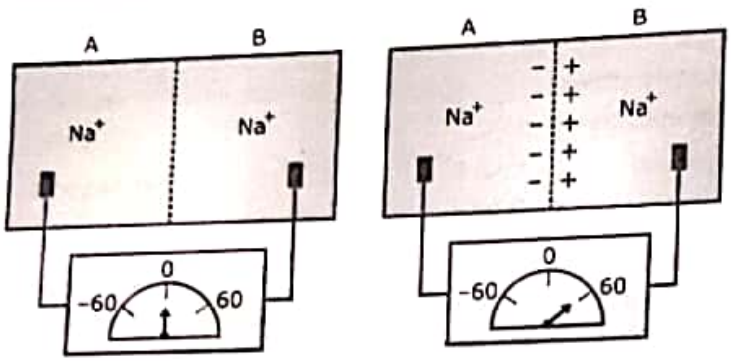
11. The diagrams depict rigid containers composed of two aqueous chambers, A and B, each containing a Na^+ solution and separated by a Na^+-permeable membrane. The panel on the left represents the distribution of Na^+ ions at rest in the absence of any electrical potential. In this scenario, the concentration of Na^+ ions in chamber A equals the concentration of Na^+ ions in chamber B ($[Na]_A = [Na]_B$). The panel on the right illustrates the effect of a +60-millivolt potential applied across the membrane (chamber B relative to chamber A). Assuming a temperature of 37°C, which of the following expressions best describes the resulting distribution of Na^+ ions between the two chambers?



a. $[Na]_A = 10\ [Na]_B$
b. $[Na]_A = 2\ [Na]_B$
c. $[Na]_A = 60\ [Na]_B$
d. $[Na]_B = 10\ [Na]_A$

Endoplasmic reticulum

1. Which of the following statement is *correct* for the endoplasmic reticulum (ER)?
   a. The SER is the site for the synthesis, modifications and sorting of proteins.
   b. The RER is the site for lipid biosynthesis.
   c. The RER is the site for the modifications and sorting of proteins.
   d. The RER is the site for protein synthesis and SER is involved in modifications and sorting of proteins.

[JNU]

2. The endoplasmic reticulum
   a. extends from the plasma membrane into the cytoplasm, thereby creating natural entrances utilized in endocytosis.
   b. extends from the inner nuclear membrane into the nucleoplasm, thereby creating a space that is called the nucleolus.
   c. extends from the outer nuclear membrane into the cytoplasm, thereby creating a space (lumen) that is separated from the cytosol.
   d. extends from the outer nuclear membrane into the cytoplasm where it forms the Golgi apparatus.

3. Membranes of the following two organelles are contiguous
   a. ER and Golgi
   b. nucleus and ER
   c. Golgi and plasma membrane
   d. Golgi and lysosomes

[IISc]

4. Which of the following statements is *true* for smooth endoplasmic reticulum?
   a. It is made up of a single lipid layer membrane structure.
   b. It is involved in protein biosynthesis.
   c. It is involved in lipid biosynthesis.
   d. It is a site for interleukin-2 biosynthesis.

[IIT]

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5. Ca^{2+} ATPase, present on sarcoplasmic reticulum, increases Ca^{2+} concentration in
   a. sarcoplasmic reticulum as compared to cytosol actively.
   b. cytosol as compared to sarcoplasmic reticulum.
   c. sacroplasmic reticulum as compared to cytosol passively.
   d. none of these

6. Which of the following are functions of RER?
   P. Splicing of polypeptide.
   Q. Formation of disulfide bond.
   R. Glycosylation of proteins during translation.
   S. Assembly of multisubunit proteins.
   a. P and Q
   b. R and S
   c. P, Q and R
   d. P, Q, R and S

7. A cell, which is very active in the synthesis and secretion of proteins, would be expected to have
   a. equal amount of SER and RER
   b. more rough than SER
   c. more smooth than RER
   d. neither SER nor RER

8. Microsomal cytochrome P450
   P. are monooxygenase
   Q. are membrane bound xenobiotic metabolizing enzymes
   R. are hemoproteins
   S. catalyzes diverse reaction such as hydroxylation, sulfoxidation, epoxidation
   a. P and Q
   b. R and S
   c. P, Q and R
   d. P, Q, R and S

9. The common precursor for synthesis of all N-linked oligosaccharide residues in glycoproteins contains a branched oligosaccharide linked to
   a. KDEL sequence
   b. GPI-anchor
   c. chaperones
   d. dolichol pyrophosphate

10. The oligosaccharide part of glycoproteins is either N-linked or O-linked. The amino acid residues through which these oligosaccharides are attached to polypeptide are
   a. Asn and Thr
   b. Asp and Ser
   c. Ser and Lys
   d. Gln and Ser [IISC]

11. Which of the following is *not* function of RER?
   a. N-linked glycosylation of proteins
   b. Folding of polypeptide chains
   c. O-linked glycosylation of proteins
   d. Specific proteolytic cleavage

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12. Which statements about the glycosylation of secreted proteins are *true*?

P. Almost all secreted proteins are glycosylated.

Q. Glycosylation occurs exclusively on the NH_2 group of asparagine.

R. An oligosaccharide is formed on the lipid dolichol and then transferred to the protein.

S. The oligosaccharide is trimmed by the removal of mannose residues.

a. P, Q
b. P and R
c. P, R and S
d. P, Q, R and S

13. Which one of the following statement is *not* true about proteins glycosylation?

a. Protein glycosylation is one of the major biosynthetic functions of SER.
b. It is a process of covalent addition of sugars to secretory proteins.
c. An oligosaccharide containing 14 sugars is transferred to the NH_2 group of asparagines residue of a protein.
d. Antibiotic tunicamycin inhibits the first step of N-linked glycosylation.

14. During N-glycosylation oligosaccharide is added to asparagine residue in the consensus sequence

a. Asn-X-Ser/Thr where, X is any amino acid except proline.
b. Tyr-Tyr-Leu-Leu
c. Tyr-X-X-Ser where, X is bulky hydrophobic amino acid.
d. Leu-Leu-Phe

Golgi complex

1. Golgi complex plays a major role in

a. protein synthesis.
b. glycosylation of lipids and proteins.
c. removal of sulfate from the carbohydrate moiety of glycolipids.
d. formation of secondary lysosomes.

2. The major functions/features of the Golgi apparatus in eukaryotic cells are

a. it carries out glycosylation of the proteins being transported.
b. it is the major protein sorting centre of the cell.
c. it forms secretory vesicles in its trans compartment.
d. all of the above.

3. [GATE]

The presence of abundant Golgi vesicles near cell plate in the middle of a plant cell at the late telophase and cytokinesis of mitosis reflects the role of Golgi complex in

a. lysosomal enzyme targeting
b. polysaccharide synthesis
c. protein packaging
d. uptake of membrane lipids

4. Choose the statement that is *not* correct for Golgi apparatus of the cell

a. degradation of xenobiotic compounds occur in Golgi apparatus.
b. Golgi apparatus has *cis* and *trans* faces.

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c. Golgi apparatus is predominant in cells that are specialized for secretion.
d. it is involved in oligosaccharide modification.

5. Which statement is *false*? [IIT]
a. The Golgi complex forms vesicles that fuse to form the endoplasmic reticulum.
b. If a lysosome bursts, its contents can seriously damage the cytoplasm of a cell.
c. Secreted proteins are initially formed by ribosomes attached to the endoplasmic reticulum.
d. The nucleolus is where ribosomes are assembled.

6. Which one of the following is *not* a function of the Golgi apparatus?
a. Protein synthesized by the ribosomes on the RER are transferred to Golgi, where it is accumulated in the sacs. These sacs may migrate to the surface of cell and discharge their content outside.
b. Protein filled sac may be retained within the cell as lysosome.
c. It is the site where glycosylation of lipid takes place.
d. It is the site where synthesis of phospholipid takes place.

7. Which of the following statements are *correct*?
P. Golgi apparatus is involved in protein sorting.
Q. Golgi body is involved in the formation of primary lysosome.
R. In Golgi body glycosylation of lipids occur.
S. Golgi apparatus is structurally and biochemically polarized.
a. Q, R and S
b. P, R and S
c. P, Q, R and S
d. P, Q and S

8. The Golgi complex
P. contains a series of stacked internal cisternae.
Q. is a polar organelle with its *cis* face close to the ER and its *trans* face close to the plasma membrane.
R. contains the same enzymes in all its parts.
S. is the site of further modification of preformed oligosaccharides to make complex oligosaccharides.
a. P and Q
b. P and R
c. P, Q and S
d. P, Q, R and S

9. Pick the *correct* statement.
a. In O-linked glycosylation, sugars are attached to the protein via O-glycosidic bonds to the carboxyl groups of Asp and Glu.
b. In O-linked glycosylation performed oligosaccharides are attached to the relevant protein.
c. In O-linked glycosylation, N-acetylgalactosamine is added via O-glycosidic bonds to the OH groups of Ser and Thr after which other sugars are added sequentially.
d. O-linked glycosylation is inhibited by the passage of the newly synthesized protein through the Golgi complex. [GATE]

10. Marker enzyme of Golgi apparatus is
a. acetyl-CoA synthetase
b. pyruvate kinase
c. galactosyl transferase
d. cytochrome oxidase

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11. You would expect a cell with an extensive Golgi apparatus to
    a. synthesize large amounts of ATP
    b. secrete large amounts of proteins
    c. synthesize large amounts of steroids
    d. synthesize excess phospholipids [IISc]

Lysosome

1. Which one of the following organelles is enclosed by a single membrane?
    a. mitochondria
    b. nucleus
    c. chloroplast
    d. lysosome [IISc]
2. If a subcellular fraction from liver tissue exhibits high level of acid phosphates activity, it is enriched in
    a. nuclei
    b. lysosomes
    c. mitochondria
    d. Golgi bodies [IISc]
3. If cells are broken up and sedimented by centrifugation, the new structure formed is
    a. centrosome
    b. lysosome
    c. microsome
    d. ribosomes
4. Which of the following is a *false* statement?
    a. All lysosomal enzymes collectively known as acid hydrolases.
    b. Lysosomes are involved in both intracellular and extracellular digestion.
    c. Lysosomes bring out digestion of both intracellular and extracellular materials.
    d. All lysosomal enzymes work efficiently at acidic pH.
5. A distinctive features of the lysosome is that it has
    a. a lower pH than the cytoplasm
    b. a reduced hydrolase activity
    c. DNA
    d. ribosome
6. Which of the following is *correct* statement?
    a. Lysosomes originate from Golgi apparatus.
    b. Lysosomal enzymes are synthesized in ribosome present on ER.
    c. Lysosomes are polymorphic and heterogenous organelles.
    d. All are correct statements.
7. The iron containing protein, transferrin, is taken up by fibroblast cells in culture. In what order will the iron atoms of this molecule follow the heterophagy pathway?
    P. Coated vesicle at cell surface.
    Q. Endosome vesicle.
    R. Primary lysosome.
    S. Residual body.
    T. Secondary lysosome.
    a. P, Q, R, T, S
    b. P, R, S, T
    c. P, R, T, S
    d. P, Q, T, R

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8. Which of the following enzymes do *not* occur in the lysosomes?
   a. Phosphatase
   b. Lipase
   c. Proteases
   d. Polymerase

9. A marker for an enzyme to go from the Golgi to the lysosome is
   a. acidic pH
   b. signal peptide
   c. phosphomannose
   d. glucosamine

10. The mitochondria serve as a marker for cytochrome oxidase and the lysosomes serve for
   a. succinic dehydrogenase
   b. catalase
   c. acid phosphatase
   d. galactosidase

11. Vacuoles present in plant cells are analogous to
   a. Golgi bodies
   b. spherosome
   c. lysosome
   d. hydrogenosome

12. Plant vacuoles
   P. have acidic pH
   Q. maintain turgor pressure
   R. perform autophagy and solute accumulation
   S. perform heterophagy
   a. P and Q
   b. R and S
   c. P, Q and R
   d. P, Q, R and S

13. In autophagy
   a. a single lysosome surrounds a single cell organelle
   b. cellular constituents are recycled
   c. internal membranes can be synthesized
   d. mitochondria are assembled

Intracellular trafficking

1. On an active ribosome, the polypeptide chain is synthesized
   a. from N-terminus to C-terminus
   b. from C-terminus to N-terminus
   c. in variable direction depending on protein
   d. from 5′ end to 3′ end

2. Which of the following is *false* statement?
   a. Svedberg unit are not additive.
   b. The greater the mass, the faster the rate of sedimentation.
   c. The greater the mass, higher the S-value.
   d. The ribosomes of a given cell compartment are different.

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3. Which of the following is the *correct* statement?

P. Free ribosomes and membrane bound ribosomes are identical.

Q. S-S bond formation occurs in the ER-lumen.

R. When cells are disrupted by homogenization, the endoplasmic reticulum is fragmented into many small closed vesicles called microsomes.

a. P and Q
b. Q and R
c. P and R
d. P, Q and R

4. Which of the following is *not* a property of the mammalian signal recognition particle?

a. It targets nascent secretory polypeptides to the rough endoplasmic reticulum.
b. It temporarily arrests translation.
c. It contains both RNA and polypeptides.
d. It contains a signal peptidase activity.

5. Which of the following is *incorrect* about signal recognition particle?

a. It is ribonucleoprotein
b. It is made up of six polypeptides
c. It contains 7S L scRNA
d. It acts as molecular chaperone

6. During the synthesis of secretory proteins the following events happen

1. the signal recognition particle (SRP) binds to the signal sequence
2. the two parts of a cytosolic ribosome binds to the mRNA
3. protein synthesis stops
4. the signal sequence is translated
5. the ribosome and protein are transferred to a translocon
6. the secretory protein folds
7. the signal sequence is cleaved by a signal peptidase
8. the SRP binds to its receptor in the ER membrane

The *correct* order for these events is

a. 4, 1, 2, 3, 6, 8, 7, 5
b. 2, 4, 1, 3, 8, 5, 7, 6
c. 1, 4, 2, 3, 6, 5, 8, 7
d. 2, 4, 1, 3, 5, 8, 7, 6

7. An mRNA coding for a secretory protein, when translated using free ribosome under *in vitro* conditions, resulted in a 40 kDa protein. The same mRNA when translated using the rough endoplasmic reticulum resulted in a 36 kDa protein. The difference in the molecular weight of the two polypeptides is due to the loss of a

a. 2 kDa peptide from N-terminus and a 2 kDa peptide from the C-terminus.
b. 1 kDa peptide from N-terminus and a 3 kDa peptide from the C-terminus.
c. 4 kDa peptide from the N-terminus.
d. 4 kDa peptide from the C-terminus.

8. Which of the following show *correct* order of the secretory pathway?

a. RER → Golgi → Secretory vesicle → Cell exterior
b. SER → Golgi → Secretory vesicle → Cell exterior
c. Golgi → SER → Secretory vesicle → Cell exterior
d. Golgi → Lysosome → SER → Secretory vesicle → Cell exterior

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9. Plant cell walls contain a proline rich protein. A plant cell is pulsed with radioactive proline and then chased in non-radioactive proline. After various periods of chase, cells are fixed and prepared for autoradiography. The labeling of cell compartments is followed

P. Cell wall
Q. ER
R. Golgi
S. Ribosomes

If the sequence of events in the plant cell is the same as that of animal cell, the order of labeling of cell compartment will be

a. P, Q, R, S
b. Q, P, R, S
c. R, S, Q, P
d. S, Q, R, P

10. Choose the *correct* statement.

P. Signal sequence present near N-terminus of polypeptide is rich in hydrophobic amino acids.
Q. BiP is a chaperone present in the lumen of RER.
R. COPI coated vesicles transport proteins in the retrograde directions from Golgi to RER.
S. Mannose-6-phosphate receptors in the membrane of the trans-Golgi bind proteins bearing mannose-6-phosphate and direct their transfer to lysosome.

a. P and Q
b. P and R
c. P and S
d. P, Q, R and S

11. Proteins are transported around the cell in membranous vesicles. These vesicles·

P. form when a section of membrane protrudes and buds off
Q. have a layer of coat protein around the inside of the vesicle
R. use their transmembrane protein called v-SNARE to find their target membrane
S. move with the help of motor proteins

a. P and Q
b. P and R
c. P, R and S
d. P, Q, R and S

12. Golgi-associated vesicles transport

a. ER resident proteins back to the ER.
b. Golgi resident proteins in both retrograde and anterograde directions.
c. cargo proteins in an anterograde direction.
d. all of the above statements are correct.

13. Which of the following vesicles does *not* have a clathrin coat when it is first formed?

a. Transport vesicles that forms from the endoplasmic reticulum.
b. Vesicles formed from coated pits on the plasma membrane.
c. Transport vesicles that takes lysosomal enzymes from the trans-Golgi reticulum.
d. Vesicles that take lysosomal enzymes to the sorting vesicle (late endosome).

14. Which of the following statement(s) is/are *false* about transport vesicles?

P. COPII vesicles transport proteins from the rough ER to the Golgi.
Q. Clathrin coated vesicles transport proteins from the *trans*-Golgi to late endosomes.
R. COPII vesicles transport proteins between Golgi cisternae and from the *cis*-Golgi back to the ER.
S. GTPase that acts as a regulatory subunit to control coat assembly for both COPI and clathrin vesicles are *ARF*.

a. P and Q
b. only R
c. R and S
d. only Q

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15. Docking and fusion of transport vesicles with their target membranes involves

P. each v-SNARE in a vesicular membrane binds to a cognate t-SNARE in the target membrane.

Q. SNARE complex is disassembled in an ATP-dependent reaction.

R. GTP-binding proteins, the Rab proteins, regulate docking of vesicles with the correct target membrane.

S. NSF-catalyzed hydrolysis of ATP then drives dissociation of the SNARE complexes.

a. P and Q
b. P and R
c. Q, R and S
d. P, Q, R and S

16. In which of the following does the inner surface of a closed membrane or vesicle become the outer surface of a closed membrane or vesicle

a. fusion of two intracellular vesicles.
b. transfer of an endoplasmic reticular membrane into Golgi membrane via vesicle.
c. exocytosis of a secretory vesicle.
d. division of a bacterial cell. [GATE]

17. Which of the following statements is *false*?

a. The SRP system translocates folded proteins into the ER.
b. Post-translational translocation of unfolded proteins into the inner mitochondrial space requires a bipartite signal sequence.
c. Nuclear export involves the movement of folded proteins through the nuclear pore complex.
d. In nuclear import, Ran-GEF helps release cargo proteins in the nucleus by inducing a conformational change that leads to release of importins from the nuclear localization signal. [JNU]

18. Choose the *correct* statements.

P. The perinuclear space is continuous with the lumen of the ER.

Q. At a nuclear pore the inner and outer nuclear membranes are fused with one another.

R. Monomeric G protein Ran participate in transport through nuclear pore.

S. Nuclear localization signal is located internally and rich in lysine and arginine.

a. P and Q
b. R and S
c. P and S
d. P, Q, R and S

19. Choose the *correct* statements.

P. Proteins are imported into peroxisomes in their fully folded state.

Q. Transport of proteins to peroxisomes occurs post-translationally.

R. Proteins are imported into peroxisomes in their fully unfolded state.

S. SKL is a tripeptide at the C-terminus acts as PTS.

a. P and Q
b. R and S
c. P, Q and S
d. P, Q, R and S

20. Choose the *incorrect* statement about mitochondrial protein targeting.

P. Matrix-targeting sequences located at the N-terminus.

Q. Matrix-targeting sequences rich in negatively charged acidic residues.

R. Matrix-targeting sequences rich in hydrophobic and positively charged basic amino acids.

S. Translocation into matrix is driven by the proton-motive force across the inner membrane and ATP hydrolysis by the Hsc70 ATPase in the matrix.

a. P and Q
b. Q only
c. P and S
d. R and S

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21. Twin membrane proteins vSNARE and tSNARE act as anchors when two vesicles fuse into one during molecular transport in the cell. These snare proteins are found in membranes of all except
a. Mitochondria
b. Golgi complex
c. Early endosome
d. Endoplasmic reticulum

22. A signal sequence KDEL is removed from an ER resident protein. Assuming that there is no change in tertiary structure of such plant protein and no other signal sequences present in protein, the changed protein will now have following fate
a. It will remain in ER and be degraded.
b. It will be targeted to Golgi apparatus.
c. It will be secreted outside the cell.
d. It will be targeted to lysosome for degradation.

23. Insulin is synthesized as a pre-pro-protein in the β cells of the pancreas and subsequently cleaved to its mature form. Its pre-peptide is cleaved off after it enters the ER lumen. To define the cellular location at which its pro-peptide is removed, antibodies specific for pro-insulin, and one that is specific for insulin was used. The experimental strategy involved using red and green fluorophores to tag the pro-insulin, and insulin antibodies respectively to double stain the cells and accordingly the following results were obtained:

Endoplasmic reticulum	Red
Mature secretory vesicles	Green
Immature secretory vesicles	Yellow
Golgi cisternae	Red

Which of the following cellular compartments is the site of pro-peptide cleavage?
a. Endoplasmic reticulum
b. Mature secretory vesicles
c. Immature secretory vesicles
d. Golgi cisternae

Cytoskeleton and Motility

1. Which of the following cell components is *not* part of the cytoskeleton of eukaryotic cells?
a. Microfilaments
b. Mitochondria
c. Intermediate filaments
d. Microtubules [JNU]

2. Actin filaments and microtubules share all of the following properties *except*
a. they are involved in cell motility.
b. they are intrinsically polar structures.
c. they can associate with motor proteins.
d. they are assembled from subunits that are heterodimers.

3. Choose the statement that is *not* correct for the cytoskeletal protein actin
a. α-actins are found in various types of muscles.
b. polymerization of pure actin in vitro requires GTP.
c. actin filaments have a slow growing *minus* end and a fast growing *plus* end.
d. cytochalasins are the inhibitors for actin polymerization.

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4. A monoclonal antibody binds to G-actin but not to F-actin, what does this tell you about epitope recognized by the antibody?
   a. The epitope in F-actin induces an enzymatic activity inactivating mAb.
   b. The epitope in F-actin is surrounded by Myosin thus no mAb interaction is observed.
   c. The epitope is likely to be a structure that is buried when G-actin polymerizes to form F-actin.
   d. A peptide fragment released from F-actin blocks the mAb interaction with the epitope.

5. Actin filaments are involved in all of the following *except*
   a. ameboid movement
   b. cytoplasmic streaming
   c. contraction of smooth muscles
   d. flagellar movement in bacteria [IISc]

6. Which of the following is/are *true* about myosin?
   P. Myosin has an actin-binding site.
   Q. Myosin has a GTPase site.
   R. Myosin has an ATPase site.
   S. Myosin walks towards the minus-end of the microtubules.
   a. P only
   b. P and Q
   c. Q and S
   d. P and R

7. Which of the following statement about myosin is *not* true?
   a. Myosin binds to polymerized actin.
   b. *In vitro*, myosin assembles spontaneously into the thin filaments.
   c. Myosin is an ATPase.
   d. Myosin has domains that interact with one another to effect its physiological functions.

8. In muscle contraction, what starts filament sliding?
   a. An increase in Ca^{2+} concentration in the cytosol.
   b. A decrease in Na^{+} concentration in the cytosol.
   c. The bending of the myosin head.
   d. The release of energy from the breakdown of ATP.

9. What would happen if ATP were suddenly unavailable after the sarcomere had begun to shorten?
   a. Contraction would proceed normally.
   b. The myosin heads would be unable to detach from actin.
   c. Troponin would bind with the myosin heads.
   d. Actin and myosin filaments would separate completely and be unable to recombine.

10. When the length of a skeletal muscle decreases during an active contraction?
    a. The A band remains the same size.
    b. The H band becomes larger.
    c. Only can bind to sites on actin molecules.
    d. All of the above.

11. Which of the following statements relating to microtubules is *not* correct?
    a. The plus-end of microtubule is the fast-growing end.
    b. Addition of short fragments of microtubules enhances polymerization.

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c. A microtubule with GDP-cap enters the shrinkage phase (catastrophe)
d. Critical concentration for polymerization is same for both plus- and minus-end.

12. Which of the following statement is *true* for skeletal muscle?
P. Actin molecules are located in the thick filaments.
Q. Acetylcholine receptors are located on the surface of the sarcoplasmic reticulum.
R. The heads of the myosin molecules attach onto the Z line.
S. Calcium ions bind to the tropoin molecule.
a. P and Q
b. R and S
c. only S
d. P, Q and R

13. Treadmilling of actin filaments refers to
a. net assembly at both plus- and minus-ends.
b. net assembly at plus-end and net disassembly at minus-end.
c. net disassembly at plus-end and net assembly at minus-end.
d. net disassembly at both plus- and minus-end.

14. The action potential of skeletal muscle
a. has a prolonged plateau phase.
b. spreads inward to all parts of the muscle via the T tubules.
c. causes the immediate uptake of Ca^{2+} into the lateral sacs of the sarcoplasmic reticulum.
d. is longer than the action potential of cardiac muscle.

15. The function of tropomyosin in skeletal muscle include
a. sliding on action to produce shortening.
b. releasing Ca^{2+} after initiation of contraction.
c. binding to myosin during contraction.
d. acting as a relaxing protein at rest by covering up the sites where myosin binds to actin.

16. The role of calcium in muscle contraction is
a. to break the cross-bridges as a cofactor in the hydrolysis of ATP.
b. to bind with troponin, changing its shape so that the actin filament is exposed.
c. to spread the action potential.
d. none of the above.

17. Order the following events observed during muscle contraction in the *correct* sequence
1. the sarcomere shortens
2. ATP is hydrolyzed
3. Ca^{2+} binds to troponin
4. ATP binds to myosin
5. myosin contracts F-actin
6. F-actin moves
7. tropomyosin moves
8. Ca^{2+} is released from the sarcoplasmic reticulum
a. 8-4-3-7-2-5-1-6
b. 4-8-7-2-3-5-6-1
c. 4-8-3-7-5-6-2-1
d. 4-8-3-7-2-5-6-1

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18. Free tubulin dimers contain two bound molecules of the nucleotide____; one of these, located on the ___tubulin subunit, is bound irreversibly and is not hydrolyzed after addition of a dimer to the end of a microtubule
    a. GTP, α
    b. ATP, α
    c. ATP, β
    d. GTP, β

19. Microtubules are *not* involved directly in which of the following processes?
    a. motion of whole cells via flagella
    b. movement of mitotic spindle
    c. amoeboid motion
    d. maintains reticular architecture of the ER

20. Plane of formation of cell plate in plant cell is governed by
    a. phragmoplast
    b. microtubules
    c. nucleus
    d. centriole

21. Which statement about microtubules is *false*?
    a. They form the spindle fibres involved in movement of chromosomes during mitosis and meiosis.
    b. They are involved in cell support and shape.
    c. They are involved in the movement of flagella.
    d. They are involved in the cyclosis.

22. Which of the following statement(s) is/are *correct*?
    P. The kinetochore is a specialized attachment site for microtubules located at the chromosome centromere; it binds to the plus end of microtubules.
    Q. A GDP cap at the end of a microtubule will cause it to depolymerize.
    R. The plus ends of flagellar microtubules point toward the basal body.
    S. Kinesin moves proteins from the minus to the plus end of microtubules.
    a. only P is correct
    b. P and R are correct
    c. P, R and S are correct
    d. P, Q and S are correct

23. Microtubule associated proteins (MAPs) are
    a. abundant only in mitochondria
    b. aggregated by colchicine
    c. involved in tubulin polymerization
    d. sensitive to cytochalasin B

24. Colchicines treated cells are arrested in
    a. S phase
    b. prophase
    c. G1 phase
    d. metaphase [IISc]

25. Colchicine is an inhibitory chemical, which
    a. prevents microtubule polymerization
    b. prevents microtubule depolymerization
    c. stops the functioning of centriole
    d. prevents attachment of spindle fibre with kinetochore

26. The core of cilium, which is composed of entirely of microtubules and their associated protein
    a. blepharoplast
    b. axoneme
    c. ciliary root let
    d. protofilament

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27. During the movement of an axoneme
a. the central pair of microtubules slide relative to each other
b. the outer ring of 9 microtubule pairs rotates around the central pair
c. movement is created by the dynamic instability of the microtubules
d. hydrolysis of ATP by dynein causes outer microtubules to slide

28. Which of the following statement is *correct*?
P. Flagella having 9+2 arrangement of microtubules.
Q. Centriole having 9+0 arrangement of microtubules.
R. Structure of kinetosome and centriole is same.
a. P and Q
b. P and R
c. Q and R
d. P, Q and R

29. The protein that is responsible for the sliding of outer microtubule doublets against one another to produce ciliary bending is
a. nexin
b. dynein
c. tubulin
d. none of these

30. Ciliary motion are generated by the sliding between
a. adjacent outer doublet microtubules
b. central pairs of microtubule and outer doublet microtubules
c. kinesin and outer doublet microtubules
d. dynein and central microtubules

31. Which of the following statement is *wrong*?
a. Motor protein kinesin and dynein transport membrane bound structures along microtubules.
b. Microtubule exhibit structural and functional polarity.
c. Microtubule-associated proteins (MAPs) organize microtubule and affect their stability.
d. MTOC (microtubule organizing center) has no role in determining the cell polarity.

32. Which of the following statement is *false*?
a. Pair of centriole present at right angle to each other.
b. Centriole has structural and functional polarity.
c. Centrioles are responsible for spindle fibres formation.
d. Kinetosome is homologous to the centriole.

33. Which of the following properties is common to all cytoskeletal motor proteins (such as kinesins, dyneins and myosins)?
a. An actin-binding domain
b. Two globular-head domains
c. The ability to bind to biological membranes
d. ATPase activity [JNU]

34. P. kinesin walks from (–) end to the (+) end
Q. dynein walks from (+) end to the (–) end
R. kinesin walks on microtubules
S. dynein walks on microtubules
Which of the choice(s) above are *true*?
a. P and Q
b. P only
c. Q and R
d. P, Q, R and S

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35. A cellular function, which is inhibited by colchicine and cytochalasin B together, but not separately, probably involves
    a. both microtubules and microfilaments
    b. intermediate filaments
    c. microtubules only
    d. no component of the cytoskeleton

36. Which of the following is *not* a characteristic of intermediate filaments?
    a. They form the nuclear lamina.
    b. They provide mechanical stability to animal cells.
    c. Their protein composition is tissue specific.
    d. They are composed of globular monomers that polymerize to form fibers.

37. You have referred one of your patients to an oncologist because of a breast tumor. The oncologist decides to treat the patient with a combination of 5-fluorouracil and taxol. The mechanism of action of the antitumor drug taxol involves
    a. Binding to intermediate filament dimers.
    b. Specific binding to actin filaments.
    c. Interference with the folic acid pathway.
    d. Stabilization of microtubules.

Extracellular matrix and Cell junctions

1. Plant cell walls contain
    a. glycosaminoglycan
    b. pectin
    c. proteoglycan
    d. peptidoglycan [JNU]

2. Which of the following is a cell adhesion molecule?
    a. integrin
    b. lysine
    c. myosin
    d. keratin [JNU]

3. Cadherin is
    a. Ca^{2+} dependent transmembrane glycoprotein.
    b. responsible for attachment of cell to extra cellular matrix.
    c. protein responsible for heterophilic interaction.
    d. structural component of gap-junction.

4. Which of the following statements about plasmodesmata is *correct*?
    a. Like gap junctions, directly connect the cytoplasms of adjacent cells.
    b. A new site of microtubule and microfilament assembly that appears during late telophase.
    c. The stack of four to six flattened sacs of Golgi apparatus.
    d. An interconnected network of fibrous proteins that is important in cell structure and for intracellular transport.

5. Which one of the following is found only in plants?
    a. Tight junctions
    b. Gap junctions
    c. Desmosomes
    d. Plasmodesmata [IISc]

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6. Desmosomes differ from tight junction because desmosomes
   a. allow molecules to pass in the intercellular space
   b. are non-communicating
   c. are present in plants
   d. lack proteins

7. Hemi desmosome differs from spot desmosome because hemi desmosome
   a. connect cell to cell
   b. connect extracellular matrix to extracellular matrix
   c. connect cell to extracellular matrix
   d. having tonofilament, made up of intermediate filament

8. Tight junctions
   a. are essential for metabolic coupling
   b. don't occur in vertebrates
   c. have the closest approach of two plasma membranes of any junction
   d. surround connexons

9. Which statement is *not* true for gap junctions? They
   a. allow the passage of proteins between cells.
   b. are essential for metabolic cooperation between cells.
   c. occur in the developing embryo.
   d. synchronize the beating of heart muscle cells.

10. In anchoring junctions, cadherins are linked to
    a. actin filaments in the cell's cytoskeleton.
    b. cell walls of adjacent cells in plants.
    c. connexons of its own and adjacent cells.
    d. extracellular matrices of adjacent cells.

11. The most important difference between gap junctions between animal cells and plasmodesmata in plants is that in plasmodesmata
    a. ionic coupling occurs
    b. two adjacent plasma membranes are fused
    c. metabolic cooperation occurs
    d. pore diameter is 1 mm [JNU]

Mitochondria and Chloroplast

1. Which of the following statements is *not* true for a chloroplast?
   a. It contains DNA as its genetic material.
   b. It produces ATP.
   c. It has an electron transport chain.
   d. It contains a transcription apparatus but no translational apparatus. [JNU]

2. Proteins made in the cytoplasm and then transported to the chloroplast
   a. are smaller upon initial translation than the final product.
   b. are made on cytoplasmic free ribosome, not on the ER.

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c. contain a hydrophobic signal cleaved at the chloroplast surface.
d. contain a general organelle signal rather than one for the plastid.

3. In chloroplast, photosynthetic reactions occur in
   a. thylakoid membrane
   b. thylakoid lumen
   c. stroma
   d. inner chloroplast membrane

4. In the presence of light, pH of lumen of thylakoid
   a. increases
   b. decreases
   c. remain same
   d. none

5. If the thylakoid are treated with detergent to solubilize lipids and then the proteins gently isolated, the following complexes will be observed, *except*
   a. ATP synthesis
   b. cytochrome oxidase
   c. light harvesting complex
   d. pheophytin

6. The endosymbiotic origin of plastids from bacteria suggested by all of the following *except*
   a. antibiotic sensitivity of plastid ribosomes
   b. circular DNA in plastids
   c. introns in plastid DNA genes
   d. ribosome size in plastids

7. Which of the following protein targeting mechanisms is common for mitochondria and chloroplast organelles?
   a. Co-translational
   b. Post-translational
   c. Co-translational and post-translational
   d. *de novo* protein synthesis [IISc]

8. Which is *incorrect* with regard to mitochondrial DNA?
   a. it is double-stranded
   b. it mutates 10 times more frequently than nuclear DNA
   c. each mitochondrion contains about 100 DNA molecules
   d. it has no introns

9. The reason that a mutation carried on human mitochondrial DNA is inherited through the mother only is that
   a. egg cells don't contain mitochondrial DNA
   b. maternal mitochondrial DNA is degraded in the zygote
   c. paternal mitochondrial DNA does not enter the egg
   d. sperm cells mitochondria lack DNA

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10. The outer mitochondrial membrane
    a. does not contain lipids
    b. is more permeable as compared to inner membrane due to presence of porin
    c. sets up a proton gradient
    d. swells and contracts readily

11. The place in the mitochondrion where pH is lowest during aerobic respiration is the
    a. matrix
    b. cytosol
    c. intermembrane space
    d. mitochondrial ribosome

12. Pumping of H^+ across the inner membrane, from the matrix space to inter membrane space in mitochondria cause
    a. lower pH in matrix than in cytosol
    b. higher pH in matrix than in cytosol
    c. both a and b
    d. none of the above

13. In the chemiosmotic hypothesis for energy generation in mitochondria
    a. electron transport carriers set up a proton gradient.
    b. a pH gradient drives the hydrolysis of ATP to ADP.
    c. ATP must be continually translocated into the mitochondria.
    d. electron transport carriers, use energy of moving electron in uphill direction to form ATP.

14. In ATP synthase, F_0 acts as
    a. H^+ channel
    b. Cl^- carrier
    c. electron carrier
    d. ATPase

15. F_1 ATPase, part of ATP synthase synthesize ATP when proton passes through it
    a. against their electrochemical proton gradient
    b. down their K^+ gradient
    c. down their electrochemical proton gradient
    d. all of the above

16. The DNA molecules within mitochondria and cholorplasts have
    a. only linear form
    b. covalently closed circular form
    c. DNA having covalently sealed end
    d. linear dsDNA with single chain breaks

17. Which statement about ATP synthesis is *false*?
    a. ATP is synthesized only in chloroplasts and mitochondria.
    b. ATP synthesis in the chloroplast occurs in the thylakoid region of this organelle.
    c. Proton motive force (proton gradient) drives the formation of ATP in mitochondria.
    d. ATP synthases are protein complexes that allow protons to cross membranes.

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18. All of the following processes occur in the mitochondria of mammalian cells *except*
    a. fatty acid biosynthesis
    b. protein synthesis
    c. DNA synthesis
    d. β-oxidation of fatty acids

19. Evidence that mitochondria originated from an endosymbiotic relationship between aerobic bacteria and ancestral eukaryotic cells includes all of the following *except*
    a. DNA in mitochondria is not packaged by histones.
    b. mitochondrial DNA is circular like prokaryotic DNA.
    c. protein synthesis in mitochondria is inhibited by antibiotics as in bacterial protein synthesis.
    d. ribosomal RNAs in mitochondria code for prokaryote-like ribosomal proteins.

20. Evidence indicating that chloroplasts were originally free-living prokaryotes that subsequently evolved a symbiotic relationship with a eukaryotic host includes all of the following *except*
    a. similarities in structure between chloroplasts and prokaryotes.
    b. presence of circular DNA in chloroplasts and prokaryotes.
    c. susceptibility of chloroplasts to prokaryotic protein synthesis inhibitors.
    d. ability of chloroplasts to synthesize all their own proteins.

Peroxisomes

1. Which enzyme catalyzes the reaction, $2H_2O_2 \longrightarrow 2H_2O + O_2$?
    a. Dehydrogenase
    b. Peroxidase
    c. Catalase
    d. Hydrolase [JNU]

2. The enzyme specific to glyoxylate cycle is
    a. isocitrate lyase
    b. succinate dehydrogenase
    c. isocitrate dehydrogenase
    d. aconitase

3. In mammalian cells, β-oxidation occurs in
    a. mitochondria
    b. peroxisome
    c. mitochondria, peroxisome and glyoxisome
    d. both a and b

4. Plant, but not animal, can convert fatty acids to sugars by a series of reaction known as
    a. glycolate cycle
    b. glycolic acid cycle
    c. glyoxylate cycle
    d. HMP pathway

5. Which of the following are *correct* about peroxisome?
    P. Present in all eukaryotes.
    Q. Divides similar to mitochondria.
    R. Peroxisomal proteins called *peroxins* are synthesized by membrane free ribosomes.
    S. Catalase acts as marker enzyme.
    a. P and Q
    b. P, Q and R
    c. P and R
    d. P, Q, R and S

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6. In peroxisome, during oxidation of fatty acids electrons and protons transferred to FAD and NAD^+. Reduced FAD finally transfer the electrons and protons to O_2 and forms

a. H_2O
b. H_2O_2
c. CH_3OH
d. none of these

7. Select the only *true* statement about peroxisomes, a cellular organelle involved in detoxification

a. peroxisomes possess their own DNA.
b. peroxisomes synthesize their own ribosomes.
c. all peroxisomal proteins are encoded by nuclear genes.
d. peroxisomes synthesize some of their own proteins. **[IISc]**

8. Which of the following are *correct* statements?

P. All proteins destined for the peroxisomal matrix bind to a cytosolic receptor.
Q. Zellweger syndrome is associated with impaired transport of proteins into the peroxisomal matrix.
R. Peroxisomal targeting sequence may be C-terminal PTS1 or N-terminal PTS2.
S. Peroxisomal targeting sequence is non-cleavable.

a. P and Q
b. P, Q and R
c. P and R
d. P, Q, R and S

9. Choose the mismatch

Organelle	Marker molecule
a. lysosomes	acid phosphatase
b. peroxisomes	catalase
c. mitochondria	cytochrome oxidase
d. SER	amino acid permease

[IISc]

Nucleus

1. Choose the *correct* statements about nucleus?

P. Nuclear pore complex is made up of about 50 to 100 different proteins called nucleoporins.
Q. Proteins found in the nucleus are imported into the nucleus contain a nuclear-localization signal.
R. Nucleus contains Ran-GTP while the cytosol contains Ran-GDP.
S. Transport of small water soluble proteins through the nuclear pore does not require energy.

a. P and Q
b. P, Q and R
c. P and R
d. P, Q, R and S

2. The nucleosome

a. contains DNA and non-histone proteins
b. has a core of histones with DNA wound around it
c. is fully responsible for DNA packaging into chromosomes
d. surrounds nuclear pores

3. The nucleosomes

a. contain about 500 base pairs of wrapped DNA
b. contain a core of histone proteins
c. are present in prokaryotes
d. can be constituted *in vitro* by mixing DNA and histone only **[JNU]**

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4. All of the following statements regarding nucleosomes are true *except*
   a. they are beaded structures, each with a histone core wrapped in a left-handed super-helix of DNA.
   b. the beads are linked by a DNA chain.
   c. each contains two molecules of histones H2A, H2B, H3 and H4.
   d. the primary structure of the histones is a characteristic of the cell type and the species.

5. Which of the following statement about histones is *false*?
   a. Histones are very similar between species.
   b. Histones have many basic amino acids.
   c. Histones are rich in lysine and arginine.
   d. Each histone has one single gene that codes for it.

6. All histones undergo post-translational modification of specific amino acids. These modifications
   P. alter the charge of the histone molecule
   Q. are permanent
   R. occur on the N-terminal tail of the histones, which are thought to extend out from the core
   S. occur at specific times during the cell cycle
   a. P, Q
   b. P, S
   c. P, R, S
   d. P, Q, R

7. Which of the following statement is *incorrect*?
   a. Histones are highly conserved protein.
   b. Histones are small basic proteins associated with acidic DNA.
   c. H1 is the linker histone.
   d. Nucleosome has one copy of each core histones.

8. Which of the following statement concerning characteristics of histones is *true*?
   a. They are acidic proteins found in the nucleus.
   b. They are covalently linked to single-stranded DNA.
   c. They are very highly conserved proteins.
   d. They are large polymeric proteins.

9. Protamines, present in fish sperm and involved in higher level of chromosomal organization are rich in which of the following amino acid?
   a. Lysine
   b. Methionine
   c. Arginine
   d. Tryptophan

10. Arginine and lysine are found in what form and it behaves in what way
    a. negative ion and alkaline
    b. positive ion and acidic
    c. negative ion and acidic
    d. positive ion and alkaline

11. Which of the following statement is *false*?
    a. DNase I hypersensitive sites are regions where the DNA is free of nucleosomes.
    b. when gene is active, the promoter is generally free of nucleosome.
    c. H1 phosphorylation occurs mainly before mitosis.
    d. DNA sequences called matrix or scaffold associated regions are generally CG-rich.

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12. The three DNA sequences, which define a chromosome, include all of the following *except*
    a. centromere
    b. enhancer
    c. origin of DNA replication
    d. telomere

13. NOR (Nucleolus Organizing Region) occurs in the region of
    a. secondary constriction
    b. primary constriction
    c. telomere
    d. centromere

14. Telomeres are present in eukaryotic genome at the chromosomal ends
    a. as selfish DNA
    b. to protect them from breakdown
    c. to encode essential genes involved in ageing
    d. to silence genes at the ends of chromosomes [JNU]

15. Which of the following statement is *false*?
    a. Telomeres contains regions with a high G content.
    b. Telomeres contain non-Watson-Crick base pairing.
    c. Telomeres contain short tandemly repeated sequence.
    d. None of the above.

16. *Correct* statements concerning heterochromatin include which of the following?
    P. It is not easily transcribed.
    Q. It occurs in the centromeres of chromosomes.
    R. It is associated with histone H1.
    S. It is relatively dispersed in chromosome and stains poorly.
    a. P and Q
    b. P, Q and R
    c. P and R
    d. P, Q, R and S

17. The common features of constitutive heterochromatin are
    P. it is permanently condensed
    Q. consists of repetitive sequences that are not transcribed
    R. it replicates late in S phase
    S. it has a reduced frequency of genetic recombination
    a. P and Q
    b. P, Q and S
    c. P and R
    d. P, Q, R and S

18. Which of the following are *correct* about polytene chromosome?
    P. Balbiani rings are present in polytene chromosome.
    Q. Cell containing polytene chromosome unable to undergo mitosis.
    R. Ecdysone, a moulting hormone induces puffing in interband region of polytene chromosome.
    S. Polytene chromosome in salivary glands of *Drosophila* show somatic pairing.
    a. P and Q
    b. P, Q and S
    c. P, Q and R
    d. P, Q, R, and S

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19. C-bands are deeply stained chromosomal regions which represents the
    a. euchromatin
    b. constitutive heterochromatin
    c. cytosine dominant region of chromosome
    d. metaphase chromosome

20. Dosage compensation describes mechanisms employed to compensate for the discrepancy between the presence of two X chromosomes in one sex but only one X chromosome in the other sex. The single X hypothesis describes the inactivation of one X chromosome in female mammals called Bar body. Choose the *correct* statement(s) about Barr body.
    P. Barr body in mammalian female is an example of facultative heterochromatin.
    Q. Two Barr bodies would be present in a human cell with 49, XXXYY karyotype.
    R. In *Drosophila*, Barr body present in female only.
    S. n-1 rule states that only one Barr body present in normal female mammalian cells.
    a. P and Q
    b. P and R
    c. P, Q and S
    d. P, Q, R and S

21. Choose the *correct* statement(s)
    P. *Xic* is a *cis*-acting locus
    Q. *Xic* contains a gene, called *Xist*, that is expressed only on the inactive X chromosome
    R. *Xist* gene codes for a noncoding RNA that is found only on inactive X chromosomes
    S. One of the two X chromosomes is inactivated at random in each cell during embryogenesis of eutherian mammals
    a. P and Q
    b. P and R
    c. P, Q and S
    d. P, Q, R and S

22. Choose the *correct* statement(s).
    P. Axial filament of the Lampbrush chromosome contains two chromatids.
    Q. Chromosomal puffs (Balbiani rings) are active sites of RNA synthesis.
    R. Nuclear proteins associated with the chromosomes are synthesized in the cytoplasm.
    S. Telocentric chromosomes during metaphase have very short or no visible *p* arms.
    a. P and Q
    b. P and R
    c. Q, R and S
    d. P, Q, R and S

23. Match the following

Column A	*Column B*
1. DNA found at the ends of linear eukaryotic chromosomes	A. centromere
2. protein structure on eukaryotic chromosomes where spindle fibers attach	B. telomere
3. DNA site where kinetochores are found	C. nucleolar organizer
4. DNA sequence that serves as the template for the synthesis and amplification of ribosomal DNA	D. kinetochore

    a. 1-A, 2-D, 3-B, 4-C
    b. 1-B, 2-D, 3-A, 4-C
    c. 1-D, 2-C, 3-A, 4-B
    d. 1-C, 2-D, 3-C, 4-B

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24. Which of the following statement are *correct* about nucleoli?
P. Nucleoli are membrane-bounded structures present in the eukaryotic nucleus.
Q. The DNA of nucleoli carries the cell's tRNA genes, which are present in clusters of multiple copies.
R. A single nucleolus always corresponds to a single nucleolus organizer region (NOR).
S. In animals and plants, the disappearance of nucleoli during mitosis correlates with cessation of ribosome synthesis.
a. P and Q
b. Q and R
c. R and S
d. P and S

25. Sex chromosome-based dosage compensation in human is brought about by
a. inactivity of one X-chromosome in females.
b. hyperactivity of single X-chromosome in males.
c. hypoactivity of both X-chromosome in females.
d. hyperactivity of autosomes in females.

26. Dosage compensation for X-chromosomes in mammals is achieved through
a. mRNA production level
b. heterochromatization
c. repressor action
d. gene deletions

27. Which among the following is a *correct* statement?
a. Human males have 23 different chromosomes whereas human females have 24 chromosomes.
b. Human males and females both have 46 different chromosomes.
c. Human females have 23 different chromosomes whereas human males have 24 chromosomes.
d. Human males and females both have 23 different chromosomes. **[JNU]**

28. Which of the following statements regarding mechanisms of dosage compensation among different species is *correct*?
P. In mammals, one of the X-chromosomes in the somatic cells of females is inactivated randomly.
Q. In marsupials, paternally derived X-chromosome is inactivated in somatic cells of females.
R. In *Drosophila*, the level of expression of genes on the X-chromosome in males is increased two fold.
a. P and Q
b. Q and R
c. P, Q and R
d. P and R

Cell signaling

1. Which of the following is *correct* statement?
P. Endocrine signaling molecules act on target cells distant from their site of synthesis.
Q. Paracrine signaling molecules affect target cells only in close proximity.
R. Autocrine signaling molecules affect the same cells from which they are released.
S. Synaptic signaling involves neurotransmitters.
a. P and Q
b. P and S
c. P, Q and R
d. P, Q, R and S

2. All of the following statements about monomeric G proteins are *true* except
a. they are regulated by GTP-GDP exchange proteins.
b. they are regulated by GTPase activating proteins.
c. they regulate vesicle fusion.
d. they help in uncoating of proteins from vesicle.

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3. The signaling system that operates over the shortest range is
   a. contact-dependent cell signaling, as in the inhibition of neighbors by differentiating nerve cells.
   b. paracrine secretion, as in the secretion of histamines by mast cells.
   c. neural signaling, as in the transmission of a nerve impulse.
   d. none of the above.

4. Signal transduction describes the ways in which external molecules cause internal changes in cells. These processes
   P. always involve the movement of ligands across the membrane.
   Q. amplify the strength of the external stimulus.
   R. rely on cell surface receptors that can interact with intracellular proteins.
   S. can trigger the production of a secondary messenger inside the cell.
   a. P, Q
   b. P, R
   c. Q, R, S
   d. P, Q, R, S

5. Which of the following biochemical reactions is most commonly utilized by living cells to propagate intracellular signals?
   a. Acylation
   b. Phosphorylation
   c. Methylation
   d. Decarboxylation [JNU]

6. Characteristics of intracellular receptors that regulate gene transcription include all of the following *except*
   a. a DNA binding site
   b. an extracellular binding site
   c. a transcription activating domain
   d. may be signaled by lipid soluble molecules

7. In eukaryotes, which of the following is *not* a second messenger?
   a. 1,2-diacyl glycerol
   b. cyclic AMP
   c. inositol 1,4,5-trisphosphate
   d. cyclins [JNU]

8. Which one of the following compounds *does not* act *secondary messenger* during signaling process?
   a. cAMP
   b. calcium ions
   c. inositol 1,4,5-trisphosphate
   d. triacylglycerol [IISc]

9. Cell surface receptors may be any of the following *except*
   a. G protein linked
   b. enzymatic receptors
   c. transcription factors
   d. ligand-gated ion channels

10. All of the following statements apply to G proteins *except*
   a. G proteins transmit a signal from the cell surface to the interior of the cell
   b. G proteins do not use second messengers but transmit the signal directly into the nucleus
   c. all G proteins have a similar structure
   d. G proteins act to amplify the signal creating a cascade response in the cell

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11. Which subunit of trimeric G-protein acts as GTPase switch protein subunit?
a. α b. β
c. γ d. $\beta\gamma$

12. G-proteins are involved in relaying signals through G-protein linked receptors. Which of the following forms of G-protein is considered to be in active state?
a. G-protein - ADP b. G-protein - ATP
c. G-protein - GDP d. G-protein - GTP

13. In the cAMP pathway, the G protein stimulates
a. phospholipase C b. adenylate
c. the endoplasmic reticulum d. calmodulin

14. Identify the *correct* order in which the extra-cellular signals are transmitted?
a. Adenylate cyclase—cAMP—protein kinase A
b. Protein kinase A—adenylate cyclase—cAMP
c. Protein kinase A—cAMP—adenylate cyclase
d. cAMP—adenylate cyclase—protein kinase A **[JNU]**

15. Match the biological functions mentioned in Column A with the enzymes given in Column B.

Column A	*Column B*
P. Diacylglycerol synthesis	1. protein kinase A
Q. CREB phosphorylation	2. ras
R. GTP hydrolysis	3. phospholipase
	4. phospholipase D
	5. protein kinase G

a. P-3, Q-1, R-5 b. P-4, Q-1, R-2
c. P-3, Q-1, R-2 d. P-3, Q-5, R-2 **[GATE]**

16. Which one of the following bacterial toxins does not have ADP-ribosyl transferase activity?
a. Cholera toxin b. Diphtheria toxin
c. Exotoxin A d. *S. aureus* toxin **[IISc]**

17. Choose the *incorrect* statement.
a. Phospholipase-C generates DAG and IP3 from phosphatidyl inositol 4,5-bisphosphate.
b. Phospholipase-C is not membrane bound enzyme.
c. DAG activates protein kinase C under increased Ca^{2+} level.
d. IP_3 induces the opening of Ca^{2+} channel present on ER.

18. Match each enzyme in column A with the product in column B.

Column A	Column B
1. Guanylcyclase	A. cAMP
2. Adenylcyclase	B. DAG and IP_3
3. Phospholipase-C	C. cGMP
4. PIP kinase	D. phosphatidylinositol 4,5-bisphosphate (PIP_2)

a. 1-C, 2-A, 3-B, 4-D b. 1-B, 2-C, 3-A, 4-D
c. 1-D, 2-C, 3-B, 4-A d. 1-D, 2-A, 3-D, 4-C

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19. Diacylglycerol activates
    a. protein kinase A
    b. protein kinase C
    c. MAP kinase [IISc]
    d. tyrosine kinase

20. Which one of the following is involved in cell signaling process?
    a. Cytochrome P450
    b. Protein kinase C
    c. Zinc fingers
    d. All of the above are involved

21. Calmodulin activates protein kinases in response to a transient increase in
    a. cAMP
    b. calcium ions
    c. DAG
    d. NO

22. Which one of the following statements concerning Ca^{2+} is *correct*?
    a. Intracellular concentration of Ca^{2+} is higher than that of the extracellular fluid.
    b. The concentration of cytosolic Ca^{2+} is transiently decreased by activation of phospholipase C.
    c. The effects of Ca^{2+} is most often mediated by calmodulin.
    d. Intracellular Ca^{2+} is mostly free. [GATE]

23. With respect to the plasma membrane, most enzymic receptors are
    a. single pass
    b. seven pass
    c. entirely on the surface
    d. multipass

24. Rhodopsin is a transmembrane protein belonging to the large family of G protein coupled receptors. It is found in the discs of rod cells of human retina. Activation of rhodopsin is due to
    a. phosphorylation of its extracellular tyrsoine residue.
    b. binding of external ligands to its extracellular loops.
    c. photoisomerization of its prosthetic group.
    d. binding of calcium ions to its transmembrane aspartic groups.

25. Find the *correct* statement.
    P. Ras is a monomeric GTP binding switch protein.
    Q. Phosphodiesterase converts cGMP into GMP.
    R. Receptor tyrosine kinase dimerise after binding with ligand.
    S. Receptor tyrosine kinase after activation stimulates Ras-MAP kinase pathway.
    a. P and Q
    b. P and R
    c. P, R and S
    d. P, Q, R and S

26. Which of the following is a component in the signaling pathway stimulated by receptor tyrosine kinases?
    a. Adenylate cyclase
    b. Adaptor proteins
    c. Autophosphorylating receptor
    d. Ras activating protein

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27. Receptor tyrosine kinases
P. have a cytoplasmic kinase domain.
Q. dimerize upon ligand binding.
R. change conformation of their cytoplasmic domains upon ligand binding.
S. become catalytically active after the conformational change.
a. P and Q
b. P and R
c. Q, R and S
d. P, Q, R and S

28. MAP Kinases
P. are activated by the *Ras* pathway.
Q. phosphorylate their targets on tyrosine residues.
R. act to stimulate or repress the transcription of target genes through the phosphorylation of transcription factors.
S. become active when phosphorylated by MEK kinases.
a. P and Q
b. P and R
c. P, Q and R
d. P, R and S

29. The Ras protein was first identified as the product of an oncogene that causes rat sarcomas, and mutations in Ras are found in many human tumors. The Ras pathway is used to respond to growth factors, and mutations that turn on these pathways in the absence of growth factors lead to cancer. Which kinds of mutations in Ras would you expect to see in cancer cells?
a. Nonsense mutations.
b. Missense mutations that decrease GTP hydrolysis.
c. Missense mutations that decrease interaction with SOS.
d. Missense mutations that decrease the interaction with MAPKKKs.

30. Which of the following is/are correct about JAK-STAT pathway?
P. JAK kinase are tyrosine kinase.
Q. STATs are transcription factors and act as substrate for JAK kinase.
R. Cytokine receptor activates JAK kinase.
S. SH2 domain in STATs is responsible for dimerization of STATs.
a. P and S
b. Q, R and S
c. P, Q, R and S
d. P, Q, R but not S

31. Cytokine receptors
P. have carboxyl terminal domain which show kinase activity.
Q. are single transmembrane polypeptide.
R. provide binding sites for non-receptor tyrosine kinase.
S. provide docking site for binding of Ras protein.
a. P and Q
b. R and S
c. Q and R
d. P and S

32. The receptor for nitric oxide is
a. intercellular
b. intracellular
c. extracellular
d. transcellular

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33. Which of the following is *not* true about NO?
    a. Acts as intracellular signaling molecule as well as neurotransmitter.
    b. Regulates the vasodilation.
    c. Synthesized from L-arginine.
    d. Induces adenylate cyclase, which catalyzes cAMP formation.

34. Which of the following statements is *false* for the nitric oxide gas?
    a. An intracellular signaling molecule.
    b. Deamination of histidine results into nitric oxide production.
    c. Stimulates guanylyl cyclase to produce cGMP.
    d. Can be produced by activated neutrophils.

35. Which of the following hormone passes through the cell membrane, binds to its intracellular receptors and activates it?
    a. Estrogen
    b. Non-epinephrine
    c. Acetylcholine
    d. Glucagons
    [JNU]

36. The receptor for which of the following hormones is a transcription factor?
    a. Insulin
    b. Glucagon
    c. Estradiol
    d. Adrenalin
    [IISc]

37. The mechanism of signal transduction by steroid hormones differs from amine and peptide hormones because
    a. steroids use small, water soluble secondary messengers.
    b. they bind with specific receptor proteins on target-cell plasma membranes.
    c. they bind to cytoplasmic or nuclear receptors and affect gene expression.
    d. they are secreted from exocrine glands.

38. Intracellular receptors were identified and have been proposed to mediate actions of steroid hormones. In an experiment to study various actions of estrogen, it was covalently coupled to albumin, a protein of 68 kD. When this protein coupled hormone was added to estrogen responsive cells, a measurable response was observed. This indicates that
    a. estrogen receptors are not found intracellularly.
    b. estrogen coupled protein is taken up by the cells by passive diffusion.
    c. estrogen receptors are also present on the cell membrane.
    d. steroid hormone coupled protein is cleaved by the cells and free estrogen is taken up by the cells by passive diffusion.
    [JNU]

39. Which of the following statements about hormones is *correct*?
    a. Steroid and peptide hormones produce different effects but use the same biochemical mechanisms.
    b. Steroid and peptide hormones produce the same effects but differ in the mechanisms that produce the effects.
    c. Steroid hormones affect the synthesis of proteins, whereas peptide hormones affect the activity of proteins already present in the cell.
    d. Steroid hormones affect the activity of certain proteins within the cell, whereas peptide hormones directly affect the processing of mRNA.

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1. The major events of mitotic prophase include all of the following *except*
   a. chromosome coiling
   b. DNA replication
   c. nuclear envelope breakdown
   d. nucleolar disaggregation

2. When the nucleus of a frog red blood cell, which does not replicate DNA, is transplanted into an enucleated frog egg, the egg goes through several cell divisions. Which of the following is the best interpretation for this phenomenon?
   a. Isolated red-blood-cell nuclei synthesize DNA.
   b. The nucleus plays no role in cell division.
   c. An enucleated frog egg can divide.
   d. The cytoplasm controls nuclear DNA replication.

3. If radioactive thymidine is used during S-phase of interphase. It will make chromosome radioactive
   a. near the centromere
   b. at both ends of chromosome
   c. in euchromatin region
   d. both in euchromatin and heterochromatin region

4. A study is done on a mammalian cell line that has a doubling time of 24 hours. These cells are synchronized in G_1 and then labeled for 2 days with BrdU (an analog of thymidine that increases the density of DNA into which it is incorporated). At the end of the labeling period, chromosomal DNA is isolated from the cells and its density analyzed by equilibrium centrifugation in cesium chloride gradients. Which of the following patterns would be expected to be seen? (H=heavy, L=light)
   a. 100% H/H
   b. 100% H/L
   c. 50% H/H, 50% H/L
   d. 50% H/H, 50% L/L

5. In the cell cycle, mitosis occurs between
   a. G1 and S phase
   b. S and G1 phase
   c. S and G2 phase
   d. G1 and G2 phase [IISc]

6. G1 phase of the cell cycle is usually absent in
   a. HeLa cells in culture
   b. early cleavage stage cells
   c. senescent fibroblast cells
   d. intestinal epithelial cells [JNU]

7. The basis for the blocking action of the alkaloid colchicine on cell division is
   a. G1 phase
   b. S phase
   c. G2 phase
   d. M phase [GATE]

8. Cyclin B breaks down
   a. during anaphase
   b. during DNA replication
   c. during cell wall formation
   d. during G2/M transition [JNU]

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9. In a dividing cell, DNA synthesis goes through only one round and is arrested before cytokinesis. The arrest is due to
   a. specific cyclin protein
   b. specific cyclin protein along with Cdc2 protein kinase
   c. Cdc2 protein kinase
   d. G protein with Cdc2 protein kinase [JNU]

10. During embryonic development, the total cell cycle time for a given cell type can change. These changes are primarily due to variation in
   a. G1 phase
   b. G2 phase
   c. M phase
   d. S phase

11. Cell cycle progression from G2 to M phase is mainly controlled by
   a. glycosyl transferases
   b. cyclins
   c. cell adhesion molecules
   d. cyclic AMP [JNU]

12. Cyclin-dependent kinase (Cdk) activity increases steadily during G2 phase as a result of
   a. phosphorylation of mitotic cyclins by Cdks.
   b. activation of mitotic Cdk-cyclins through the activity of a phosphatase.
   c. phosphorylation of Cdks at a point not located inside the catalytic site of the cdk-cyclin complex.
   d. transient increase in the cytosolic GTP concentration.

13. The decline of MPF (M phase promoting factor or M phase kinase) at the end of mitosis I caused by
   a. the destruction of protein cyclin dependent kinase
   b. decrease in synthesis of cyclin
   c. the enzymatic destruction of cyclin
   d. synthesis of DNA

14. Biochemical mechanisms that regulate progression through the cell cycle include
   a. phosphorylation
   b. dephosphorylation
   c. protein degradation
   d. all of the above

15. Cell cycle is controlled by
   a. proteolysis of cyclin dependent kinase
   b. phosphorylation of cyclins
   c. dephosphorylation of cyclin
   d. proteolysis of cyclins

16. Which of the following amino acids is the likely target for a kinase activity?
   a. Tyrosine
   b. Phenylalanine
   c. Glutamine
   d. Methionine [JNU]

17. Among phosphyorylases and kinases
   a. both require ATP as the source of phosphate.
   b. only kinases require ATP as the source of phosphate.
   c. only phosphorylases require ATP as the source of phosphate.
   d. both require Pi as the source of phosphate. [JNU]

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18. M-phase kinase
P. can cause G2-blocked *Xenopus laevis* oocytes to enter M-phase.
Q. undergo proteolytic degradation by APC.
R. contains a regulatory subunit that is a cyclin.
S. phosphorylates target proteins at specific times in the cell cycle.
a. P, Q
b. P, R
c. R, S
d. P, Q, R, S

19. Which statement about cyclins are *true*?
P. There are several families of cyclins that differ in amino acid sequence and in their properties.
Q. Cyclins are present at a constant steady-state level throughout the cell cycle.
R. Cyclins are degraded rapidly when the cell enters mitosis.
S. Cyclins regulate the activity of the M phase kinase.
a. P and Q
b. P and R
c. P, R and S
d. P, Q, R and S

20. Which of the following does *not* occur when a cell enters M phase?
a. Chromatin condenses.
b. Histone H1 is dephosphorylated.
c. The nuclear envelope, the endoplasmic reticulum and the Golgi break down.
d. The spindle is formed.

21. How many DNA molecules are present in the nucleus of human somatic cell in G2 stage of cell cycle?
a. 23
b. 46
c. 69
d. 92

22. Entry into M phase is *not* allowed if
a. the cell is not big enough
b. sufficient nutrients are not available
c. DNA replication is not complete
d. mitotic cyclin is over expressed

23. Find out the *false* statement.
a. Kinesin, a motor protein helps in movement of chromatids or chromosomes.
b. Mitogens are substances that promotes cell proliferation.
c. Anaphase promoting complex ubiquitinates M-phase cyclin for proteasomal degradation.
d. Cyclin dependent kinase acts as regulatory subunit in M-phase kinase or maturation promoting factor.

24. Cell cultured in colchicines undergo complete cell cycle, except for the fact that chromosomes don't migrate to the poles during anaphase. One can conclude that
a. colchicines does not affect the microtubules.
b. DNA replication doesn't necessarily result in chromosome separation.
c. the centromere represents under replicated DNA.
d. the spindle is made after interphase.

25. If a sperm cell of a mouse contains 1.8 picogram of DNA per nucleus, the mass of DNA expressed in the nucleus of a kidney cell would be
a. 1.8 picogram
b. 0.9 picogram
c. 3.6 picogram
d. 7.2 picogram

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26. Cytokinesis in animal cells is caused by
P. the sliding movements of a band of microtubules around the circumference of the cell.
Q. the contraction of a band of actin filaments around the circumference of the cell.
R. the movement of the mitotic spindle fibers.
S. endocytosis of the plasma membrane around the equator of the cell.
a. only P
b. only Q
c. Q and R
d. P and S

27. During anaphase as chromosome moves poleward
a. polymerization of kinetochore microtubules occur.
b. depolymerization of kinetochore microtubules occur.
c. depolymerization of polar spindle fibers occur.
d. none of the above.

28. After meiosis I, the resultant daughter cells have
a. twice as much DNA as a haploid gamete.
b. as much DNA as a haploid gamete.
c. four times as much DNA as a haploid gamete.
d. all of the above.

29. Synapsis means
a. pairing of homologous chromosome during meiosis II.
b. pairing of any two chromosome.
c. pairing of homologous chromosome during meiosis I.
d. pairing of non homologous chromosomes during meiosis I.

30. Reassortment, occurs as a consequence of random orientation of pairs of homologous chromosome and recombination of genes on the same chromosome occurs respectively in
a. metaphase I and prophase I of meiosis.
b. prophase I and prophase II of meiosis.
c. prophase II and metaphase I of meiosis.
d. prophase I and metaphase I of meiosis.

31. In the meiosis type cell division, chromatids segregates during
a. never divides
b. anaphase I
c. interkinesis
d. anaphase II

32. Crossing over at the time of meiosis I will cause
a. independent assortment of genes.
b. recombination of linked genes.
c. both independent assortment and recombination of genes present on non-homologues chromosome.
d. linkage between genes.

33. If the numbers of bivalents are 10 in prophase I. What is the number of chromosomes during anaphase II?
a. 10
b. 20
c. 40
d. 30

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34. Match the following

Column A	Column B
1. The stage of meiosis in which chiasmata become visible	A. diplonema
2. The stage of meiosis in which homologous chromosomes pairs	B. zygonema
3. The stage of meiosis in which lampbrush chromosomes are visible	C. pachynema
4. The stage of meiosis in which crossing over actually occurs	

a. 1-A, 2-B, 3-A, 4-C
b. 1-B, 2-C, 3-B, 4-C
c. 1-C, 2-A, 3-C, 4-A
d. 1-B, 2-B, 3-C, 4-C

35. If haploid number of chromosomes in human is 23 and the amount of nuclear DNA in a sperm is 1C, state how many chromosome a human cell has in the metaphase of mitosis and prophase I of melosis respectively?

a. 46 and 23
b. 46 and 46
c. 23 and 46
d. none of these

36. The DNA content of a diploid cell in the G1 phase of the cell cycle is measured. If this DNA content is X, then the DNA content of the same cell at metaphase of meiosis I and metaphase of meiosis II would be respectively

a. 0.25X and 2X
b. 2X and X
c. 4X and X
d. 2X and 0.25X

37. If spermatogenesis is normal and all cells survive, how many sperms will result from 50 primary spermatocytes and 50 spermatids, respectively?

a. 200 and 200
b. 200 and 50
c. 50 and 50
d. 100 and 50

38. Sporogenic meiosis will occur

a. at the time of spore formation
b. after spore formation
c. at the time of gamete formation
d. at the time of zygote formation

39. Which of the following statements are *correct*?

P. Centrosomes are replicated before M phase begins.
Q. Polar microtubules attach end-to-end and are therefore continuous from one spindle pole to the other.
R. Microtubule polymerization and depolymerization and microtubule motor proteins are all required for chromosome movement.
S. Microtubules nucleate at the centromeres and then connect to the kinetochores, which are structures at the centrosome regions of chromosomes.

a. P and Q
b. R and S
c. P and R
d. Q and S

40. Recombination during meiosis is caused by

a. crossing-over alone
b. crossing-over and independent assortment
c. Independent assortment only
d. chiasma formation

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41. What would be the result of meiosis in commercial bananas, which are triploid?
    a. Aneuploid gametes
    b. Triploid gametes
    c. 50% diploid and 50% haploid gametes
    d. No gametes at all

42. A male child resembles their parents but is not exactly like them. This is because of
    a. variations produced by crossing over at the time of gamete formation.
    b. variations produced by chance distribution of chromosome to the two poles, when spermatids convert into spermatozoa.
    c. both a and b.
    d. none of these.

43. In flowering plants, meiosis occurs at the time of
    a. gamete formation
    b. formation of pollen grains
    c. formation of spore mother cells
    d. none of these

44. How many meiotic divisions are required to produce 12 pollen grains in *Cyperaceae*?
    a. 2
    b. 3
    c. 6
    d. 12 [JNU]

45. Which statement is *incorrect*?
    a. In meiotic division one DNA replication is followed by two round of segregation.
    b. Astral type mitosis is present in most of animal cells.
    c. Two sister chromatids are held together with help of cohesin protein.
    d. Pluripotent cells have ability to give rise to whole organism.

46. Which of the following will produce a non-mosaic XYY male?
    a. Meiotic non-disjunction in Y-chromosome.
    b. Meiotic non-disjunction in X-chromosome.
    c. Meiotic non-disjunction in both X and Y-chromosome.
    d. Meiotic non-disjunction in Y-chromosome in zygote.

47. The circumferential band of microtubules that appears and forms a ring around the entire plant cells just below the plasma membrane is called the
    a. cell plate
    b. phragmoplast
    c. pre-prophase band
    d. contractile ring

48. Which of the following statements is/are *false*?
    P. The muscle like sliding of actin and myosin filaments in the contractile ring generates the force required for cleavage.
    Q. In many unicellular organisms mitosis occur without breakdown of the nuclear envelope.
    R. The mechanism of cytokinesis in plant cells is fundamentally different form animal cells.
    a. only P
    b. P and Q
    c. P, Q and R
    d. none of these

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49. The most convincing experiments that established the chromosomal basis of genes and gene order involved
a. distributive pairing of chromosomes.
b. genetic crossing over paralleled that of physical crossing over between homologous chromosomes.
c. karyotyping of chromosomes.
d. non-disjunction of chromosomes. [IISc]

50. You are studying the cell cycle parameters of an organism that can exist as haploid, diploid and tetraploid forms. You estimate the length of the S-phase of the cell cycle that represents the time the cells spend in duplicating its genome for all the three forms of the organism. You find that
a. the duration of the S-phase for the haploid form is the shortest and longest for the tetraploid form that has four times as much DNA to be duplicated.
b. the duration of the S-phase for the haploid form is the longest and shortest for the tetraploid form that has four times less DNA to be duplicated.
c. the S-phase duration is constant for all the three forms.
d. the S-phase duration has no correlation with the ploidy of the genome as chromosomes and genomes have a defined order and timing of replication which depends on epigenetic modifications and overall organization of the genome that may be different for all the three forms of the organism. [JNU]

Cancer

1. p53 protein is associated with all of the following *except*
a. tumor suppression
b. programmed cell death
c. transcription
d. post-transcription modifications [JNU]

2. Which of the following statements *correctly* describes p53?
a. It is a membrane receptor that binds to a cell growth hormone.
b. Cell division stops until p53 binds to DNA and repairs the damage.
c. p53 is an example of an oncogene, and its expression causes uncontrolled cell division.
d. Activation of p53 leads to inhibition of the S-phase cyclin-CDK complex.

3. Retroviruses are capable of causing cancer because they
a. produce a very high number of progeny viruses per infected cell.
b. often contain point mutations in their *pol* gene.
c. transduce mutant versions of cellular genes that normally regulate cell growth.
d. infect cells more efficiently than other viruses. [JNU]

4. Which of the following statements is *not* correct for an oncogene?
a. They cause cellular transformation.
b. They are often transduced by retroviruses.
c. Generally they are growth regulatory proteins.
d. They are always localized in the nucleus. [JNU]

5. Proto-oncogenes are
a. oncogenes found in transforming retroviruses.
b. oncogenes present in protozoa.
c. genes encoding oncogene related proteins in extinct organisms.
d. cellular genes encoding proteins related to viral oncogenes. [IISc]

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6. Formation of a retinoblastoma eye tumor requires
   a. a germinal mutation in one RB allele, then a somatic mutation in the other allele.
   b. a somatic mutation that turns on the RB gene.
   c. a germinal mutation in one RB allele.
   d. a germinal mutation in the dominant RB allele.

7. Which of the following is mismatch?
   a. Epstein-Barr Virus
   b. HTLV I
   c. Herpes virus
   d. HIV

   P. Burkitt's lymphoma
   Q. adult T-Cell leukemia
   R. Kaposi sarcoma
   S. Retinoblastoma

8. Chronic myelogenous (myeloid) leukemia is caused by a translocation that creates
   a. a proto-oncogene.
   b. a fusion protein that acts like a transcription factor.
   c. a fusion protein that deregulates the cell cycle of myeloid white blood cells.
   d. a protein that increases growth factor production.

9. Which of the following statements are *correct*?
   P. Gatekeeper genes regulate mitosis and apoptosis.
   Q. Caretaker genes control mutation rates of gatekeeper genes.
   R. Proto-oncogenes normally regulate the cell cycle.
   S. Oncogenes are often overexpressed in cancer cells.
   a. P and Q
   b. P, R and S
   c. Q, R and S
   d. P, Q, R and S

10. Which of the following is *not* a property of cancer cells?
    a. Lack of density dependent inhibition of cell proliferation.
    b. Loss of contact inhibition of movement.
    c. Fail to undergo apoptosis.
    d. Increased requirement of extra cellular growth factors.

11. One difference between a cancer cell and a normal cell is that
    a. the cancer cell is unable to synthesize DNA
    b. the cell cycle of the cancer cell is arrested at the S phase
    c. cancer cells continue to divide even when they are tightly packed
    d. cancer cells cannot function properly because they suffer from density-dependent inhibition

12. Choose the *incorrect* statement.
    a. *c-abl*, a proto-oncogene encode a tyrosine kinase.
    b. Translocation of *c-abl* from chromosome 9 to chromosome 22 result in chronic mylogenous leukemia.
    c. All retrovirus are oncovirus.
    d. p53 gene located on chromosome 17 of human.

13. How does the Philadelphia chromosome activate an oncogene?
    a. It puts a proto-oncogene downstream of a strong promoter.
    b. It fuses a proto-oncogene with another protein and alters its activity.
    c. It deletes its opposing tumor suppressor.
    d. It deletes a regulatory region for the gene.

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14. *Oncogenes* incorporated into viral genomes can differ from their normal, cellular counterparts (*proto-oncogenes*) in which of the following ways

P. They can be expressed from much stronger promoters than the normal, cellular promoter.
Q. They can contain point mutations in regulatory domains that result in a loss of protein regulation.
R. They can be present in many tandem copies, as opposed to the single copy present in the cell.

a. only P
b. P and Q
c. only R
d. Q and R

15. The normal cellular counterparts of oncogenes are important for the following functions *except*
a. promotion of cell cycle progression
b. inhibition of apoptosis
c. promotion of DNA repair
d. promotion of nuclear transcription **[IISc]**

16. It was found that normal stem cells isolated from brain have a doubling time of 24 hours, whereas tumor-derived cells have a doubling time of 12 hours. Both these cell types were grown in the presence of radioactive thymidine for 24 hours, following which they were grown in normal media. At the end of a week, radioactivity per cell will be
a. double in stem cells as compared to cancer cells.
b. half in stem cells as compared to cancer cells.
c. same in both stem and cancer cells.
d. none in stem cells and high levels in cancer cells. **[IISc]**

17. Comparative genomic hybridization is a technique to look for certain type of genetic changes in cancer cells by comparing the DNA from the cancer cells with that of corresponding normal cells. Which types of genetic changes are detected here?
a. Base mismatch
b. Gene deletion and amplification
c. Loss of single base
d. Promotor methylation **[IISc]**

18. Some oncogenic retroviruses can transform normal mammalian cells to cancer cells in vitro. This transformation of a beast cell is generally associated with
a. conversion of the host DNA into viral RNA.
b. presence of viral particles in the host's cell membrane.
c. integration of the viral genome into the host's nuclear DNA.
d. release of numerous viral particles and concomitant death of host cells. **[IISc]**

19. Which of the following statements is *false*?
a. Hereditary retinoblastoma results from overexpression of a cellular oncogene.
b. Translocation of c-*myc* gene is found in many patients with Burkitt's lymphoma.
c. Viral integration into the cellular genome may convert a proto-oncogene into an oncogene.
d. Oncogenic retroviruses carry viral oncogenes.

20. How is the translocation between chromosomes 8 and 14 thought to cause Burkitt lymphoma?
a. It deletes the tumor suppressor gene.
b. It fuses a proto-oncogene with another protein and alters its activity.
c. It puts a proto-oncogene downstream of a strong promoter.
d. It deletes a regulatory region for the gene.

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21. Which of the following is *least* likely to cause a proto-oncogene to become an oncogene?
    a. A gene is incorporated into a retroviral genome.
    b. A gene is moved into centromeric heterochromatin, silencing its transcription.
    c. A gene is moved close to an enhancer, causing excess product to be made.
    d. A gene is truncated, yielding a protein with modified activity.

22. The tumor suppressor protein p53 can be activated in response to various cellular stress. What describes most precisely the mechanism of p53 activation after DNA damage?
    a. ATM phosphorylates p53 and disrupts the p53-MDM2 interaction.
    b. Chk1 phosphorylates p53 and disrupts the p53-MDM2 interaction.
    c. p14ARF phosphorylates p53 and disrupts the p53-MDM2 interaction.
    d. ATM and ATR bind MDM2 and disrupt the p53-MDM2 interaction.

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Fill in the blanks in the following statements.

Eukaryotic cell : Structure and function

1. Lipid molecules in biological membranes are arranged as a continuous double layer called the ....., which is about 5 nm thick.
2. The most abundant lipids in cell membranes are .....
3. All the lipids found in membranes are said to be ..... because they have one hydrophilic end and one hydrophobic end.
4. The hydrophilic end of a phospholipid consists of a .....head group, and the hydrophobic end contains two .....tails.
5. When amphipathic molecules are placed in an aqueous environment, they tend to aggregate so as to bury their hydrophobic ends and expose their hydrophilic ends to water, giving rise to two different kinds of structures, either spherical .....or planar ....., with the hydrophobic tails sandwiched between the hydrophilic head groups.
6. Synthetic bilayers containing defined lipids or mixtures of lipids can be made either as spherical vesicles, called ....., or as planar bilayers, called ..... membranes.
7. Special ER-membrane-bound enzymes called ..... catalyze the rapid flip-flop of specific phospholipids from the cytosolic monolayer where they are made to the opposite monolayer.
8. Artificial lipid bilayers made from a single type of phospholipid change from a liquid state to a rigid crystalline state or *vice versa* at a characteristic temperature; this change of state is called a .....
9. Eukaryotic plasma membranes contain especially large amounts of ....., which enhances the mechanical stability of the lipid bilayer.
10. Sugar-containing lipids called ..... are found only in the outer half of the bilayer and their sugar groups are exposed at the cell surface.
11. Glycolipids that contain sialic acid are called ..... One example is ..... to which cholera toxin binds.
12. Proteins that extend across the bilayer and are exposed to an aqueous environment on both sides of the membrane are called ..... proteins.
13. While still in the ER, the transmembrane segment of some single-pass transmembrane proteins is cleaved off and a ..... is added, leaving the protein bound to the noncytoplasmic surface of the membrane solely by this anchor.
14. .....proteins can be released from membranes by gentle procedures, such as extraction by a salt extraction by a salt solution, whereas ..... proteins can be removed only by totally disrupting the bilayer with detergents or organic solvents.
15. In ..... proteins, the polypeptide crosses the membrane only once, whereas in .... proteins, the polypeptide chain crosses multiple times.
16. The outer membrane of *E. coli* is penetrated by various .....,which allow hydrophilic solutes of up to 600 daltons to diffuse across the outer lipid bilayer.
17. The most useful agents for disrupting hydrophobic associations and destroying the bilayer are ....., which are small amphipathic molecules that tend to form micelles in water.
18. The technique of using a membrane-impermeant radioactive or fluorescent labeling reagent to determine the sidedness of membrane proteins is termed.....
19. .....is a long fibrous molecule composed of a complex of two very large polypeptide chains, arranged in a filamentous meshwork on the cytoplasmic surface of the red blood cell membrane.

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20. ..... is the anion channel in red cells that is responsible for the exchange of HCO_3^- for Cl^- when CO_2 from the tissues is delivered to the lungs.

21. In ..... electron microscopy, cells are frozen in liquid nitrogen and the resulting block of rice is fractured with a knife, splitting the lipid bilayer into its two monolayers.

22. The purple membrane of the bacterium Halobacterium halobium is a specialized patch in the plasma membrane containing a single species of protein molecule,..... , which converts light energy into a proton and voltage gradient that in turn drives production of ATP.

23. Like membrane lipids, membrane proteins are able to rotate about an axis perpendicular to the plane of the bilayer (..... diffusion), many are able to move laterally in the membrane (..... diffusion), and they do not tumble (.....) across the bilayer.

24. Direct evidence that some plasma membrane proteins are mobile in the plane of the membrane was provided in 1970 by an elegant experiment using hybrid cells called ..... , which were produced artificially by fusing mouse and human cells.

25. The lateral diffusion rates of membrane proteins that contain a chromophore or bind a fluorescent ligand can be quantitated using a technique called .....

26. The carbohydrate-rich zone at the surface of most eukaryotic cells is known as the ..... or .....

27. Proteins that recognize specific sugar residues are called ..... .

28. .....are a family of cell-cell adhesion molecules that contain a carbohydrate-binding domain exposed at the cell surface.

29. Specific proteins called ..... proteins must be present in order for cell membranes to be permeable to small polar molecules such as ions, sugars, and amino acids.

30. There are two major classes of membrane transport proteins: ..... proteins, which bind specific solutes and change conformation to transfer the solute across the membrane; and ..... proteins, which form water-filled pores that allow specific solutes to cross the membrane down their electrochemical gradients.

31. Two general transport processes control the entry of solutes into cells: ..... transport requires no energy input by the cell, whereas ..... transport pumps specific solutes across a membrane against an electrochemical gradient.

32. The concentration gradient for a charged solute and the membrane potential constitute the ..... gradient for that solute.

33. Small hydrophobic molecules that make membranes more permeable to specific inorganic ions are called ..... There are two kinds, known as ..... carriers and .....

34. Most animal cells take up glucose from the extracellular fluid by passive transport through glucose carriers that operate as .....

35. The transport of sugars into intestinal cells occurs by inward ..... of Na^+ ions along with the sugar molecules.

36. The band 3 protein of the human red blood cell, which is an .....carrier, couples transport of Cl^- in one direction with HCO_3^- transport in the other and is thus a good example of an ..... transporter.

37. The .....is inhibited by ouabain.

38. The network of tubular sacs in muscle cells that stores Ca^{2+} by means of an ATP-driven Ca^{2+} pump is called the .....

39. Ion-driven carriers are said to mediate ..... active transport, whereas transport ATPases are said to mediate ..... active transport.

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40. In the epithelial cells of the gut Na^+-linked symporters in the apical (absorptive) domain of the plasma membrane and Na^+-independent transport proteins in the basolateral domain bring about the ..... of absorbed solutes.

41. Thin fingerlike projections called ..... on the apical surfaces of kidney and intestinal epithelial cells increase their absorptive area by as much as 25 times.

42. The transport ATPases in the bacterial plasma membrane belong to a diverse family of transport proteins known as the ..... because each member contains a highly conserved ATP-binding cassette.

43. Over expression of the ..... in human cancer cells can make these cells simultaneously resistant to a variety of chemically unrelated cytotoxic drugs that are widely used in cancer chemotherapy.

44. .....proteins form hydrophilic pores across membranes; almost all such proteins in eukaryotic plasma membranes are concerned with inorganic ion transport and are therefore referred to as .....

45. Three kinds of perturbation that can cause gated ion channels to open or close are ....., ..., and .....

46. The uneven distribution of ions on either side of the plasma membrane gives rise to a voltage across the membrane known as the .....This voltage depends crucially on the existence of ..... channels, which make most animal cells much more permeable to K^+ than to Na^+.

47. The equilibrium condition, in which there is no net flow of ions across the plasma membrane, defines the ....., which is expressed quantitatively by the ..... equation.

48. Every neuron consists of a .....(containing the nucleus), usually one long ..... to conduct signals toward distant targets, and several shorter ....., which extend like antennae to receive signals from other nerve cells.

49. An ....., or nerve impulse, is a traveling wave of electrical excitation that can carry a message without attenuation from one end of a neuron to the other.

50. In nerve and skeletal muscle cells a depolarizing stimulus causes ..... to open, allowing a small amount of ..... to enter the cell down its electrochemical gradient.

51. In many nerve cells ..... help bring the activated plasma membrane back to its original negative potential by allowing an efflux of .....

52. The axons of many vertebrate neurons are insulated by a ....., which greatly increases the rate at which an axon can conduct an action potential.

53. A revolutionary technique that has made it possible to study the behavior of single channels in cell membranes is called ..... recording.

54. Transmission of neuronal signals across a ..... is indirect: a small signal molecule known as a ..... is released from the nerve terminal and diffuses to the target cell where it provokes an electrical change by binding to ..... ion channels.

55. ..... such as acetylcholine open Na^+ channels, which causes depolarization of the postsynaptic membrane, whereas ..... such as GABA open Cl^- channels, which suppresses depolarization.

56. A .....is the specialized chemical synapse between a motor neuron and a skeletal muscle cell.

57. ..... and ..... summation of individual excitatory and inhibitory postsynaptic potentials (PSPs) provide the means by which the rate of firing of many presynaptic neurons jointly control the ..... PSP in the body of a single postsynaptic cell.

58. Action potentials are initiated at the ....., a unique region of each neuron where voltage-gated Na^+ channels are plentiful.

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59. ..... channels repolarize the membrane after each action potential to prepare the cell to fire again; ..... channels ensure that the firing rate is proportional to the strength of the depolarizing stimulus over a very broad range.

60. .....channels help to decrease the response of the cell to an unchanging, prolonged stimulation; a process called .....

61. Whereas occasional single action potentials leave no lasting trace, a short burst of repetitive firing causes ....., such that subsequent single action potentials in the presynaptic cells evoke a greatly enhanced response in the postsynaptic cells.

62. The complex network of protein filaments that extends throughout the cytoplasm is called the .....

63. In most cells, the minus ends of microtubules are stabilized by embedding them in a structure called the ....., which generally lies next to the nucleus, at center of the cell.

64. The intermediate filament protein ..... is found in both smooth and striated muscle cells.

65. The fast-growing end of a microtubule is defined as the ..... and the other as the .....

66. When depolymerized microtubules in a cell are observed to repolymerize, the new microtubules grow out from the centrosome to form a starlike structure called an ..... and then elongate toward the cell periphery.

67. The ..... is the major microtubule organizing center in almost all animal cells.

68. Surrounding each centriole pair is a specialized region of the cytoplasm termed the ....., or ....., and it is the part of the centrosome that nucleates microtubule polymerization.

69. The most versatile modifications of microtubules are conferred by the binding of proteins called ....., which serve both to stabilize microtubules against disassembly and to mediate their interaction with other cell components.

70. ..... are microtubule-dependent motor proteins involved in organelle transport and mitosis and are closely related to the motor protein in cilia and flagella.

71. ..... are a diverse family of microtubule-dependent motor proteins that are involved in organelle transport, in mitosis, in meiosis, and in transport of synaptic vesicles along axons.

Cell signaling

1. Regardless of the nature of the signal, the target cell responds by means of a specific protein called a ....., which binds the signaling molecule and then initiates a response in the target cell.

2. Signaling molecules that a cell secretes may act as ....., affecting only cells in the immediate environment of the signaling cell, in a process called .....

3. When a nerve impulse reached the nerve terminals at the end of the axon, it stimulates the terminals to secrete a chemical signal called a ....., which is delivered rapidly to the postsynaptic target cell in a process termed .....

4. ..... cells secrete their signaling molecules, called ....., into the bloodstream, which carries the signal to target cells throughout the body.

5. Cells can send signals to other cells of the same type and to themselves in a process called .....

6. ....., which are derived from fatty acids (mainly arachidonic acid), are released to the cell exterior, where they influence cells in their immediate neighborhood.

7. ..... are specialized cell-cell junctions that can form between closely apposed plasma membranes, directly connecting the cytoplasm of the joined cells.

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8. Acetylcholine indirectly causes smooth muscle cells in blood vessel walls to relax by inducing the epithelial cells to make and release the gas ....., which then signals the smooth muscle cells to relax.

9. Steroid hormones, thyroid hormones, and retinoids diffuse across the plasma membrane and bind to specific receptors that are structurally related and constitute the .....

10. ..... hormones are all made from cholesterol.

11. The ..... hormones, which are made from the amino acid tyrosine, act to increase metabolism in a wide variety of cell types.

12. The ....., which are made from vitamin A, play important roles as local mediators in vertebrate development.

13. ..... receptors are involved in rapid synaptic signaling between electrically excitable cells.

14. ..... receptors act indirectly to regulate the activity of a separate plasma-membrane-bound target protein, which can be an enzyme or an ion channel.

15. ..... receptors, when activated, either function directly as enzymes or are associated with enzymes.

16. ..... functionally couple G-protein-linked receptors to their target enzymes or ion channels in the plasma membrane.

17. ..... is synthesized from ATP by a plasma-membrane-bound enzyme ....., and it is rapidly and continuously destroyed by one or more .....

18. A trimeric G protein that activates adenylyl cyclase is called a .....

19. ....., which mediate some of the actions of adrenaline and noradrenaline, are coupled by G proteins to the activation of adenylyl cyclase.

20. ....., which is responsible for the disease symptoms of cholera, is an enzyme that catalyzed the transfer of ADP ribose from intracellular NAD^+ to the α-chain of a stimulatory G protein.

21. While the β-adrenergic receptors are functionally coupled to adenylyl cyclase by G_s, the α_2-adrenergic receptors are coupled to this enzyme by an .....

22. ....., made by the bacterium that causes whooping cough, catalyzes the ADP ribosylation of α_i.

23. Cyclic AMP exerts its effects in animals cells mainly by activating the enzyme ....., which catalyzes the transfer of the terminal phosphate group from ATP to specific serines or threonines of selected proteins.

24. The dephosphorylation of phosphorylated serines and threonines is catalyzed by four groups of .....

25. Some extracellular signaling mólecules stimulate the incorporation of radioactive phosphate into ....., a minor phospholipid in cell membranes.

26. The enzyme activated by diacylglycerol is called ..... because it is Ca^{2+}-dependent.

27. ..... functions as a multipurpose intracellular Ca^{2+} receptor, mediating many Ca^{2+}-regulated processes.

28. Olfactory cells in the lining of the nose recognize odorants by means of specific G-protein-linked .....

29. Cyclic-nucleotide-gated ion channels are involved in signal transduction in vertebrate vision, where the crucial cyclic nucleotide is .....

30. In ..... in the vertebrate retina the phototransduction apparatus is in the outer segment, which contains a stack of discs, each formed by a closed sac of membrane in which photosensitive ..... molecules are embedded.

31. In vision the activated photoreceptor binds to the trimeric G-protein transducin, causing the α subunit to dissociate and activate ....., which hydrolyzes cyclic GMP.

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32. ..... use cyclic GMP as an intracellular mediator in the same way that some G-protein-linked receptors use cyclic AMP, except the linkage between ligand binding and cyclase activity is direct rather than via a trimeric G protein.

33. Most growth factors bind to ....., which are transmembrane tyrosine-specific .....

34. ..... recognize phosphorylated tyrosines and enable proteins that contain them to bind to intracellular signaling proteins that have been transiently phosphorylated on tyrosines.

35. The large ..... of ..... contains two other subfamilies:

(1) the Rho and Rac proteins and (2) the Rab family.

36. The ..... help relay signals from receptor tyrosine kinases to the nucleus to stimulate cell proliferation or differentiation.

37. Many serine/threonine kinases are involved in the cascades that relay signals from receptor tyrosine kinases downstream to the nucleus, but the family of ....., which contains at least five members, seems to play an especially important role.

38. Members of the ..... of nonreceptor protein tyrosine kinases each contain SH2 and SH3 domains and are all located on the cytoplasmic side of the plasma membrane.

39. Members of the TGF-β superfamily of extracellular signaling proteins bind to receptors that are the first ..... to be identified.

Cell cycle, Cell division and Cancer

1. One of the key proteins of the cell-cycle control system is ....., which induces downstream processes by phosphorylating selected proteins on serines and threonines.

2. There are two main classes of cyclins: ....., which bind to *Cdk* during G_2 and are required for entry into mitosis, and ....., which bind to *Cdk* during G_1 and are required for entry into S phase.

3. ..... is the activity identified in *Xenopus* egg cytoplasm that, when injected into a G_2-phase oocyte, forces the oocyte into M phase to complete its maturation.

4. Whereas most proteins in sea urchin eggs accumulate continuously after fertilization, ... shows a periodic pattern; it accumulates steadily during each interphase until the metaphase–anaphase transition, and then it is suddenly destroyed.

5. Each ..... mutant is typically deficient in a gene product required to get the cell past the specific point in the cell cycle at which the mutant cells arrest.

6. The fission yeast homologue of the protein kinase subunit of MPF is encoded by a gene called .....

7. In budding yeast the G_1 checkpoint, called ....., is the major size checkpoint.

8. A safety device that operates in mammalian cells to restrain them from entering S phase with damaged DNA depends on a protein called ....., which accumulates in the cell in response to DNA damage and halts the cell-cycle control system in G_1.

9. The mutant form of a proliferation gene that results in the excessive cell proliferation characteristic of cancer is called an ....., and the normal proliferation gene is called a .....

10. The antiproliferation genes found in normal cells are often referred to as .....

11. The ..... gene was identified originally through studies of a cancer that occurs in the eyes of children who lack a functional copy of the gene.

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Chapter 06

Respiration

Glycolysis and Fermentation

1. Glucokinase is characterized by which of the following statements?
   a. It has a much higher Michaelis constant for glucose than hexokinase.
   b. It phosphorylates most hexoses.
   c. It is an allosteric enzyme.
   d. It is found in most cells.

2. Glucose-6-phosphate inhibits which of the following enzymes?
   a. Glucokinase
   b. Hexokinase
   c. Phosphorylase Kinase
   d. Fructose-1,6-bisphosphatase **[IISc]**

3. If ^{32}P-labeled inorganic phosphate were introduced to RBCs undergoing glycolysis then which of the following glycolytic intermediate would be radiolabeled?
   a. Fructose-1,6-bisphosphate
   b. 1,3-bisphosphoglycerate
   c. Glyceraldehyde-3-phosphate
   d. Glucose-6-phosphate

4. Phosphofructokinase is characterized by which of the following statements?
   P. It is a major regulatory enzyme in glycolysis.
   Q. ATP is a substrate of the enzyme.
   R. ATP is a negative modulator of the enzyme.
   S. Citrate is a positive modulator of the enzyme.
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

5. Conversion of glucose to glucose-6-phosphate (G6P) requires ATP; yet critically ill patients are given glucose solutions intravenously instead of G6P. The reason for not giving G6P directly is
   a. G6P is degraded very fast in the blood before it enters the cells.
   b. commercial preparations of G6P are always contaminated with toxic chemicals.
   c. high cost of G6P.
   d. cells cannot take up G6P. **[IISc]**

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6. Given below are four-enzymatic reactions involved in glycolysis. In which of the following steps is ATP generated?
   a. 2-phosphoglycerate to phosphoenol pyruvate.
   b. Glucose-6 phosphate to fructose-6-phosphate.
   c. Phosphoenolpyruvate to pyruvate.
   d. Glyceraldehyde-3 phosphate to 1,3-bis-phosphoglycerate. [GATE]

7. Which of the following enzymes in glycolysis catalyzes a reaction that is essentially non-reversible?
   a. Enolase
   b. Phosphofructokinase
   c. Triose phosphate isomerase
   d. Phosphohexose isomerase

8. For each glucose molecule that is broken down in glycolysis there is a net gain of
   a. 1 ATP molecule
   b. 2 ATP molecules
   c. 3 ATP molecules
   d. 4 ATP molecules [IISc]

9. In anaerobic glycolysis, 2 mol of inorganic phosphate are used per mole of glucose consumed. Which of the following enzymes catalyzes the uptake of inorganic phosphate?
   a. Hexokinase
   b. Phosphofructokinase
   c. Pyruvate kinase
   d. Glyceraldehyde 3-phosphate dehydrogenase

10. Which of the following enzyme-catalyzed reactions has a product containing a newly formed high-energy phosphate bond?
   a. The phosphorylation of glucose.
   b. 2-phosphoglycerate to phosphoenolpyruvate.
   c. 3-phosphoglycerate to 2-phosphoglycerate.
   d. Phosphoenolpyruvate to pyruvate.

11. In glycolytic pathway
   P. acetyl-CoA acts as allosteric inhibitor of pyruvate kinase.
   Q. enolase activity is inhibited by fluoride.
   R. citrate acts as allosteric inhibitor of phosphofructokinase.
   S. hexokinase requires divalent metal ion such as Mg^{2+} or Mn^{2+} for activity.
   a. P, Q and R
   b. P, Q, R and S
   c. Q, R and S
   d. R and S

12. The glycolytic pathway requires which of the following as allosteric regulatory enzymes?
   a. Glucokinase, phosphofructokinase and pyruvate kinase.
   b. Hexokinase, phosphofructokinase and pyruvate kinase.
   c. Hexokinase, glyceraldehyde 3-phosphate dehydrogenase and enolase.
   d. Phosphofructokinase, enolase and pyruvate kinase.

13. Which of the carbon atoms of pyruvate would be labeled during glycolysis of glucose having ^{14}C at C-1?
   a. carboxylate carbon
   b. carbonyl carbon
   c. methyl carbon
   d. none of the above [GATE]

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14. Glyceraldehyde-3-phosphate dehydrogenase, an enzyme of the glycolytic pathway, was modified with iodoacetic acid that resulted in the loss of enzyme activity. This is due to modification of active site residue
a. lysine
b. glutamic acid
c. cysteine
d. cystine [IISc]

15. Fermentation
a. in some organisms can produce ethanol.
b. allows ATP to continue to be synthesized in the absence of oxygen.
c. recycles NADH to NAD^+ so that glycolysis can continue.
d. all of the above are correct.

16. Fluoride is an inhibitor of glycolysis. It inhibits the glycolytic enzyme
a. phosphofructokinase
b. glucokinase
c. enolase
d. lactate dehydrogenase [JNU]

17. The oxidation of 1 mol of glucose by anaerobic glycolysis yields a net of
a. 2 mol of lactate and 2 mol of ATP.
b. 2 mol of lactate, 2 mol of NADH and 2 mol of ATP.
c. 2 mol of lactate, 2 mol of NAD^+ and 6 mol of ATP.
d. 2 mol of acetyl-CoA and 2 mol of ATP.

18. The end product of fermentation of molasses by yeast is
a. pyruvate
b. methyl alcohol
c. ethyl alcohol
d. lactate [IISc]

19. The final electron acceptor in lactic acid fermentation is
a. NAD^+
b. pyruvate
c. oxygen
d. lactic acid

20. At the end of glycolysis X is the net energy gain from one glucose molecule via Y. But there is also energy stored in the form of Z.

	X	Y	Z
a.	1 ATP	oxidative phosphorylation	NADH
b.	2 ATPs	oxidative phosphorylation	NADH
c.	2 ATPs	substrate-level phosphorylation	NADPH
d.	2 ATPs	substrate-level phosphorylation	NADH

Krebs cycle

1. The function of the TCA cycle is characterized by all of the following statements *except*
a. it generates reduced NAD^+ and reduced FAD.
b. it generates guanosine triphosphate.
c. it catalyzes the complete oxidation of acetate to carbon dioxide and water.
d. it causes the net synthesis of oxaloacetate from acetyl-CoA.

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2. Which of the reactions of the tricarboxylic acid cycle listed below results in the formation of a high-energy phosphate compounds?
   a. Isocitrate dehydrogenase
   b. Succinyl coenzyme A synthetase
   c. Succinate dehydrogenase
   d. Citrate synthetase

3. Most CO_2 from catabolism of glucose is released during
   a. glycolysis
   b. the krebs cycle
   c. lactate fermentation
   d. oxidative phosphorylation

4. All of the enzymes of the TCA cycle are located in the mitochondrial matrix *except*
   a. citrate synthetase
   b. α-ketoglutarate dehydrogenase
   c. succinate dehydrogenase
   d. fumarase

5. The release of carbon dioxide results from which of the following reactions in the TCA cycle?
   P. Isocitrate to α-ketoglutarate
   Q. Malate to oxaloacetate
   R. α-ketoglutarate to succinyl-CoA
   S. Succinate to fumarate
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

6. Substrate-level phosphorylation in the citric acid cycle depends directly on the energy of the
   a. thioester bond of succinyl CoA.
   b. oxidative decarboxylation of isocitrate to alpha-ketoglutarate.
   c. formation of citrate from oxaloacetate and acetyl-CoA.
   d. FAD-dependent oxidation of succinate to fumarate.

7. Suppose [4-^{14}C] oxaloacetate is fed to mitochondria. After one turn of the citric acid cycle, which carbon(s) of succinate would be labeled?
   a. None
   b. Equally distributed between C-1 and C-4
   c. Equally distributed between C-2 and C-3
   d. C-4

[GATE]

8. If 1 mol of acetyl coenzyme A labeled with ^{14}C in both carbon positions of the acetate moiety goes through one turn of the TCA cycle, how many moles of $^{14}CO_2$ would be produced?
   a. 0
   b. 1
   c. 2
   d. 4

9. Which C-atom of glucose was tagged with ^{14}C so that after one cycle of glycolysis and Krebs cycle none of the radioactive C-atom is found in intermediates?
   a. C-6
   b. C-1
   c. C-3
   d. C-2

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10. Using glucose, labelled with ^{14}C in its aldehyde group, via glycolysis and alcoholic fermentation, where would the carbon label be at the end of alcoholic fermentation?
    a. First carbon of ethyl alcohol
    b. Methyl carbon of pyruvate
    c. Carbon of aldehyde group
    d. Second carbon of ethyl alcohol

11. Which of the following statement is *incorrect*?
    a. During fermentation final electron acceptor is organic compound.
    b. Cardiolipin reduces the permeability of inner mitochondrial membrane to protons.
    c. Malate-asparate shuttle results in the formation of 32 ATP during aerobic respiration.
    d. PFK-2 catalyzes the formation of fructose 1,6 bisphosphate.

12. Glucose is metabolized through glycolysis and the Krebs cycle. How do amino acids and fats get metabolized?
    a. There are separate cycle and pathways for such metabolism.
    b. They are broken down, modified, and fed into various points in the same pathway as is used by glucose.
    c. They are first converted to glucose molecules.
    d. Only the amino group on the amino acids can be used for energy, the rest is just degraded, while fats are only stored.

Oxidative phosphorylation

1. Inside an active mitochondrion, most electrons follow which pathway?
    a. Glycolysis → NADH → oxidative phosphorylation → ATP → oxygen
    b. Krebs cycle → $FADH_2$ → electron transport chain → ATP
    c. Electron transport chain → Krebs cycle → ATP → oxygen
    d. Krebs cycle → NADH → electron transport chain → oxygen **[MKU]**

2. Which one of the following statement *correctly* describes the relationship between the Krebs cycle and electron transport pathways?
    a. The Krebs cycle releases H^+ used by the electron transport.
    b The electron transport pathway obtains electrons from the CO_2 produced by the Krebs cycle.
    c. ATP produced by the Krebs cycle drives electron transport and formation of a H^+ gradient.
    d. NADH produced by the Krebs cycle is used to make ATP by electron transport.

3. All of the following electron carriers are components of the mitochondrial electron transport chain *except*
    a. nicotinamide-adenine dinucleotide.
    b. nicotinamide-adenine dinucleotide phosphate.
    c. flavin mononucleotide.
    d. flavin-adenine dinucleotide.

4. All of the following statements about NAD^+ are true *except*
    a. NAD^+ is reduced to NADH during both glycolysis and the Krebs cycle.
    b. NAD^+ has more chemical energy than NADH.
    c. NAD^+ is reduced by the action of enzymes called dehydrogenases. **[MKU]**
    d. NAD^+ can receive electrons for use in oxidative phosphorylation.

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5. Electrons from NADH enter the mitochondrial electron transport chain at the level of
   a. coenzyme Q
   b. NADH-coenzyme Q reductase (complex I)
   c. coenzyme Q-cytochrome c reductase (complex III)
   d. cytochrome c oxidase (complex IV)

6. Which of the following processes does *not* involve cytochrome c?
   a. Oxidative phosphorylation
   b. Electron transport
   c. TCA cycle
   d. Apoptosis [JNU]

7. Which statement about oxidative phosphorylation is *not* true?
   a. The five assemblies of proteins and cofactors involved include the electron transferring complexes I-IV and ATP synthase.
   b. The mobile carrier ubiquinone links complexes I and II with complex III and cytochrome C links complex III with complex IV.
   c. For every two electrons transferred from NADH to Q by complex I, two protons are translocated to the intermembrane space.
   d. FAD, FMN, iron-sulphur clusters and copper atoms are other co-factors that participate in electron transfer. [GATE]

8. During respiration, NADH donates two electrons to the carrier known as ubiquinone. When this happens, ubiquinone
   a. becomes oxidized
   b. passes the electrons directly to O_2 which is reduced to water
   c. pumps protons across the inner mitochondrial membrane
   d. all of the above

9. Which of the following statements about ubiquinol are *correct*?
   a. It is the mobile electron carrier between cytochrome *c* oxidoreductase and cytochrome *c* oxidase.
   b. It is an integral membrane protein.
   c. Its oxidation involves the transfer of two electrons to the Fe-S center of cytochrome reductase.
   d. It is a lipid-soluble molecule.

10. Which of the following statements about the structure and properties of cytochrome *c* are *incorrect*?
   P. It is an integral membrane protein.
   Q. It contains very little α-helix or β-pleated sheet secondary structure.
   R. It lacks a heme.
   S. It has retained a highly conserved conformation throughout evolution.
   T. It is soluble in water.
   a. P, Q, R
   b. P, Q, R, S
   c. P, R, S, T
   d. Q, R, S, T

11. Under normal conditions, as electrons flow down the electron transport chain of the mitochondria
   a. NADH and $FADH_2$ are oxidized
   b. the pH of the matrix increases

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c. an electrochemical gradient is formed
d. all of the above

12. Sub-mitochondrial particles are isolated by sonication of intact mitochondria and supplied with external NADH. According to the chemiosmotic hypothesis
a. ATP will be made on the inside of the particles
b. NADH will become reduced
c. the pH will increase on the inside
d. the pH will decrease on the inside

13. Which of the following is/are *true*?
P. NADH yields less ATP than $FADH_2$.
Q. More NADH is produced in the Krebs cycle than in glycolysis.
R. The final acceptor of electrons is ATP synthase.
S. ETC pumps H^+ into the intermembrane space.
a. P only
b. Q and R
c. Q and S
d. P, Q and S

14. Which of the following can serve as an inhibitor of electron transport?
a. Puromycin
b. Actinomycin D
c. Malonate
d. Cyanide **[JNU]**

15. The cyanide resistant respiration in plants leads to production of high amounts of
a. ATP
b. heat
c. phosphocreatine
d. high energy glyceraldehyde phosphate

16. $NADH \rightarrow Q \rightarrow \overset{\otimes}{Cyt\ b} \rightarrow Cyt\ c_1 \rightarrow Cyt\ c \rightarrow \overset{\otimes}{Cyt\ (a+a_3)} \rightarrow O_2$

Sequence of electron transfer in oxidative phosphorylation is given above. Which of the following pair of inhibitors block the electron transfer in the steps marked with ⊗?
a. Rotenone and CO
b. Antimycin-A and CO
c. Antimycin-A and DCMU
d. DCMU and CO **[GATE]**

17. Rotenone, which is used as fish poison and insecticide, blocks mitochondrial electron transport by
a. inhibiting the interaction between oxygen and the terminal electron carrier.
b. inhibiting the reduction of cytochrome c.
c. inhibiting the transfer of electrons through the NADH-CoQ reductase complex.
d. inhibiting electron transfer at all nonheme iron-sulfur protein centers.

18. Carbon monoxide inhibits mitochondrial electron transport by
a. binding to hemoglobin in the erythrocytes and so blocking the transport of oxygen to tissues.
b. binding to the oxygen - binding site of cytochrome oxidase.
c. blocking electron transport at the level of the cytochrome b-cytochrome c_1 complex.
d. combining with coenzyme Q and preventing its interaction with the nonheme iron-sulfur protein center of complex II.

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19. ATP synthetase is a multifunctional enzyme with a subunit constitution of $\alpha_3\beta_3\gamma\delta\varepsilon$. The pair of constituent subunits taking part in nucleotide binding and catalysis is

a. $\alpha_1\beta_1$
b. $\alpha_2\beta_2$
c. $\alpha_3\beta_3$
d. $\gamma\delta$

[GATE]

20. The chemiosmotic coupling hypothesis of oxidative phosphorylation proposes that ATP is formed because
a. of a change in the permeability of the inner mitochondrial membrane toward ADP
b. of the formation of high-energy bonds in mitochondrial proteins
c. ADP is pumped out of the matrix into the intermembrane space
d. a proton gradient forms across the inner membrane

21. The initial step in the biosynthesis of ATP by chemiosmosis in the mitochondrion is the
a. accumulation of sufficient inorganic phosphate by active transport
b. removal of phosphate from glucose-6-phosphate
c. pumping of protons into the inter mitochondrial space
d. pumping of electrons into the matrix

22. Some of the free energy released in the mitochondrial electron transport chain can be harnessed to the formation of ATP. How many moles of ATP can be formed per mole-pair of electrons transferred from NADH to oxygen?

a. 0
b. 1
c. 2
d. 2.5

23. In aerobic cellular respiration, which generates more ATP, substrate level phosphorylation or chemiosmosis?
a. substrate level phosphorylation
b. chemiosmosis
c. both generate the same amount of ATP
d. neither generates any ATP

24. If valinomycin, an antibiotic produced by *Streptomyces* is added to actively respiring mitochondria,
a. the yield of ATP increases.
b. the rate of oxygen consumption decreases.
c. heat is released and the voltage gradient across the inner mitochondrial membrane decrease.
d. all of the above.

25. The uncoupling of oxidative phosphorylation in a mitochondrial system describes which of the following actions?
a. The phosphorylation of ADP to ATP accelerates.
b. The phosphorylation of ADP continues but oxygen uptake stops.
c. The phosphorylation of ADP stops but oxygen uptake continues.
d. Oxygen uptake stops.

26. Uncoupling of oxidative phosphorylation implies that
a. the ATPase activity of mitochondria is abolished.
b. the mitochondria ceases to oxidize succinate.
c. ATP formation ceases but respiration continues.
d. ATP formation continues but respiration ceases.

[JNU]

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27. Thermogenin, the natural uncoupler in brown fat mitochondria, generates heat based on its ability to
a. inhibit electron transport by binding to b-type cytochromes.
b. allow protons to re-enter into the mitochondrial matrix.
c. inhibit ATP production by binding to the ATP synthase.
d. block electron transport to flavoproteins.

28. An uncoupler of oxidative phosphorylation such as dinitrophenol (DNP)
a. inhibits electron transport and ATP synthesis.
b. allows electron transport to proceed without ATP synthesis.
c. inhibits electron transport without impairment of ATP synthesis.
d. specifically inhibits cytochrome b.

29. Injection of dinitrophenol (DNP) into a rat causes an immediate increase in its body temperature, because
a. DNP acts as an inhibitor of mitochondrial ATPase.
b. DNP blocks the electron transport chain.
c. DNP inhibits succinate dehydrogenase.
d. DNP uncouples electron transport from oxidative phosphorylation. **[GATE]**

30. If both oligomycin and dinitrophenol are added to a mitochondrial preparation in the presence of substrate, inorganic phosphate, Mg^{2+} and ADP, then
a. both oxygen uptake and phosphorylation of ADP would cease.
b. oxygen uptake would be reduced, but phosphorylation of ADP would continue.
c. oxygen uptake would be high, but phosphorylation of ADP would cease.
d. none of the above.

31. Dicyclohexylcarbodiimide (DCCD) is a reagent that reacts with Asp residue. The addition of DCCD blocks the ATP synthesizing activity of F_0F_1-ATPase by inhibiting the
a. flow of protons by F_0
b. binding of ADP with β-subunits
c. release of ATP from β-subunit
d. protonation and deprotonation of α-subunit

32. It is possible for a variety of substrates to use a common pathway for the transfer of electrons to oxygen because
a. the substrates are oxidized in the mitochondria.
b. many of the substrates are oxidized by enzymes linked to oxidized NAD^+ or FAD.
c. the substrates are all oxidized by the same enzymes.
d. the electrons from all substrates are transferred to the common acceptor, ATP.

33. Which of the following statement is *correct*?
P. Electron transport generates the proton motive force.
Q. P/O ratio need not be an integral number.
R. Proton motive force in mitochondria is largely due to voltage gradient across the membrane.
S. Total of $10H^+$ are translocated from the matrix across the inner membrane per electron pair moving from NADH to O_2.
a. P and Q
b. P and R
c. P, Q and R
d. P, Q, R and S

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Pentose phosphate pathway

1. In the pentose phosphate pathway
   a. only the C-1 carbon of glucose is oxidized to CO_2.
   b. all the carbons of glucose is oxidized to CO_2.
   c. no decarboxylation occurs.
   d. C-4 and C-5 of glucose is oxidized to CO_2.

2. In pentose phosphate pathway
   a. transaldolase catalyzes the transfer of C_3 units.
   b. transketolase catalyzes the transfer of C_2 units.
   c. transketolase activity is dependent on TPP.
   d. all of the above statements are correct.

3. Which of the following statement is/are correct about oxidative phase of pentose phosphate pathway?
   P. Irreversible phase
   Q. Oxidative decarboxylation
   R. Transketolase and transaldolase activity
   S. Substrate level phosphorylation
   a. P and R
   b. P and Q
   c. R and S
   d. Q and S

4. Ribose-5-phosphate is formed in the pentose phosphate pathway from
   a. isomerization of ribulose-5-phosphate.
   b. reaction of fructose-6-phosphate and glyceraldehyde-3-phosphate.
   c. reaction of xylulose-5-phosphate and erythrose-4-phosphate.
   d. none of the above.

5. Choose the *correct* statement
   a. NADPH maintains the concentration of reduced glutathione in RBC.
   b. G6PD deficiency is associated with resistance to malaria.
   c. Wernicke–Korsakoff syndrome is associated with disorder in HMP shunt.
   d. all of the above statements are correct.

6. Dehydrogenase enzymes of the hexosemonophosphate shunt pathway are
   a. NAD specific
   b. NADP specific
   c. FAD specific
   d. TPP specific [JNU]

7. In the pentose phosphate pathway, glucose is first converted to ribulose-5-phosphate by oxidative decarboxylation. Then, ribulose-5-phosphate undergoes
   a. further oxidation
   b. reduction
   c. further decarboxylation
   d. only rearrangements of carbon skeleton

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Chapter 07

Photosynthesis

Photosynthesis : General features

1. Chlorophyll in the membrane of the thylakoid is
   a. all directly active in photosystem I and II.
   b. embedded in the lipid via hydrophobic side chain.
   c. not associated with membrane protein.
   d. present entirely as *Chl* a.

2. The pigment molecules responsible for photosynthesis are located in the
   a. cytoplasm of the cell.
   b. stroma of the chloroplast.
   c. thylakoid membrane of the chloroplast.
   d. all of the above.

3. Antenna pigments
   P. are bound specifically in pigment-protein complexes.
   Q. donate electrons directly to the electron transport chain.
   R. absorb light energy and transfer it efficiently to the reaction center.
   S. often include carotenoids and/or xanthophylls.
   a. P and Q
   b. P and R
   c. P, R and S
   d. P, Q, R and S

4. Consider the following pigments
   1. *Chl* a
   2. *Chl* b
   3. *Chl* c
   4. Carotene
   5. Biliprotein
   6. Xanthophyll

   The characteristic photosynthetic pigments present in chlorophyceae would include
   a. 1, 2, 3, 4
   b. 1, 3, 4, 5
   c. 2, 3, 5, 6
   d. 1, 2, 4, 6

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5. A carotenoid-less mutant plant was grown under normal sunlight, it will experience
   a. increased photosynthetic rate
   b. increased chlorophyll biosynthesis
   c. reduced photorespiration
   d. increased chlorophyll oxidation and necrosis [JNU]

6. Photooxidation of chlorophyll is prevented by
   a. carotenoid
   b. anthocyanin
   c. phycobilin
   d. fucoxanthin [GATE]

7. Antenna pigments transfer light energy to a reaction center
   a. by means of a process known as fluorescence
   b. by means of a process known as inductive resonance (direct transfer of energy)
   c. by the conduction of heat
   d. a, b and c are correct

8. Which pigment will have higher absorbance in red region?
   a. *Chl* a
   b. *Chl* b
   c. both have same absorbance
   d. carotene

9. Some bacteria contain extensive internal membranes that are involved in specific functions. One example of such membranes is chromatophores, which are
   a. photosynthetic membranes where light is converted to chemical energy in the form of ATP.
   b. convolutions of the inner membrane that extend into the interior of the mitochondrion.
   c. internal membranes where inorganic nitrogen containing compounds oxidize to generate ATP.
   d. interior compartment of chloroplast where fixation of CO_2 occurs. [GATE]

10. Colors of light most useful in photosynthesis are
    a. green, yellow and orange
    b. red, violet and blue
    c. infrared, red and yellow
    d. red, white and blue

11. Thomas Engelmann illuminated a filament of algae with light that passed through a prism, thus exposing different segments of the algal filament to different wavelengths of light. He added aerobic bacteria and found that these bacteria congregated in the areas illuminated by red and blue light. If you run the same experiment without passing light through a prism, what would you predict?
    a. There would be no difference in results.
    b. The number of bacteria would decrease along the entire length of the filament.
    c. The bacteria would be relatively evenly distributed along the length of the filament.
    d. The number of bacteria would increase along the entire length of the filament.

12. Photosynthetic yield will be more in
    a. continuous red light
    b. continuous blue light
    c. intermittent white light
    d. green light

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13. The individual (or individuals) who demonstrated, using isolated chloroplasts, that splitting of water by light could reduce a dye solution as well as release oxygen is/are
    a. R. Hill
    b. Ruben and Kamen
    c. F.F. Blackman
    d. Melvin Calvin

14. When DCPIP (Hill reagent) accepts electron from H_2O, its colour changes to
    a. blue
    b. red
    c. colourless
    d. none

15. The experiment material used by Van Neil, to prove that O_2 comes out from water was
    a. *Chlorella pyrenoidosa*
    b. Scenedesmus
    c. blue green algae
    d. *Chromatium vinosum*

16. The function of water in photosynthesis is to
    a. absorb light energy
    b. supply electrons in the light-dependent reactions
    c. transport H^+ ions in the light-independent reactions
    d. provide O_2 for the light-independent reactions

17. If photosynthesizing green algae (oxygenic photosynthesis) are provided with CO_2 labeled with an isotope of oxygen (^{18}O that is heavy but not radioactive), later analysis will show that all of the following compounds produced by the algae contain the ^{18}O label *except*
    a. PGA
    b. RuBP
    c. glucose
    d. O_2

18. Which statement about oxygenic photosynthesis is *false*?
    a. Oxygenic photosynthesis is a type of photosynthesis in which carbon dioxide and water, in the presence of light and chlorophyll, is converted into glucose and oxygen.
    b. Oxygenic photosynthesis is responsible, at least in part, for the oxygen present in today's earth's atmosphere.
    c. Oxygenic photosynthesis occurs in the cyanobacteria, algae and plants.
    d. Oxygenic photosynthesis is the only type of photosynthesis in existence today.

19. Which organism does *not* evolve oxygen in photosynthesis?
    a. anabaena
    b. funaria
    c. higher plants
    d. rhodospirillum

20. In anoxygenic photosynthesis, the green and purple bacteria do not use the following one as electron source
    a. H_2O
    b. H_2
    c. H_2S
    d. S (elemental sulphur) **[GATE]**

21. *Chlorella* is a suitable plant for an astronaut in space travel because
    a. it provides oxygen to the astronaut and carbon dioxide released is utilized for its food manufacture.
    b. it supplies abundant food to the astronaut.
    c. it is a single celled autotrophic plant and slow growing.
    d. its growth is quite rapid during space flight.

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22. Discovery of *Emerson effect* showed the existence of
    a. photorespiration
    b. light and dark reaction in photosynthesis
    c. photophosphorylation
    d. two distinct pigment systems

23. Which of the following most precisely designates the position of the photosystems in higher plants?
    a. Membranes surrounding the chloroplast.
    b. The thylakoid space or compartments.
    c. Membranes of the thylakoid that form the grana.
    d. Stroma of the chloroplast.

24. In chloroplast, PSI present on the
    a. appressed membrane of the thylakoid
    b. non-appressed membrane of the thylakoid
    c. lumen of the thylakoid
    d. inner membrane of chloroplast

25. The two photosystem I and II during non-cyclic photophosphorylation acts in
    a. parallel
    b. series only
    c. parallel and series only
    d. none

26. The mechanism of ATP formation both in chloroplast and mitochondria is explained by
    a. relay pump theory of Godlewski
    b. Cholodny went's model
    c. chemiosmotic theory
    d. Munch's pressure/mass flow model

27. Phototrophs trap sunlight whereas chemotrophs derive their energy from the oxidation of a fuel. On this basis, which among the following observations is *true*?
    a. *Spirulina* sp. are chemotrophs whereas hydrogen bacteria are phototrophs.
    b. Both *Spirulina* sp. and hydrogen bacteria are phototrophs.
    c. Both *Spirulina* sp. and hydrogen bacteria are chemotrophs.
    d. *Spirulina* sp. is a phototroph while hydrogen bacteria are chemotrophs. [JNU]

28. A scientist who worked on a single-celled green plant (an alga) treated it with ultraviolet light and produced mutant cells that did not contain chloroplasts. To keep these cells alive, the scientist had to
    a. provide them with high light intensity.
    b. keep them in the dark.
    c. provide them with a sugar solution.
    d. give them chlorophyll.

29. Which of the following is *correct* concerning the evolution of photosystem II in cyanobacteria?
    a. It made photosystem I in these organisms unnecessary for photosynthetic fixation of carbon dioxide.
    b. It provided these organisms with an almost inexhaustible supply of electrons from water.
    c. It allowed these organisms to use any electron donor to replace electrons lost from excited chlorophyll *a* molecules.
    d. It allowed these organisms to generate ATP for the fixation of carbon dioxide into sugars without chemiosmosis.

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30. Reaction center of *Rhodopseudomonas viridis*

P. absorbs light of >800 nm.

Q. transfers electrons from QH_2 to a cytochrome.

R. special pair chlorophyll molecules reduced by electrons from H_2O.

S. special pair chlorophyll molecules reduced through cytochrome.

a. P and Q
b. R and S
c. P, Q and S
d. Q, R and S

Light reactions

1. Assimilatory powers produced during photosynthesis are
   a. RUBP and Rubisco
   b. H_2O and O_2
   c. ATP and NADPH
   d. $C_6H_{12}O_6$ and PGAL

2. In photosynthesis, the first step is
   a. conversion of PG to PGA
   b. carboxylation of RUBP
   c. electron release by chlorophyll *a*
   d. photolysis of water

3. Photochemical reaction occurs in
   a. lumen of thylakoid
   b. stroma of chloroplast
   c. membrane of thylakoid
   d. inner membrane of chloroplast

4. Which of the following pigment is involved in photochemical reaction in pigment system?
   a. *Chl* b
   b. *Chl* a
   c. carotene
   d. xanthophylls

5. Consider the following events in photochemical reaction by chlorophyll during light reaction
   1. charge separation
   2. absorption of quantum of energy
   3. ejection of electron
   4. transfer of light energy

   a. 4, 2, 1, 3
   b. 4, 2, 3, 1
   c. 2, 4, 1, 3
   d. 2, 4, 3, 1

6. What happens in chloroplast in the presence of light?
   a. Hydrogen's protons accumulate in thylakoid spaces
   b. Water is being split
   c. Electrons are moving
   d. All of the above

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7. During non-cyclic photophosphorylation, the reduction of chlorophyll *a* to its original forms is effected by
   a. electron stored within cytochrome *a*.
   b. electron released following irradiation of cytochrome *b*.
   c. electron carried by NADH.
   d. electron released by photolysis of water.

8. Oxygen evolving complex, involved in photolysis of water contain
   a. Mn^{2+}, Cl^-, K^+
   b. Mg^{2+}, Cl^-, Ca^{2+}
   c. Mn^{2+}, Cl^-, CO
   d. Mn^{2+}, Cl^-, Ca^{2+}

9. During photosynthesis, the protons created by the splitting of H_2O
   a. increases the concentration of protons in the lumen.
   b. are transferred to photosystem I.
   c. freely dissipate through the thylakoid membrane.
   d. are pumped out by the b_6f complex to generate a proton gradient across the thylakoid membrane.

10. Noncyclic photophosphorylation in photosynthesis activates
    a. PSII only
    b. PSI then PSII
    c. PSII then PSI
    d. PSI and PSII simultaneously

11. During non-cyclic photophosphorylation, plants and cyanobacteria produce
    a. ATP
    b. NADPH
    c. NADPH and ATP
    d. NADH and ATP [JNU]

12. Consider the following
    1. cytochrome b_6
    2. cytochrome *f*
    3. plastocyanin
    4. plastoquinone

    What is the sequence of these in the photo-induced electron carriers between PSII to PSI in photosynthesis?
    a. 4, 1, 2, 3
    b. 3, 4, 1, 2
    c. 1, 2, 3, 4
    d. 2, 3, 4, 1

13. Electron carriers in non-cyclic flow of electrons
    a. pass electrons from a more negative to a more positive reduction potential.
    b. pass electrons from higher free energy to lower free energy.
    c. pass electrons from a more positive to a more negative reduction potential.
    d. both a and b.

14. Cooperation of the two photosystems of the chloroplast is required for
    a. ATP synthesis
    b. reduction of $NADP^+$
    c. oxidation of the reaction center of photosystem I
    d. generation of a proton-motive force

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15. The effect of herbicide, which blocks electron transport from photosystem II to photosystem I will result in
a. enhancement of dark reaction
b. failure to oxidize NADPH
c. lack of reduction of ferredoxin
d. both b and c

16. Which is the strongest reducing agent in photosynthetic electron-transfer reactions?
a. plastoquinone
b. P680*
c. P700*
d. P700

17. Photoexcited P_{680} and P_{700}
a. have the same chemical properties of the ground state molecules
b. are strong oxidants-taking electrons from donor molecules
c. are strong reductants-giving electrons to acceptor molecules
d. transfer excited state energy to another molecule

18. In the presence of light, pH of the lumen of thylakoid
a. increases
b. decreases
c. remain same
d. none

19. What is the step in photosynthesis that contributes the greatest number of protons to the generation of a concentration gradient across the thylakoid membrane?
a. the Q cycle
b. reduction of $NADP^+$ to NADPH
c. oxidation of water to O_2
d. none of the above contribute to the proton concentration gradient

20. The proton motive force develops across the thylakoid membrane in the presence of light is result of
a. only pH gradient
b. only voltage gradient
c. sum of both pH and voltage gradient
d. none of the above

21. If the thylakoid are treated with detergent to stabilize lipids and then the proteins gently isolated, the following complexes will be observed, *except*
a. ATP synthase
b. cytochrome oxidase
c. light harvesting complex
d. pheophytin

22. Which one of the following compound in non-cyclic photophosphorylation in light reaction contains copper?
a. cytochrome
b. ferridoxin
c. plastoquinone
d. plastocyanin

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23. A mutation that inactivates the cytochrome b/*f* complex would
    a. inhibit movement of electrons from PSII to PSI
    b. inhibit movement of electrons from PSI to PSII
    c. inhibit reduction of quinone
    d. promote formation of NADPH

24. In non-cyclic light reactions of photosynthesis, electrons from chlorophyll molecules in photosystem I are used in the formation of NADPH. What is the source of such electrons?
    a. From light, which is converted directly into an electron.
    b. From ferredoxin.
    c. From mitochondria, which release the electrons at the end of the electron transport chain.
    d. From photosystem II, which splits water molecules.

25. All of the following are *false* for non-cyclic electron transport in photosynthetic light reactions *except*
    a. the components of electron transport are arranged asymmetrically across the thylakoid membrane.
    b. electrons flow from low energy water to reduce high energy $NADP^+$ due to absorption of photons of light (by PSII and PSI).
    c. electron transport establishes a proton gradient across the thylakoid membrane.
    d. electron transport between PSII and PSI produces ATP by substrate phosphorylation.

26. Reduction of NADP into NADPH during light reaction occurs in
    a. lumen of thylakoid
    b. stroma
    c. membrane of thylakoid
    d. stroma lamellae

27. Photosynthetic pathway for liberation of one molecule of O_2 requires 8 quanta. How many ATP and NADH forms during this process
    a. 2 ATP + 2 NADPH
    b. 9 ATP + 1 NADPH
    c. 2 ATP + 1 NADPH
    d. 4 ATP + 0 NADH

28. The reactions of photosynthesis are always presented as light dependent and light independent. Which of the following is not required (not part of the input) of the light dependent reactions of photosynthesis?
    a. Carbon dioxide
    b. $NADP^+$
    c. ADP
    d. A structurally intact chloroplast

29. Which of the following is not required by the light independent reactions?
    a. NADPH
    b. ATP
    c. RuBP–a five carbon sugar
    d. oxygen

30. The synthesis of ATP in the chloroplast is associated with
    a. non-cyclic electron flow
    b. an accumulation of H^+ in the thylakoid lumen
    c. electron flow between photosystem II to photosystem I
    d. all of the above

31. The energy in photosynthesis for attaching a phosphate group to ADP comes from
    a. breaking down glucose
    b. protons moving from the thylakoid compartment

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c. taking a phosphate off a sugar
d. all of the above

32. The chemiosmotic process in chloroplasts involves the
a. establishment of a proton gradient.
b. oxidation of water to produce ATP.
c. movement of water by osmosis into the thylakoid space from the stroma.
d. reduction of carbon dioxide to glucose by ATP and NADPH.

33. Formation of ATP by ATP synthase present on thylakoid membrane occurs during
a. downhill movement of protons from thylakoid lumen to stroma.
b. downhill movement of protons from stroma to thylakoid lumen.
c. uphill movement of protons from stroma to thylakoid lumen.
d. uphill movement of protons from thylakoid lumen to stroma.

34. Which of the following statements about cyclic photophosphorylation is *not* correct?
a. It does not involve NADPH formation.
b. It uses electrons supplied by photosystem I.
c. It involves a substrate level phosphorylation.
d. It does not generate oxygen. **[GATE]**

35. Which of the following statements about cyclic photophosphorylation is *not* correct?
a. It does not lead to the synthesis of NADPH.
b. It uses electrons supplied by photosystem II.
c. It does not generate oxygen.
d. It leads to pumping of protons via cytochrome *bf* complex. **[GATE]**

36. In the chemiosmotic hypothesis at ATP synthesis by chloroplast
a. addition of ATP synthetase to membrane vesicles will allow the synthesis of ATP if only ADP is added.
b. ATP synthetase must be very close to the two photosystems in the lamella.
c. proton ionophores have no effect on ATP synthesis.
d. the pH of the stroma rises due to proton pumping. **[JNU]**

37. Which statement about the light reactions of photosynthesis is *false*?
a. Electron from chlorophyll *a* are boosted to a higher energy level.
b. As electrons are lost from chlorophyll *a* and they are replaced through the photolysis of water.
c. The photolysis of water releases protons into the interior of chloroplast membrane sacs (thylakoids), thus generating a proton gradient that can drive ATP synthesis.
d. Photophosphorylation results in the oxidation of $NADP^+$ to NADPH.

38. Assume the thylakoid membrane within a chloroplast is punctured so that the thylakoid space is no longer separated from the stroma. Which of these processes would be most affected?
a. The splitting of water.
b. The absorption of sunlight by pigments.
c. The flow of electrons from PSII to PSI. **[MKU]**
d. The synthesis of ATP.

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Calvin cycle

1. Which of the following statements about the Calvin cycle are *true*?

P. It regenerates the ribulose 1,5-bisphosphate consumed by the rubisco reaction.

Q. It forms glyceraldehyde 3-phosphate, which can be converted to fructose 6-phosphate.

R. It requires ATP and NADPH.

S. It consists of enzymes, which can be activated through reduced thioredoxin.

a. P and Q
b. Q and R
c. P, R and S
d. P, Q, R and S

2. The total requirement of ATP and NADPH for each molecule of CO_2 fixed and reduced in photosynthesis in the Calvin cycle is

a. 2 ATP and 2 NADPH
b. 2 ATP and 3 NADPH
c. 3 ATP and 2 NADPH
d. 4 ATP and 3 NADPH

3. Chloroplasts are disrupted and the stroma separated from the lamellae. The isolated stroma will fix CO_2 if it is supplied with

a. oxygen
b. carotenoid
c. light
d. ATP + NADPH

4. In Calvin cycle, Rubisco incorporates CO_2 into ribulose 1,5-bisphosphate which rapidly splits into

a. glyceraldehyde-3-phosphate
b. 2,3-phosphoglyceric acid
c. 3-phosphoglycerate
d. 1,3-diphosphoglycerate

[GATE]

5. Which of the following is the most abundant protein in leaves?

a. chlorophyll a/b binding protein
b. ATP synthase
c. ribulose-1,5 bisphosphate carboxylase
d. globulins

[GATE]

6. If radioactive carbon dioxide ($^{14}CO_2$) is used in Calvin cycle then which carbon atom of 3-phosphoglycerate will appear radioactive?

a. Carbon-3
b. Carbon-1
c. Carbon-2
d. None of these

7. The first carboxylation product of a C_3 plant is

a. phosphoglyceric acid
b. oxaloacetic acid
c. malic acid
d. phosphoglyceraldehyde

8. Which of the following statement is *correct*? Rubisco

[JNU]

a. catalyzes the fixation of CO_2
b. can function as an oxygenase
c. is the most abundant protein on earth
d. all of the above

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9. Rubisco
   a. catalyzes the carboxylation of CO_2 to ribulose 1,5-bisphosphate.
   b. initiates photorespiration when the CO_2/O_2 ratio is low.
   c. catalyzes the reduction of two molecules of PGAL to form glucose.
   d. a, b but not c.

10. If asked to pinpoint precisely the site of synthesis of 3-phosphoglyceraldehyde in plants, you would have to say that it takes place within the
   a. mesophyll
   b. grana
   c. stroma
   d. chloroplast

11. Which one of the following statements regarding photosynthesis is *correct*?
   a. Light is absorbed by chlorophyll during Calvin cycle.
   b. Cyclic phosphorylation produces neither oxygen nor NADPH.
   c. The site of light reaction is the stroma of the chloroplast.
   d. NADPH and ATP are generated during the light independent phase to power Calvin cycle. **[IISc]**

12. During photosynthesis, the final product of the Calvin cycle is
   a. ribulose bisphosphate
   b. phosphoglycerate
   c. pyruvate
   d. glyceraldehyde-3-phosphate **[IISc]**

13. Three molecules of RuBP in C_3 cycle are regenerated from
   a. 4 molecules of glyceraldehyde-3-phosphate
   b. 5 molecules of 3-phosphoglyceric acid
   c. 5 molecules of glyceraldehyde-3-phosphate
   d. 6 molecules of glyceraldehyde-3-phosphate

14. The final product of the Calvin cycle is
   a. RuBP
   b. phosphoglyceric acid
   c. phosphoglyceraldehyde
   d. glucose

15. Production of one molecule of 3-phosphoglyceraldehyde requires how many turns of the Calvin cycle?
   a. 1 b. 2
   c. 3 d. 6

16. Calvin cycle represents which of the following phenomenon?
   a. oxidative carboxylation
   b. substrate level phosphorylation
   c. dark respiration
   d. reductive carboxylation

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17. Which of the following is *not* a regulatory factor of carbon fixation?
    a. The level of acetyl-CoA in the chloroplast stroma.
    b. The levels of ATP and NADPH in the stroma.
    c. pH of the stroma.
    d. Magnesium level in the stroma.

18. Which of these statements about thioredoxin is *correct*?
    a. It contains a heme that cycles between two oxidation states.
    b. Its oxidized form predominates while light absorption is taking place.
    c. It activates some biosynthetic enzymes by reducing disulfide bridges.
    d. It activates some degradative enzymes by reducing disulfide bridges.

19. Synthesis of sucrose during photosynthesis occurs in
    a. chloroplast
    b. cytosol
    c. vacuoles
    d. amyloplast

20. Which of the following statements is *correct* with respect to the synthesis of starch and sucrose in plants?
    a. Starch is synthesized in the mitochondria and sucrose in the cytosol.
    b. Starch is synthesized in the cytosol and sucrose in the chloroplast.
    c. Starch is synthesized in the chloroplast and sucrose in the mitochondria.
    d. Starch is synthesized in the chloroplast and sucrose in the cytosol. [JNU]

Photorespiration, C_4 and CAM pathway

1. Inhibition of photosynthesis in the presence of O_2 in C_3 plants is called
    a. Pasteur effect
    b. Warburg effect
    c. Decker effect
    d. hexose monophosphate shunt

2. Photorespiration
    a. occurs because oxygen rather than carbon dioxide links to the rubisco enzyme in the Calvin cycle.
    b. occurs more in C_4 than in C_3 plants under identical conditions.
    c. describes the uptake of carbon dioxide and the release of oxygen in chloroplasts.
    d. all of the above.

3. In photorespiration, glycolate and glyoxylate are produced sequentially in the following organelles. Choose the *correct* sequence.
    a. chloroplast and mitochondria
    b. chloroplast and peroxisome
    c. peroxisome and mitochondria
    d. peroxisome and chloroplast [GATE]

4. When *Rubisco* acts as an oxygenase
    a. phosphoglycerate and phosphoglycolate are produced
    b. phosphoenolpyruvate is oxidized
    c. net carbon fixation is enhanced
    d. it must mean that the plant is deprived of CO_2

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5. Which of the following statements best supports the fact that photorespiration, commonly occurs in C_3 plants?
   a. C_3 plants do not possess Kranz anatomy.
   b. C_3 plants have usually high CO_2 compensation species.
   c. C_3 plants are less efficient in photosynthesis.
   d. C_3 plants are characterized by RuBP oxygenase activity under high oxygen supply.

6. C_4 plants are able to minimize photorespiration because C_4 plants
   a. do not carry out the Calvin cycle.
   b. use a more efficient enzyme to initially fix CO_2.
   c. exclude Calvin cycle.
   d. none.

7. The enzyme phosphoenolpyruvate carboxylase catalyses the combination of CO_2 and phosphoenolpyruvate to form oxaloacetic acid in
   a. pea
   b. sugarcane
   c. pinus
   d. apple **[JNU]**

8. PEP carboxylase is
   a. involved in at least some CO_2 fixation in both C_3 and C_4 plants.
   b. catalyzes the reaction fixing CO_2 into pyruvic acid in bundle sheath cells.
   c. capable of fixing CO_2 more efficiently at lower atmospheric CO_2 concentrations than RuBP carboxylase.
   d. the most abundant enzyme in the world.

9. In C_4 pathway, the site for oxidative decarboxylation is
   a. cytoplasm of bundle sheath
   b. chloroplast of mesophyll cell
   c. chloroplast of bundle sheath
   d. cytoplasm of mesophyll cell

10. Conversion of pyruvate into PEP by use of ATP in C_4 pathway occurs in
   a. mesophyll cell's cytoplasm
   b. mesophyll cell's chloroplast
   c. bundle sheath cell's cytoplasm
   d. bundle sheath cell's chloroplast

11. Consider the following compounds
   1. phosphoenolpyruvate
   2. malic acid
   3. pyruvic acid
   4. oxaloacetic acid

   What is the correct sequence of these intermediate compounds as they form in C_4 dicarboxylic acid pathway following the PEP-carboxylation reaction?
   a. 1, 2, 3, 4
   b. 2, 4, 3, 1
   c. 4, 1, 2, 3
   d. 4, 2, 3, 1

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12. Total energy requirement for fixing one CO_2 by combined C_4 and C_3 cycle is
    a. 5 ATP + 4 NADPH
    b. 3 ATP + 4 NADPH
    c. 5 ATP + 2 NADPH
    d. 8 ATP + 2 NADPH

13. In which of the following cells of a C_3 leaf would you find the highest concentration of enzyme rubisco?
    a. epidermal cell
    b. mesophyll cell
    c. bundle sheath cell
    d. b and c

14. Which of the following is *true* of C_4 photosynthesis?
    a. It seems to be advantageous in cooler climates.
    b. It has high compensation point for CO_2.
    c. Similar to the CAM photosynthesis pathway.
    d. All of the above.

15. By decreasing photorespiration and efficiently carrying out photosynthesis, C_4 plants can
    a. reduce stomatal opening and thus reduce leaf temperature.
    b. reduce stomatal opening and thus reduce water loss.
    c. increase stomatal opening and thus increase water loss.
    d. increase stomatal opening and thus increase transpiration rates. [IISc]

16. C_4 plants are abundant in
    a. temperate region with more humid conditions
    b. temperate region with more dry conditions
    c. tropical region with more humid conditions
    d. tropical region with more dry condition

17. A C_3 mustard plant, was grown at 300 ppm of CO_2 in 14hr light/10hr dark cycles, it was transferred to 1000 ppm CO_2. This will lead to (other environmental parameters remaining identical)
    a. increase in photosynthetic rate
    b. decrease in photosynthetic rate
    c. increase in respiration rate
    d. no change [JNU]

18. CAM plants have an adapted photosynthetic process that primarily allows them to
    a. survive more efficiently in humid conditions.
    b. survive in arid conditions.
    c. fix carbon dioxide during the day and night.
    d. prevent the enzyme rubisco from being denatured.

19. Which of the following plant shows diurnal change in malic acid content in vacuoles and popularly known as Dark fixer?
    a. C_4 plants
    b. C_3 plants
    c. CAM plants
    d. plant showing hexose monophosphate shunt

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20. In CAM plants, CO_2 acceptor in the night is
a. RUBP
b. PEP
c. OAA
d. PGA

[JNU]

21. In Crassulacean Acid Metabolism (CAM) plants
a. phosphoenolpyruvate carboxylase reduces CO_2 to make malic acid.
b. stomata open only during the day.
c. vascular tissue is surrounded by a bundle sheath.
d. rubisco is activated only at night.

22. Which of the following incorporate CO_2 into organic acids, and then release it for use in the Calvin cycle?
a. C_3 plants
b. C_4 plants
c. C_3 and CAM plants
d. C_4 and CAM plants

23. Which of the following plant species have highest photosynthetic yield?
a. Species possessing CAM pathway
b. Species possessing C_3 pathway
c. Species possessing C_4 pathway
d. Same for all

24. At low CO_2 concentration, photosynthesis is limited by
a. the carboxylation capacity of rubisco
b. ability of Calvin cycle enzyme to regenerate RuBP
c. ATP + NADPH
d. all of the above

25. Which of the following statements about C_3 and C_4 plants are *correct*?
P. In normal air, the quantum yield of C_3 plants is higher than C_4 plants.
Q. In C_4 plants, the quantum yield does not show a temperature dependence.
R. In C_3 plants, increase in atmospheric CO_2 concentrations above the compensation point stimulates photosynthesis over a wide concentration range.
S. In C_4 plants, due to presence of the CO_2 concentrating mechanisms, photorespiration is low.
a. Q, R and S
b. P, Q and R
c. P, R and S
d. P, Q, R and S

26. According to British plant physiologist F.F. Blackman's law of limiting factor, at any given time, photosynthesis can be limited by
a. light
b. CO_2 concentration
c. both light and CO_2 concentration
d. either by light or by CO_2

27. At CO_2 compensation point, net efflux of CO_2 from plant is
a. maximum
b. zero
c. minimum
d. none of the above

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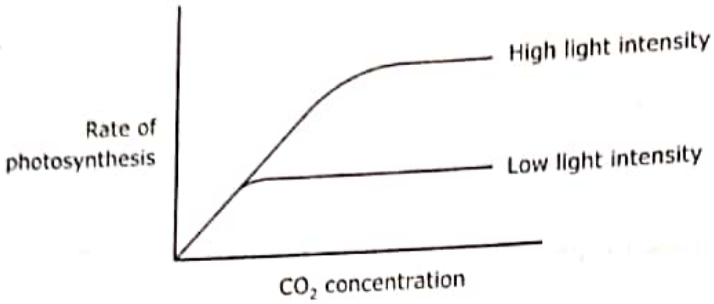
28. The diagram given below shows the relation between light intensity and CO_2 concentration on rate of photosynthesis



Which of the following explains the diagram *correctly*?

a. At low intensity, the light dependent reaction is rate limiting but at high light intensity, the light independent reaction involving CO_2 is rate limiting.
b. The number of chlorophyll molecules prevents the rate of photosynthesis.
c. Photosynthesis is independent of light intensity.
d. Chlorophyll respond equally to light intensities and CO_2 concentrations, thereby limiting the rate of photosynthesis.

29. In shade loving plants, their light compensation point is reached at low irradiance because

a. leaves have more chlorophyll per reaction center.
b. these plants have greatly reduced respiration rates.
c. shade plant is capable of much lower rates of photosynthesis than the sun plants.
d. shade loving plants synthesize pigments such as anthocyanins to promote photosynthesis.

30. The quantum yield of photosynthetic carbon fixation in a C_3 plant and C_4 plant is studied as a function of leaf temperature. Following are some statements based on this study.

P. At lower temperature, the quantum yield of C_3 plant is lower than C_4 plant.
Q. In C_4 plant, quantum yield does not show a temperature dependence.
R. Since the photorespiration is low in C_4 plants because of CO_2 concentrating mechanism, quantum yield is not much affected by change in atmospheric CO_2 concentration.
S. At higher temperature, the quantum yield of C_3 plant is lower than C_4 plant.

Which one of the following combination of above statements is *correct*?

a. P, Q and S
b. P, R and S
c. P, Q and R
d. Q, R and S

[CSIR]

Transport of photoassimilate

1. Sieve tube elements

a. are the conducting elements of the phloem.
b. have sieve plates with pores normally blocked with P-protein.
c. must lose their plasma membranes to become fully functional.
d. all of the above.

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2. Transport of the photoassimilate is thought to occur by pressure flow through the sieve tubes of the phloem from a source to a sink which of the following statements about phloem transport is *false*?
   a. Water enters the sieve tubes by osmosis.
   b. Sieve tubes in a source have a low hydrostatic pressure.
   c. Water and solutes move through the sieve tubes along a pressure gradient.
   d. Solutes are swept along by the bulk flow of water in the sieve tube.

3. Which statement about phloem transport is *correct*?
   a. Gravity influence phloem transport.
   b. Phloem transport occurs unidirectionally.
   c. Sugar is transported in phloem as non-reducing sugar.
   d. Ca^{2+} is most abundant cation in phloem sap.

4. Translocation of carbohydrates occur mainly as
   a. glucose
   b. fructose
   c. maltose
   d. sucrose

5. Which of the following observations provides direct support for Munch's hypothesis?
   a. Transpiration occurs from open stomata on leaves and stems during the day.
   b. Turgor pressure in the sieve tubes of a fully expanded tomato leaf is higher than the turgor pressure in the sieve tubes of a developing tomato fruit.
   c. Carbohydrates move within the plant from sink to source.
   d. The osmotic potential of the sieve tubes in a root tip is more negative than that of the sieve tubes in the leaf.

6. Two aphids, named Alice and Bob, on the same plant always have their lunch in different places. Alice inserts her stylet into the phloem of a minor vein on a large, mature leaf, while Bob prefers to probe the phloem of the young growing tissue near the shoot apical meristem.
   a. Alice's phloem is under higher pressure than Bob's.
   b. The sap of Bob's phloem has more negative is than that of Alice's phloem.
   c. The phloem sap flows from Bob to Alice.
   d. Alice is having lunch at a source, while Bob has lunch at a sink.

7. Which of the following is not a difference between sugar translocation in phloem and water movement in the xylem?
   a. Sugar translocation is a metabolically active process, while water movement is entirely passive.
   b. Pressure is positive in sieve tubes, while xylem is usually under tension.
   c. Sieve tubes are living cells, while mature xylem is dead.
   d. All of these are correct.

8. Regarding sugar translocation, which of the following choices is *correct*?
   a. Sugars are always translocated downward, or towards the base of the plant.
   b. A source is a site where sugars are produced, while sinks are sites where sugars are utilized.
   c. The transport sugar in all plants is sucrose.
   d. Sucrose moves through the phloem, against a concentration gradient.

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9. Starch is insoluble in water but still it is stored in large quantity in potato root because
   a. it is synthesized in potato root.
   b. it is translocated in the form of sugar from leaves.
   c. soil microorganisms deposit it in the tuber.
   d. it is useful for human.

10. Translocation of sugars through phloem is an active process, requiring metabolism. The input of energy is mainly needed for which of the following processes?
   a. Loading of sugars into phloem at the source.
   b. Storage of sugars at the sink.
   c. The pumping of water at the source.
   d. Pumping of water at the sink.

11. Translocation of organic materials in plant is best explained by
   a. active transport
   b. transpiration pull
   c. imbibition theory
   d. mass flow hypothesis

12. Which one of the following statements about transport of nutrients in phloem is *false*?
   a. Solute particles can be actively loaded into phloem at the source.
   b. Companion cells control the rate and direction of movement of phloem sap.
   c. Differences in osmotic concentration at the source and sink generate a hydrostatic pressure gradient.
   d. A sink is that part of the plant where a particular solute is consumed or stored.

13. Arrange the following five events in an order that explains the bulk flow of substances in the phloem.
   1. Water diffuses into the sieve tube elements.
   2. Leaf cells produce sugar by photosynthesis.
   3. Solutes are actively transported into sieve elements.
   4. Sugar is transported from cell to cell via the apoplast and/or symplast.
   5. Sugar moves down the stem.

   a. $2 \rightarrow 1 \rightarrow 4 \rightarrow 3 \rightarrow 5$
   b. $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$
   c. $2 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 5$
   d. $4 \rightarrow 2 \rightarrow 1 \rightarrow 3 \rightarrow 5$

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Chapter 08

Molecular Genetics

DNA replication

1. Characteristic unique to DNA is
   a. denaturation and renaturation
   b. polymer complex
   c. replication
   d. resistance to temperature change

2. Radioactive thymidine when added to the medium surrounding living mammalian cells gets incorporated into the newly synthesized DNA. Which of the following types of chromatin is expected to become radioactive if cells are exposed to radioactive thymidine as soon as they enter the S-phase?
   a. neither heterochromatin nor euchromatin but only the nucleus
   b. heterochromatin
   c. euchromatin
   d. both heterochromatin and euchromatin

3. A bacterial culture growing in a medium containing $^{15}NH_4Cl$ is switched to a medium containing $^{14}NH_4Cl$ for three generations (resulting into eight fold increase in its population), what is the molar ratio of hybrid DNA (^{15}N- ^{14}N) to light DNA (^{14}N- ^{14}N) at this point?
   a. 2/6 or 0.33
   b. 3/5 or 0.60
   c. 4/4 or 1.00
   d. 1/7 or 0.14

4. *E. coli* cells that have been cultured in a light medium containing ^{14}N-ammonium chloride as the sole nitrogen source were transferred to a heavy medium containing ^{15}N-ammonium chloride. After three generations of growth in the heavy medium, what will be the proportion of dual light strands (LL), dual heavy strands (HH), and hybrid light- plus-heavy strands (LH) in the DNA duplexes?
   a. 3/4 LH to 1/4 HH
   b. 3/4 HH to 1/4 LH
   c. 3/4 HH to 1/2 LH
   d. 3/4 LL to 1/4 LH

5. In the Meselson and Stahl experiment, *E. coli* cells grown on heavy nitrogen were transferred to light nitrogen. What % of DNA can be expected to be constituted of light nitrogen after 3 generation of multiplication?
   a. 25
   b. 50
   c. 75
   d. 100 **[JNU]**

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6. *E. coli* cells grown on ^{15}N medium are transferred to ^{14}N medium and allowed to grow for two generations (two cell replications). DNA extracted from these cells is centrifuged. What density distribution of DNA would you expect in this experiment?
   a. One high-density and one low-density band.
   b. One intermediate-density band.
   c. One high-density and one intermediate-density band.
   d. One low-density and one intermediate-density band

7. Isotopes used for proving semiconservative replication of DNA are
   a. N^{14} and P^{31}
   b. N^{14} and N^{15}
   c. N^{14} and C^{14}
   d. C^{14} and P^{31}

8. If DNA is synthesized in an *in vitro* reaction containing equimolar amounts of substrates (adenine, thymine, guanine and cytosine) from a template with [A+G/C+T] = 0.97, the value of [A+G/C+T] in the product will be close to
   a. 0.25
   b. 0.5
   c. 0.8
   d. 1.01

[JNU]

9. A solution contains DNA polymerase I, Mg^{2+} salts of dATP, dGTP, dCTP, and dTTP, and an appropriate buffer. Which of the following DNA molecules would serve as a template for DNA synthesis when added to this solution?
   a. A single-stranded closed circle.
   b. A single-stranded closed circle base-paired to a shorter linear strand with a 3'-terminal hydroxyl.
   c. A single-stranded closed circle base-paired to a shorter linear strand with a 3'-terminal phosphate.
   d. A double-stranded closed circle.

10. In a reaction containing dNTPs and DNA polymerase in buffer, on which of the following molecules DNA synthesis would occur?
   a. A single-stranded closed circular DNA.
   b. A double-stranded closed circular DNA.
   c. A single-stranded closed circular DNA paired with a short oligonucleotide with a 3' terminal phosphate.
   d. A single-stranded closed circular DNA paired with a short oligonucleotide with a 3' hydroxyl group.

11. Any DNA molecule that has the ability to replicate autonomously is called
   a. plasmid
   b. chromosome
   c. genome
   d. replicon

[MKU]

12. In genetics, an origin is
   a. a protein that binds to DNA at the point where replication begins.
   b. a protein that binds to RNA at the point where translation begins.
   c. the point in DNA where transcription starts.
   d. the point in DNA where replication starts.

13. Replication of DNA in eukaryotes commences from
   a. one end of chromatid extending to the other end.
   b. both ends of the chromatid simultaneously.
   c. the centromere to either of the ends of chromatids.
   d. several sites of the DNA of the chromatid simultaneously.

[JNU]

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14. How does the DNA of the circular *E. coli* chromosome replicate?
a. unidirectionally: it starts at one point on the chromosome and replicates in one direction.
b. bidirectionally: it starts at one point on the chromosome and replicates in both directions.
c. multidirectionally: it starts at several points on the chromosome proceeding in both directions.
d. all the above statements are true.

15. DNA gyrase
P. has the ability to introduce negative supercoils into DNA.
Q. requires free energy from ATP hydrolysis.
R. is a tetramer of two different subunits – GyrA and GyrB subunit.
S. activity inhibited by quinolone antibiotics, such as nalidixic acid, norfloxacin and ciprofloxacin.
a. P and Q
b. P, Q and R
c. P, Q, R and S
d. Q and R

16. Synthesis of DNA was shown to be template dependent and semi-conservative. This would predict that
a. no organism can have a single-stranded genome.
b. RNA cannot serve as the genome of any organism.
c. all single stranded genomes would be synthesized via a double-stranded intermediate.
d. template independent DNA polymerases do not exist in nature. **[IISc]**

17. *Correct* statements describing DNA polymerase I include which of the following?
a. It can add deoxyribonucleoside triphosphates to the 3'-hydroxyl end of a DNA primer chain.
b. It can act as an endonuclease in DNA repair.
c. It can act as a 5' to 3' - exonuclease.
d. It is the major polymerizing enzyme in the replication process.

18. Which of the following statements about DNA polymerase I is *correct*?
a. It functions as a DNA repair enzyme but is not involved in the DNA replication process.
b. It requires a template and a primer to polymerize deoxyribonucleoside monophosphates.
c. It joins together Okazaki fragments to complete the lagging strand during DNA replication.
d. It produces Okazaki fragments linked to RNA primer chains.

19. Which of the following property is *not* associated with DNA polymerase I?
a. 5' → 3' exonuclease activity
b. 3' → 5' exonuclease activity
c. 5' → 3' endonuclease activity
d. 5' → 3' polymerase activity

20. The fact that DNA polymerase I from *E. coli* has a 5'–3' exonuclease activity
a. implies that the enzyme has multiple subunits.
b. implies that DNA polymerase I can use both DNA and RNA as primers.
c. makes the enzyme able to detect thymine dimers in double - stranded DNA.
d. enables the enzyme to play an important role in DNA replication. **[GATE]**

21. In prokaryotes, the lagging primers are removed by
a. 3' to 5' exonuclease
b. DNA ligase
c. DNA polymerase I
d. DNA polymerase III

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22. The replicative polymerase in *E. coli* is
    a. DNA polymerase I
    b. DNA polymerase II
    c. DNA polymerase III
    d. DNA polymerase IV
    [IISc]

23. The large (Klenow) fragment of *E. coli* DNA polymerase I contains which of the following activities?
    a. Reverse transcriptase activity and nick-translation activity.
    b. Polymerase activity and 3′–5′-exonuclease activity.
    c. Polymerase activity and nick-translation activity.
    d. 5′–3′-exonuclease activity.
    [MKU]

24. The function of *E. coli* DNA polymerase II in the cell is to
    a. initiate replication at the origins.
    b. carry out 'fill in' reaction at the Okazaki fragments after RNA primer removal.
    c. synthesis of leading strand only.
    d. replication restart at the stalled replication fork.
    [IISc]

25. During replication of the *E. coli* chromosome, Okazaki fragments are produced from
    a. only one of the strands of the circular genome.
    b. both the strands of the circular genome.
    c. one of the strands in one generation and the other strand in the next generation.
    d. both the strands of the circular genome provided that the heavy nitrogen (^{15}N) is present in the medium.
    [GATE]

26. During DNA replication, the synthesis of the leading strand occurs in the 5′→3′ direction along with the movement of the replication fork. Which one of the following statements is *true* about the synthesis of the lagging strand?
    a. DNA polymerase II, which has an additional activity of polymerization in 3′→5′ direction, is utilized.
    b. DNA polymerase I is utilized, which can carry out DNA synthesis in the absence of any primer.
    c. synthesis of the lagging strand occurs in the small fragments by DNA polymerase in 5′→3′ direction using a *de novo* synthesized DNA primer.
    d. synthesis of the lagging strand occurs in small fragments in 5′→3′ direction utilizing RNA primers.

27. Okazaki fragments
    a. require the activity of only a DNA polymerase for synthesis.
    b. require only RNA polymerase activity for synthesis.
    c. are made when DNA is exposed to UV radiation.
    d. are composed of both DNA and RNA.

28. Short chains of nucleic acid can be isolated from cells in which DNA is undergoing replication. These segments, known as Okazaki fragments, have which of the following properties?
    [JNU]
    a. They are double-stranded.
    b. They contain covalently linked RNA and DNA.
    c. They are DNA-RNA hybrids.
    d. They arise from the nicking of the sugar-phosphate backbone of the parental DNA chain.

29. Okazaki fragments are used to elongate
    a. the leading strand towards the replication fork.
    b. the lagging strand towards the replication fork.

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c. the leading strand away from the replication fork.
d. the lagging strand away from the replication fork.

30. Approximately how many Okazaki fragments are synthesized during one round of replication of the *E. coli* genome?
a. 5000 to 10000
b. 4×10^6
c. 2500 to 5000
d. 2

31. During replication of DNA, which one of the following enzyme polymerizes the Okazaki fragments? **[IISc]**
a. DNA polymerase I
b. DNA polymerase II
c. DNA polymerase III
d. RNA polymerase I

32. Which enzyme removes the RNA primer and fills in deoxyribonucleotides in prokaryotic replicons?
a. DNA polymerase III
b. DNA polymerase II
c. DNA polymerase I
d. exonuclease MFI

33. Which of the sequences listed below best describes the order in which the following enzymes participate in the replication of DNA in bacteria?
1. 5′ → 3′ polymerase activity of DNA polymerase I
2. 5′ → 3′ exonuclease of DNA polymerase I
3. DNA polymerase III
4. DNA ligase
5. primase

a. 5, 3, 1, 2, 4
b. 3, 2, 1, 5, 4
c. 5, 3, 4, 2, 1
d. 5, 3, 2, 1, 4

34. Which of the following statements regarding the supercoiling of duplex DNA is *correct*?
a. DNA replication promotes positive supercoiling downstream from the replication fork.
b. DNA polymerase III can replicate positively supercoiled duplex DNA.
c. DNA topoisomerases unwind duplex DNA into single-stranded DNA chains.
d. DNA topoisomerases unwind positively supercoiled duplex DNA without cutting the phosphosugar backbone.

35. The bacterial enzyme that changes positively supercoiled DNA into negatively supercoiled DNA is
a. DNA helicase
b. DNA gyrase
c. single strand binding protein
d. polymerase

36. DNA replication undergoing in an *E. coli* cell resulted into two circular DNAs interlocked with each other; this may be due to a defective gene encoding
a. primase
b. DNA topoisomerase I
c. DNA polymerase
d. DNA topoisomerase II

37. DNA polymerization requires the presence of template and of a free 3′-OH end. This end may be generated by
P. synthesis of an RNA primer at the origin or at Okazaki fragment start points.
Q. nicking one strand of duplex DNA followed by strand displacement.
R. by loop formation at the 3′ end (self-priming).
S. binding of a terminal nucleotide-carrying protein to the 5′ end of template.

a. P and Q
b. P and R
c. R and S
d. P, Q, R and S

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38. The exonuclease activity of the DNA polymerases functions to
    a. remove the RNA primer sequences
    b. proofread the new DNA strand and remove inappropriate nucleotides
    c. maximize the fidelity of DNA replication
    d. all of the above

39. In prokaryotes, the lagging strand primers are removed by
    a. 3′ to 5′ exonuclease
    b. DNA ligase
    c. DNA polymerase I
    d. DNA polymerase III

40. T4 DNA ligase
    a. requires ATP
    b. joins double-stranded DNA fragments with an adjacent 3′-phosphate and 5′-OH
    c. requires NADH
    d. joins single-stranded DNA

41. *E. coli* DNA ligase uses NAD^+ as a cofactor. During the actual ligation reaction (joining of the 5′ phosphate and 3′ hydroxyl to form a phosphodiester bond), the leaving group is
    a. NAD^+
    b. NMN
    c. ATP
    d. AMP

42. DNA ligase of *E. coli* requires which of the following cofactors?
    a. FAD as an electron acceptor.
    b. $NADP^+$ as a phosphate donor.
    c. NAD^+ to form an active adenyl enzyme.
    d. NAD^+ as an electron acceptor.

43. Which of the following is used in rolling circle DNA replication but not in normal cellular DNA replication?
    a. an endonuclease
    b. an exonuclease
    c. primase
    d. DNA ligase

44. The effects of various inhibitors of DNA synthesis have been investigated in an *in vitro* replication system containing *E. coli* enzymes. In such a system, replication of bacteriophage M13 DNA, a 6.4-kb molecule, is sensitive to rifampicin, an inhibitor of host-cell RNA polymerase, whereas replication of the *E. coli* cellular DNA is not inhibited by rifampicin. What do these results imply about the enzymes involved in replication of M13 DNA and *E. coli* DNA?
    P. Rifampicin inhibits DNA polymerase responsible for DNA replication of M13 DNA.
    Q. RNA primers required for replication of M13 DNA is synthesized by *E. coli* RNA polymerase.
    R. Rifampicin inhibits helicase involved during DNA replication of M13 DNA.
    S. Insensitivity of *E. coli* cellular DNA synthesis to rifampicin suggests that synthesis of RNA primers for replication of cellular DNA is catalyzed by primase rather than RNA polymerase.
    a. P and S
    b. R and S
    c. Q and S
    d. P and R

45. Which of the following DNA Pol has proofreading property in animal cells?
    a. DNA Pol α
    b. DNA Pol δ
    c. DNA Pol β
    d. all of these

46. Which of the following statement is *correct*?
    a. Meselson and Stahl used radioactive isotope of ^{15}N.
    b. Klenow fragment in DNA Pol I has 5′ → 3′ exonuclease property.

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c. DNA replication is semi-conservative as well as semi-discontinuous.
d. All the statements are correct.

47. Choose the *incorrect* statement.
a. Mitochondrial DNA replicates by D-loop formation.
b. Rolling circle replication is also known as σ-replication.
c. DNA Pol I is made up of single polypeptide.
d. DNA Pol III has ability of nick translation.

48. Under laboratory conditions, it takes 20 min for *E. coli* to divide. However, 40 min are required for *E. coli* chromosome to replicate. Nevertheless each daughter cell gets a full complement of chromosome because
a. replication in *E. coli* occurs in bidirectional mode.
b. homologous recombination leads to duplication of the genome.
c. completion of the lagging strand synthesis takes place only after the cell division.
d. second round of initiation of DNA replication begins before the completion of the first round. **[IISc]**

49. Choose the *correct* statement.
a. Both circular and linear DNA chromosomes after replication are separated by type II topoisomerase.
b. In *E. coli*, DNA Pols I, II and III have proofreading properties.
c. Telomerase is chemically ribonucleoprotein.
d. All the statements are correct.

50. Which of the following statements about topoisomerases is *incorrect*?
a. They alter the linking numbers of topoisomers.
b. They break and reseal phosphodiester bonds.
c. They require NAD^+ as a cofactor to supply the energy to drive the conversion of a supercoiled molecule to its relaxed form.
d. They form covalent intermediates with their DNA substrates.

51. Choose the mismatch

a. enzyme synthesizes the leading strand of eukaryotic DNA	1. DNA polymerase-ε
b. enzyme removes positive superhelical turns from DNA	2. helicase
c. enzyme maintains the ends of eukaryotic chromosomes	3. telomerase
d. enzyme synthesizes the lagging strand of eukaryotic DNA	4. DNA polymerase-δ

52. Eukaryotic DNA synthesis is inhibited by
a. aphidicoline
b. cyclohexamide
c. chloramphenicol
d. ampicillin **[IISc]**

53. What is the role of gyrase in DNA replication?
a. To relax or prevent supercoiling of the double helix.
b. To separate the two template strands of DNA at each replication fork.
c. To synthesize a short RNA primer for each of the new DNA strand.
d. To protect single-stranded regions of DNA from damage. **[MKU]**

54. Action of topoisomerase leads to change in
a. linking number of single-stranded linear DNA.
b. linking number of double-stranded linear DNA.
c. linking number of closed circular single stranded DNA.
d. linking number of closed circular double stranded DNA. **[IISc]**

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55. Which of the following statement about topoisomerases are *correct*?

P. Type I topoisomerases act by making a transient break in one strand of DNA.

Q. Type II topoisomerases act by introducing a transient double-strand break.

R. DNA gyrase is a type II topoisomerase.

S. DNA gyrase removes the positive supercoiling and introduce negative supercoiling.

a. P and Q
b. P, Q and R
c. P, R and S
d. P, Q, R and S

56. All of the following statements about telomerase are correct *except*

a. the RNA component acts as a template for the synthesis of a segment of DNA.
b. it adds telomeres to the 5'-ends of the DNA strands.
c. it provides a mechanism for replicating the ends of linear chromosomes.
d. it recognizes a G-rich single strand of DNA.

57. All of the following cell types contain the enzyme telomerase which protects the length of telomeres at the end of chromosomes, *except*

a. germinal
b. somatic
c. haemopoetic
d. tumor

58. Reverse transcriptases are encoded by retroviruses. The only enzyme from the mammalian cells with reverse transcriptase like activity is

a. telomerase
b. ribonuclease
c. ribozyme
d. caspase

[IISc]

59. Which of the following statement is *incorrect*?

a. The protein that binds to the 9-mers in the *oriC* site of *E. coli* is DnaA.
b. The enzyme that replicates DNA in mitochondria is called DNA pol γ.
c. The origin of replication site in the *E. coli* chromosome is about 245 bp long.
d. None of the above.

60. Mitochondrial DNA is replicated from

a. a single *ori* site bidirectionally.
b. two different *ori* sites in the same direction.
c. two different *ori* sites at different times in opposite directions.
d. many sites bidirectionally, like nuclear chromosomes.

61. Okazaki fragments were prepared from an *E. coli* strain harboring pBR322 plasmid. These fragments were then hybridized to single stranded DNA molecules prepared by carefully separating the two strands of the *E. coli* chromosome and the plasmid. The Okazaki fragments would anneal to

a. one strand of the chromosome but both the strands of the plasmid.
b. both the strands of the chromosome but only one strand of the plasmid.
c. both the strands of the chromosome and the plasmid.
d. one strand each of the chromosome and the plasmid.

62. Telomerase adds 6 nucleotide repeats to the ends of chromosomes called telomeres. The telomeric repeat sequence of *Tetrahymena* is TTGGGG. The telomeric repeat sequence of yeast is GTGTGT. Propagation of *Tetrahymena* chromosomes in yeast will result in extension as follows

a. TTGGGG TTGGGG TTGGGG
b. GTGTGT GTGTGT GTGTGT
c. GTGTGT TTGGGG GTGTGT
d. TTGGGG GTGTGT TTGGGG

[IISc]

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63. A mutant of *E. coli* with a heat sensitive DNA ligase (25°C permissive, 37°C non-permissive) has been used to show that DNA synthesis is discontinuous. Examination of DNA replication in the presence of [^{3}H]-Thymidine in the mutant would demonstrate the accumulation of short segments of

a. unlabeled DNA at 25°C and at 37°C
b. unlabeled DNA at 25°C but not at 37°C
c. radioactive DNA at 37°C but not at 25°C
d. radioactive DNA at 25°C but not at 37°C **[IISc]**

Genome complexity

1. Which of the following genome feature increases as the complexity of an organism increases?

P. Genome size
Q. Number of genes
R. Density of genes in the genome
S. Average size of individual genes

a. P and Q
b. P and S
c. P, Q and S
d. R and S

2. Approximately what percentage of human genome contains protein coding genes?

a. ~2%
b. ~20%
c. ~50 %
d. ~80% **[IISc]**

3. Noncoding DNA in eukaryotic cells may include

a. introns
b. pseudogenes
c. spacer DNA
d. all of the above

4. Scientists estimate that greater numbers of proteins are produced in humans than the number of predicted genes. This is because

a. the relationship between genes and proteins is not known.
b. primary RNA transcripts can be alternatively spliced to make different mRNA.
c. DNA replication occurs throughout life, so that genes are produced as a person gets older.
d. both the coding and the template strand of a single gene is transcribed.

5. When the human genome draft sequence was released, which of the following observation was least expected?

a. The large amount of repetitive DNA.
b. The size of the total genome.
c. The size of individual chromosomes.
d. The small number of protein-coding genes. **[JNU]**

6. The following table shows the genome size, number of genes and chromosome number for a variety of organisms.

Organism	Genome size (approximately)	Number of genes (approximately)	Chromosome number
E. coli bacterium	4369000 bp	4289	N = 1
Baker's yeast	12069000 bp	6200	2N = 32
Amoeba	290000000000 bp	data not available	500–1000 (possibly polyploid)
Rodent	3399900000 bp	data not available	2N = 64
Rhesus monkey	3399900000 bp	data not available	2N = 42
Fruit fly	137000000 bp	14000	2N = 8
Humans	3200000000 bp	30000	2N = 46

From this data it is possible to conclude that

a. there is more non-coding DNA in humans than in the bacterium.
b. as chromosome number increases, so do the number of genes.
c. the rodent and the monkey will have the same number of genes.
d. the genome size relates to the evolutionary position of an organism.

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Transposable elements

1. Transposons
   a. insert into DNA by homologous recombination.
   b. cannot be transferred by phage mediated transduction.
   c. contain the equivalent of insertion (IS) elements.
   d. can insert into plasmids but not the bacterial chromosome. [JNU]

2. Movement of a segment of DNA from one site of the genome to another is called
   a. mutation
   b. cleavage
   c. reversion
   d. transposition [MKU]

3. All of the following are true about transposition *except*
   a. transposons move from one location to a different one within a chromosome.
   b. both the donor and target sites must be homologous.
   c. transposons have insertion sequences that are recognized by transposases.
   d. the transposon may either be excised and moved, or be replicated with the replicated piece moving.

4. IS elements
   P. are all identical.
   Q. encode proteins required for transposition.
   R. are flanked by direct repeats.
   S. cause a duplication of host DNA upon integration.
   a. P and Q
   b. P and S
   c. P, R and S
   d. Q, R and S

5. Which one of the following statements about retrotransposons is *not* correct?
   a. They are mobile genetic elements residing in the genomes of all classes of organisms.
   b. They are transposons that behave much like retroviruses and have a retrovirus like structure.
   c. The reverse transcriptase makes a cDNA copy of the retrotransposon.
   d. The retrotransposon RNA is packed into virus particles. [IISc]

6. L1 elements
   P. are part of the viral super family of retro-transposons.
   Q. found in far more abundant in AT-rich, gene-poor, and low-recombining regions of the genome.
   R. are approximately 300 nucleotides in length.
   S. contain an open reading frame with homology to reverse transcriptase.
   a. P and Q
   b. P and S
   c. Q and S
   d. P, Q and R

7. Alu elements
   a. are part of the viral superfamily of retrotransposons.
   b. are autonomous retrotransposons.
   c. characterized by the action of the Alu (*Arthrobacter luteus*) restriction endonuclease.
   d. Alu elements encode proteins and do not depend on LINE for their retrotransposition.

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8. The Alu element found in the human genome
   a. a 7SL scRNA homologue.
   b. constituting ~10 percent of the total DNA.
   c. ~300 bp long repetitive element.
   d. all of the above.

9. Which of the following are genes found in all retroviruses?
   P. *gag*
   Q. *pol*
   R. *env*
   S. *onc*
   a. P and Q
   b. P and S
   c. R and S
   d. P, Q and R

[JNU]

Satellite DNA

1. The term *satellite* DNA refers to
   a. extrachromosomal DNA fragments that are found close to (orbiting) the full-length chromosomes.
   b. mitochondrial DNA, which is circular in nature.
   c. long tandem repeats of simple DNA sequences.
   d. mobile DNA elements such as transposons and insertion sequences.

2. Satellite DNA consists of
   a. extrachromosomal DNA
   b. short repetitive nucleotide sequences
   c. ribosomal RNA genes
   d. single gene regions

3. Highly repetitive sequences are normally located in eukaryotic chromosome at/near
   a. centromeric region
   b. euchromatin region
   c. both euchromatin and heterochromatin region
   d. nuclear organizing region

4. Simple sequence DNA
   P. reassociates with the intermediate component of a $Cot_{1/2}$ curve.
   Q. consists of individual short repeated sequences that are dispersed throughout the genome.
   R. accounts for approximately 10% of mammalian genome.
   S. has a characteristic buoyant density based on its nucleotide composition.
   a. P and Q
   b. P and S
   c. R and S
   d. P, Q and R

5. Highly repetitive sequences in DNA are characterized by
   a. 2 to 10 base pairs repeated in tandem 10^5 to 10^7 times in a genome.
   b. a slowness to reanneal as compared to unique sequences.
   c. unique sequences transcribed together.
   d. their absence in human DNA.

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6. Minisatellite repeats
   P. contain fewer individual repeats per cluster than satellite repeats.
   Q. have a core repeat of 10-15 nucleotides.
   R. frequently vary in cluster size between individuals in a population.
   S. are not active in DNA recombination.
   a. P and Q
   b. P and S
   c. R and S
   d. P, Q and R

7. Microsatellites are used more commonly than minisatellite as DNA markers because
   a. minisatellites are present in too many locations within the genomes.
   b. restriction enzymes can be used to type microsatellites but not minisatellite.
   c. there are very few microsatellites in eukaryotic genomes so they are easily identified and analyzed.
   d. microsatellites are present throughout eukaryotic genomes and are easily amplified using PCR.

8. Human genomic DNA is digested into fragments approximately 1 kb in size, denatured and then renatured. Which of the following statements is *true*?
   a. All fragments will renature at the same rate.
   b. Fragments composed largely of repetitive DNA sequences will renature fastest.
   c. Fragments composed largely of non-repetitive DNA sequences will renature fastest.
   d. Fragments with high A : T content will renature fastest. [IISc]

9. Which of the following are steps in a reassociation experiment?
   a. Melting the DNA.
   b. Reannealing the DNA.
   c. Shearing the DNA into fragments.
   d. All of the above.

10. Which of the following statement is *incorrect*?
    a. Unequal crossing over changes gene organization but not total gene number.
    b. Histone genes and rRNA genes are typically organized in tandem units in eukaryotic genomes.
    c. Organelle genomes are mostly circular and dsDNA.
    d. $Cot_{1/2}$ values are affected by genome size but not by the complexity of genome.

Gene families

1. Which of the following would be found in tandem arrays in all organisms?
   a. Origins
   b. mRNA genes
   c. rRNA genes
   d. Transcription terminators

2. Which of the following is the most likely mechanism for the origin of multigene families?
   a. Endosymbiosis
   b. Gene duplication
   c. Convergent evolution of dissimilar genes
   d. Horizontal gene transfer

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3. Gene duplication
   P. has occurred many times in eukaryotic genomes.
   Q. is critical for generating new genes.
   R. creates gene families.
   S. occurs only within gene clusters.
   a. P and Q
   b. P and S
   c. P, Q and S
   d. P, Q and R

4. Amplification of genes involves
   a. removal of histones from the DNA to allow transcription of the gene.
   b. multiple duplications of the gene via replication.
   c. multiplication of extrachromosomal elements only.
   d. invertebrate genomes only.

5. Globin genes
   a. are derived from a common ancestral gene.
   b. are organized into a single cluster.
   c. are organized differently in all higher eukaryotes.
   d. are all expressed simultaneously.

6. How are the eight human globin genes organized?
   a. They are randomly distributed on the human chromosomes.
   b. They are in two clusters, one on chromosome 11 and the other on chromosome 16.
   c. They are in a single cluster on the X chromosome.
   d. They are in four clusters, alpha, beta, gamma and delta.

7. DNA sequence rearrangements are involved in all of the following processes *except*
   a. immunoglobulin gene expression in mammals.
   b. mating-type switching in yeast.
   c. antigen switching in trypanosomes.
   d. intron splicing in ciliates.

8. What is the fate of *most* duplicated genes?
   a. Gene inactivation.
   b. Gain of a novel function through subsequent mutation.
   c. They are transferred to a new organism using lateral gene transfer.
   d. They become orthologs.

9. Which of the following best describes pseudogenes?
   a. Two functional genes within an organism that arose from the duplication of one gene.
   b. Genes that share the same ancestral sequence, but are found in different organisms.
   c. Sequences of DNA that are very similar to functional genes, but do not produce a functional product.
   d. Sequences of DNA that are very similar to inactive genes, but do produce a functional product.

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Transcription

1. Which of the following features of transcription is similar to that of replication?
   a. No primer is required for polymerization.
   b. Polymerization does not have proofreading activity.
   c. Newly synthesized strand starts falling off the template before complete RNA is synthesized.
   d. RNA synthesis requires DNA topoisomerase action. [GATE]

2. Mark the *correct* statement.
   a. Transcription is a process that yields, semi-conservatively, two identical strands of DNA.
   b. DNA-dependent DNA polymerases are multi-subunit enzymes responsible for DNA transcription.
   c. Bacterial mRNAs are mostly polycistronic.
   d. Sigma factors guide the post-transcriptional modification of eukaryotic hnRNAs to mRNAs.

3. Synthesis of mRNA on DNA template is
   a. unidirectional
   b. bidirectional
   c. bidirectional with the help of primer
   d. unidirectional with the help of primer

4. If a segment of an mRNA molecule has the sequence 5′ GUACCGAUCG 3′, which of the following could have been the template DNA molecule?
   a. 5′ GCUAGCCAUG 3′
   b. 5′ GUACCGAUCG 3′
   c. 5′ CATGGCTAGC 3′
   d. 5′ CGATCGGTAC 3′

5. Which of the following processes does *not* have a mechanism that checks to make sure that the right monomer is incorporated into the polymer?
   a. Protein synthesis
   b. Double stranded rolling circle DNA synthesis
   c. RNA synthesis
   d. Excision repair DNA synthesis [GATE]

6. Which of the following enzymes does *not* require a primer?
   a. RNA dependent DNA polymerase.
   b. DNA dependent DNA polymerase.
   c. DNA dependent RNA polymerase.
   d. *Taq* DNA polymerase. [JNU]

7. Which of these substrates is best suited for measuring RNA synthesis by RNA polymerase
   a. γ^{32}P ATP
   b. α^{32}P dATP
   c. γ^{32}P dATP
   d. α^{32}P CTP [GATE]

8. DNA-dependent RNA polymerase of *E. coli* is characterized by which of the following properties?
   P. It requires a DNA template, Mg^{2+} and all four ribonucleoside triphosphates and forms a phosphodiester bond and pyrophosphate.
   Q. It is a polymeric protein.
   R. It requires a protein factor (σ factor) for initiation of transcription.
   S. It requires a primer with a free 3′-hydroxyl group.
   a. P and Q
   b. P and S
   c. R and S
   d. P, Q and R

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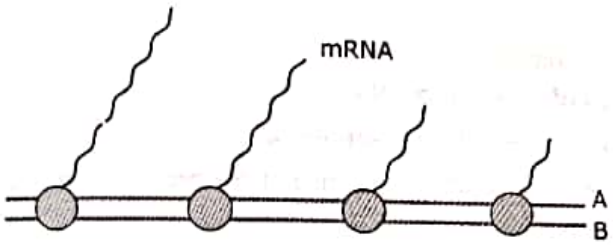
9. Similarities between RNA polymerase and DNA polymerase activities in prokaryotes include
P. the requirement for a template.
Q. the requirement for a primer.
R. synthesis of the nascent chain in the 5′ to 3′ direction.
S. the 3′ to 5′ exonuclease editing function.
a. P and Q
b. P and R
c. P and S
d. P, Q and R

10. Which one of the following statements about *E. coli* RNA polymerase is *false*?
a. The holoenzyme includes the sigma factor.
b. The core enzyme includes the sigma factor.
c. It requires Mg^{2+} for its activity.
d. It requires Zn^{2+} for its activity.

11. Which one of the following statements is *incorrect*?
a. There are two α-subunits in the *E. coli* RNA polymerase.
b. There is one β-subunits in the *E. coli* RNA polymerase.
c. *E. coli* has 3 different RNA polymerases.
d. The β subunit of *E. coli* RNA polymerase is inhibited by rifampicin.

12. The following figure represent the transcription of strand A of a DNA



Which of the following assignment of the DNA strand polarities is *correct*?
a. 5′ → 3′ is A and 3′ → 5′ is B
b. 3′ → 5′ is B and 3′ → 5′ is A
c. 3′ → 5′ is A and 5′ → 3′ is B
d. 5′ → 3′ is A and 5′ → 3′ is B

13. The prokaryotic RNA polymerase holoenzyme has the subunit structure
a. $\alpha\alpha\beta\beta'\sigma$
b. $\alpha\beta\sigma$
c. $\alpha\beta$
d. $\alpha\beta\beta$ **[GATE]**

14. Which of the following statement about transcription is *correct*?
a. RNA synthesis occurs in the 3′ to 5′ direction.
b. The RNA polymerase enzyme moves along the sense strand of the DNA in 5′ to 3′ direction.
c. The RNA polymerase enzyme moves along the template strand of the DNA in 5′ to 3′ direction.
d. The transcribed RNA is complementary to the non template strand.

15. The DNA-binding protein that initiates the transcription of bacterial genes is called a/an
a. operator
b. promoter
c. repressor
d. sigma factor

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16. Sigma subunit of RNA polymerase
    a. specifies the site for transcription.
    b. initiates replication.
    c. provides contact between ribonucleotides and DNA template.
    d. terminates transcription.

17. Consider the following statement.
    A factor called 'sigma factor' is involved in the transcription process during protein synthesis. This factor
    P. interacts with the core enzyme RNA-polymerase and forms the complete active enzyme.
    Q. recognizes the start signal on DNA and directs the binding of the enzyme to the promoter region.
    R. along with the rho factor helps RNA to dissociate from RNA-polymerase and thus helps to terminate transcription.
    Which of these statements are *correct*?
    a. P and Q
    b. P and R
    c. Q and R
    d. P, Q and R

18. Binding of a given sigma factor depends on the
    a. recognition of the length and spacing of promoter consensus sequences.
    b. interaction with core enzyme.
    c. length of the transcription unit.
    d. distance of the translational start codon.

19. Which of the following is *true* about sigma factors?
    a. They are made by eukaryotes to regulate synthesis of different mRNAs.
    b. They are necessary for prokaryotic RNA polymerase to recognize promoters.
    c. They are necessary for prokaryotic RNA polymerase to elongate RNA, but not for promoter recognition.
    d. They are involved in initiation of prokaryotic translation.

20. What is the role of the promoter region of a gene?
    a. it is the site where the ribosome binds to the mRNA.
    b. it is the site where RNA polymerase binds to the DNA.
    c. it is the site where DNA polymerase binds to DNA.
    d. it is the site where Shine–Dalgarno sequence present.

21. DNA contains sequence of bases known as promoters which
    a. are recognized by the core enzyme of RNA polymerase.
    b. occur in the chain just before the message sequence that defines the messenger RNA.
    c. are single-stranded regions of the duplex DNA.
    d. are transcribed into RNA sequences.

22. RNA polymerase recognizes sites on DNA known as promoters. All of the following statements regarding these sites are correct *except*
    a. a sequence of DNA base-pairs known as the pribnow box is located 5-10 bases upstream from the first base to be copied into RNA.
    b. all pribnow box sequences are variants of the sequence TATAAT.
    c. a second sequence, the –35 sequence, in many promoters, is located about 35 bases downstream from the pribnow box.
    d. the σ subunit of prokaryotic RNA polymerase is involved in recognition of the promoter sequence.

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23. Pribnow box is centered at
a. +10 position
b. −10 position
c. −35 position
d. +35 position

24. Rho-dependent and rho-independent transcription termination mechanisms operate in prokaryotes. Rho independent termination mechanism involves
a. binding of the rho protein upstream of the termination element.
b. no protein factors and only RNA secondary structure and run of 'U's.
c. presence of UGA or UAA stop codons.
d. binding of accessory factors at termination signal. **[GATE]**

25. Rho-dependent termination
a. requires the RNA involved to take on a specific secondary structure before termination can occur.
b. requires that a specific protein bind to the RNA and cause termination.
c. requires that the RNA be cut before transcription terminates further downstream.
d. only occurs on tRNA genes in eukaryotes.

26. What is the role of the *Rho* protein in termination of transcription?
a. It is a subunit of RNA polymerase that binds to RNA hairpins and stalls transcription.
b. It is a DNA-binding protein that blocks the movement of RNA polymerase down the template.
c. It is a helicase that actively breaks base pairs between the template and transcript.
d. It is a nuclease that degrades the 3' ends to RNA transcripts.

27. Which of the following statements regarding the termination of transcription of DNA is *correct*?
a. The rho protein binds tightly to RNA polymerase and inactivates the polymerization activity.
b. Termination sites of DNA contain an inverted-repeat base sequence that results in the RNA formed adopting a stem and loop configuration.
c. Each termination site contains a unique base where transcription stops.
d. All prokaryote termination sites contain the *stem and loop* promoting sequence and also require the participation of the rho protein to halt RNA synthesis.

28. Which one of the following is a common feature found in the 3' terminal segment of prokaryotic mRNAs?
a. Poly-T-stretch
b. cap
c. Shine-Dalgarno sequence
d. Stem loop structure

29. Mode of action of rifampicin in *E. coli* is through
a. cell division
b. initiation of transcription
c. RNA polymerase binding to DNA template **[GATE]**
d. inhibition of the oxidation potential

30. Which of the following agents inhibits the initiation of RNA synthesis in prokaryotes by binding to the beta subunit of RNA polymerase?
a. α-amanitin
b. actinomycin D
c. rifamycin SV
d. puromycin

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31. Wild type *E.coli* was plated on a Rifampicin containing medium and incubated at 37°C. Majority of the cells died; however, some colonies appeared after a few days. What is the most likely explanation for this observation?

a. Degradation of Rifampicin
b. Mutation in DNA polymerase III
c. Efflux of Rifampicin
d. Mutation in the β subunit of RNA polymerase

[IISc]

32. Which one of the following statements about eukaryotic RNA polymerases I, II and III is *false*?

a. RNA Pol II is very sensitive to α-amanitin.
b. RNA Pol II is located in the nucleoplasm.
c. RNA Pol III transcribes the genes for tRNA.
d. The carboxyl end of RNA Pol II contains a short sequence of only seven amino acids which is called the CTD and which may be phosphorylated.

33. In eukaryotes the ribosomal RNA genes are transcribed by

a. reverse transcriptase
b. RNA dependent RNA polymerase
c. RNA polymerase I
d. RNA polymerase III

[GATE]

34. There are three kinds of RNA polymerases (I, II, III) in eukaryotic cells, each specific for one class of RNA molecule. Which of the following is a *correct* match?

a. RNA pol I–rRNA; RNA pol II–tRNA
b. RNA pol II–mRNA; RNA pol III-rRNA
c. RNA pol I–rRNA; RNA pol II-mRNA
d. RNA pol I-tRNA; RNA pol III-rRNA

[GATE]

35. A new antibiotic was discovered which strongly inhibited mRNA precursor transcripts and snRNA transcripts. This antibiotic was predicted to be an inhibitor of

a. RNA polymerase I
b. RNA polymerase II
c. RNA polymerase III
d. Helicase

[GATE]

36. A promoter of a typical eukaryotic gene is composed of

a. a binding site for the sigma factor
b. a binding site for TATA binding protein
c. a binding site for transcription factor IID
d. a binding site for transcription factor IIB

[IISc]

37. About 25-35 bp upstream of the transcriptional start site in most plant genes is a short sequence, which serves as the site of assembly of the transcriptional initiation complex called

a. CAAT box
b. GC box
c. TATA box
d. repressor

[JNU]

38. The TATA box plays a key role in assembling active transcription complexes by

a. binding with TATA box-binding protein
b. binding with DNA protein
c. binding with RNA protein
d. it does not bind with any protein

[GATE]

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39. Which one of the following transcription factors bind to TATA box?
a. TFIID
b. TFII B
c. TFIIIA
d. TFII E
[IISc]

40. Which of the following is *true* about interaction between TBP and TATA sequence?
a. TBP interacts with minor groove and bends DNA by about 80°.
b. TBP interacts with major groove and bends DNA by about 80°.
c. TBP interacts with minor groove with no significant bending.
d. TBP interacts with major groove with no significant bending.
[JNU]

41. Which one of the following statements about the primary transcripts in eukaryotes is *incorrect*?
a. They are usually longer than the functional RNA.
b. They contain no modified bases.
c. They contain nucleotide sequences which are complementary to the DNA template.
d. They contain the TATA box sequence.

42. Changes in RNA pol III promoters result in changes in the sequence of the gene product. This is because
a. RNA pol III initiates transcription at various positions.
b. the promoter elements are located within the coding region of the gene.
c. promoters are located far upstream of the gene transcribed.
d. polymerase is unable to bind to the promoter.
[IISc]

43. Which of the following statements is *true*?
a. RNA Pol II only transcribes protein-coding genes.
b. The TATA box has a role in transcription efficiency but not in positioning the start of transcription.
c. TBP binds to the TATA box.
d. Enhancers lie 100-200 bps upstream from the transcription start site.

44. General transcription factors are factors required for the correct initiation of transcription by RNA polII in eukaryotes. A set of five general transcription factors, denoted TFIIB, TFIID, TFIIE, TFIIF and TFIIH, is responsible for promoter recognition and for unwinding the promoter. Choose the *correct* statements about the general transcription factors.
P. TFIID is a complex composed of the TBP and TBP-associated factors (TAFs).
Q. TBP is highly conserved through evolution and appears to be required for transcription by all three eukaryotic RNA polymerases.
R. TBP binds with the minor groove.
S. TFIIH is the protein kinase that phosphorylates the CTD of RNA polII.
a. P, R and S
b. Q, R and S
c. P, Q and S
d. P, Q, R and S

45. Phosphorylation of the RNA polymerase II C-terminal domain (CTD) correlates with
P. the kinase activity of TFIIH.
Q. the presence of specific TAF proteins in TFIID.
R. the switch from an initiating polymerase to an elongation polymerase.
S. the release of initiation factors TFIIA, TFIIB, and TFIID.
a. P and Q
b. P and R
c. P, R and S
d. P, Q, R and S

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46. Transcription termination of mRNA genes in eukaryotes occurs
   a. at polyadenylation sites by the action of a termination factor.
   b. by the formation of a strong hairpin structure in the vicinity of polyadenylation site.
   c. termination factor bound to the termination site in the vicinity of polyadenylation site.
   d. at pause sites following the polyadenylation sites. [IISc]

47. Enhancers can be located
   P. upstream of the promoter
   Q. downstream of the promoter
   R. within the promoter
   S. far from the promoter
   a. P and Q
   b. P and R
   c. P and S
   d. P, Q and S

48. Which of the following is *not* true of transcriptional coactivator proteins?
   a. Coactivator proteins bridge interactions of activator and GTFs.
   b. All coactivators bind to specific DNA sequence in the promoter.
   c. Coactivator proteins provide critical histone modification activity.
   d. Coactivator proteins provide critical histone modification and remodeling function and bridge interaction between activator and transcription machinery. [JNU]

RNA processing

1. Which of the following is *not* true of RNA processing?
   a. Exons are excised and hydrolyzed before mRNA moves out of the nucleus.
   b. The presence of introns may facilitate crossing over between regions of a gene that code for polypeptide.
   c. RNA splicing may be catalyzed by spliceosomes.
   d. A primary transcript is often much longer than the final RNA molecule that leaves the nucleus.

2. Most eukaryotic pre-mRNAs are matured by which of the following modifications to their ends?
   a. Capping at the 3′ end, cleavage and polyadenylation at the 5′ end.
   b. Addition of a GMP to the 5′ end, cleavage and polyadenylation to create the 3′ end.
   c. Addition of a guanine residue to the 5′ end, cleavage and polyadenylation to create the 3′ end.
   d. Addition of a GMP to the 5′ end, polyadenylation, then cleavage to create the 3′ end.

3. Post-transcriptional modification of prokaryote RNA molecules includes
   P. cleavage of primary transcripts to form functional molecules of rRNA and tRNA.
   Q. addition of a CCA 3′ terminus of tRNA molecules.
   R. methylation of bases using S-adenosyl-methionine as methyl donor.
   S. addition of a cap structure to the 5′ end of mRNAs.
   a. P and Q
   b. P and R
   c. P and S
   d. P, Q and R

4. *Correct* statements concerning eukaryotic mRNA include which of the following?
   P. It is formed from nuclear RNA precursors of widely variable size known as hnRNA.
   Q. It has a 7-methylguanosine at the 3′ end of the chain.
   R. It is prepared from precursor RNA by splicing enzymes, which remove introns and join exons.
   S. It has a poly-A-tail consisting of 150 to 200 nucleotides at the 5′ end.
   a. P and Q
   b. P and R
   c. P and S
   d. P, Q and R

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5. Mature eukaryotic mRNAs have a 5′ cap that is residue of
   a. 2-methylguanosine
   b. 3-methylguanosine
   c. 5-methylguanosine
   d. 7-methylguanosine

6. Which one of the following regions in the mature mRNA is not found in the gene for the mRNA?
   a. start codon
   b. leader sequence
   c. poly A tail
   d. exon

[IISc]

7. The poly A end of eukaryotic mRNAs originates from
   a. transcription of corresponding poly T region of the respective gene.
   b. addition of a pre-synthesized poly-A-tail to the 3′ end of the primary transcript.
   c. sequential addition of A residues at the 3′ end of the primary transcript.
   d. by as yet an unknown mechanism.

[JNU]

8. Addition of the poly-A-tail to a pre-mRNA
   P. involves a two-step transesterification mechanism.
   Q. requires a conserved AAUAAA sequence.
   R. proceeds by the stepwise assembly of a multi component complex.
   S. is carried out by a template-dependent RNA polymerase.
   a. P and Q
   b. P and R
   c. Q and R
   d. Q, R and S

9. Which one of the following statements is *not* correct about eukaryotic mRNAs?
   a. It contains a poly-A-tail of up to 250 residues.
   b. The poly-A-tail is coded by the DNA template.
   c. The primary nuclear transcripts are much larger than the cytoplasmic mRNAs and contain introns.
   d. A guanosine residue is attached to the 5′ terminus in an unusual 5′-5′ linkage.

[JNU]

10. Mark all *true* statements.
   P. Exons are found in the same order in the genome.
   Q. All cells in the human body have the same set of genes.
   R. All cells in the human body express the same set of genes.
   S. All cells in the human body splice mRNAs for each gene in the same way.
   a. P and S
   b. P and Q
   c. R and S
   d. Q and R

11. Broker, Chow, Roberts and P. Sharp hybridized mRNA from adenovirus to denatured adenovirus DNA and examined the resultant duplex molecules by electron microscopy. Which of the following did they observe?
   a. mRNA and DNA formed a continuous duplex throughout the length of the RNA.
   b. mRNA and DNA formed a continuous duplex throughout the length of the DNA.
   c. On one strand there were looped out regions of single-stranded RNA.
   d. On one strand there were looped out regions of single-stranded DNA.

12. The co-linearity of gene and product clearly seen in prokaryote system is obscured in eukaryotes because
   a. DNA sequences coding for proteins are interrupted by non-coding sequences that are not transcribed.
   b. the sequence of bases in the mRNA is modified in the cytoplasm prior to interaction with the ribosomes.
   c. Ribosomes in eukaryotes only translate exon regions of the mRNA.
   d. The coding sequence in DNA is interrupted by DNA sequences that are transcribed and then spliced out of the mRNA.

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13. What is the purpose of storing information in exons separated by introns?
    a. It keeps the exons safe from infecting viral DNA.
    b. Eukaryotes are able to then use the information in one DNA sequence in alternative ways.
    c. It allows for multiple copies of instructions to be hidden within several chromosomes.
    d. There appears to be no significance to this arrangement.

14. Splicing of pre-mRNA
    P. within the nucleus
    Q. is self catalyzed process
    R. involves snRNP
    S. is similar to pre tRNA intron splicing
    a. P and Q
    b. P and R
    c. R and S
    d. P and S

15. Spliceosome, a complex of snRNA, protein and pre-mRNA is found in
    a. eukaryotic cell
    b. prokaryotic cell
    c. both eukaryotic and prokaryotic cell
    d. only in plant cell

16. During the processing of introns, a single snRNP complex catalyzes both the cleavage of the RNA and the joining of the cut ends. What would be the consequence if these two processes were catalyzed by separate enzymes not associated in a single complex?
    a. The rate of RNA processing would be much faster.
    b. The cell would be unable to identify the correct cleavage sites.
    c. The exons might not be joined in the correct sequence.
    d. Exons instead of introns would be cleaved from the RNA molecule.

17. The branch site nucleotide
    P. is always an A
    Q. is located randomly within the intron
    R. initiates the first step of splicing by attacking the 3′ splice site
    S. is covalently joined to three other intron nucleotides after the first step of splicing
    a. P and Q
    b. P and R
    c. P and S
    d. P, Q and R

18. Spliceosome mediated splicing of GU-AG intron
    a. requires ATP
    b. involves transesterification reaction
    c. involves the nucleophilic attack of an OH group on the sugar phosphate backbone
    d. all of the above

19. Mark the *true* statement about snRNAs.
    a. snRNAs are strictly localized to the nucleus.
    b. Most snRNAs are very abundant.
    c. Some snRNAs can base pair with intron consensus sequences.
    d. All of the above.

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20. Which one of the following statement *correctly* describes the splicing process in most eukaryotic pre-mRNAs?
a. In a two-step reaction, the spliceosome removes the exon as lariat and joins the two introns together.
b. Splicing requires conserved sequences, which are the 5'-splice site, the 3'-splice site, the branch point and the polypurine tract.
c. The U1 snRNP initially binds to the 5'-splice site, U2 to the branch point sequence and then the three snRNP, U4, U5 and U6 can bind.
d. In the first step of splicing the G at the 3'-end of the intron is joined to the 2' hydroxyl group of the A residue of the branch point sequence to create a lariat.

21. Introns are released as lariats during splicing of
a. pre-mRNA transcripts and group II intron containing transcripts.
b. pre-mRNA transcripts only.
c. group I intron containing transcripts.
d. tRNA precursors.

22. Alternate splicing means that
a. the same gene can code for several different proteins.
b. several different genes can code for the same protein.
c. gene expression can be regulated at the level of transcription.
d. pieces of DNA can move around within the genome.

23. What is the molecular activity of the sex-lethal protein of *Drosophila*?
a. Sex-lethal protein is a transcription factor that activates female-specific genes.
b. Sex-lethal protein controls the splicing of specific mRNAs such that certain proteins required for female development are produced only in females.
c. Sex-lethal protein is a repressor of the transcription of male-specific genes.
d. Sex-lethal protein is equivalent to testosterone in mammals, in determining male development.

24. Alternative splicing
P. uses a completely different mechanism than 'constitutive' splicing.
Q. allows generation of protein isoforms (variants that differ by addition or deletion of a small number of amino acids) from a single gene.
R. involves the use of different 5' and 3' splice sites.
S. is used to make different proteins in different tissues and at different developmental stages.
a. P and Q
b. P and R
c. P and S
d. Q, R and S

25. In which of the following splicing events, the intermediates do *not* form the lariat or the branched structure?
a. trans splicing
b. group I intron splicing
c. group II intron splicing
d. hnRNA splicing **[GATE]**

26. The introns in tRNA
P. are removed by a two-step transesterification mechanism.
Q. have a sequence complementary to the anticodon.
R. all form a similar secondary structure.
S. are removed by protein enzymes (endonuclease and ligase).
a. P and Q
b. P and R
c. P and S
d. Q, R and S

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27. Group I introns can use various forms of guanine including
P. GMP
Q. GDP
R. GTP
S. ddGMP (2',3'-dideoxy-GMP)
a. P and Q
b. P and R
c. P and S
d. P, Q and R

28. In mammalian cells, RNA editing
P. can change the coding capacity of a gene after transcription.
Q. creates a stop codon in the middle of the apo-B mRNA in intestinal cells but not in liver cells.
R. usually makes many changes in each mRNA.
S. can change specific nucleotides by deamination.
a. P and Q
b. P and R
c. P, Q and S
d. Q, R and S

29. Which ribonucleases are involved in producing mature tRNA in *E. coli*?
a. RNases A, D, E and F
b. RNases D, E, F and H
c. RNases D, E, F and P
d. RNases A, D, H and P

30. The footprinting assay can be used to
a. map the position of nucleosomes in chromatin.
b. determine the level of expression of a specific transcript.
c. determine the DNA-binding site for a transcription factor.
d. map the ends of a transcript.

31. Which of the following is *not* a post-transcriptional modification?
a. splicing
b. 5' capping
c. 3' polyadenylation
d. glycosylation [IISc]

Prokaryotic gene regulation

Lac operon

1. Genes whose products are constantly needed for cellular activity are called
a. regulator genes
b. structural genes
c. housekeeping genes
d. smart genes

2. The products of constitutive genes may be
P. involved in catabolic pathways
Q. involved in anabolic pathways
R. continuously synthesized
S. involved in the synthesis of structural elements in the cell
a. P and Q
b. P and R
c. P and S
d. P, Q, R and S

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3. Which of the following processes is *not* an example of allosteric regulation?
   a. Regulation of phosphofructokinase activity by fructose 2,6-bisphosphate.
   b. Inactivation of nitrogenase by ADP ribosylation.
   c. Regulation of the *lac* operon by allolactose in *E. coli*.
   d. Catabolite repression by CAP in *E. coli*.

4. In bacteria, the genes that code for the enzymes of a metabolic pathway are usually arranged consecutively to form a functional unit called
   a. an induction system
   b. an end-product repression system
   c. an operon
   d. a constitutive enzyme system

[IISc]

5. A series of genes that are transcriptionally regulated in *trans* by DNA-binding proteins are called a/an
   a. transcriptor
   b. repressor
   c. promoter
   d. operon

6. Operons
   a. are of approximately uniform in size
   b. do not bind proteins
   c. are found in some eukaryotic genes
   d. are shorter and smaller in lower eukaryotes than higher eukaryotes

[GATE]

7. The following are basic properties of a prokaryotic operon. The operon
   P. usually consists of multiple genes that are organized between a promoter and a terminator.
   Q. is usually under autogenous transcriptional control.
   R. member genes are coordinately regulated.
   S. exclusively contains genes that encode proteins involved in stress tolerance.
   a. P and Q
   b. P and R
   c. P and S
   d. Q, R and S

8. Which of the following statements about the repressor of the lactose operon of *E. coli* are *true*?
   P. It is the gene product of the regulatory gene.
   Q. It binds to the operator region of the *lac* operon.
   R. It blocks the transcription of structural genes.
   S. It combines with allolactose, and the combination will not bind to the operator.
   a. P and Q
   b. P and R
   c. P and S
   d. P, Q, R and S

9. Choose the *incorrect* pair

a. the DNA sequence to which RNA pol binds	1. promoter site
b. the DNA sequence to which sigma factors bind	2. enhancer sequence
c. the DNA sequence that codes for a protein	3. structural gene
d. the DNA sequence to which repressors bind	4. operator site

10. Which of the following genes is *not* transcribed from the promoter for β-galactosidase?
    a. *lac Z*
    b. *lac I*
    c. *lac Y*
    d. *lac A*

[JNU]

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11. Which of the following compounds is a gratuitous inducer of β-galactosidase in *E. coli*?
    a. glucose
    b. allolactose
    c. isopropylthiogalactoside
    d. lactose

12. Induction of β-galactosidase activity by IPTG is the result of
    a. stimulation of *lac* repressor function.
    b. IPTG binding to *lac* operon and inducing transcription.
    c. IPTG binding to lac I gene product and inhibiting its activity.
    d. inhibition of β-galactosidase degradation. [JNU]

13. If there is a deletion mutation in the operator for the *Lac* operon, the expression of *Lac* structural gene will be
    a. permanently stop
    b. constitutively expressed
    c. not expressed
    d. resistant to catabolite expression [IISc]

14. The *lac operon* is negatively regulated by the repressor encoded by the *lacI* gene. A mutation in *LacI* that prevents the binding of lactose will result in
    a. a constitutive phenotype that will be recessive.
    b. an uninducible phenotype that will be dominant.
    c. a constitutive phenotype that will be dominant.
    d. an uninducible phenotype that will be recessive. [IISc]

15. Bacteria utilize glucose first, even if other sugars are present, through a mechanism called
    a. operon repression
    b. enzyme repression
    c. catabolite repression
    d. glucose utilization [IISc]

16. Catabolite activator protein (CAP) acts as
    a. apo repressor
    b. co-repressor
    c. apo inducer
    d. inducer

17. cAMP regulates the *lac* operon by
    a. binding to the operator to turn on transcription.
    b. binding to the *lac* repressor to prevent transcription.
    c. combining with the catabolite activator protein to form a complex that turns on transcription after binding near the promoter.
    d. none of the above.

18. Full expression of the *lac* operon requires
    a. lactose and cAMP
    b. allolactose and cAMP
    c. lactose
    d. allolactose [IISc]

19. A loss-of-function mutation in the gene that encodes the catabolite activator protein of *E. coli* would have what effect on expression of the *lac* operon?
    a. Expression would be very high in the presence of lactose and off in its absence regardless of the presence or absence of glucose.
    b. Expression would be low in the presence of lactose and off in its absence, regardless of the presence or absence of glucose.
    c. Expression would be low in the presence of glucose and very high in its absence, regardless of the presence or absence of lactose.
    d. Expression would be off in the presence of glucose and low in its absence, regardless of the presence or absence of lactose.

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20. Which of the following sequence of events occurs when *E. coli* are released from catabolite repression by transfer to low glucose medium?
   a. cAMP levels rise, cAMP binds to CAP, cAMP-CAP complex binds to a site on DNA and activates transcription.
   b. cAMP levels rise, cAMP binds to CAP, cAMP-CAP complex binds to a site on DNA and represses transcription.
   c. cAMP levels rise, cAMP binds to CAP, cAMP-CAP complex is removed from a site on DNA and activates transcription.
   d. cAMP levels fall, cAMP is removed from CAP, CAP binds to a site on DNA and represses transcription.

21. In cells that have not been previously exposed to lactose, how does lactose enter the cells to induce *lac* operon?
   a. IPTG would induce the expression of *lac* operon instead of lactose.
   b. Lactose would enter by diffusion across the membrane without the need for permease.
   c. Some other transport protein facilitates the entry of lactose.
   d. Basal level of expression of lac operon, leading to the low level synthesis of permease.

22. A positively controlled operon can be identified by mutation in the regulatory gene leading to **[IISc]**
   a. constitutive expression
   b. very high levels of expression
   c. inducibility of the operon
   d. repression of the operon

[IISc]

23. A mutation in the repressor of the *lac* operon in *E. coli*, preventing binding of the repressor to the operator, should result in
   a. constitutive expression of the lac operon genes.
   b. lack of or reduced expression of the *lac* operon genes under all conditions.
   c. expression of the genes only when lactose is present.
   d. expression of the genes only when lactose is absent.

[JNU]

24. In case of inducible *lac* operon, the transcription is inhibited by
   a. the binding of repressor to operator
   b. the presence of lactose
   c. the interaction of repressor with lactose
   d. the interaction of lactose with operator

[JNU]

25. Synthesis of β-galactosidase will be constitutive in a strain with the genotype
   a. $I^+Z^+Y^+$
   b. $I^-Z^+Y^+$
   c. $I^+Z^-Y^+$
   d. $I^+Z^+Y^-$

[JNU]

26. Which of the following regulators are said to act in *cis*?
   a. The *lac* repressor and the *lac* operator.
   b. The lac operator and mammalian transcription factors.
   c. The *lac* operator and mammalian enhancers.
   d. Mammalian transcription factors and enhancers.

27. Which of the following partial diploids will express β-galactosidase constitutively?
   a. F' $lacO^c\ lacZ^+$ / $lacO^+\ lacZ^+$
   b. F' $lacI^-\ lacZ^+$ / $lacI^+\ lacZ^+$
   c. F' $lacI^+\ lacZ^+$ / $lacI^-\ lacZ^+$
   d. F' $lacO^c\ lacZ^-$ / $lacO^+\ lacZ^+$

[JNU]

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28. A merozygote resulting from recombination of two *E. coli lac* mutants produced half maximal level of β-galactosidase without the inducer and maximal level of β-galactosidase with the inducer. The genotype of the merozygote is

a. $I^- O^+ Z^-$ and $I^+ O^+ Z^+$
b. $I^+ O^c Z^-$ and $I^+ O^+ Z^+$
c. $I^- O^c Z^+$ and $I^+ O^+ Z^+$
d. $I^+ O^c Z^-$ and $I^- O^+ Z^+$

[GATE]

29. In an operon, positively controlled gene expression occurs because

a. repressor is inactivated by activator binding to initiate transcription.
b. RNA polymerase is able to bind to the promoter and transcribed the gene on its own.
c. activator is needed for high level transcription initiation.
d. repressor removal facilities RNA Pol binding to the promoter to facilitate transcription.

[IISc]

30. A scientist is using an ampicillin-sensitive strain of bacteria that cannot use lactose because it has a nonfunctional gene in the *lac* operon. She has two plasmids. One contains a functional copy of the affected gene of the *lac* operon, and the other contains the gene for ampicillin resistance. Using restriction enzymes and DNA ligase, she forms a recombinant plasmid containing both genes. She then adds a high concentration of the plasmid to a tube of the bacteria in a medium for bacterial growth that contains glucose as the only energy source. This tube (+) and a control tube (−) with similar bacteria but no plasmid are both incubated under the appropriate conditions for growth and plasmid uptake. The scientist then spreads a sample of each bacterial culture (+ and −) on each of the three types of plates indicated below.

	Glucose Medium	Glucose Medium with Ampicillin	Glucose Medium with Ampicillin and Lactose
Bacterial strain with added plasmid (+)	#1	#2	#3
Bacterial strain with no plasmid (−)	#4	#5	#6

If no new mutations occur, it would be reasonable to expect bacterial growth on which of the following plates?

a. 1 and 2 only
b. 3 and 4 only
c. 4, 5 and 6 only
d. 1, 2, 3 and 4 only

Tryptophan operon

1. Tryptophan operon in *E. coli* is an example of

a. inducible operon
b. positively regulated operon
c. repressible operon
d. all of the above

2. Which one of the following statements about the tryptophan (trp) operon is *true*?

a. The RNA product of the trp operon is very stable.
b. The trp repressor is a product of the trp operon.
c. The trp repressor, like the lac repressor, is a tetramer of identical subunits.
d. The trp repressor binds to tryptophan.

3. True statements about the bacterial tryptophan (*trp*) operon include which of the following?

P. The *trp* operon has two kinds of regulation one by an operator and the other by an attenuator.
Q. The repressor complexes with the co-repressor and on binding to the operator blocks transcription.

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When tryptophan is in short supply, the leader sequence containing the attenuator cannot be passed and the stalled ribosome modifies the mRNA so that the structural genes are transcribed.

5. The co-repressor is the catabolite activator protein.
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

4. The expression of the *trp* operon in *E. coli* is regulated in part by the availability of the amino acid tryptophan. This regulatory process is referred to as
   a. attenuation
   b. translational read-through
   c. antitermination
   d. nonsense suppression

5. The leader region (*trpL*) of the tryptophan biosynthetic operon codes for RNA that may function as the operon's
   a. corepressor
   b. holorepressor
   c. aporepressor
   d. attenuator

6. A mutant of *E. coli* was isolated that synthesized greater than normal levels of tryptophan synthesizing enzymes in the absence of tryptophan. It was found to make normal *trp* repressor and also had normal *trp* operator. Which of the following mutations might result in the observed phenotype?
   a. Deletion in the *trp* promoter.
   b. Deletion just after the promoter-operator region.
   c. Deletion just before the coding sequences for the *trp* enzymes.
   d. Deletion in the rho-independent transcription termination signal. **[GATE]**

7. Attenuator in the *trp* operon of *E. coli* refers to
   a. the operator.
   b. a leader sequence of the operon offering site for RNA polymerase binding.
   c. a leader sequence in the trp mRNA with many codons for tryptophan.
   d. a non-transcribed region of the operon. **[JNU]**

8. The diagram below is a schematic showing the structure of the leader region of the *trp* operon.

Which of the following mutations would lead to an increase in expression of *trp* genes?
   a. Changing the two UGG codons to GGG.
   b. A mutation in sequence 3 that decreased pairing with sequence 4 without affecting 2 : 3 pairing.
   c. A mutation in sequence 3 that decreased pairing with sequence 2 without affecting 3 : 4 pairing.
   d. A mutation in the *rho* gene. **[JNU]**

Eukaryotic gene regulation

1. Which of the following statements about enhancers is *incorrect*?
   a. They function as promoters.
   b. They function in any orientation.
   c. They function even when located many base pairs away from the promoter.
   d. They function in tissue specific manner.

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2. In the RNAi regulatory pathway, the DICER enzyme cleaves
   a. RNA polymerase into non-functional pieces.
   b. single-stranded DNA into repetitive sequences.
   c. double-stranded RNA into short strands.
   d. mRNAs of genes to be repressed.

3. Function of an enhancer during transcription of an eukaryotic gene is to
   a. initiate transcription.
   b. stimulate the rate of transcription.
   c. elongate transcription.
   d. enhance mRNA stability. [JNU]

4. Enhancer elements are segments of DNA that
   P. are required for the transcription of certain genes.
   Q. activate transcription of a gene irrespective of their relative position in the DNA chain.
   R. have no polarity (i.e. they can be inverted).
   S. are nonspecific for tissue or species.
   a. P and Q
   b. P and R
   c. P, Q and R
   d. P, Q, R and S

5. In a plant species, loss of gene A causes meristem to enlarge whereas loss of gene B causes meristem to shrink. In a mutant plant where both genes are lost, shrunken meristem is observed. In a biochemical pathway involving A and B products, which one of the following is *correct*?
   a. A promotes B
   b. B promotes A
   c. A inhibits B
   d. B inhibits A

6. Which one of the following factors is involved in the epigenetic control of gene expression in eukaryotes?
   a. RNA polymerase II
   b. DNA methyl transferase
   c. TFIID
   d. TFIIH [IISc]

7. What is the chemical basis of gene imprinting?
   a. Phosphorylation of DNA
   b. Methylation of DNA
   c. Oxidation of DNA
   d. Glycosylation of DNA

8. Histone acetylation increases transcription of gene because
   a. it increases the DNA–histone interaction.
   b. the acetyl groups on histones are recognized by the RNA pol.
   c. histone acetylation loosens the DNA–histone complex, thereby making it more accessible to RNA pol.
   d. histone acetylation induces DNA bending which is recognized by RNA pol. [IISc]

9. One of the post-replication modification in DNA is the methylation of the bases. In this context which of the following statements is *incorrect*?
   a. Methylation protects cell's own DNA from digestion by its own restriction endonucleases.
   b. Methylation may result in the localized conversion of B-DNA to Z-DNA.
   c. Methyl groups are added predominantly to cytosines in bacterial cells and to adenines in eukaryotes.
   d. Methylation of bases is important in some aspects of DNA repair in bacteria.

10. Which of the following is an epigenetic factor for gene expression in eukaryotes?
    a. Recombination
    b. DNA methylation
    c. Protein phosphorylation
    d. DNA protein interaction [JNU]

11. The yeast two-hybrid system is designed to identify which of the following?
    a. All of the components of a multiprotein complex.
    b. Human proteins that are required for binding RNA polymerase.
    c. Two proteins that directly interact with one another.
    d. Two proteins that are involved in the same metabolic pathway.

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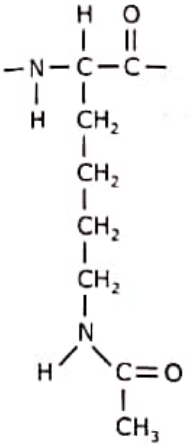
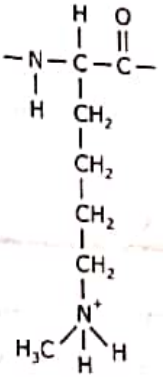
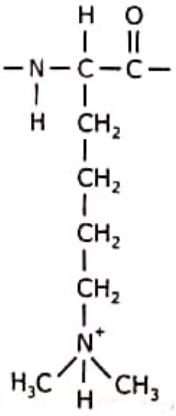
12. siRNA
a. forms a complex in the spliceosome.
b. recruits histone acetyl transferases to the nucleus.
c. forms a complex with RISC proteins to inhibit translation or cause degradation of complementary mRNA.
d. is not transmitted to daughter cells after cell division.

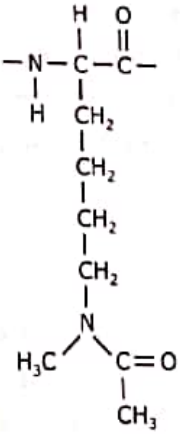
13. The most common modification of histone tails includes
a. acetylation and methylation of lysine and phosphorylation of threonine.
b. acetylation and methylation of lysine and phosphorylation of serine.
c. acetylation and methylation of arginine and phosphorylation of serine.
d. acetylation and methylation of arginine and phosphorylation of threonine.

14. Post-translational modification(s) of the amino-terminal Lys residues in histone H3 is important for regulation of transcription. Which of the following structures is not a likely modification of the Lys side chain in histone H3? [JNU]

a.


b.


c.


d.


[JNU]

15. In the presence of low intracellular iron concentrations, the iron response element binding protein (IRE-BP) has which one of the following actions?
a. It destabilizes transferrin receptor mRNA.
b. It blocks translation of ferritin mRNA.
c. It stabilizes ferritin mRNA.
d. It allows translation of ferritin mRNA. [JNU]

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16. Prokaryotic and eukaryotic genes have many common features, but also important differences. What is the best description of three important differences between prokaryotic and eukaryotic genes?

a. Prokaryotes genes are often larger, are polycistronic and often contain enhancers.

b. Eukaryotic genes are often larger, are polycistronic and contain enhancers.

c. Eukaryotic genes are often larger, are monocistronic and contain enhancers.

d. Prokaryotic genes are often larger, are polycistronic and contain introns.

Genetic switch

1. The mechanism regulating the decision between lysis and lysogeny for bacteriophage λ has been described as a *genetic switch*. Which of the following statement is *false* of this system?

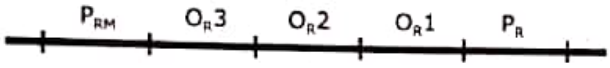
a. The *cI* and *Cro* genes are transcribed in opposite directions.

b. The mRNAs encoding *cI* and *Cro* protein are synthesized using the same strand of DNA as template.

c. Of the three O_R sites in the λ genome, *Cro* has the greatest affinity for O_R3.

d. Of the three O_R sites in the λ, *cI* has greatest affinity for O_R1.

2. One of the operator regions of the λ genome is diagrammed in the following figure. Which regulatory proteins bind to this region of the genome and to which of these operators do they bind during (P) lysogeny and (Q) the lytic phase of the viral life cycle?



a. P-cI binds to O_R1 and O_R2, Q-Cro binds to O_R3.

b. Q-cI binds to O_R1 and O_R2, P-Cro binds to O_R3.

c. P-cI binds to O_R1, Q-Cro binds to O_R2 and O_R3.

d. P-cI binds to O_R2, Q-Cro binds to O_R1 and O_R3.

3. Choose the *correct* statement(s).

P. *N* is required to express the delayed early genes.

Q. Lysogeny requires the delayed early genes *cII-cIII*.

R. Lytic cycle requires the immediate early gene *cro*.

S. C-terminal domain of lambda repressor is responsible for dimerization.

a. P and Q
b. P and R
c. P, Q and R
d. P, Q, R and S

4. Which of the following is/are *correct*? In lysogeny

P. delayed early gene products cII and cIII are necessary to initiate transcription at the promoter P_{RE}.

Q. cII acts directly at the promoter and cIII protects cII from degradation.

a. only P
b. only Q
c. both P and Q
d. none of these

5. A lysogen of *E. coli* becomes resistant to further infection by bacteriophage lambda because

a. *E. coli* no longer contains receptors on its cell surface.

b. one copy of phage is already present inside the cell.

c. the presence of repressor in the cell.

d. *E. coli* is dead.

[IISc]

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6. Which of the following is *not* true for lysogenic bacterium?
   a. Prophage is integrated into the bacterial genome.
   b. Single integrated prophage imparts immunity against infection by another phage particle of the same type.
   c. λ repressor the product of *cI* gene inhibits lytic cycle.
   d. Product of cro gene induces the synthesis of repressor.

7. When bacteriophage lambda infects a sensitive bacterium, one of the first messenger RNA species synthesized is very short, beginning at a site P_L and extending just through an adjacent gene N. After the appearance of the gene N protein, messages become much longer, still beginning at P_L but extending far beyond gene N. The N gene encodes
   a. an antiterminator acting just beyond gene N.
   b. a new sigma factor acting on a promoter beyond gene N.
   c. an activator for a promoter beyond gene N.
   d. an antirepressor that removes a protein repressor bound at gene N.

DNA binding motifs

1. C_2H_2 zinc finger
   a. contains one Zn^{2+} ion bonded with two cystine and two histidine.
   b. contains four conserved cystines in contact with Zn^{2+}.
   c. a structural motif does not bind with DNA.
   d. like leucine zipper is required for dimerization.

2. Which of the following structural motifs is common in prokaryotic DNA binding proteins?
   a. Homeodomain
   b. Helix-turn-helix
   c. Helix-loop-helix
   d. Leucine zipper

[JNU]

3. Zinc finger protein and helix-turn-helix proteins are
   a. types of DNA-binding proteins.
   b. involved in the control of translation.
   c. subunits of RNA polymerases.
   d. members of metal binding proteins.

[JNU]

4. Choose the correct common sequence motif of Zn finger proteins from the choices given below. X stands for any amino acid
   a. X_3- Cys-X_{2-4}-Cys-X_{1-2}-His-X_{3-4}-His-X_4
   b. X_3-Cys-Cys-X_{2-4}-His-His-X_4
   c. X_3-Cys-Cys-His-His-X_4
   d. X_3-Cys-X_{2-4}-His-X_{1-2}-His-X_{3-4}-Cys-X_4

[GATE]

5. 'Zinc fingers' are important in cellular function because they are
   a. the catalytic site of many metabolic enzymes.
   b. structural motifs in many DNA-binding proteins.
   c. structures with high redox potential.
   d. characteristic of palindromic stretches of unique DNA sequence.

[IISc]

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6. Choose the mismatch

a. Steroid hormone receptors	1. zinc finger
b. Homeotic proteins (homeoproteins)	2. helix-turn-helix
c. *Lac* repressor and *Trp* repressor	3. helix-turn-helix
d. λ repressor	4. leucine zipper

7. Homeodomain proteins

P. form a structure with three α-helices.

Q. contact DNA primarily through α-helix 3 and an N-terminal arm.

R. are similar in structure to prokaryotic helix-turn-helix proteins like the λ-repressor.

S. are always present in the cell nucleus.

a. P and Q
b. P and R
c. P, Q and R
d. P, Q, R and S

8. Which statements about leucine zipper proteins are *true*?

P. They bind DNA through conserved leucine residues.

Q. They are similar to HLH proteins in that they have adjacent DNA binding and dimerization domains.

R. Jun can homodimerize but Fos cannot.

S. Fos/Jun binds different DNA sequences from Jun/Jun.

a. P and Q
b. Q and R
c. P, Q and R
d. P, Q, R, S

9. Which of the following statements is *true* with respect to homeobox?

a. It is a conserved DNA sequence found in genes that code for DNA binding motif homeodomain.
b. It is a conserved DNA sequence found in genes that code for proteins that regulate homeostasis.
c. It is a conserved DNA sequence found in genes that code for proteins found in glycolytic pathway.
d. It is a conserved DNA sequence found in genes that code for proteins involved in segmentation. [JNU]

Genetic code

1. The genetic code refers to the

a. number of chromosomes in the diploid cells of a species.
b. nucleotide sequences that correspond to common amino acids.
c. amino acid sequence of cellular proteins.
d. ratio of Mendelian inheritance.

2. The genetic code is

a. degenerate in that many triplets code for more than one amino acid.
b. read in the direction of 3'-5'.
c. generally referred to as universal, since it is nearly the same in all organisms.
d. degenerate for all the amino acids. [GATE]

3. If a genetic code is degenerate, it means that

a. a given base triplet can code for more than one amino acid.
b. there is no punctuation in the code sequences.
c. the third base in a codon is not important in coding.
d. a given amino acid can be coded for by more than one base triplet.

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4. Translation of a synthetic polyribonucleotide containing the repeating sequence CAA in a cell free protein synthesizing system produced three homopolypeptides: polyglutamine, polyasparagine and polythreonine. If the codons for glutamine and asparagine are CAA and AAC, respectively, which of the following triplets is a codon for threonine?

a. AAC
b. CAA
c. CAC
d. ACA

5. In a heteropolymer experiment using 1/2C : 1/4A : 1/4G, how many different triplets will occur in the synthetic RNA molecule and also how often will the most frequent triplet occur?

a. 27 triplets, most frequent will be CAG.
b. 9 triplets, most frequent will be CCC.
c. 8 triplets, most frequent will be CAG.
d. 27 triplets, most frequent will be CCC.

6. Which of the following amino acids has the greatest number of codons?

a. proline
b. leucine
c. tryptophan
d. aspartic acid

7. What is one conclusion that can be drawn from the observation that the genetic code is nearly identical in all cells on earth?

a. Life on earth evolved only once.
b. Only four types of nitrogenous bases occur in DNA.
c. Higher forms of life evolved from lower forms.
d. Natural selection.

8. At one time the genetic code was believed to be universal, but today we know that there are some variations in the code. This implies that

a. the word 'code' is inappropriate.
b. different codons can code for the same amino acid.
c. different amino acids can be encoded by the same codon.
d. the genetic code can evolve.

[IISc]

9. In the genome of a eukaryotic organism, if all the four bases occur randomly, then which amino acid pair will occur least frequently in its proteome?

a. Methionine and tryptophan
b. Arginine and serine
c. Proline and tryptophar
d. Glycine and methionine

[IISc]

10. Which of the following group of amino acids have maximum codons?

a. Methionine, tryptophan
b. Leucine, serine, valine
c. Leucine, serine, arginine
d. Tryptophan, valine, proline

11. Which of the following amino acids is specified by only a single codon?

a. Glutamine
b. Tryptophan
c. Asparagine
d. Isoleucine

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12. If genetic code is tetraplet then what is the possible number of codons which code 20 types of amino acids?
    a. 261
    b. 64
    c. 256
    d. 43

13. An organism isolated from outer space has the genetic material similar to DNA. Chemical analysis showed that there are six bases in the genetic material and the proteins coded by the genome are made up of 40 different types of amino acids. The most likely number of bases in each codon of the organism is
    a. 2
    b. 4
    c. 3
    d. 5

[NSc]

14. To clone a gene corresponding to a protein with partial amino acid sequence. Met - Trp - Cys - Trp (number of codons for Met = 1, Cys = 2, trp = 1), the number of oligonucleotides that need to be designed to screen c-DNA library is
    a. 2
    b. 4
    c. 5
    d. 8

15. The wobble hypothesis refers to the less stringent base pairing specificity of the
    a. 5'-end base of the codon
    b. 3'-end base of the anticodon
    c. middle base of the anticodon
    d. 5'-end base of the anticodon

16. A synthetic mRNA of repeating sequence 5'-CACACACACACACAC... is used for a cell-free protein synthesizing system like the one used by Nirenberg. If we assume that protein synthesis can begin without the need for an initiator codon, what product or products would you expect to occur after protein synthesis?
    a. One protein, consisting of a single amino acid.
    b. Three proteins, each consisting of a different, single amino acid.
    c. Two proteins, each with an alternating sequence of two different amino acids.
    d. One protein, with an alternating sequence of two different amino acids.

17. In an experimental condition, *in vitro* translation of repeating sequence of CAA produced three polypeptides, polyglutamine, polyasparagine and polythreonine. If the codon for glutamine and threonine are CAA and ACA respectively, what will be the codon for asparagine?
    a. AAC
    b. CAC
    c. CCA
    d. ACC

Ribosomes and tRNAs

1. Which base(s) in the anticodon contribute to the degeneracy in codon reading?
    a. first
    b. second
    c. third
    d. first and second

2. The wobble hypothesis explains
    a. how different types of tRNAs are charged with a single type of amino acid.
    b. how multiple stop codons can associate with the same release factor.
    c. how multiple tRNAs can associate with the same codon.
    d. how one tRNA can associate with multiple codons.

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3. Which statement about the codon-anticodon interaction is *false*?
   a. It is antiparallel and can include non-standard base pairs.
   b. Inosine in the 5'-anticodon position can pair with A, C or U in the 3'-codon position.
   c. Inosine in the 3'-anticodon position can pair with A, C or U in the 5'-codon position.
   d. G in the 5'-anticodon position can pair with C or U.

4. 'Isoaccepting tRNAs' are
   P. multiple tRNAs that recognize synonym mRNA codons (third base degeneracy).
   Q. multiple tRNAs that react with the same codon.
   R. multiple tRNAs representing the same amino acid.
   S. multiple tRNAs recognized by the same aminoacyl-tRNA synthetase.
   a. P and Q
   b. Q and R
   c. R and S
   d. P, Q, R and S

5. Transfer RNA specifies amino acid, during proteins synthesis. tRNA is charged with amino acid at 3' end and its anticodon pairs with codon of mRNA. Select the *incorrect* statement out of the following.
   a. Information is transmitted between 3' end of tRNA and anticodon.
   b. Codon-anticodon interaction occurs in A site of ribosome.
   c. Each tRNA has specific anticodon.
   d. Each codon is triplet.

[JNU]

6. Characteristics of prokaryotic ribosomes include which of the following?
   P. They are composed of two subunit particles with sedimentation coefficients of 30S and 50S.
   Q. They are composed of about 65% RNA and about 35% protein.
   R. They can be split into their component subunits by treatment with a Mg^{2+} chelator.
   S. They have all of their protein in the 50S subunit.
   a. P and Q
   b. Q and R
   c. P, Q and R
   d. P, Q, R and S

7. How many polypeptide chains can be formed simultaneously by a given ribosome?
   a. one
   b. up to 30
   c. variable, depending on the length of the mRNA
   d. variable, depending both on length of mRNA and temperature

8. Three of the four eukaryotic rRNAs are synthesized from a single transcription unit consisting of the rDNA. Which one of the following does *not* belong to this group?
   a. 5.8S
   b. 5S
   c. 18S
   d. 28S

[JNU]

9. Wobble base pairs function during protein synthesis by allowing a non-Watson Crick base pair to be formed between?
   a. G at the 5' end of the anticodon and U at the 3' end of the codon.
   b. G at the 3' end of the anticodon and U at the 5' end of the codon.
   c. A at the 5' end of the anticodon and G at the 3' end of the codon.
   d. A at the 3' end of the anticodon and G at the 5' end of the codon.

[MKU]

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Protein synthesis

1. A genetic unit that codes for amino acid sequence of a complete functional polypeptide could be termed as
   a. recon
   b. intron
   c. exon
   d. cistron [IISc]

2. N → C polarity of a polypeptide is colinear with the
   a. 3′ → 5′ polarity of its mRNA
   b. 3′ → 5′ polarity of its tRNA
   c. 5′ → 3′ polarity of its tRNA
   d. 5′ → 3′ polarity of its mRNA

3. An aminoacyl tRNA synthetase is responsible for
   a. formation of a peptide bond.
   b. binding of mRNA to ribosomes.
   c. attaching an amino group to an organic acid.
   d. joining an amino acid to tRNA. [GATE]

4. The number of aminoacyl-tRNA synthetases is
   a. determined by the number of existing tRNAs.
   b. determined by the number of translatable mRNA codons.
   c. determined by the number of utilized amino acids.
   d. determined by the individual organism species.

5. All of the following statements about amino acid activation are correct *except*
   a. aminoacyl-tRNA synthetases transfer the amino acids to the A on the 3′ end of tRNA via an aminoacyl-adenosine monophosphate intermediate.
   b. the aminoacyl-tRNA synthetase is specific for a given amino acid.
   c. there may be more than one kind of tRNA for a given amino acid.
   d. the activation step requires the hydrolysis of the terminal phosphate group of guanosine triphosphate.

6. Activation of amino acids for peptide synthesis utilizes ATP, generating pyrophosphate, rather than P_i, as a product. The metabolic advantage of producing pyrophosphate rather than P_i is that
   a. hydrolysis of the α–β phosphodiester linkage releases more energy than does hydrolysis of the β–γ linkage.
   b. ATP is readily resynthesized from the product AMP by the adenylate kinase reaction.
   c. subsequent pyrophosphate hydrolysis by pyrophosphatase pulls the reaction in favor of the product.
   d. the pyrophosphate ion buffers over a wider range than phosphate.

7. Accuracy in the charging of tRNA by aminoacyl-tRNA synthetase is accomplished by
   P. chemical proofreading (noncognate aminoacyl-adenylate is hydrolyzed).
   Q. kinetic proofreading (noncognate aminoacyl- adenylate dissociates rapidly).
   R. the specificity of the amino acid acylation reaction.
   S. a post synthetic event, i.e. only the correct aminoacyl- tRNA fits into the ribosomal A-site.
   a. P and Q
   b. Q and R
   c. P, Q and R
   d. P, Q, R and S

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8. To verify that mRNAs are translated from the 5'- to 3'-end, a cell-free protein synthesis system was programmed with β-globin mRNA and pulse labeled with radioactive leucine for a brief period. The system synthesized β-globin chains and the full length chains made during the labeling period were isolated, digested by trypsin and the peptide fragments were analyzed for the level of radioactivity. Which of the following statements is *correct*?

a. The C-terminal position of the peptide will correlate with a higher level of radioactivity.
b. The N-terminal position of the peptide will correlate with a higher level of radioactivity.
c. The mid-position of the peptide will correlate with a higher level of radioactivity.
d. There will be uniform distribution of the level of radioactivity. **[JNU]**

9. Which of the following statements about the ribosomal peptidyl transferase activities are *correct*?

P. The peptidyl transferase activity resides in the large (50S or 60S) ribosomal subunit.
Q. It facilitates transfer of the C-terminus of the peptide chain from the peptidyl-tRNA to the N-terminus of the aminoacyl-tRNA in the A-site.
R. It facilitates transfer of the N-terminus of the aminoacyl-tRNA from the A-site to the C-terminus of the peptidyl-tRNA in the P-site by deacylation of aminoacyl-tRNA.
S. It hydrolyses GTP in order to allow translocation of the ribosome.

a. P and Q
b. Q and R
c. P, Q and R
d. P, Q, R and S

10. GTP is required by which of the following steps in protein synthesis?

a. Aminoacyl-tRNA synthetase activation of amino acids.
b. Attachment of ribosomes to endoplasmic reticulum.
c. Translocation of tRNA-nascent protein complex from A to P sites.
d. Attachment of mRNA to ribosomes.

11. Formylmethionine tRNA is present in

a. yeast
b. bacteria
c. insects
d. mammals **[IISc]**

12. All of the following factors are required to initiate translation of mRNA in prokaryotes *except*

a. formylmethionine-tRNA,
b. 30S ribosomal subunits
c. guanosine triphosphate
d. EF-Tu

13. GUG codes for valine in both prokaryotes and eukaryotes but when GUG is initiation codon, this codes

a. methionine
b. valine
c. tryptophan
d. none of these

14. At initiation the two ribosomal subunits combine with mRNA and

a. threonine charged tRNA
b. methionine charged tRNA
c. serine charged tRNA
d. proline charged tRNA

15. During translation in prokaryotes, the small ribosomal subunit

P. initially binds to the mRNA at shine Dalgarno sequence.
Q. associates with the mRNA after binding to a protein bound to the mRNA.
R. binds to the mRNA through base pairing between the rRNA and the mRNA.
S. initially binds to the mRNA through base pairing between the tRNA and the mRNAs.

a. P and R
b. Q and R
c. P, Q and R
d. P, R and S

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16. Shine–Dalgarno sequence of mRNA is helpful in
    a. recognition of 50S ribosomal subunit.
    b. recognition of 60S ribosomal subunit.
    c. recognition of 5S rRNA.
    d. recognition of 30S subunit of ribosome.

17. 'Kozak' is associated with
    a. transcription
    b. DNA replication
    c. DNA repair
    d. translation [IISc]

18. Initiation of protein synthesis from most eukaryotic mRNA requires
    a. a 3' poly-A tail
    b. a 5' cap
    c. a 3' cap
    d. both a poly-A tail and a 5' cap

19. In the protein synthesis, tRNA carrying the initiating amino acid enters in which site of ribosome?
    a. 'A' site
    b. 'P' site
    c. anticodon site
    d. recognition site

20. Which of the following protein factors are required for the addition of an amino acid residue to a growing polypeptide chain during protein biosynthesis in *E. coli*?
    a. IF-1, IF-2 and IF-3
    b. factor G
    c. factor R
    d. factors Tu and Ts

21. During protein synthesis in *E. coli*, the initiating methionine and the internal methionine codons of a given mRNA are distinguished by
    a. fmet-tRNA$_f$ and met-tRNA$_m$ respectively.
    b. fmet-tRNA$_f$-IF-2 versus met-tRNA$_m$-EF-Tu respectively.
    c. fmet-tRNA$_f$ and 30S ribosomal subunit versus met-tRNA$_m$ and 50S ribosomal subunit respectively.
    d. fmet-tRNA$_f$-IF-2 versus fmet-tRNA$_f$-EF-Tu respectively.

22. During peptide bond synthesis which of the following step does *not* require an input of energy?
    a. an amino acid activation.
    b. the formation of the 70S initiation complex of prokaryotes.
    c. the binding of the aminoacyl-tRNA to the A site on the ribosome.
    d. transpeptidation.

23. Synthesis of peptide bond is catalyzed by
    a. A site of ribosome
    b. P site of ribosome
    c. 23S rRNA
    d. tRNA [JNU]

24. What is the function of initiation factor IF-3?
    a. If bound to the 40S subunit, it facilitates the association of the 40S and 60S subunits.
    b. If bound to the 30S subunit, it prevents the association of the 30S and 50S subunits.
    c. If bound to the 30S subunit, it allows the 16S rRNA of the 30S subunit to interact with the Shine-Dalgarno sequence of the mRNA.
    d. It directs the initiator tRNA to enter the partial P-site on the 30S subunit bound to mRNA.

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25. Which of the following statement about chain termination in peptide synthesis is *true*?
a. The formation of peptide bonds by the ribosomal mRNA complex continues until a stop codon on mRNA is reached.
b. The formation of peptide bonds by the ribosomal mRNA complex continues until a tRNA with an anticodon for UAA, UAG, or UGA interacts with the A site on the ribosome.
c. In prokaryotes, peptide bond formation ceases when a formylmethionine-$tRNA_f$ interacts with the A site on the ribosome.
d. Peptide bond formation ceases when the ribosome reaches the 5′ end of the mRNA.

26. The translation of mRNA into the amino acid sequence of a polypeptide in prokaryotes is terminated at the end of the message by one of three chain termination codons in the mRNA. The stop codon is recognized by
a. a specific uncharged tRNA.
b. a specific aminoacyl-tRNA.
c. a specific ribosomal RNA.
d. a release factor.

27. *E. coli* release factor1 (RF1) recognizes which codons?
a. UAA only
b. UGA only
c. UGA and UAA
d. UAG and UAA

28. During protein biosynthesis in an eukaryotic cell culture, Dr. Subramanian observed that the binding of a factor called as e*RF*-2 to a genetic codon on the mRNA strand resulted into
a. hydrolysis of the terminal peptidyl-tRNA bond.
b. a free polypeptide and tRNA.
c. 70S ribosome dissociating into 30S and 50S subunits.
d. recognizes the termination codon.

29. Which of the following occurs when the eukaryotic translational machinery encounters the UAG codon?
a. The ribosome can recognize the UAG codon as a translational stop codon and does not allow any tRNA to bind the mRNA.
b. The termination factors recognize this codon and translation ends.
c. This codon is recognized by the corresponding anti-codon of an empty tRNA molecule that is not linked to an amino acid.
d. The non-recognition of this codon by any tRNA ultimately causes the translational machinery to stop.

[IISc]

30. What is the E-site of the ribosome?
a. The site where eukaryotic mRNA is processed.
b. An exciting term made up by your instructor.
c. The site where tRNA exits the prokaryotic ribosome.
d. The site where the endonuclease *Eco*RI restricts the ribosome.

31. What is the *correct* order of the following steps in protein synthesis?
1. A peptide bond is formed.
2. The small ribosomal subunit is loaded with initiation factors, mRNA, and initiation aminoacyl-tRNA.
3. The intact ribosome slides forward three bases to read a new codon.
4. The primed small ribosomal subunit binds with the large ribosomal subunit.
5. Elongation factors deliver aminoacyl-tRNA to bind to the A site.

a. 1, 2, 5, 4, 3
b. 2, 3, 4, 5, 1
c. 4, 5, 1, 2, 3
d. 2, 4, 5, 1, 3

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32. Which of the following is *not* characteristic of eukaryotes?
    a. Ribosomes are larger than those of prokaryotes.
    b. All mRNAs specify a single protein or a polyprotein.
    c. No specific ribosome binding sites exist in mRNA comparable to those of the Shine-Dalgarno sequences in prokaryotes.
    d. N-formylmethionine initiates all nascent polypeptide chains.

33. Which one of the following statements *correctly* describes initiation of protein synthesis in *E. coli*?
    a. The initiator tRNA binds to the Shine-Dalgarno sequence.
    b. Three initiation factors are involved and IF2 binds to GTP.
    c. The intermediate containing IF1, IF2, IF3 initiator tRNA and mRNA is called the 30S initiation complex.
    d. binding of the 50S subunit releases IF1, IF2, GMP and PP_i.

34. Peptide bond synthesis requires an input of energy during
    P. an amino acid activation.
    Q. the formation of the 70S initiation complex of prokaryotes.
    R. the binding of the aminoacyl-tRNA to the A site on the ribosome.
    S. the movement of the peptidyl-tRNA to the P site and the associated movement of the mRNA.
    a. P and Q
    b. Q and R
    c. P, Q and R
    d. P, Q, R and S

35. How many energy bonds are expended in the formation of a peptide bond?
    a. 2
    b. 4
    c. 3
    d. 6

[GATE]

36. Which two of the following statements about initiation of eukaryotic protein synthesis are *true*?
    P. Eukaryotes use a mRNA scanning method to locate the correct start codon.
    Q. There are at least nine eukaryotic initiation factors (eIFs).
    R. Eukaryotic initiation uses *N*-formylmethionine.
    S. The 80S initiation complex completes the initiation process and contains the initiator tRNA base paired to the start codon in the A site.
    a. P and Q
    b. Q and R
    c. P, Q and R
    d. P, Q, R and S

37. Choose the mismatch
    a. peptidyl transferase – peptide bond formation
    b. wobble position – third position of codon
    c. aminoacyl-tRNA – amino acid activation synthetase
    d. Poly-U – homopolymer of methionine

38. Which of the following is/are *correct* statements?
    P. Structurally different tRNA molecules that recognize the same amino acid are called isoaccepting tRNA.
    Q. Shine-Dalgarno sequence on a bacterial mRNA molecule is recognized by rRNA.
    R. Proofreading reduces the error rate of incorporation of incorrect amino acids into a polypeptide chain during translation.
    S. A series of hydrophobic amino acids called signal sequences at the N terminus of a protein that targets that protein for delivery to the cell membrane or beyond.

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a. P and Q
b. R and S
c. Q and R
d. P, Q, R and S

39. Which of the following statements about the diphtheria toxin is *correct*?
a. It is cleaved on the surface of susceptible eukaryotic cells into two fragments, one of which enters the cytosol.
b. It binds to peptidyl transferase and inhibits protein synthesis.
c. It reacts with ATP to phosphorylate eIF2 and prevents the insertion of the Met-tRNA into the P site.
d. It reacts with NAD^+ to add ADP-ribose to eEF2 and prevents movement of the peptidyl-tRNA from the A to the P site.

40. Which of the following is *not* an antibacterial antibiotic?
a. Tetracycline
b. Streptomycin
c. Nystatin
d. Nalidixic acid

[JNU]

41. An antibiotic which inhibits translation in both prokaryotes and eukaryotes is
a. chloromycetin
b. puromycin
c. actinomycin-D
d. tetracycline

42. All aminoglycoside antibiotics inhibit protein synthesis by
a. binding to small ribosomal subunit.
b. binding to large ribosomal subunit.
c. binding to both small and large ribosomal subunits.
d. inactivating eEF-2.

43. Chloramphenicol inhibits
a. cell wall synthesis in bacteria.
b. protein synthesis on 70S ribosome.
c. protein synthesis on 80S ribosome.
d. DNA replication.

44. An antibiotic that resembles the 3' end of a charged tRNA molecule is
a. streptomycin
b. penicillin
c. tetracycline
d. puromycin

45. The following antibiotics affect the bacterial protein synthesis with their site of action. Which of the following combinations is *correct*?

Antibiotic	*Site of action*
P. Streptomycin	1. Aminoacyl tRNA association with ribosome
Q. Tetracyclin	2. Transpeptidation
R. Fusidic Acid	3. Translocation
S. Chloramphenicol	4. Misreading of mRNA

a. P-2; Q-3; R-4; S-1
b. P-4; Q-1; R-3; S-2
c. P-1; Q-4; R-3; S-2
d. P-4; Q-1; R-2; S-3

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46. Which of the following statements about antibiotic inhibitors of protein synthesis is *correct*?
P. Chloramphenicol inhibits the peptidyl transferase activity of the large ribosomal subunit in eukaryotes.
Q. Cycloheximide inhibits the peptidyl transferase activity of the large ribosomal subunit in eukaryotes.
R. Puromycin blocks protein synthesis in prokaryotes only.
S. Streptomycin inhibits protein synthesis by binding to 30S ribosomal subunits.
a. P and Q
b. Q and R
c. Q and S
d. P, Q, R and S

47. Diptheria toxin inhibits protein synthesis by
a. causing formation of ADP-EF2 complex
b. release of peptidyl-tRNA from the P site
c. binding to factor eEF-2
d. inhibiting peptide bond formation [GATE]

48. Chaperone proteins help in
a. protein folding and assembly
b. protein stability
c. both of the above
d. none of the above [IISc]

49. Which of the following statements is *correct* in case of chaperone proteins?
a. These do not prevent aggregation.
b. They cleave incorrect S-S bonds.
c. Act on fully synthesized polypeptides.
d. Are involved in the transport of proteins across mitochondria and endoplasmic reticulum. [GATE]

50. What is ubiquitin?
P. A subunit of the electron transport chain coenzyme Q, ubiquinone.
Q. A generic name for the most abundant protein in chloroplasts, Rubisco.
R. A polypeptide involved in marking proteins for degradation.
S. A polypeptide that modifies a fraction of H2A histone molecules during the cell cycle.
a. P and Q
b. Q and R
c. R and S
d. P, Q, R and S

51. Which of the following statements is *not* true for ubiquitin mediated proteolysis?
a. Ubiquitinylated proteins are degraded in the lysosome.
b. Ubiquitin is polypeptide of 76 amino acids.
c. Ubiquitination occurs at a specific amino acid of the target proteins.
d. Following ubiquitination, proteins are degraded by proteasomes. [JNU]

52. Which of the following is involved in the majority of the ATP-dependent cytosolic degradation of proteins in eukaryotes?
a. Cathepsins
b. Calpains
c. Lysosome
d. 26S proteosomes [IISc]

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53. Which of the following post-translational modifications of proteins does *not* occur in the lumen of the endoplasmic reticulum?
a. Glycosylation
b. Formation of disulfide bonds
c. Folding and formation of quaternary structure
d. Ubiquitination

54. The term protein splicing refers to [JNU]
a. removal of intervening sequences between the genes.
b. splicing out of introns from RNA.
c. removal of intervening protein sequences from the translated protein.
d. joining (splicing) of two different gene products to generate a novel protein. [GATE]

DNA recombination

1. In bacteria, which enzyme binds single-stranded DNA, denatures double-stranded DNA and matches the single-stranded DNA with complementary denatured DNA?
a. RecA
b. RecBCD
c. UvrABC
d. UvrD

2. A triple-stranded intermediate is postulated to occur during
a. eukaryotic DNA replication
b. RecA-mediated DNA recombination
c. excision repair
d. thymidine dimer photoreactivation

3. In bacteria, which enzyme unwinds double-stranded DNA until it meets a *chi* site and then cuts one strand of the DNA?
a. RecA
b. RecBCD
c. UvrABC
d. UvrD

4. All of the following statements regarding genetic recombination events are correct *except*
a. site-specific recombination is characterized by an exchange between two identical DNA sequences.
b. the processes of transformation and transduction require extensive homology between the interacting sequences.
c. DNA replication is required for genetic recombination to occur.
d. during genetic recombination events, the DNA molecules are broken and then rejoined.

5. Branch migration of a Holliday junction is necessary to extend the repair/recombination segment into homoduplex regions of the interacting double strands. The migration is achieved by
a. the ATP-hydrolysis-dependent activity of the RecA protein.
b. the ATP-hydrolytic action of the RuvAB helicase complex.
c. the activity of the RecBCD protein complex.
d. the tension in the DNA molecule caused by coiling. [JNU]

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6. The Rec A protein of *Escherichia coli* has several functions. Which of the following statements is false?
   a. Rec A acts as a co-protease.
   b. Rec A protein has an affinity for single-stranded DNA.
   c. Rec A binds pyrimidine dimers and initiates their repair.
   d. Rec A is required for homologous recombination.

7. Lambda phage integrates its double-stranded DNA into the bacterial genome where it becomes an integral part of the bacterial chromosome as a prophage. Phage DNA integration is accomplished by
   a. restriction of the bacterial genome with *Lam*RI followed by integration of the phage DNA facilitated by integrase.
   b. site-specific recombination between the phage *attP* and the bacterial *attB* sequences.
   c. by single-strand assembly (linear DNA is incorporated into the bacterial genome).
   d. homologous recombination (precise assimilation of identical DNA).

[JNU]

8. In a mating between two strains of yeast that differ at a sigle locus (wild type × mutant), 15 percent of the tetrads contain wild-type and mutant spores in the ratio of 3:1 or 1:3. The most likely explanation for this observation is
   a. gene conversion
   b. deletion of the mutant allele
   c. allelic exclusion
   d. reversion

9. Gene conversion is observed when
   a. during meiosis, one allele is replaced by another allele.
   b. during meiosis, the expected 2:2 ratio of alleles is replaced with a 4:0 ratio.
   c. during meiosis, the expected 2:2 ratio of alleles is replaced with a 3:1 ratio.
   d. all of the above.

10. In the Meselson-Radding model of meiotic recombination, explain how the two DNA molecules interact at the beginning of homologous recombination?
   a. Single-strand nicks appear at equivalent positions in each molecule.
   b. A specialized topoisomerase produces single-strand breaks in both DNA molecules.
   c. The two DNA molecules initiate recombination without breakage.
   d. A single-strand nick occurs in one molecule creating a free end that invades the other molecule to displace one of its strands.

[JNU]

DNA repair

1. Photoreactivation
   a. repairs dimers in DNA using an endonuclease.
   b. uses light to activate repair enzymes.
   c. removes alkylating agents from bases.
   d. deaminates bases.

2. Photoreactivation in DNA repair requires
   P. an endonuclease
   Q. pyrimidine dimers
   R. activation by ultraviolet light
   S. an enzyme that cleaves -C-C- bonds
   a. P and Q
   b. Q and R
   c. R and S
   d. Q and S

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3. Enzyme DNA photolyase which is involved in direct repair of damaged DNA like thymine dimer
   a. can work only in the presence of light
   b. can work in dark
   c. is independent of light
   d. none of these

4. Nucleotide excision repair of DNA in *E. coli*
   a. replaces both strands of the DNA in the damaged region.
   b. uses high energy phosphate bonds.
   c. utilizes RNA polymerase to make a primer.
   d. requires uvrABC excinuclease.

[MKU]

5. Xeroderma pigmentosum is an autosomal recessive mutation that affects the mechanism for repair of ultraviolet light-induced thymine dimer formation in DNA. Which of the following enzymes is most likely to be defective?
   a. DNA polymerase I
   b. DNA polymerase II
   c. DNA ligase
   d. enzymes involved in nucleotide excision repair

6. Which of the following abnormalities in DNA can be corrected by excision repair systems?
   P. Intrastrand pyrimidine dimers
   Q. Alkylated bases
   R. Cytosine deamination
   S. Missing base-pairs
   a. P and Q
   b. Q and R
   c. P and R
   d. P, Q, R and S

7. How does nucleotide-excision repair differ from base-excision repair?
   a. Base-excision repair recognizes and removes single damaged bases, whereas nucleotide-excision repair is more general, recognizing many different kinds of lesions that distort the DNA molecule.
   b. Nucleotide-excision repair reverses the chemical reaction that caused the lesion, whereas base excision repair removes the damaged bases and replaces them with normal ones.
   c. Only the base is removed in base-excision repair, whereas the entire nucleotide is removed in nucleotide-excision repair.
   d. Base-excision repair requires no protein components and can occur by simple absorption of UV light, whereas nucleotide-excision repair requires several enzymes.

8. The most common modification of DNA is methylation. In prokaryotes, this serves to
   P. identify damaged DNA for repair.
   Q. distinguish the strands after replication and allow/disallow continued replication.
   R. identify methylated foreign DNA for recombinational incorporation into the genome.
   S. protect its own DNA but not foreign DNA from restriction by endonuclease.
   a. P and Q
   b. Q and R
   c. R and S
   d. P, Q and S

9. In *Escherichia coli*, which of the following repair systems is most error-prone?
   a. photoactivation
   b. excision repair
   c. recombinational repair
   d. SOS repair

[JNU]

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10. The SOS repair mechanism is activated by which of the following?
    a. 5-bromouracil
    b. acridine orange
    c. hydroxylamine
    d. thymidine dimers

Gene mutation

1. Which statement about mutation is *false*?
    a. Mutations can happen at a rate of $\sim 10^{-5}$.
    b. A change in the nucleotide sequence of a gene is a point mutation.
    c. A change in the nucleotide sequence of a gene doesn't necessarily change the product of that gene.
    d. Most of mutation are beneficial to the organisms that have them.

2. Mutations which do not cause any functional change in the protein are known as
    a. non-sense mutations
    b. missense mutations
    c. backward mutations
    d. silent mutations

3. A change in DNA that creates UAA codon at a site previously occupied by another codon, would result in one of the following mutations
    a. Amber
    b. Ochre
    c. Opal
    d. Polar

4. A single nucleotide pair is inserted near the 5′ end of a protein-coding DNA sequence. The most likely effect will be production of
    a. a protein with a single altered amino acid
    b. a protein with an almost completely altered sequence
    c. a greatly truncated protein
    d. no protein

5. A recessive mutation is one which [JNU]
    a. is not expressed
    b. is expressed only when heterozygous
    c. is expressed only when homozygous or hemizygous
    d. is eliminated by natural selection

6. Thymine dimer formation during replication of DNA is caused due to [JNU]
    a. gamma radiation
    b. UV radiation
    c. X-rays
    d. IR radiation

7. Which of the following processes would not contribute to genetic variation within a bacterial population?
    a. transduction
    b. transformation
    c. mutation
    d. meiosis

8. Spontaneous mutations
    a. are caused by chemicals such as acridine orange and caffeine.
    b. are caused by physical agents such as ultraviolet light or X-rays.
    c. are the result of errors in the base pairing of nucleotides during replication.
    d. occur at a rate higher than the rate of induced mutations.

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9. A DNA mutation that results in no change in protein product produced is termed a
   a. missense mutation
   b. nonsense mutation
   c. silent mutation
   d. frameshift mutation

10. Which of the following changes to a DNA molecule is least likely to result in a deleterious mutation?
   a. Insertion of a transposable element in a coding region.
   b. Deletion of a base pair in a coding region.
   c. Change of a base pair in the first codon of a coding region.
   d. Change of the third base pair of a codon.

11. Mutations often occur as a result of base substitutions. The most common cause of base substitutions is
   a. tautomeric shifts
   b. meiotic errors
   c. base insertions
   d. base deletions

12. A tautomeric shift causing the substitution of one purine for a pyrimidine is called
   a. transversion
   b. translocation
   c. transition
   d. inversion

13. When a mutational event leads to the replacement of one codon for another coding for the same amino acid, the resulting mutation is known as
   a. frameshift
   b. neutral
   c. missense
   d. silent

14. When bacteria are spread on a plate containing an antibiotic, colonies of resistant bacteria appear on the plate. The Darwinian explanation for this is that
   a. the antibiotic induced the resistance.
   b. resistant cells pre-existed in the population and were selected by the antibiotic.
   c. no genetic change is required to exhibit antibiotic resistance.
   d. the antibiotic is mutagenic. **[IISc]**

15. Using site-directed mutagenesis four mutants of a protein have been generated. Which of the following missence mutations has the largest difference in terms of number of atoms between the wild type and the mutant?
   a. Ser → Cys
   b. Tyr → Phe
   c. Lys → Ala
   d. Arg → Lys **[IISc]**

16. Which of the following is most likely to lead to a loss of gene function?
   a. A missense mutation in the open reading frame.
   b. A change from T to C in the promoter region.
   c. A frameshift mutation in the coding region.
   d. A sequence change in the 3′ untranslated region.

17. A bacterial protein coding gene contains a terminator codon in the middle of the coding region, yet expression of the gene in the bacterium produces a functional protein. Translation of the gene probably requires
   a. the excision of an intron
   b. a suppressor tRNA
   c. ribosomes that lack 5S RNA
   d. an mRNA with no ribosome binding site

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18. In a bacterial cell, a mutation in an aminoacyl-tRNA synthetase leads to charging of the entire $tRNA^{ser}$ population with alanine. Which of the following describes the result of using these aminoacyl tRNAs for protein synthesis in the cell?
    a. Proteins synthesized using the alanyl-$tRNA^{ser}$ will contain neither alanine nor serine.
    b. Proteins synthesized using the alanyl-$tRNA^{ser}$ will contain only serine where alanine would normally occur.
    c. Proteins synthesized using the alanyl-$tRNA^{ser}$ contain only alanine where serine would normally occur.
    d. Proteins synthesized using the alanyl-$tRNA^{ser}$ will randomly contain either alanine or serine where serine would normally occur.

19. In genetics, suppression of a mutation refers to
    a. restoration of the original phenotype due to a second mutation.
    b. restoration of the original DNA sequence by mutation.
    c. prevention of expression of the mutant gene by metabolic regulation.
    d. appearance of the recessive phenotype in a heterozygous diploid.

20. When nonsense mutations occur in the reading frame of mRNA, protein synthesis gets terminated at the nonsense mutation to deliver a truncated polypeptide. However, in certain bacterial strains, this does not happen; these bacterial cells are able to synthesize full-length polypeptide. This phenomenon is due to
    a. compensatory frameshift mutation that occurs elsewhere in the mRNA.
    b. involvement of suppressor tRNAs.
    c. polypeptide splicing at the broken point.
    d. post-transcriptional editing of the nonsense mutation. **[IISc]**

21. A protein that is normally localized to nucleus has been mutated in such a way that its localization is no longer nuclear, but cytosolic instead. The function of the protein, however, is 99% of the wild-type protein. Which of the followings could be said about the mutation?
    a. It is a frameshift mutation.
    b. It is a nonsense mutation.
    c. It is a silent mutation.
    d. It is a small deletion.

22. Which of the following is *true*?
    P. Transversional mutations are more harmful than transitional mutations.
    Q. Frameshift mutations may lead to premature termination.
    R. Missense mutations are more deleterious than nonsense mutations.
    S. G → C mutation is a transversional mutation.
    a. P only
    b. P and R
    c. Q and R
    d. P, Q and S

23. If natural selection in a particular environment favored genetic systems that permitted the production of daughter cells that were genetically dissimilar from the mother cells, then one should expect selection for which of the following?
    1. Polynucleotide polymerase with low mismatch error rates.
    2. Polynucleotide polymerases without proofreading capability.
    3. Batteries of efficient polynucleotide repair enzymes.
    4. Polynucleotide polymerases with proofreading capability.
    5. Polynucleotide polymerases with high mismatch error rates.
    a. 1 and 4
    b. 2 and 5
    c. 3 and 5
    d. 1, 4 and 5

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24. Which of the following chemical mutagens is likely to cause GC → AT transitions?
a. 5-bromouracil
b. 2-aminopurine
c. Acridine orange
d. Hydroxylamine

25. Sites where mutations occur at rates higher than normal are known as
a. suppressor sites
b. hotspots
c. mutator sites
d. cistrons

26. Choose the *incorrect* statement(s).
1. Intercalating agents cause primarily frameshift mutation.
2. Luria-Delbruck fluctuation test provided the first experimental evidence supporting the hypothesis that mutations occur spontaneously.
3. Tautomer is an alternative isomeric form of a base due to the movement of one hydrogen atom from one position to another.
4. Missense type of mutation results when a base substitution in a codon leads to the formation of another codon calling for the insertion of a different amino acid.
5. Suppressor mutations cause the reversion of a mutant phenotype to the wild type even though they are not back mutations.

a. 1 and 2
b. Only 4
c. 4 and 5
d. None of these

27. Match the following

Column A	*Column B*
P. Acridine orange	a. base analog
Q. 5-BU	b. intercalating agent
R. Ethyl methane sulfonate	c. deaminating agent
S. HNO_2	d. alkylating agent

a. P-b, Q-d, R-a, S-c
b. P-b, Q-a, R-d, S-c
c. P-b, Q-d, R-c, S-a
d. P-a, Q-b, R-c, S-d

28. Choose the mismatch
a. clastogen – causes fragmentation of chromosome
b. oncogen – induces tumor formation
c. teratogen – result in developmental abnormalities
d. alkylating agent – removes amino group from nitrogenous base

29. If GUG codon is used as a starting codon then it codes for
a. valine
b. cysteine
c. methionine
d. leucine

30. 5-Bromouracil can cause a transitional mutation in the base sequence of DNA because it
a. bonds irreversibly with A.
b. may resemble C and bond with G.
c. is analog of T and pairs with A.
d. resembles U and would not be incorporated into DNA.

31. Acridines act as mutagens to duplex DNA, causing which of the following mutations?
a. Transitional mutation
b. Transversional mutation
c. Frameshift mutation
d. Single amino acid substitution in the gene product

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32. Which one of the following situations could be termed as intergenic suppression of a mutation in a coding sequence?
   a. A non-sense mutation in the codon leading to lack of interaction with cognate tRNA anticodon.
   b. The amber mutation in the codon recognized by the altered anticodon of a tRNA.
   c. A stop codon recognized by the ribosomes.
   d. A frameshift mutation skipping an otherwise stop codon. [JNU]

33. Nonsense mutations differ from deletion mutations in that
   a. nonsense mutations are non-reversible.
   b. nonsense mutations can be suppressed.
   c. protein chain termination does not take place in nonsense mutations.
   d. no transcription takes place when there is a nonsense mutation. [IISc]

34. A missense mutation is said to occur when
   a. a base pair substitution occurs in the DNA generating a stop codon and a truncated protein.
   b. a base pair substitution occurs in the DNA so that one amino acid is changed to another amino acid in the protein.
   c. a base pair substitution occurs in the DNA but the encoded amino acid and protein is unchanged.
   d. a base pair is inserted into the DNA and a shift in the reading frame occurs so that all subsequent encoded amino acids are altered.

35. Deamination of 5-methyl cytosine leads to the formation of
   a. thymine
   b. uracil
   c. cytosine
   d. hydroxymethyl cytosine [IISc]

36. Mutations that are expressed only under certain environmental conditions are called
   a. transposition
   b. conjugation
   c. recombination
   d. conditional mutations [MKU]

37. An intercalating mutagen, ICR-191; induced LacZ mutation gives rise to spontaneous Lac^+ revertants at a frequency of 10^{-8}. The reversion frequency can be enhanced by treatment with
   a. nitrous acid
   b. hydroxylamine
   c. ICR-191
   d. 5-bromodeoxyuridine [JNU]

38. The Ames test is a mass screening approach used for the detection of
   a. toxins
   b. mutagenic carcinogens
   c. lactose intolerance
   d. phenylketonuria [IISc]

39. A chemical that resembles one of the bases and can mistakenly get incorporated into DNA/RNA is called
   a. deoxyribonucleotide
   b. ribonucleotide
   c. base analog
   d. nitrogen base [MKU]

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40. Suppressor mutation results in restoration of the wild type phenotype. The suppression of the mutant phenotype is usually brought about
    a. by misreading of the mutant codon and incorporation of a correct amino acid.
    b. by insertion of another normal copy of the gene.
    c. by reversion of the mutation to wild type.
    d. by deletion of the mutant gene.

[JNU]

41. With respect to the wild type strain, a silent mutant will have
    a. same genotype and phenotype.
    b. same genotype but different phenotype.
    c. different genotype but same phenotype.
    d. different genotype and different phenotype.

[JNU]

42. A method to detect whether two mutations are located on the same gene or different genes is
    a. generalized transduction
    b. complementation analysis
    c. *hfr* mapping
    d. karyotyping

[IISc]

43. Ethidium bromide acts as a mutagen by
    a. substituting adenine by its structural analogue.
    b. chemical modification of base.
    c. production of interstrand cross-links in DNA.
    d. intercalating between DNA bases interfering with proper base stacking. [JNU]

44. The fluctuation tests done by Luria and Delbruck showed that
    a. antibiotics induce the development of resistance in bacteria.
    b. the growth of bacteria fluctuates based on the concentration of antibiotics in the media.
    c. the concentration of antibiotics fluctuates in response to the number of bacteria in a sample.
    d. phage T1 does not induce mutation in wild type *E. coli*.

45. Given below is the result of a complementation test for six independent mutants (1 to 6).

1	2	3	4	5	6	
0	+	0	+	+	0	1
	0	+	+	+	+	2
		0	+	+	0	3
			0	0	+	4
				0	+	5
					0	6

'+' represents complementation; '0' represents non-complementation.

Based on the above, which one of the following conclusions is *correct*?

a. The mutations can be ordered in a single cistron as 1-3-5-2-4-6.
b. All mutations belong to a single cistron, but their order cannot be determined.
c. There are three cistrons, mutations 1, 3 and 6 represent one cistron, 4 and 5 represent the second cistron and 2 represents the third cistron.
d. There are three linkage groups, mutations 1, 3 and 6 represent linkage group A, 4 and 5 represent linkage group B, and 6 represents linkage group C. [NET]

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46. A chemist synthesized three new chemical compounds, M1, M2 and M3. The compounds were tested for their mutagenic potential and were found to be highly mutagenic. Tests were made to characterize the nature of mutations by allowing the reversion with other mutagens. The following results were obtained:

Mutations produced by	Reversed by			
	2 Aminopurine	Nitrous acid	Hydroxyl amine	Acridine orange
M1	No	No	No	No
M2	Yes	Yes	No	No
M3	No	No	No	Yes

Which one of the following conclusions drawn regarding the nature of mutations by the compounds is correct?

a. M1 – transversion, M2 – insertion, M3 – deletion.
b. M1 – transition, M2 – transversion, M3 – insertion.
c. M1 – insertion, M2 – transition, M3 – transversion.
d. M1 – transversion, M2 – transition, M3 – insertion. [NET]

47. The Ames test of mutagenicity is characterized by which of the following statements?

P. The test uses a mutant strain of bacteria that requires histidine in the medium to grow but that reverts to wild type at a slow rate.
Q. The mutant bacterial strain used reverts back to wild type on withdrawal of histidine from the medium.
R. Mutagens markedly enhance the rate at which the bacterial strain reverts to wild type.
S. Mutagenicity (in the Ames test) and carcinogenicity (defined by animal studies) are not correlated.

a. P and Q
b. P and R
c. R and S
d. P, Q, R and S

48. Which of the following is correct about dominant-negative mutation?

a. Loss of function mutation that completely abolishes the activity of a gene.
b. Either reduces or abolishes the activity of the gene.
c. Blocks gene activity, causing a loss of function even in presence of normal gene product.
d. Increases the activity of the gene or makes it active in inappropriate circumstances. [JNU]

49. In a study of arginine biosynthesis in yeast, four mutant haploids requiring arginine (Arg^-) were isolated. The Arg^- haploids were fused in pairwise combinations to form diploids, whose requirement for arginine was tested. The results of the tests were that all diploid combinations yielded arginine prototrophs. How many different Arg genes are represented among the four mutants?

a. One
b. Two
c. Three
d. Four [JNU]

50. Mutations in the human genome can cause hereditary disease and cancer. Which statement does most accurately describe a point mutation?

a. A nucleotide change that causes a single amino acid substitution.
b. A deletion, insertion or replacement of a single nucleotide in the genome.
c. A mutation that does not cause a change in amino acid.
d. Any nucleotide change that does not cause shift of reading frame.

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Fill in the blanks in the following statements

1. Enzymes called ..... couple each amino acid to its appropriate tRNA molecule to create an ..... molecule.
2. A ..... contains two binding sites for tRNA molecules; the ....., or P-site, holds the tRNA molecule that is linked to the growing end of the polypeptide chain, and the ....., or A-site, holds the incoming tRNA molecule that is charged with an amino acid.
3. Peptide bond formation is catalyzed by ....., a catalytic activity that is thought to be mediated by the major ..... molecule in the large ribosomal subunit.
4. Proteins called release factors bind to ....., codons in the A-site on the ribosome, causing peptidyl transferase to hydrolyze the bond that links the nascent polypeptide to the tRNA molecule.
5. An RNA sequence can be translated in any of three ....., each of which will specify a completely different polypeptide chain.
6. The initiation process of protein synthesis is complicated, involving a number of steps catalyzed by proteins called .....
7. In all cells, a special ..... molecule, recognizing the ..... codon AUG and carrying the amino acid ....., provides the amino acid that begins a protein chain.
8. Most spontaneous changes in DNA are quickly erased by a correction process called .....; only rarely do DNA maintenance procedures fail and allow a permanent sequence change, which is called a .....
9. The genes encoding the ....., which are discarded during blood clotting, accumulate mutations without being selected against.
10. Two common spontaneous changes in DNA are ....., which results from disruption of the N-glycosyl linkages of adenine and guanine to deoxyribose, and ....., which converts cytosine to uracil.
11. DNA repair involves three steps: recognition and removal of the altered portion of the DNA strand by enzymes called ....., resynthesis of the excised region by ....., and sealing the remaining nick by the enzyme .....
12. One major repair pathway, called ....., involves a battery of enzymes called ....., each of which recognizes an altered base in DNA and catalyzed its removal.
13. The ..... pathway in capable of removing almost any type of DNA damage that creates a large change in the DNA double helix.
14. In *E. coli* any block in DNA replication caused by DNA damage produces a signal that induces the ....., which allows replication through the block, hence giving the cell a chance for survival.
15. The enzyme responsible for DNA synthesis in both replication and repair is called .....
16. The active region of a chromosome involved in replication is a Y-shaped structure called a .....
17. The enzyme that seals nicks in the DNA helix during DNA synthesis and repair is called .....
18. During DNA replication, the daughter strand that is synthesized continuously is called the ....., and the strand that is synthesized discontinuously is known as the .....
19. If DNA polymerase adds an incorrect nucleotide to the 3' terminus, a separate catalytic domain containing a 3'-to-5' ..... activity removes the mismatched base.
20. Initiation of DNA synthesis on the lagging strand requires short ..... made by an enzyme called ....., which uses ribonucleotide triphosphates as substrates.

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21. The unwinding of the DNA helix at the replication fork is catalyzed by a ....., which uses the energy from ATP hydrolysis to move unidirectionally along DNA.

22. ....., which aid DNA unwinding, bind to single-stranded DNA in such a way that the bases are still available for templating reactions.

23. The DNA primase molecule is linked directly to the DNA helicase to form a unit called a ....., which moves down the lagging strand at the fork, synthesizing RNA primers as it goes.

24. If DNA polymerase makes a mistake, thus creating a pair of incorrectly hydrogen-bonded bases, the mistake is corrected by a special ..... system that uses methylation to distinguish new strands from old.

25. For bacteria and yeasts, as well as for several viruses that grow in eukaryotic cells, replication bubbles have been shown to form at special DNA sequences called .....

26. ..... can be viewed as reversible nucleases that create either a transient single-strand break (type I) or a transient double-strand break (type II).

27. In ....., genetic exchange occurs between homologous DNA sequences, most commonly between two copies of the same chromosome.

28. At the site of exchange, a strand of one DNA molecule has become base-paired to a strand of the second DNA molecule to create a ..... between the two double helices.

29. Two single-stranded, complementary DNA molecules come together to form a fully double-stranded helix by ....., which is thought to start with a slow ..... step.

30. The ..... is required for chromosome pairing in *E. coli*; it binds to single-stranded DNA and promotes its pairing with homologous, double-stranded DNA.

31. Once synapsis has occurred, the short heteroduplex region where the strands from two different DNA molecules have begun to pair is enlarged through a reaction called .....

32. A central intermediate in general recombination is the ....., which is also called a ...., after its discover.

33. Occasionally during recombination between two slightly different copies of the same gene (alleles), one allele is replaced by the other in a process known as .....

34. Mobile DNA sequences and some viruses enter and leave a target chromosome by ..... genetic recombination.

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Chapter 09

Recombinant DNA technology

Enzymes

Which of the following is *not* true about restriction endonuclease?

a. Restriction enzymes work in presence of Mg^{2+}.
b. Type II restriction endonucleases do not require ATP for restriction activities.
c. It is present in both eukaryotes and prokaryotes.
d. Each restriction enzyme only recognizes the same palindromic sequences regardless of source of DNA.

2. A restriction enzyme *Alu*I with a 4-bp recognition site produces restriction fragments that should have average size of

a. 4096 bp
b. 256 bp
c. 16 bp
d. 436 bp

3. Identify the mismatch

a. Alkaline phosphatase	1. removes phosphate group present at 5' end of DNA
b. DNA polymerase I	2. nick translation
c. S1 endonuclease	3. cleaves only single strand DNA
d. DNase 1	4. cleaves only dsDNA

4. Restriction enzyme digestion of a 4 kb PstI-digested DNA molecule with EcoRI yields two fragments of 1 kb and 3 kb each. Digestion of the same molecule with HindIII yields fragments of 1.5 kb and 2.5 kb. Finally, digestion with EcoRI and HindIII together yields fragments of 0.5 kb, 1 kb and 2.5 kb. Based on this information, the 0.5 kb fragment is flanked by which of the following enzyme sites

a. PstI-HindIII
b. EcoRI-HindIII
c. EcoRI
d. EcoRI-PstI **[JNU]**

5. A linear DNA fragment is (100%) labeled at one end and has 3 restriction sites for *Eco*RI. If it is partially digested by *Eco*RI so that all possible fragments are produced, how many of these fragments will be labeled and how many will not be labeled?

a. 4 labeled; 6 unlabeled
b. 4 labeled; 4 unlabeled
c. 3 labeled; 5 unlabeled
d. 3 labeled; 3 unlabeled **[GATE]**

6. Restriction endonucleases hydrolyzes polynucleotide from

a. only the 5' end
b. either terminal
c. at an internal phosphodiester bond
d. a phosphodiester bond within a specific sequence

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7. Bacteria protect themselves from viruses by fragmenting viral DNA upon entry with
   a. methylase
   b. endonucleases
   c. ligases
   d. exonucleases

8. Which of the following tools of recombinant DNA technology is *incorrectly* paired with its use?
   a. Restriction enzyme–production of RFLPs.
   b. DNA ligase–enzyme that cuts DNA, creating the sticky ends of restriction fragments.
   c. DNA polymerase–used in a polymerase chain reaction to amplify sections of DNA.
   d. Reverse transcriptase–production of cDNA from mRNA.

9. Match List I (Enzyme) with List II (Characteristic/Activity) and select the *correct* answer using the codes given below the Lists

List I (Enzymes)	List II (Characteristic/Activity)
A. Terminal transferases	1. stable above 90°C
B. Polynucleotide kinases	2. cleave the ends of linear DNA
C. Taq DNA polymerases	3. adds phosphate to 5' OH end of DNA or RNA
D. Exonucleases	4. adds a number of nucleotides to 3' end of DNA/RNA
	5. regulate the level of super coiling of DNA molecule

   a. A-5, B-4, C-3, D-2
   b. A-4, B-3, C-1, D-2
   c. A-1, B-4, C-5, D-2
   d. A-2, B-3, C-4, D-1

10. Restriction endonucleases are
   a. used for *in vitro* DNA synthesis.
   b. synthesized by bacteria as part of their defense mechanism.
   c. present in mammalian cells for degradation of DNA when the cell dies.
   d. used in genetic engineering for ligating two DNA molecules.

11. Which of the following sequences along a double-stranded DNA molecule may be recognized as a cutting site for a particular restriction enzyme?
   a. A A G G
      T T C C
   b. A G T C
      T C A G
   c. G G C C
      C C G G
   d. A C C A
      T G G T

12. Restriction-modification systems of bacteria exists to
   a. protect bacteria from invading foreign DNA.
   b. promote conjugation.
   c. help the bacterial chromosome to replicate.
   d. encourage recombination of new genetic material.

13. Which of the following processes require energy?
   a. Ligation
   b. Transformation
   c. Restriction digestion
   d. Hybridization

[GATE]

14. The restriction endonuclease Eco52I recognizes the sequence C/GGCGG and cuts between the first C and the first G, indicated by the slash. DNA cut by which of the following enzymes (given with their recognition sequences and cut sites) could be cloned into a plasmid digested with Eco52I?
   a. *EcoRI* (G/AATTC)
   b. *XmaIII* (C/GGCGG)
   c. *SmaI* (CCC/GGG)
   d. *SacII* (CCGC/GG)

[GATE]

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15. T4 Polynucleotide kinase is used for
a. labeling 3' ends of DNA
b. labeling 5' ends of DNA
c. creating blunt ends of DNA
d. dephosphorylation of DNA

16. You have cut the genome of a double-stranded viral genome with a restriction endonuclease and electrophoresed the products on an agarose gel. You observe only one band on the gel, equivalent to the size of the genome. This is because [GATE]
a. there are no introns in the genome.
b. the introns contain the recognition sites and have already been spliced out.
c. all of restriction fragments are too small to detect.
d. restriction endonucleases do not cut RNA, and this virus has a dsRNA genome. [GATE]

17. A southern transfer of *E. coli* DNA after complete digestion with EcoRI was probed with labeled cDNA probe of a gene which occurs only once in the *E. coli* genome. If the gene contains one EcoRI cleavage site near its center, the number of radioactive bands you are most likely to find on autoradiography would be
a. 0
b. 1
c. 2
d. 3 [GATE]

18. The substrate for restriction enzyme is
a. single stranded RNA
b. partially double stranded RNA
c. cell wall proteins
d. double stranded DNA [GATE]

19. If you discovered a bacterial cell that contained no restriction enzymes, which of the following would you expect to happen?
a. The cell would be unable to replicate its DNA.
b. The cell would create incomplete plasmids.
c. The cell would be easily infected and lysed by bacteriophages.
d. The cell would become an obligate parasite.

20. Mung bean nuclease could be used for
a. DNA synthesis.
b. nucleotide hydrolysis.
c. trimming single stranded regions in DNA.
d. removal of phosphate groups from the ends of the DNA. [GATE]

21. DNA of a bacterium is *not* cleaved by its own restriction enzymes because the recognition DNA sequences are
a. methylated
b. deleted
c. bound by inhibitory proteins
d. not accessible to restriction enzymes [IISc]

22. Two restriction enzymes A and B have four and eight base pairs as their recognition sites respectively. The ratio of the number of fragments that they will generate on restriction digestion of a genomic DNA of *E. coli* is approximately
a. 4 : 8
b. 8 : 4
c. 1 : 64
d. 1 : 256 [GATE]

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23. Restriction endonucleases are enzymes that
P. cleave the 5′ terminal nucleotides from duplex DNA molecules.
Q. make sequence-specific cuts in both strands of duplex DNA molecules.
R. promote circularization of the duplex DNA molecule by removal of the 5′ terminal nucleotides.
S. generate 3′-hydroxyl and 5′-phosphate ends after cutting in DNA strands.
a. P and Q
b. P and R
c. Q and S
d. P, Q, R and S

24. Correct statements regarding the enzyme reverse transcriptase include which of the following?
P. It requires a primer.
Q. It requires an RNA template.
R. It synthesizes an RNA-DNA hybrid molecule.
S. It can be extracted from retroviruses.
a. P and Q
b. P and R
c. Q and R
d. P, Q, R and S

25. If the linear double stranded genomic DNA of bacteriophage T4 is heated and cooled, one gets both linear and circular molecules because,
a. all linear DNA on heating and cooling attain circular shape.
b. the DNA molecules are circularly permuted.
c. the DNA molecule has cohesive terminal ends.
d. the RNA component of the genomes gets hydrolyzed to result in the circularization. [IISc]

26. Terminal transferase is used
a. to add base at the 3′ end of the DNA.
b. to add base at the 5′ end of the DNA.
c. to carry out nick translation.
d. to transfer phosphate at the 3′ end of the DNA. [GATE]

27. Which statement about restriction enzymes are *false*?
a. Restriction enzymes cut DNA at specific sequence called recognition sites.
b. A restriction enzyme always cut DNA to leave the same sequence at the ends.
c. Some restriction enzymes cut the two DNA strands at slightly different points within their recognition site to make a 'sticky' end.
d. Restriction enzymes are exonucleases rather than endonucleases.

28. The DNA of temperate phage P4 is linear, double stranded, 11.5 kb long and has cohesive ends. Digestion with BamHI yields fragments 6.4, 4.1 and 1.0 kb in length. The partial digestion with the same enzyme yields fragments 10.5, 7.4, 6.4, 4.1, and 1.0 kb in length. Circular P4 DNA made with DNA ligase can be digested with BamHI to yield fragments in the DNA. What is the order of fragments in DNA?
a. 6.4- 1.0- 4.1
b. 4.1-1.0
c. 1.0- 6.4- 4.1
d. 4.1- 1.0- 6.4

29. A gene that does not contain introns has an EcoRI site just down stream of the translational termination codon in the 3′ untranslated region. The size of the gene is 4 kbp. The length of 3′ UTR is 2 kbp, the 5′ UTR is 1 kbp and that the open reading frame is 1 kbp. Upon probing a southern blot of an EcoRI digested genomic DNA with the radioactively labeled fragment containing only the open reading frame, the size of the band detected by hybridization would be
a. 1 kbp
b. > 2 kbp
c. 0.5 kbp
d. 1.5 kbp [IISc]

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30. Which of the following sequences is most likely to be a restriction enzyme recognition site?
a. CGGCTT
b. CGCCGC
c. GTAATG
d. GTCGAC
[JNU]

31. In order to clone an eukaryotic gene in pBR322 plasmid vector, the desired DNA fragment was produced by PstI cleavage and incubated with PstI digested pBR322 (PstI cleavage site lies within the ampicillin resistant gene) and ligated. Mixture of ligated cells were used to transform *E. coli* and plasmid containing bacteria were selected by their growth in tetracycline containing medium. Which type of plasmid/s will be found?
a. Circular pBR322 plasmid containing the target gene and resistant to only tetracycline.
b. Circular pBR322 plasmid containing the target gene and resistant to tetracycline only and recircularized pBR322 plasmid resistant to both ampicillin and tetracycline.
c. Circular pBR322 plasmid containing the target gene and resistant to only tetracycline, recircularized pBR322 resistant to both ampicillin and tetracycline and concatemerized pBR322 resistant to both ampicillin and tetracycline.
d. Circular pBR322 plasmid containing the target gene and resistant to both ampicillin and tetracycline. [NET]

Vector

1. To be a cloning vector, a plasmid does *not* require
a. an origin of replication.
b. an antibiotic resistance marker.
c. a restriction site.
d. to have a high copy number.
[GATE]

2. In recombinant DNA methods, the term vector refers to
a. the enzyme that cuts DNA into restriction fragments.
b. the sticky end of a DNA fragment.
c. a plasmid used to transfer DNA into a living cell.
d. a DNA probe used to identify a particular gene.

3. Which of the following is a desirable characteristic for a cloning plasmid?
a. a site at which replication can be initiated.
b. one or more unique restriction endonuclease sites.
c. one or more antibiotic-resistance or drug resistance genes.
d. all of the above.

4. A multiple cloning site
a. contains many copies of a cloned gene.
b. allows flexibility in the choice of restriction enzymes for cloning.
c. allows flexibility in the choice of organism for cloning.
d. contains many copies of the same restriction enzyme site.

5. Many plasmids have ampicillin marker. This implies
a. the plasmids contain genes for ampicillin biosynthesis.
b. ampicillin is required for bacterial growth after transformation.
c. the plasmid contains the gene encoding β-lactamase.
d. ampicillin is essential for cell survival.
[GATE]

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6. BAC, which can be used to clone large DNA fragments, is derived from
   a. ColE plasmid
   b. F plasmid
   c. 2μ plasmid
   d. Mu phage

[GATE]

7. A cloning vector consisting of *cos* site inserted in a plasmid, used to clone DNA fragments of lambda phage is
   a. phagemid
   b. cosmid
   c. plasmid
   d. YAC

8. Which of the following is not component of yeast artificial chromosome?
   a. centromere
   b. telomere
   c. origin of replication
   d. cos site

9. Identify the *incorrect* statement for cosmids.
   a. cosmids are hybrid between a phage DNA and bacterial plasmid
   b. cosmids are basically a plasmid that carries a cos site
   c. cosmids can be used to clone DNA of 30-40 kb
   d. cosmids produce plaques

10. Bacterial artificial chromosomes (BACs), cosmids, phages, plasmids, and yeast artificial chromosomes (YACs) are all commonly used cloning vectors that differ in their cloning capacities, with a range from approximately 100 bp to 3000 kb. Which of the following is the proper order for these vectors in terms of increasing cloning capacity?
   a. BAC, cosmid, phage, plasmid, YAC
   b. YAC, BAC, cosmid, phage, plasmid
   c. plasmid, phage, cosmid, BAC, YAC
   d. plasmid, cosmid, phage, BAC, YAC

11. Choose the mismatch
   a. phagemid — 1. part of M13 genome with plasmid DNA
   b. P1-derived artificial chromosome — 2. combine features of P1 vector and BACs
   c. shuttle vector — 3. yeast episomal plasmids
   d. Ti plasmid — 4. *Agrobacterium rhizogenes*

12. Match List I (cloning vector) with List II (character) and select the *correct* answer using the codes given below the Lists

List I (Cloning vector)	List II (Character)
A. Yeast episomal plasmid (YE_p)	1. integrates with host chromosome for replication
B. Yeast integrative plasmid (YI_p)	2. carries a chromosomal origin of replication
C. Yeast replicative plasmid (YR_p)	3. carries origin of replication of the 2μ plasmid

   a. A-3, B-2, C-1
   b. A-2, B-1, C-3
   c. A-3, B-1, C-2
   d. A-2, B-3, C-1

13. pBR322 which is frequently used as a vector for cloning gene in *E. coli* is
   a. an original bacterial plasmid
   b. a modified bacterial plasmid
   c. a viral genome
   d. a transposon

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14. Phage T7 promoter containing plasmids are used for over expression of cloned genes because of
a. their convenient size
b. their single stranded nature
c. exquisite specificity of T7 RNA polymerase to phage promoters
d. T7 infects *E. coli* and lysogenizes the cell [GATE]

15. Yeast artificial chromosomes (YAC) is used for
a. cloning large segments of DNA
b. cloning only yeast genomic sequences
c. cloning of only cDNA sequences
d. all DNA except plant DNA sequences [GATE]

16. For cloning a DNA fragment larger than 100 kb, which of the following vector system would be suitable?
a. plasmid
b. cosmid
c. yeast artificial chromosome
d. lambda bacteriophage [IISc]

17. Expression vectors contain a sequence, not normally found in other vector that is known as
a. a ribosome-binding site
b. an *ori* site
c. a multiple-cloning site
d. an antibody-resistant marker

18. What is the difference between a PAC and a BAC?
a. One has ampicillin and another has kanamycin resistance marker.
b. One is plasmid-based vector and the other is derived from a yeast chromosome.
c. One is derived from the *E.coli* F-plasmid and other is derived from bacteriophage P1.
d. One is derived from bacteriophage P1 and other is derived from bacteriophage B1. [MKU]

DNA cloning

1. For a plasmid to be a cloning vector, the minimum numbers of elements required are
a. origin of replication, multiple cloning site, selection marker.
b. origin of replication, multiple cloning site, selection marker, promoter.
c. origin of replication, multiple cloning site, selection marker, translation start site.
d. origin of replication, multiple cloning site, promoter. [JNU]

2. What is the most logical sequence of steps for splicing foreign DNA into a plasmid and inserting the plasmid into a bacterium?
P. Transform bacteria with recombinant DNA molecule.
Q. Cut the plasmid DNA using restriction enzymes.
R. Extract plasmid DNA from bacterial cells.
S. Hydrogen-bond the plasmid DNA to nonplasmid DNA fragments.
T. Use ligase to seal plasmid DNA to nonplasmid DNA.
a. P, Q, S, R, T
b. Q, R, T, S, P
c. R, Q, S, T, P
d. R, S, T, P, Q

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3. Eukaryotic genes may not function properly when cloned into bacteria because of
   a. inability to excise introns.
   b. destruction by native endonucleases.
   c. failure of promoter to be recognized by bacterial RNA polymerase.
   d. all of the above.

4. The presence of a plasmid in a bacterial culture is usually determined by
   a. blue-white screening.
   b. growth in the presence of an antibiotic.
   c. a restriction enzyme digests.
   d. agarose gel electrophoresis.

5. Blue-white selection is used
   a. to test for the presence of a plasmid in bacteria.
   b. to reveal the identity of a cloned DNA fragment.
   c. to express the product of a cloned gene.
   d. to test for the presence of a cloned insert in a plasmid.

6. Infection of *E. coli* by bacteriophage λ is normally detected by
   a. resistance of the bacteria to an antibiotic.
   b. the growth of single bacterial colonies on an agar plate.
   c. the appearance of areas of lysed bacteria on an agar plate.
   d. restriction digest of the bacterial DNA.

[GATE]

7. Which of the following statements gives a correct explanation for the use of vectors containing drug resistance genes in the cloning of recombinant DNA (cDNA) molecules?
   a. The products of the drug resistance genes protect the cDNA from destruction by the host cells.
   b. The drug resistance genes provide additional base sequences that enable the vector to accommodate larger inserts of cDNA.
   c. Entry of the vector containing the cDNA and the drug resistance genes into the host cell renders the later identifiable as it is now resistant to antibiotic drugs.
   d. The cloned cDNA imparts drug resistance upon any cellular system with which it is used.

8. What is the approximate length of DNA between two COS sites that can be packaged by a lambda packaging extract?
   a. 10 kb
   b. 15 kb
   c. 25 kb
   d. 45 kb

[MKU]

9. Which statement *correctly* describes sequential steps in cDNA cloning?
   a. Reverse transcription of mRNA, second strand synthesis, cDNA end modification, ligation to vector.
   b. mRNA preparation, cDNA synthesis using reverse transcription, second strand synthesis using terminal transferase, ligation to vector.
   c. mRNA synthesis using RNA polymerase, reverse transcription of mRNA, second strand synthesis, ligation to vector.
   d. Double stranded cDNA synthesis, restriction enzyme digestion, addition of linkers, ligation to vector.

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PCR

1. The PCR is used to
   a. amplify a small amount of DNA
   b. cleave bacteria plasmids
   c. seal sticky ends
   d. identify target plasmids

2. The polymerase enzyme used in PCR is
   a. DNA polymerase I
   b. Taq polymerase
   c. reverse transcriptase
   d. DNA polymerase III

3. The first step in the PCR is
   a. denaturation
   b. primer extension
   c. annealing
   d. cooling

4. Polymerase chain reaction is considered as a revolutionary technology because all of the following, *except*
   a. it enables an unlimited production of a DNA fragment *in vitro*
   b. it is a highly sensitive technology
   c. its experimental protocol is simple
   d. it enables the direct production of a synthetic gene that did not exist before **[JNU]**

5. PCR amplification cycle involves
   P. template denaturation
   Q. primer annealing
   R. DNA polymerization
   S. reaction mixture containing target DNA, primer, thermostable DNA polymerase and dNTP
   a. P, R and S
   b. P, Q and R
   c. Q, R and S
   d. P, Q, R and S

6. The amplification product of a PCR reaction is defined by the sequence of the PCR primers. Several general rules follow to design primers. Select the following rules commonly used to design primers
   P. length of primers.
   Q. T_m of both primers.
   R. G+C content of both primers.
   S. complementarity between the primers.
   a. P and Q
   b. P, Q and R
   c. Q, R and S
   d. P, Q, R and S

7. Choose the *correct* statement(s) about thermostable DNA polymerase used in PCR
   P. *Taq* Pol has no proof reading ability.
   Q. *Taq* Pol has 5′ → 3′ exonuclease activity.
   R. *Pfu* Pol has 3′ → 5′ exonuclease activity.
   S. *Tli* Pol has no 3′ → 5′ exonuclease activity.
   a. P and Q
   b. P, Q and R
   c. Q, R and S
   d. P, Q, R and S

8. Which of the following would be eliminated by *hot start* PCR?
   a. Aerosol contamination from the barrel of pipetors.
   b. Addition of a nucleotide to the terminal end of PCR products.
   c. Infidelity of DNA copying by Taq DNA polymerase.
   d. Formation of primer-dimers. **[MKU]**

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9. The annealing temperature, at which the primers attach to the template, can be calculated by determining the melting temperature (T_m) of the primer template hybrid. What will be T_m of the primer?

5'-AGACTCAGAGAGAACCC-3'

a. 50°C
b. 52°C
c. 102°C
d. 43°C

Statement for linked answer question 10 and 11:

A 200 µl of polymerase chain reaction mixture has 100 template DNA molecules and the reaction was performed for 10 cycles.

10. How many molecules of amplicons will be generated?

a. 1.024×10^4
b. 1.024×10^5
c. 2.048×10^4
d. 2.048×10^5

11. How many molecules of amplicons will be present in 0.1 µl of reaction?

a. 102.4
b. 1024
c. 51.2
d. 512

[GATE]

Protein expression

1. Efficient expression of a heterologous protein product is influenced by

a. transcriptional efficiency
b. copy number of the plasmid
c. codon bias
d. all of the above

[GATE]

2. Which of the following is *not* a potential problem associated with expressing a eukaryotic, protein coding nuclear gene in prokaryotic cells?

a. Lack of an intron-splicing mechanism in prokaryotes.
b. Differences in the translation initiation codons used by eukaryotic cells and prokaryotic cells.
c. Stability of mRNA in prokaryotic cells.
d. Differences in transcriptional signals between eukaryotic cells and prokaryotic cells.

3. Which of the following conditions prevent efficient expression of foreign gene cloned in *E. coli*?

P. The foreign gene might contain introns
Q. Foreign gene might contain sequences that act as termination signals in *E. coli*
R. Codon bias
S. Nature of nitrogenous bases in nucleotides

a. P and Q
b. P and R
c. P, Q and R
d. P, Q, R and S

4. A reporter gene

a. acts as repressor
b. allows gene expression to be readily measured
c. enhances mRNA stability
d. interacts with RNA polymerase

[IISc]

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5. Most common reporter gene whose product can be directly visualized in transformed cells is
   a. neomycin phosphotransferase II (NPTII)
   b. chloramphenicol acetyl transferase (CAT)
   c. β-galactosidase
   d. green fluorescent protein (GFP)

6. Which of the following statement is *false* about reporter genes?
   a. Replace the coding region of a gene of interest with a coding region that is easily assayed.
   b. Replace the promoter region of a gene of interest with a promoter region that is easily assayed.
   c. Can be used to measure the activity of a promoter.
   d. Can be used to determine when and where a promoter is active.

7. Which of the following is *incorrect* about reporter gene?
   a. gene whose phenotype can be assayed in a transformed organism.
   b. β-galactosidase gene is an example of reporter gene.
   c. test gene that is fused to the upstream region of the cloned gene.
   d. none of the above.

8. The principle behind the yeast two-hybrid system is
   a. the detection of protein and DNA interaction from two yeast hybrid strains.
   b. the detection of protein-protein interactions by assembling a functional transcription factor from two fusion proteins.
   c. the detection of protein-protein interactions by studying the hybridization of two cDNA sequences.
   d. the detection of protein-protein interactions in a pair of hybrid yeast strains. **[MKU]**

DNA library

1. Shotgun approach is used for the construction of
   a. cDNA library
   b. genomic library
   c. both a and b
   d. none of the above

2. Which one of the following statements about genomic libraries are *false*?
   a. Genomic libraries are made from cDNA.
   b. Genomic libraries must be representative if they are to contain all the genes in an organism.
   c. Genomic libraries must contain a minimum number of recombinants if they are to contain all the genes in an organism.
   d. The DNA must be fragmented to an appropriate size for the vector that is used. **[GATE]**

3. The term *cDNA library* means
   a. collection of cDNA clones by an individual researcher.
   b. compilation of cDNA sequences in the database.
   c. pool of cDNAs generated from a specific tissue inserted into an appropriate vector that can be used as a source of the cDNA of interest.
   d. it is manual for cDNA research. **[JNU]**

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4. A gene cannot be isolated from a human genomic DNA library by functional complementation in *E. coli* because of
   a. non-functional promoter
   b. the absence of splicing machinery
   c. coupled transcription and translation
   d. codon bias [GATE]

5. What clones should a eukaryotic cDNA library contain?
   a. clones that represent every fragment of DNA in approximately equal frequencies.
   b. clones that represent one copy of every coding region.
   c. clone representing transcribed DNA in approximately equal frequencies.
   d. clones representing transcribed DNA in frequencies that reflect their level of expression.

Engineering plants and animals

1. Identify the *incorrect* statement for the model plant *Arabidopsis thaliana*
   a. its mutants can be easily produced and characterized.
   b. its genome sequence is known.
   c. all genes encompassing its genome have already been identified.
   d. the molecular genetics of its flowering has been extensively studied. [JNU]

2. *Agrobacterium tumefaciens* is often used as vehicle to introduce foreign DNA into plants. The agrobacterial T-DNA in plant cells can be found as
   a. an autonomously replicating nuclear plasmid.
   b. a mitochondrial plasmid.
   c. a chloroplast plasmid.
   d. integrated into the plant genome. [IISc]

3. The following are useful to introduce genes into crop plants except
   a. Ti plasmid
   b. particle gun
   c. breeding
   d. auxin [GATE]

4. Which one of these statements about the applications of gene cloning is *false*?
   a. Large amounts of recombinant protein can be produced by gene cloning.
   b. DNA fingerprinting is used to detect proteins bound to DNA.
   c. Cloned genes can be used to detect carriers of disease-causing genes.
   d. Gene therapy attempts to correct a disorder by delivering a good copy of a gene to a patient.

5.

Group I	Group II
P. biolistic	1. gene Pulser
Q. *Agrobacterium*	2. PDS 1000He
R. electroporation	3. micromanipulator
S. microinjection	4. silicon carbide
	5. *vir* operons 6 *rol* C

   a. P-2, Q-5, R-1, S-3
   b. P-2, Q-1, R-6, S-4
   c. P-5, Q-6, R-2, S-4
   d. P-1, Q-5, R-4, S-2 [GATE]

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6. The length of each boarder sequence in Ti-plasmid is about
   a. 25 million base pairs
   b. 200 kilo base pairs
   c. 25 kilo base pairs
   d. 25 base pairs [GATE]

7. The essential component of Ti plasmid required for integration into plant genome is
   a. origin of replication
   b. tumor inducing gene
   c. nopaline utilization gene
   d. all of the above [GATE]

8. *Agrobacterium tumefaciens* is an effective vector for use with
   a. corn
   b. rice
   c. wheat
   d. soyabean [IISc]

9. What role does opines play in crown gall diseases?
   a. Source of carbon, nitrogen and energy for the *Agrobacterium.*
   b. Transfer of T-DNA to plant cells.
   c. Attachment of a*grobacterium* to the plants.
   d. Induction of the expression of *vir* genes. [IISc]

10. A plant genetic engineer wishes to transfer and express a gene from sunflower into beans. Which of the following would be the vector of choice?
   a. Lambda phage
   b. pBR322 plasmid
   c. Ti plasmid
   d. maize streak virus [GATE]

11. Genetically engineered male sterile crop plants may be produced by inserting
   a. BT toxin gene
   b. barnase gene
   c. lectin gene
   d. chitinase gene [JNU]

12. The term *transfection* refers to which of the following?
   a. Synthesis of mRNA from a DNA template.
   b. Synthesis of protein based on an mRNA sequence.
   c. Introduction of foreign DNA into a cell.
   d. The process by which a normal cell becomes malignant.

13. Which of the following methods for the production of transgenic mice involves the initial creation of a cell chimera (a mixture of cells from distinct sources?)
   a. Pronuclear microinjection
   b. Retroviral transduction
   c. Transfection of ES cells
   d. All the above

14. Which of the following methods for studying loss of gene function does not involve any modification of the genome?
   a. RNA interference by injection of double stranded RNA.
   b. Expression of an integrated antisense transgene.
   c. Gene knockout by homologous recombination.
   d. None of the above.

15. The major difference between transgenic mice and knockout mice is that
   a. transgenic mice always employ the use of cloned genes derived from other species.
   b. transgenic mice have foreign genes that integrate at targeted loci through homologous recombination.
   c. transgenic mice have a functional foreign gene added to their genome.
   d. knockout mice always have a unique phenotype. [MKU]

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16. Match the gene of interest for various aspects of crop improvement

Gene insert	*Aspects of crop improvement*
P. bar	1. Tolerance to heavy metals
Q. *vip3A*	2. Nutritional improvement with increased vitamin A
R. *β-lac*	3. Insect resistance
S. *gsh-II*	4. Herbicide resistance
	5. Delayed ripening
	6. Resistance to fungal infection

a. P-4, Q-3, R-5, S-6
b. P-4, Q-3, R-2, S-1
c. P-2, Q-4, R-5, S-3
d. P-4, Q-2, R-6, S-1 [GATE]

Sequencing/Blotting/Electrophoresis/Labelling

1. Dideoxy DNA sequencing exclusively depends on one of the followings
   a. termination
   b. ATP
   c. plasmid vector
   d. vector primer [JNU]

2. The genome sequencing of rice is important because
   a. the rice genome is very unique and contains genes not found in other plants.
   b. the rice genome is very large compared to the DNA of other grains and so more genes will be identified.
   c. it can identify genes associated with disease resistance, growth capacity, etc.
   d. it has a rapid life cycle and has many identifiable mutations.

3. An analysis of chromosomal DNA, using the Southern blot technique, involves the following five major steps
   1. autoradiography
   2. blotting
   3. cleavage
   4. electrophoresis
   5. hybridization

   Which of the following sequences of steps best illustrates this technique?
   a. 1, 2, 3, 4, 5
   b. 1, 3, 2, 4, 5
   c. 3, 5, 2, 4, 1
   d. 3, 4, 2, 5, 1

4. Which is not a step in the Southern blotting procedure?
   a. Ligation of the DNA into a vector.
   b. Separation of the DNA fragments on a gel.
   c. Transfer of the DNA fragments to a nitrocellulose membrane.
   d. Hybridization of the membrane with a labeled probe.

5. Which of the following types of information *cannot* be determined from the traditional Northern blotting technique?
   a. The size of an mRNA species.
   b. The half-life of an mRNA species.
   c. The strand of DNA that is transcribed into mRNA.
   d. The amino acid sequence of the protein coded by an mRNA species.

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6. Which of the following would *not* be possible to address using a Northern Blot?
   a. Location of restriction sites in a particular gene.
   b. Spatial expression of a particular gene.
   c. Temporal expression of a particular gene.
   d. mRNA size.

[MKU]

7. Choose the mismatch

a. Reporter molecule	1. acts as shuttle vector
b. Maxam-Gilbert method	2. chemical modification of bases
c. Dideoxy terminators	3. Sanger method
d. Biotin	4. non-radioactive label

8. There are two different ways to determining the nucleotide sequence of a nucleic acid: the chemical sequencing (Maxam-Gilbert) method and the enzymatic sequencing (Sanger) method. The basic principle/ advantage of the Sanger method is
   a. the differential interaction of the bases with particular dyes.
   b. extension of a synthetic primer and reliable termination of DNA repair synthesis.
   c. the correlation of restriction sites with the end-label of the DNA.
   d. the ability to 'sequence' both strands of the DNA duplex simultaneously.

9. In agarose gel electrophoresis
   a. DNA migrates towards the negative electrode.
   b. supercoiled plasmids migrate slower than their nicked counterparts.
   c. larger molecules migrate faster than smaller molecules.
   d. ethidium bromide can be used to visualize the DNA.

10. Which of the following are not valid methods of labeling duplex DNA?
    a. 5'-end labeling with polynucleotide kinase.
    b. 3'-end labeling with polynucleotide kinase.
    c. 3'-end labeling with terminal transferase.
    d. nick translation.

11. Which of the following methods would give you the most precise and accurate information about where and when a given gene is expressed?
    a. In *situ* hybridization
    b. DNA microarray
    c. Protein microarray
    d. Reporter gene fusion including introns

[MKU]

12. Which of the following methods can *not* be used to introduce a specific mutation at a predetermined site in a DNA sequence?
    a. Primer extension on a single stranded template using a primer that incorporates a base mismatch.
    b. Primer extension on a single stranded template using an error-prone DNA polymerase.
    c. PCR amplification using a double stranded template with one primer incorporating a base mismatch.
    d. None of the above.

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13. Which of the following is/are useful marker(s) for genetic or physical mapping of human chromosomes?

P. Restriction fragment length polymorphisms.

Q. Expressed sequence tags.

R. Short tandem repeat polymorphisms.

S. Sequence tagged sites.

a. P and Q
b. P, Q and R
c. Q, R and S
d. P, Q, R and S

14. Which of the following type(s) of polymorphism are commonly detected by using the polymerase chain reaction (PCR) and specific oligonucleotide primers?

P. Restriction fragment length polymorphisms.

Q. Expressed sequence tags.

R. Short tandem repeat polymorphisms.

S. Sequence tagged sites.

a. P and Q
b. P, Q and R
c. Q, R and S
d. P, Q, R and S

15. A certain purified DNA sample was cut with two restriction endonucleases E1 and E2. The following results were obtained from agarose gel electrophoresis

Sample cut with E1 alone: two bands of size 35 kb and 15 kb.

Sample cut with E2 alone: two bands of size 40 kb and 10 kb.

Sample cut simultaneously with E1 and E2: three bands of size 35 kb, 10 kb and 5 kb.

From these data, it can be inferred that the DNA has

a. two sites for E1 and one site for E2
b. one site for E1 and two sites for E2
c. one site each for E1 and E2
d. three sites for E1 and one site for E2

Applications of recombinant DNA technology

1. In order to identify the person who committed a crime. Forensic experts will need to extract DNA from the tissue sample collected at the crime scene, and conduct one of the following procedures for DNA finger-printing analysis

a. cut the DNA and hybridize with specific micro-satellite probes.

b. cut the DNA and subclone the fragments.

c. determine the sequence of the subclones.

d. b followed by c.

[GATE]

2. The DNA fingerprinting process involves

a. chain terminators
b. degenerate oligonucleotides
c. VNTR loci
d. RFLPs

3. Restriction fragment length polymorphism (RFLP) is

a. the technique used to *fingerprint* patterns of inheritance.

b. the difference in the restriction maps between the two alleles in a diploid cell.

c. the difference in the restriction maps between two individuals of one species.

d. the difference in the restriction maps between two individuals of two species.

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4. Which of the following *could not* possibly give rise to a restriction fragment length polymorphism (RFLP)?
   a. A missense mutation within the protein coding region of a gene.
   b. A silent mutation within the protein coding region of a gene.
   c. A single base change within the intron sequence of a gene.
   d. An error in RNA splicing that mistakenly removes an exon during RNA processing.

5. RFLP analysis can be used to distinguish between alleles based on differences in
   a. restriction enzyme recognition sites between the alleles.
   b. the amount of DNA amplified from the alleles during PCR.
   c. the ability of the alleles to be replicated in bacterial cells.
   d. the proteins expressed from the alleles.

6. Positional cloning refers to
   a. using a selection procedure to clone a cDNA.
   b. isolating a gene by PCR using primers from another species.
   c. isolating a gene from a specific tissue in which it is being expressed.
   d. mapping a gene to a chromosomal region and then identifying and cloning a genomic copy of the gene from the region.

7. Which one of the following genes is defective in patients suffering from severe combined immunodeficiency syndrome (SCID)?
   a. CFTR
   b. Adenosine deaminase
   c. Ribonucleotide reductase
   d. α_2-microglobulin **[IISc]**

8. A mouse in which one particular gene has been replaced by its inactivated form generated *in vitro* is called
   a. transgenic mouse
   b. knockout mouse
   c. nude mouse
   d. mutant mouse **[JNU]**

9. The principle of the yeast two-hybrid system is
   a. the detection of protein-protein interactions by assembling a functional transcription factor from two fusion proteins.
   b. the detection of protein-protein interactions in a pair of hybrid yeast strains.
   c. the detection of protein-protein interactions by studying the hybridization of two cDNA sequences.
   d. the detection of protein-protein interactions between phage coat protein and target proteins.

10. Which of the following is *not* one of the objectives of the Human Genome Project?
    P. Create a detailed genetic map of every human chromosome, with an average of 2–5% recombination frequency between markers.
    Q. Obtain a detailed physical map of every human chromosome, based on overlapping recombinant DNA molecules cloned as yeast artificial chromosomes.
    R. Determine the sequence of all expressed human genes by cDNA cloning and sequencing.
    S. Determine the complete DNA sequence of each human chromosome.
    a. P and Q
    b. P, Q and R
    c. Q, R and S
    d. P, Q, R and S

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Fill in the blanks in the following statements

1. In ....., a DNA fragment that contains a gene of interest is inserted into the purified DNA genome of a self-replicating genetic element—generally a virus or a plasmid.
2. The ..... used for gene cloning are small circular molecules of double-stranded DNA derived from larger plasmids that occur naturally in bacterial cells.
3. A plasmid that contains one fragment from cleaved genomic DNA is said to contain a .....; an extensive collection of such plasmids is said to constitute a .....
4. A clone that contains a DNA copy of an mRNA is called a ....., and the entire collection of clones derived from one mRNA preparation constitutes a .....
5. .....can be used to enrich a particular nucleotide sequence prior to cDNA cloning by hybridizing cDNA molecules from a cell type that makes the protein of interest with a large excess of mRNA molecules from a closely related cell type that does not make it.
6. Once one genetically mapped gene has been cloned, the clones in a genomic DNA library that correspond to neighboring genes can be identified using a technique called .....
7. The technique called ..... allows the DNA from a selected region of a genome to be amplified a billionfold, provided that at least part of its nucleotide sequence is already known.
8. ..... can provide a powerful solution to the problem of defining the function of an unknown gene through the specific abnormalities in a mutant organism in which the gene is defective.
9. Recombinant DNA technology has made it possible to start with a particular gene, make mutations in it, and then create mutant cells or organisms, a process commonly referred to as .....
10. Regions of a protein with special functions can be identified by making ..... with an easily reporter protein and then following the behavior of the reporter protein in a cell.
11. Fusing a protein to a short peptide of 8 to 12 amino acids that can be recognized by an antibody is called .....
12. A ..... mutation is one in which a mutant gene eliminates the activity of its normal counterpart in the cell.
13. If a cloned gene is engineered so that the complementary DNA strand is transcribed instead of the usual strand, it will produce ..... molecules that have a sequence complementary to the normal RNA transcripts.
14. Animals that have been permanently altered so that they pass foreign genes on to their progeny are called ..... organisms, and the foreign genes are called .....

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Chapter 10

Classical Genetics

Mendel's principle

1. Mendel's law of segregation, as applied to the behavior of chromosomes in meiosis, means that
   a. pairing of homologs will convert one allele into the other, leading to separation of the types.
   b. alleles of a gene separate from each other when homologs separate in meiosis I.
   c. genes on the same chromosome will show 50% recombination.
   d. alleles of a gene will be linked and passed on together through meiosis.

2. The phenomenon of independent assortment refers to
   a. expression at the same stage of development.
   b. unlinked transmission of genes in crosses resulting from being located on different chromosomes.
   c. association of an RNA and a protein implying related function.
   d. independent location of genes from each other in an interphase cell.

3. One of Mendel's pure strains of pea plants had green peas. How many different kinds of eggs could such a plant produce with regard to pea color?
   a. One
   b. Two
   c. Three
   d. Four

4. List all the different gametes produced by the following individuals
   1. AA BB Cc
   2. aa Bb Cc
   a. 1. ABC, ABc ; 2. aBC, aBc, abC, abc
   b. 1. AbC, ABc ; 2. ABC, aBc, abC, aBC
   c. 1. ABC, ABC ; 2. aBC, aBc, abc, aBC
   d. 1. ABC, ABc ; 2. aBC, ABc, abc, abC

5. If a diploid cell contains six chromosomes, how many possible random arrangements of homologous could occur during Metaphase-I?
   a. 4
   b. 8
   c. 6
   d. 64

6. Determine the frequency of only one genotype, BBLl, in the offspring of dihybrid parents (BbLl).
   a. 1/4
   b. 1/8
   c. 1/16
   d. 1/2

7. If an individual of genotypes AaBbCcDd is test crossed, how many different phenotypes can appear in the progeny?
   a. 16
   b. 8
   c. 32
   d. 64

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8. If individuals of genotype AaBbCc are intercrossed, how many different phenotypes can appear in their offspring?
   a. 16 b. 8
   c. 32 d. 64

9. If individuals of genotype AaBbCc are intercrossed, how many different genotypes can occur in their progeny?
   a. 8 b. 64
   c. 27 d. 9

10. If individuals of genotype AaBbCc are intercrossed, how many different F2 phenotypes can appear assuming complete codominance at all loci?
   a. 8 b. 64
   c. 27 d. 9

11. A man is heterozygous *B/b* for one autosomal gene and he carries a recessive X-linked allele *d*. What proportion of his sperm will be *bd*?
   a. 1/4 b. 1/8
   c. 1/16 d. 1/2

12. A trihybrid cross is made between two yeast, both with genotype *A/a*, *B/b*, *C/c*. What proportion of the offspring will be of genotype *a/a*, *b/b*, *c/c*?
   a. 1/8 b. 1/32
   c. 1/64 d. 1/74

13. When heterozygous black pigs are intercrossed then what is the chance of the first two offspring being black?
   a. 46 b. 56
   c. 66 d. 76

14. Medical geneticists usually abbreviate the normal β-globin gene as b, and the abnormal gene (in this case) as b^{o}. Neither of your patient's parents has β-thalassemia. Which of these described the most likely genotypes of both parents?
   a. One is b^{o} and one is b b. One is $b^{o}b^{o}$ and one is bb
   c. Both are $b^{o}b^{o}$ d. Both are bb^{o}

[JNU]

15. If a man of blood group *AB* marries a woman of blood group *A* whose father was of blood group *O*, to what different blood groups can this man and woman expect their children to belong?
   a. A, AB, B b. A, AB
   c. AB, O d. A, O, B

16. In the human ABO blood system, the alleles A and B are dominant to O. What will be the number of different possible genotype?
   a. 4 b. 8
   c. 6 d. 12

17. A brown mouse is mated with two female black mice. When each female has produced several litters of young, the first female has had 48 black and the second female has had 14 black and 11 brown young. Deduce the pattern of inheritance of coat color and the genotypes of all the parents (B—Black; b—brown).
   a. male Bb; female 1, BB; female 2 Bb
   b. male BB; female 1, BB; female 2 Bb
   c. male Bb; female 1, BB; female 2 bb
   d. none of the above

[JNU]

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18. Three genes (P, Q and R) are found of three different chromosomes. For the following diploid genotypes, describe all of the possible gamete combinations and their predicted ratios.

P. AABbCC

Q. AaBBCc

R. Aabbcc

a. P. 1ABC: 1AbC Q. 1ABC : 1ABc : 1aBC : 1aBc R. 1Abc : 1abc

b. P. 1ABC: 1Abc Q. 1abc : 1Abc : 1aBC : 1aBc R. 1abc : 1abc

c. P. 1AbC: 1Abc Q. 1ABC : 1aBC : 1aBc R. 1Abc : 1abc

d. P. 1ABC: 1Abc Q. 1ABC : 1Abc : 1aBC R. 1Abc : 1Abc

19. A true-breeding pea plant with round and green seeds is crossed to a true-breeding plant with wrinkled and yellow seeds. Round and yellow seeds are the dominant traits. The F1 plants are allowed to self-fertilize. What are the following probabilities for the F2 generation?

P. An F2 plant with wrinkled, yellow seeds

Q. An F2 plant with round, yellow seeds

a. P. 2/16 Q. 9/16

b. P. 3/16 Q. 7/16

c. P. 3/16 Q. 9/16

d. cannot be determined from the information

20. In Guinea pig, black coat color is a dominant trait and white is recessive trait. A black female is test crossed, producing six black offspring. The probability that a heterozygous black would do this by chance alone is approximately

a. 50%

b. 25%

c. 1%

d. cannot be determined from the information **[JNU]**

21. What are the minimum progeny population sizes allowing for random union of all kinds of gametes from AaBbCc parent?

a. 64

b. 32

c. 16

d. 8

22. A red-flowered tall parent plant (P1) was crossed to a true breeding red-flowered dwarf plant (P2) and half of the progenies obtained was red and tall and the other half red and dwarf. In the next generation, half of all these progenies segregated only for flower color and the other half segregated only for height. The genotype of the P1 is

a. heterozygous for color and heterozygous for height.

b. homozygous for color and heterozygous for height.

c. heterozygous for color and homozygous for height.

d. homozygous for color and homozygous for height. **[IISc]**

23. According to classical genetics, which of the following statements is *true*?

a. Recessive alleles are detected by the phenotype of the F1 generation.

b. The closer two genes are, the more frequently they recombine.

c. Genes on different autosomes segregate independently.

d. Gene on sex chromosomes segregate with the same pattern as autosomal genes.

24. Assuming the comparable chromosomes in different individuals are genetically dissimilar because of different alleles. How many unique zygotic combinations are possible following fertilization in an organism where n = 3 (Assuming that no crossing over occurs)?

a. 8

b. 16

c. 64

d. 216

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25. In pea plants, the allele for green color of seeds (*G*) is dominant over that for yellow color (*g*) while the allele for round seeds (*R*) is dominant over that for wrinkled seed (*r*). The results of an experimental cross with such garden pea plants are tabulated below:

Speed phenotype	*Number*
Green and round	32
Green and wrinkled	28
Yellow and round	12
Yellow and wrinkled	9

The parental genotypes are most likely to be:

a. *GgRr* and *Ggrr*
b. *Ggrr* and *GgRR*
c. *GgRr* and *GgRr*
d. *GgRR* and *ggRr*

26. Coat colors of dogs depend upon the action of at least two genes. At one locus a dominant epistatic inhibitor of coat color pigment (A^-) prevents the expression of color alleles at another independently assorting locus, producing white coat color. When the recessive condition exists at the inhibitor locus (aa), the alleles of the hypostatic locus may be expressed, aaB^- producing black and aabb producing brown. When dihybrid white dogs are mated together, determine the phenotypic proportions expected in the progeny.

a. 9 : 7
b. 9 : 3 : 4
c. 12 : 3 : 1
d. 13 : 3

27. Short hair in rabbits is governed by a dominant gene L and long hair by its recessive allele l. Black hair results from the action of the dominant genotype B- and brown from the recessive genotype bb.

P. In crosses between dihybrid short-haired, black and homozygous short-haired, brown rabbits, genotypic ratio will be 1 : 1 : 1 : 1.
Q. In crosses between dihybrid short-haired, black and homozygous short-haired, brown rabbits, phenotypic ratio will be 1 : 1.
R. Expected phenotypic ratio from the cross LlBb × Llbb will be 3 : 3 : 1 : 1.
S. Expected genotypic ratio of LlBb will be 1/4.

a. P and Q
b. P, R and S
c. R and S
d. P, Q, R and S

28. In a cross between two individuals with the genotypes AaBbccDdEeFf and AaBbCCDDeeff, the probability that an offspring will be heterozygous at all these loci is

a. 0
b. 1/16
c. 1/32
d. 1/64 [JNU]

29. Two pink-flowered four-o'clocks are crossed to each other. Flower color is incompletely dominant and giving phenotypic ratio of 1 red : 2 pink : 1 white. What are the following probabilities?

P. The first three plants with while flower.
Q. A plant with either white *or* pink flower.

a. P. 1/64 Q. 3/4
b. P. 3/16 Q. 1/64
c. P. 3/64 Q. 9/16
d. P. 1/64 Q. 1/64

30. Which of the following is a mismatch?

a. Phenotypic F2 ratio of 1 : 4 : 6 : 4 : 1 — polygenic inheritance.
b. Phenotypic F2 ratio of 9 : 3 : 4 — recessive epistasis.
c. Phenotypic F2 ratio of 1 : 2 : 1 — codominance.
d. Phenotypic F2 ratio of 3 : 1 — partial dominance.

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Linkage and Mapping

1. If AABB is crossed to aabb and the F1 is test crossed. What % of the test cross progeny will be aabb if the two genes are
   P. Unlinked?
   Q. Completely linked (no crossing over at all)?
   R. 10 map units apart?
   S. 24 map units apart?
   a. P – 0%, Q – 0%, R – 38%, S – 45%
   b. P – 25%, Q – 50%, R – 45%, S – 38%
   c. P – 0%, Q – 25%, R – 45%, S – 38%
   d. P – 25%, Q – 0%, R – 38%, S – 45%

2. Two genes are located on the same chromosome and are known to be 12 map units apart. An *AABB* individual is crossed to an *aabb* individual to produce *AaBb* offspring. The *AaBb* offspring are then crossed to *aabb* individuals. If this cross produces 1000 offspring, what are the predicted numbers of offspring with each of the four genotypes: *AaBb*, *Aabb*, *aaBb* and *aabb*?
   a. 60 *Aabb*, 440 *aaBb*, 440 *AaBb* and 60 *aabb*.
   b. 0 *Aabb*, 0 *aaBb*, 440 *AaBb* and 440 *aabb*.
   c. 440 *Aabb*, 440 *aaBb*, 60 *AaBb* and 60 *aabb*.
   d. 60 *Aabb*, 60 *aaBb*, 440 *AaBb* and 440 *aabb*.

3. Two different traits affecting pod characteristics in garden pea plants are encoded by genes found on chromosome 5. Narrow pod is recessive to normal pod; yellow pod is recessive to green pod. A true-breeding plant with narrow, green pods was crossed to a true-breeding plant with normal, yellow pods. The F1 offspring were then test crossed to plants with narrow, yellow pods. The following results were obtained 144 narrow green pods, 150 normal yellow pods, 11 narrow yellow pods, 9 normal green pods. How far apart are these two genes?
   a. 1 mu
   b. 12 mu
   c. 6.4 mu
   d. 32.4 mu

4. D, F, and J are three genes in *Drosophila*. The recombination frequencies for two of the three genes are shown in

Gene pair	*Recombination frequency*
D–F	50%
D–J	25%
F–J	?

   1. Genes D and F could be located on different chromosomes.
   2. Genes D and F located very near to each other on the same chromosome.
   3. Genes D and F located far from each other on the same chromosome.
   4. Frequency of crossing over between any two linked genes will be higher if they are recessive.
   5. Frequency of crossing over between any two linked genes dependent on how many alleles there are.
   6. Frequency of crossing over between any two linked genes dependent on the distance between them.

   Which of the following statements are *correct*?
   a. 1, 3 and 4
   b. 1, 2, 4 and 5
   c. 1, 3 and 6
   d. 3, 5 and 6

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5. Consider the following autosomal *Drosophila*, traits caused by recessive alleles: bent wings (bn^-), short legs (sh^-) and orange eyes (or^-). You cross two true breeding lines to produce F1 flies, all of which have the wild type phenotype - strait wings (bn^+), long legs (sh^+) and red eyes (or^+). F1 females are then mated to triply homozygous males with bent wings, short legs, and orange eyes. Among 100 progeny from this cross you observe the following phenotypes:

Phenotype	*Number*
Strait wings, long legs and red eyes	10
Bent wings, short legs and orange eyes	14
Strait wings, short legs and red eyes	26
Bent wings, long legs and orange eyes	30
Strait wings, long legs and orange eyes	8
Bent wings, short legs and red eyes	6
Strait wings, short legs and orange eyes	2
Bent wings, long legs and red eyes	4

Which of the following statements are *correct* with respect to above data?

P. The smallest class of F2 progeny represents the double-crossover class.

Q. The order of genes are *sh – or – bn*.

R. The distance between *sh* and *or* is 30 cM.

S. The distance between *sh* and *or* is 20 cM.

a. Q and S
b. P and S
c. P, Q and R
d. P, Q and S

Sex determination and Sex-linked inheritance

1. Why would you predict that half of the human babies born will be males and half will be females?
   a. Because of the segregation of the X and Y chromosomes during male meiosis.
   b. Because of the segregation of the X chromosomes during female meiosis.
   c. Because all eggs contain an X chromosome.
   d. Because, on average, one-half of all eggs produce females.

2. Hemophilia is a recessive X-linked trait in humans. If a heterozygous woman has children with a normal man, what are the odds of the following combinations of children?

   P. An unaffected daughter or son

   Q. Two out of five offspring that are affected

   a. P = 1/4 and Q = 52.6%
   b. P = 3/4 and Q = 26.3%
   c. P = 3/4 and Q = 50%
   d. P = 81/256 and Q = 26.3%

3. Which of the following is *not* true about autosomal dominant traits?
   a. Every affected person should have at least one affected parent.
   b. Males and females should be often equally affected.
   c. An affected person has a 50% chance of transmitting the dominant allele to each offspring.
   d. All the daughters of an affected male will be affected but none of the sons.

4. Albinism is a recessive human trait. If a normal couple produces an albino child, what is the probability that their next child will be albino?
   a. 1/4
   b. 1/8
   c. 1/16
   d. 1/64

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5. When a man with hypertrichosis marries a normal woman, what percentage of their sons would be expected to have hairy ears?

a. 50%
b. 100%
c. 0%
d. 25%

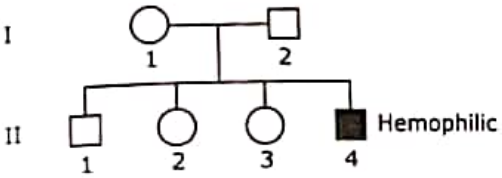
6. Human blood types are genetically determined. The table below shows the symbols used to represent two of the alleles for blood types and gives a description of each allele.

Two alleles controlling human blood type

Symbol	*Allele description*
I^A	produces antigen A on red blood cells
I^B	produces antigen B on red blood cells

In homozygous individuals, two I^A alleles result in blood type A and two I^B alleles result in blood type B. The I^A and I^B alleles are codominant, resulting in blood type AB in individuals heterozygous for the two alleles. A male and a female both have blood type AB. If they have a child, what is the probability that the child will also have blood type AB?

a. 1/4
b. 1
c. 3/4
d. 1/2

7. 

Hemophilia is a recessive sex-linked disease. From the information in the pedigree

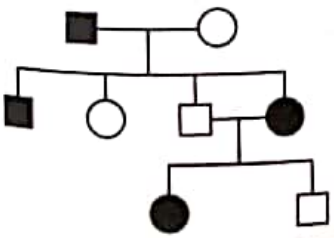
P. I1 must be a carrier.
Q. II3 has a 50% chance of being a carrier of hemophilia (Hh).
R. If II3 marries a hemophilic man, 25% of their child are expected to be hemophilic.
S. If II3 marries a hemophilic man, the probability that her first child will be normal is 75%.

a. P and Q
b. P, Q and R
c. R and S
d. P, Q, R and S

8. Following are four modes of inheritance

P. X-linked recessive
Q. X-linked dominant
R. Autosomal recessive
S. Autosomal dominant

Which of the above modes of inheritance can explain the pedigree shown below?



a. P and R
b. Q and R
c. Only S
d. R and S

[NET]

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9. Cystic fibrosis is a hereditary disease that affects the respiratory and digestive systems. Cystic fibrosis occurs when two recessive genes (cc) are present. A person with one allele for cystic fibrosis is called a carrier (Cc) of the disease. If the mother is a carrier of the disease and the father is homozygous dominant, what are chances that their child will be a carrier of cystic fibrosis?

a. 25%
b. 50%
c. 75%
d. 100%

[MKU]

10. Which of the following is/are *correct* about sex-linked recessive inheritance?

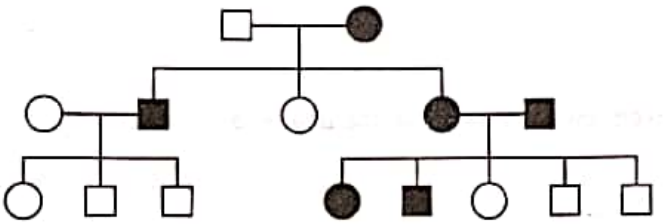
P. Most affected individuals are male.
Q. Affected females come from affected fathers and affected or carrier mothers.
R. The sons of affected females should be affected.
S. The trait does not skip generations.

a. P and S
b. P, Q and R
c. Q, R and S
d. P, Q, R and S

11. A woman who is heterozygous for both phenylketonuria mutation and for X-linked hemophilia mutation has a child with a phenotypically normal man who is also heterozygous for a phenylketonuria mutation. What is the probability that the child will be affected by both?

a. 1/8
b. 1/4
c. 1/16
d. 3/8

12.

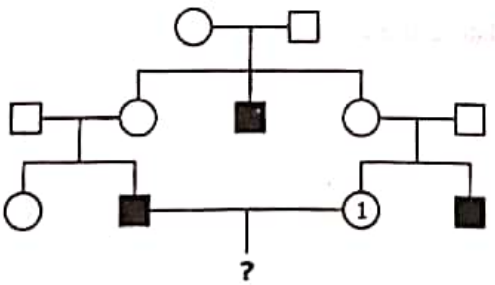


Which of the statements below are *true* regarding the trait of interest in the above pedigree?

P. If it is autosomal recessive, all the children of the affected daughter would be affected as well.
Q. If it is X-linked dominant, the daughter of the affected son would also be affected.
R. If it is X-linked recessive, all the children of the affected daughter would be affected.
S. It is likely autosomal dominant, the affected individuals would be heterozygous.

a. P and Q
b. P, Q and R
c. Q, R and S
d. P, Q, R and S

13. The pedigree shows inheritance of an X-linked recessive trait. (Assume the trait is completely penetrant).



Which of the following statements are *correct* with respect to above pedigree?

P. The probability that the female designated 1 is a carrier for the trait is 1/2.
Q. If the child indicated by **?** is a boy, the probability he will be affected by the trait is 1/2.

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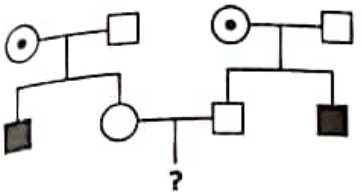
R. If the child indicated by ? is a girl, the probability she will be affected by the trait is 1/4.
S. If the child indicated by ? is a boy, the probability he will be affected by the trait is 1/4.

a. Q and S
b. P and S
c. Q and R
d. P, R and S

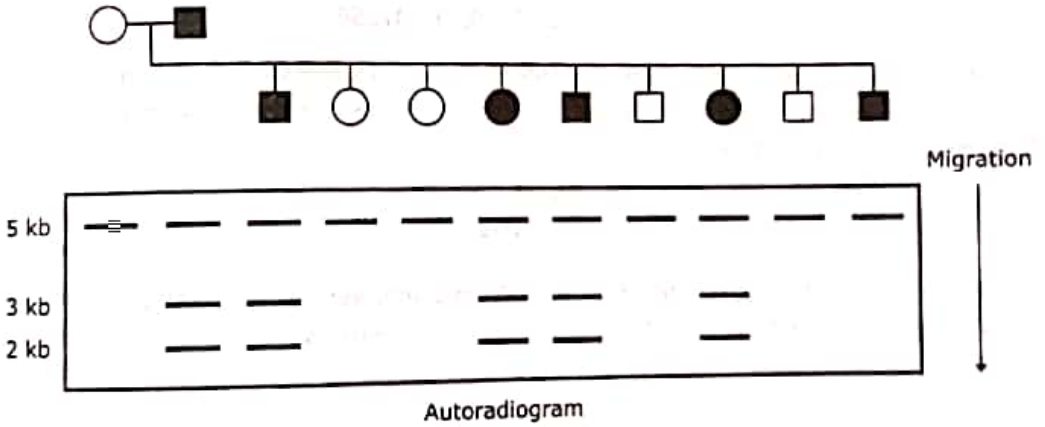
14. Consider the following pedigree showing inheritance of an X-linked recessive trait. (Assume the trait is completely penetrant and no new mutations arise in the individuals shown).



What is the probability that a child indicated by ? wll be affected by the trait?

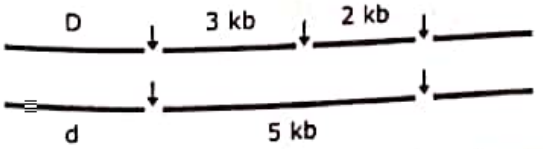
a. 1/2
b. 1/4
c. 1/8
d. 1/64

15. DNA studies are performed on a large family that shows a certain *autosomal dominant disease*. A DNA sample from each family member is digested with the restriction enzyme *Taq*I and subjected to gel electrophoresis. A Southern blot is then performed, with the use of a radioactive probe consisting of a part of human DNA cloned in a bacterial plasmid. The autoradiogram, aligned with the family pedigree, is as follows. Affected members are shown in black.



Autoradiogram

P. The probe detects three bands of 5, 3 and 2 kb.
Q. Five of the six affected persons have the 3 and 2 kb bands.
R. A likely arrangement in the father is as follows



(where D is the disease allele and an arrowhead represents the restriction-enzyme target site)
S. The last son most likely represents a crossover between the disease locus and the RFLP locus, resulting in a chromosome with D and the 5 kb fragment.

Which of the above statements are *correct*?

a. P and Q
b. P, R and S
c. Q, R and S
d. P, Q, R and S

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Quantitative inheritance

1. Which of the following is/are not feature(s) of quantitative trait?

   P. Characters of degree

   Q. Continuous variation

   R. Polygenic control

   S. Discontinuous variation

   a. P and S
   b. Q and R
   c. only S
   d. R and S

2. Which of the following are *correct*?

   P. Classical Mendelian traits are qualitative in nature.

   Q. Qualitative traits show discontinuous variations.

   R. Qualitative traits are polygenic traits.

   S. Qualitative traits are referred to as metric traits.

   a. P and Q
   b. P and S
   c. Q and R
   d. R and S

3. Wheat grain color is controlled by four loci and following quantitative inheritance.

   P. Calculate the number of different F2 phenotypes in a certain stock of wheat.

   Q. Calculate the number of F2 as extreme as one parent or the other.

   a. P – 9, Q – 1/16
   b. P – 16, Q – 1/256
   c. P – 8, Q – 1/32
   d. P – 9, Q – 1/256

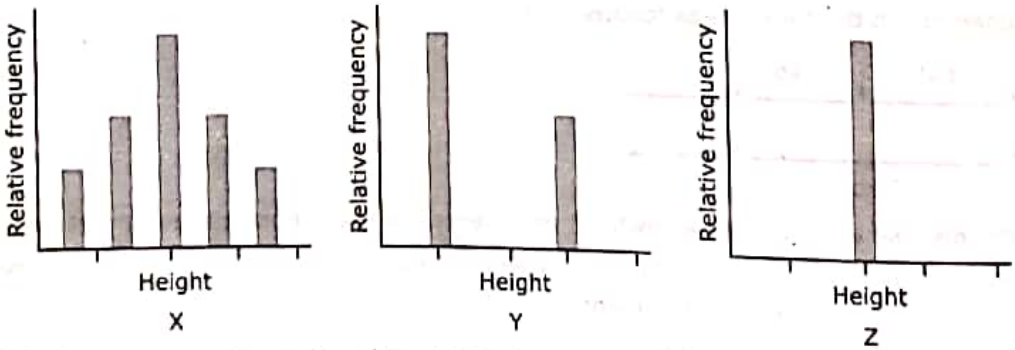
4. Two pure lines of corn have mean cob lengths of 9 and 3 inches, respectively. Two non-allelic genes are involved in this trait. All exhibit additive gene action. Crossing these two lines is expected to produce a progeny with mean cob length (in inches) of

   a. 12.0
   b. 7.5
   c. 6.0
   d. 2.75

5. In a plant, height varies from 6 to 36 cm. When 6 cm and 36 cm plants were crossed all F1 plants were 21 cm. In the F2 generation, a continuous range of heights was observed. Most were around 21 cm, and 3 of 200 were 6 cm. Find out how many gene pairs are involved in this mode of inheritance?

   a. 3
   b. 4
   c. 2
   d. 5

6. The phenotypes of three experimental populations of plants are shown in the following graphs.



   The three populations X, Y and Z represent, respectively, the:

   a. F1, F2 and F3 generations
   b. P, F2 and F2 generations
   c. F2, P and F1 generations
   d. F3, F1 and F2 generations

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Cytogenetics

1. If four chromosomes synapse into a cross-shaped configuration during meiotic prophase, the organism is heterozygous for a
   a. pericentric inversion
   b. deletion
   c. translocation
   d. paracentric inversion [JNU]

2. Philadelphia chromosome is generated by translocation between
   a. chromosome 18 and chromosome 6
   b. chromosome 22 and chromosome 9
   c. chromosome 22 and chromosome 3
   d. chromosome 16 and chromosome 4 [JNU]

3. Retinoblastoma is caused by loss of both copies of the RB gene in the chromosome band
   a. $13p11$
   b. $13q11$
   c. $13q14$
   d. $21q14$ [JNU]

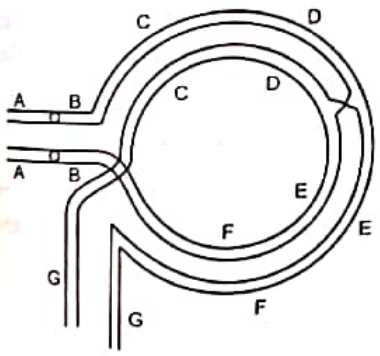
4. Position effect is the result of
   a. mutations
   b. deletions
   c. inversions
   d. transversions

5. The bridge-fragment configuration at anaphase-1 is characteristic of
   a. translocation heterozygote.
   b. paracentric inversion heterozygote.
   c. pericentric inversion heterozygote.
   d. duplication heterozygote.

6. A mechanism that can cause a gene to move from one linkage group to another is
   a. translocation
   b. inversion
   c. crossing over
   d. duplication

7. Pseudodominance may be observed in heterozygotes for
   a. a deletion
   b. a duplication
   c. a reciprocal translocation
   d. more than one of the above

8. Following is the diagram of a paracentric inversion heterozygote ABCDEFG/ABFEDCG involved in recombination during meiosis I:



The consequence of this recombination will be the formation of

P. a dicentric and an acentric chromosome in meiosis I as the chiasmata gets terminated.
Q. no dicentric or acentric chromosome but appearance of deletion and duplication in both the chromosomes
R. all non-viable gametes.
S. non-viable gametes from crossover products.

Which of the above statements are *correct*?

a. P and Q
b. P and R
c. P and S
d. Q and R [NET]

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9. In a trisomic individual the number of chromosomes is
   a. $2n - 1$
   b. $2n + 2$
   c. $2n + 3$
   d. $2n + 1$

10. A person with Klinefelter syndrome is considered a
   a. monosomic
   b. triploid
   c. trisomic
   d. deletion heterozygote

11. If the garden pea has 14 chromosomes in its diploid complement, how many double trisomics could theoretically exist?
   a. 6
   b. 9
   c. 16
   d. 21 [GATE]

12. The condition in which there is one too many or one too few chromosomes is called
   a. aneuploidy
   b. polytene
   c. polyploidy
   d. monoploidy

13. Most cases of Down's syndrome are caused by the presence of a third copy of chromosome 21 associated with the chromosome 21 pair. This genetic condition, known as trisomy 21, is caused by
   a. a frame-shift mutation
   b. chromosome nondisjunction
   c. fragile X syndrome
   d. chromosome translocation [MKU]

14. Which statements about Down's syndrome are *correct*?
   P. The frequency increases dramatically in mothers over the age of 40.
   Q. The cause is a non-disjunction when chromosomes do not separate during the first meiotic division.
   R. Affected individuals have an extra autosome.
   S. The long time lag between onset of meiosis in ovarian tissue and its completion (at ovulation) is most likely the reason for increased incidence in older mothers.
   a. P and S
   b. P, Q and R
   c. Q, R and S
   d. P, Q, R and S

15. People with Klinefelter syndrome have 47 chromosomes, including three sex chromosome (XXY). What is the term to describe the aberration that occurs during meiosis that results in abnormal chromosome number?
   a. Crossing over
   b. Non-disjunction
   c. Independent assortment
   d. Pairing of homologous chromosome [IISc]

16. The genes *abcde* are determined to be closely linked on the *E. coli* chromosome. Three random short deletions are created in the region, resulting in the removal of various genes are shown below. Which of the following gene orders is possible based on the deletion analysis?
   Deletion 1: Genes bde are lost
   Deletion 2: Genes ac are lost
   Deletion 3: Genes abd are lost
   a. *abcde*
   b. *acbed*
   c. *bdeac*
   d. *cabde*

17. *Datura* plants have been regenerated from anther cultures, endosperm culture and embryo culture. Their respective ploidy levels will be
   a. n, 2n and 2n
   b. n, 3n and 2n
   c. n, 2n and 3n
   d. 2n, 2n and 2n [IISc]

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Chapter 11

Prokaryotes and Viruses

Bacterial cell structure

Which statement is *true* of both prokaryotic and eukaryotic cells?

a. Prokaryotic cells are generally much larger than eukaryotic cells.
b. Eukaryotic cells have ribosomes and prokaryotic cells do not.
c. Both have DNA as their primary genetic material.
d. Eukaryotic cells have plasma membranes and prokaryotic cells do not.

2. Acid fast staining of gram positive mycobacterium is due to

a. presence of dipicolinic acid in cell wall.
b. presence of mycolic acid in cell wall.
c. presence of teichoic acid in cell wall.
d. presence of diaminopimelic acid in cell wall.

3. Which of the following statement is *correct*?

a. Acid-fast staining is due to presence of mycolic acid.
b. Action of lysozyme in gram negative bacteria results in formation of protoplast.
c. Gram staining is due to chemical nature of cell wall.
d. All of the above.

4. The cell walls of gram positive bacteria contain two modified sugars, N-acetylglucosamine (NAG) and N-acetylmuramic acid (NAM). They are covalently linked by

a. α-1,4-glycosidic bond
b. β-1,6-glycosidic bond
c. α-1,6-glycosidic bond
d. β-1,4-glycosidic bond

[GATE]

5. Which of the following is essential for living cell?

a. Flagella
b. Capsule
c. Cell wall
d. Cytoplasmic membrane

6. Which of the following is *not* true about capsules and slime layers?

a. They consist of secreted material lying outside of the bacterial cell wall.
b. They can prevent desiccation of bacteria cells.
c. They are required for bacteria to grow normally in culture.
d. They help bacteria resist phagocytosis by macrophages.

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7. Periplasm is
   a. found in both gram-negative and gram-positive bacteria.
   b. found to contain specific bacterial lipids.
   c. space between two plasma membrane.
   d. an artifact from microscopic aberrations.

8. Which is *true* for bacteria?
   a. Bacteria lack DNA.
   b. Mitochondria and nuclear membrane is present.
   c. Nucleoid is the region which contains DNA.
   d. RNA acts as genetic material.

9. Pili represent
   a. extra chromosomal genetic elements.
   b. protoplasmic outgrowths of donor cells.
   c. small flagella.
   d. special bacterial cilia.

10. Streaming of protoplasm is absent in
    a. parenchyma and collenchyma cells.
    b. bacterial cells and vessels.
    c. cells of higher plants.
    d. cells of hydrilla.

11. Teichoic acid present in the cell wall of gram positive bacteria makes a covalent bond with
    a. N-acetyl muramic acid
    b. N-acetyl glucosamine
    c. cell membrane lipid
    d. both a and c

12. Encapsulated bacteria
    a. are sometimes more virulent than their non-encapsulated counterparts.
    b. are more susceptible to phagocytes destruction.
    c. have more signal for protein synthesis than non-encapsulated bacteria.
    d. both a and c are correct.

13. Bacterial flagella impart motility to the cell by
    a. undulating movement
    b. rotatory movement
    c. gliding movement
    d. both a and b

14. Which of the following is a fact that argues against a symbiotic origin of eukaryotic flagella from bacterial flagella?
    a. Bacterial flagella typically contain flagellin; eukaryotic flagella contain tubulin.
    b. Bacterial flagella rotate; eukaryotic flagella do not rotate.
    c. Bacterial flagella are directly energized by electrons and protons from the respiratory system of the plasma membrane; eukaryotes are not.
    d. More than one of the above argue against the symbiotic theory.

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Bacillus anthracis capsules consist of which one of the following?

a. γ-glutamyl polypeptide
b. Hyaluronic acid
c. Repeating polysaccharide capsule of glucose and glucuronic acid
d. Sialic acid polymers

Bacterial growth and Cultivation

The best definition of the generation time of a bacterial population is the

a. time it takes for a bacterial population to double.
b. time it takes for the lag phase.
c. length of the exponential phase.
d. maximum rate of doubling.

Bacteria broth cultures eventually stop growing and enter stationary phase because they

a. deplete essential nutrients
b. accumulate toxic product
c. become too crowded
d. both a and b

3. A bacterial culture had an initial cell density of 10^3 cells/ml. In 6 hours the cell density reached 10^6 cells/ml. Given the formula for the number of generations,

$n = (\log_{10} N_t - \log_{10} N_0)/0.301$

The number of generations (n) the cells have undergone is

a. 3 b. 10
c. 15 d. 20 **[GATE]**

4. A bacterium, having doubling time of 10 minutes, fills a cylindrical vessel completely in 3 hours. How much time will it take to fill half of the vessel?

a. 80 minutes b. 90 minutes
c. 150 minutes d. 170 minutes **[GATE]**

5. A bacterial culture has an initial cell density of 0.5×10^3 cells/ml. Its generation time is 20 min. The cell density at the end of 1hr 40 min is

a. 1.5×10^3 cells/ml b. 16×10^3 cells/ml
c. 1.6×10^3 cells/ml d. 160×10^3 cells/ml

6. A bacterial strain with a doubling time of 4 hours is mixed in equal proportion with another strain with a doubling time of 9 hours, and the growth is monitored during the log phase which lasts for about three days. The observed doubling time will

a. decrease with time b. increase with time
c. remain constant at about 6.5 hours d. fluctuate between 4 and 9 hours **[IISc]**

Chemolithotrophs are those bacteria which can utilize

a. inorganic material as the energy source.
b. light as the energy source.
c. organic compound as the electron source.
d. crude oil as carbon source. **[JNU]**

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8. Bacteria that must have organic molecules both for energy and as a source of carbon are called
   a. photoautotrophs
   b. photoheterotrophs
   c. chemoheterotrophs
   d. chemoautotrophs

9. Which of the following is an example of chemolithoautotroph?
   a. sulphur-oxidizing bacteria
   b. hydrogen bacteria
   c. nitrifying bacteria
   d. all of these

10. The group of organisms which uses light as the energy source and CO_2 as the principal carbon source is called
    a. photoheterotrophs
    b. chemoautotrophs
    c. chemoheterotrophs
    d. photoautotrophs

[GATE]

11. Chemoautotrophic metabolism involves
    P. the use of inorganic carbon dioxide as the sole source of carbon.
    Q. sunlight as the source of energy.
    R. the use of organic carbon compounds as the source of carbon.
    S. energy derived from the oxidation of inorganic or organic molecules.
    a. P and Q
    b. P and S
    c. P, Q and R
    d. R and S

12. Match the column

P. Psychrophile	1. Can grow at 0–7°C, optimum growth temperature between 20°C and 30°C
Q. Psychrotroph	2. Grow well at 0°C
R. Mesophile	3. Optimum growth around 20–45°C
S. Halophile	4. Require right salt concentration

    a. P-2, Q-1, R-3, S-4
    b. P-1, Q-2, R-4, S-3
    c. P-1, Q-2, R-3, S-4
    d. P-2, Q-1, R-4, S-3

13. Which of the following statement(s) is/are *correct*?
    P. Mannitol salt agar is an example of selective media used to select out one species of microorganism.
    Q. Those bacteria that live at pH level from 5 to 8.5 are known as neutrophiles.
    R. *Mycobacterium leprae* can be cultivated on solid media containing agar.
    a. P and Q
    b. P, Q and R
    c. Q and R
    d. R only

14. Choose the correct pair
    a. Tryptic soy agar — Supportive media
    b. Blood agar — enriched media
    c. MacConkey agar — selective as well as differential media
    d. Beef extract — defined media

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15. In the penicillin-enrichment method, auxotrophic mutants are isolated based on the following principle
a. both the auxotroph and prototroph grow but at different rates in the minimal medium.
b. only the prototroph grows and gets killed while the auxotroph remains unaffected.
c. while the prototroph grows readily, auxotroph grows with a lag phase.
d. only the auxotroph growth is supported by minimal medium. [IISc]

16. What properties is/are *not* expected to be significantly different between hot-spring bacteria (that live at 120 degrees Celsius) and regular intestinal *E. coli*?
P. Number of cysteines in the proteins.
Q. Number of methionines in the protein.
R. Molecular weight of the protein.
S. GC richness of the genomic DNA.
T. Richness of saturated fatty acids in plasma membranes.
a. P and T
b. Q and R
c. R and S
d. P and R

17. Import of vitamin B_{12} into Gram-negative bacterial cells is facilitated by a special class of proteins in the outer membrane known as
a. integrins
b. adhesins
c. tubulins
d. porins [IISc]

18. The Entner-Doudoroff pathway degrades one molecule of glucose to two pyruvate molecules and yields
a. one ATP and one NADH
b. one ATP, one NADH and one NADPH
c. one ATP and two NADH
d. two ATP and two NADH

19. Which of the following differences between BGA and purple photosynthetic bacteria is *not* correct?

Properties	*BGA*	*Purple photosynthetic bacteria*
a. Photosynthetic pigment	chl *a*	Bchl
b. O_2 production	oxygenic	anoxygenic
c. Carbon source	CO_2	organic or CO_2
d. Assimilatory power	NADPH + ATP	NADPH + ATP

Bacterial groups

1. Currently bacterial phylogeny is based on
a. GC content analysis
b. DNA-DNA hybridization analysis
c. 16S rRNA analysis
d. DNA melting temperature analysis [JNU]

2. The group of bacteria that don't have cell walls are the
a. archaebacteria
b. mycobacteria
c. mycoplasma
d. both a and b are correct

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3. Which of the following is *not* true for mycoplasma?
   a. It is pleomorphic.
   b. Its membrane contains sterol.
   c. It is penicillin resistant.
   d. It is non-self reproducing.

4. All the members of the genus *Bacillus* are known for
   a. their ability to break down sulfur containing compounds.
   b. their ability to form spores.
   c. their ability to live in the absence of oxygen.
   d. the capsules they possess.

5. All the following characteristics apply to spirochetes *except*
   a. they form spores.
   b. they do not stain easily in the laboratory.
   c. certain species cause human disease.
   d. they move freely by moving axial filaments.

6. Mycoplasmas are different from other prokaryotes by
   a. presence of chitin in cell walls.
   b. presence of murein in cell walls.
   c. presence of proteins in cell walls.
   d. absence of cell wall itself.

7. The spirochete responsible for syphilis is [GATE]
   a. *Borrelia*
   b. *Spirocheta*
   c. *Leptospira*
   d. *Treponema*

8. Which of the following statement is *false*? [IISc]
   a. Chlamydias are obligate intracellular parasite.
   b. Rickettsias are sensitive to antibiotics.
   c. Chlamydias are not sensitive to interferon.
   d. Chlamydias have no ribosomes.

9. The chalamydiae are distingulshed by all the following characteristics *except*
   a. they are tiny bacteria.
   b. their developmental cycle includes an elementary body and a reticulate body.
   c. they cause Rocky Mountain spotted fever and typhus.
   d. they multiply only in living cells.

10. A highly aerobic and metabolically versatile organism used in oil-spill-clearing is
   a. *Mycobacterium smegmatis*
   b. *Azotobacter vinelandii*
   c. *Pseudomonas cepacia*
   d. *Leuconostoc mesenteroides*

[GATE]

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All the following apply to members of the genus *Pseudomonas except*
a. they are gram-positive cocci.
b. they excrete extracellular pigments that diffuse into the local environment.
c. they cause blue-green rot in eggs.
d. they use nitrate instead of oxygen as a final electron acceptor.

12. Actinomycetes
a. are gram negative bacteria
b. bear spores
c. are intracellular obligate parasite
d. lack peptidoglycan cell wall

13. Clostridia are anaerobes and they form ATP by 'stickland' reaction. For this they use
a. palmitic acid
b. pyruvic acid
c. ethanol
d. amino acids [JNU]

14. Identify the *correct* statement.
a. Cyanobacteria are the highest evolved algae.
b. Dominant pigment of blue green algae is haemoerythrin.
c. Sexual reproduction in cyanobacteria is isogamous.
d. No spindle formation occurs in nostoc cell at the time of division.

15. Heterocyst of cyanobacteria
a. is specialized for oxygenic photosynthesis
b. forms spore
c. is specialized for gamete formation
d. is specialized for N_2 fixation

Archaebacteria

1. Which one of the following sequences has helped in identifying eukaryotes, eubacteria and archaebacterial cell types?
a. signature sequence
b. signal sequence
c. Shine-Dalgarno sequence [GATE]
d. amino acid sequence

2. Which of the following statement is *not* true? In archaebacteria
a. cell wall is not made up of peptidoglycan
b. cell membrane have branched chain hydrocarbons
c. cell wall have both D- and L- form of amino acids
d. first amino acid to initiate polypeptide chain is methionine

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3. Archaeal cells usually do not contain peptidoglycan, rather contain pseudopeptidoglycan which is mainly composed of
   a. N-acetylmuramic acid and L-amino acids.
   b. N-acetyltalosaminuronic acid and D-amino acids.
   c. N-acetylmuramic acid and D-amino acids.
   d. N-acetyltalosaminuronic acid and L-amino acids. [GATE]

4. Cell membrane of archeobacteria differs from eubacteria due to presence of
   a. ether linkage instead of ester linkage between fatty acid/aliphatic chain and glycerol.
   b. branched hydrocarbon chain instead of unbranched hydrocarbon chain.
   c. ribitol instead of glycerol in phospholipid.
   d. both a and b.

5. In archaebacteria
   a. T or TψC arm of tRNA is without thymine
   b. *N*-formylmethionine is carried by the initiator tRNA
   c. cell wall contains D-amino acid
   d. cell wall is sensitive to lysozyme [IISc]

6. Choose the *correct* statement(s).
   P. Chlorophyll based photosynthesis is absent in archaea.
   Q. Initiator tRNA in eubacteria carry *N*-formyl-methionine.
   R. Methanogenic bacteria only belong to archaea.
   S. DNA dependent RNA polymerase in eubacteria is rifampicin sensitive.
   a. P and S
   b. Q, R and S
   c. Q and S
   d. P, Q, R and S

Bacterial genome

1. Which of the following statement is *correct*?
   a. Bacteria chromosome has many origin of replication.
   b. Bacterial chromosome is present mostly in positively supercoiled structure.
   c. Bacterial chromosome is a circular ssDNA.
   d. Bacterial chromosome has no centromere and telomere.

2. A plasmid is
   a. a fragment of a chromosome that is transferred during bacterial conjugation.
   b. a piece of DNA carrying a multiple cloning site.
   c. a circular dsDNA which can replicate independently.
   d. a piece of DNA carrying gene essential for bacteria.

3. Which of the following statements regarding plasmid is *incorrect*?
   a. They are mostly circular dsDNA, with multiple copies of pseudogenes.
   b. A bacterial cell can harbor more than one type of plasmids.
   c. Replication of plasmid may be unidirectional or bidirectional.
   d. θ-replication in plasmid is more common form of replication then rolling circle replication.

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4. Which of the following statement is *incorrect*?
   a. Plasmid is present in all bacteria.
   b. F-factor is an example of conjugative plasmid.
   c. Plasmids are also present in some eukaryotic organisms.
   d. Plasmid of unknown function is known as cryptic plasmid.

5. The copy number of a plasmid is the
   a. number of copies of the plasmid per cell immediately after cell division.
   b. number of plasmids that take part in conjugation.
   c. number of plasmids containing *OriV*.
   d. number of identical genes on plasmid.

6. F' plasmids
   a. are responsible for high frequency recombination.
   b. carry some chromosomal genes.
   c. are those plasmids that have never been incorporated into a bacterial chromosome.
   d. none of these.

7. Plasmids do which of the following?
   a. Direct synthesis of conjugation pili.
   b. Provide resistance to certain antibiotics.
   c. Induce the formation of tumors in plants.
   d. All of these are functions of plasmids.

Gene transfer and recombination

1. A bacterial cell that is mostly haploid, but is diploid for some regions of the genome is called .....
   a. heterodiploid
   b. pseudodiploid
   c. half diploid
   d. merodiploid

   [MKU]

2. Which of the following processes would not contribute to genetic variation within a bacterial population?
   a. Transduction
   b. Transformation
   c. Mutation
   d. Meiosis

3. During conjugation
   a. cell to cell contact is required
   b. naked DNA is transferred
   c. bacteriophage mediates DNA transfer
   d. only plasmid is transferred from donor to recipient

4. Which one of the following statements is the most likely explanation for the rapid spread of drug resistant bacterial strain?
   a. Drug induced mutation that produces resistant strain.
   b. Genetic variability that results from increased recombination of homologous chromosome during meiosis.
   c. Plasmid mediated exchange of resistance genes.
   d. Gene conversion that results in evolution of resistance genes.

   [IISc]

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5. *E. coli* cells containing F plasmid are capable of carrying out the following important process
   a. transduction process for transferring genes.
   b. induce mutations in the chromosome.
   c. conjugation, promoting horizontal gene transfer.
   d. promote increased replication and cell division. [IISc]

6. If two populations of bacteria cultured in a U-shaped tube are separated by a membrane filter (which does not allow phage particles to pass), but recombination takes place anyway, the mechanism of genetic exchange is
   a. specialized transduction
   b. site-specific recombination
   c. conjugation
   d. transformation

7. In an $F^+ \times F^-$ cross
   a. the F^+ cell becomes an Hfr cell.
   b. the F^- cell becomes an Hfr cell.
   c. the F^+ cell becomes an F^- cell.
   d. the F^- cell becomes an F^+ cell. [IISc]

8. When cells conjugate for longer periods of time, which of the following will occur?
   a. Fewer genes will be transferred.
   b. More genes will be transferred.
   c. The same number of genes will be transferred.
   d. It is impossible to tell because conjugation time and number of genes transferred are not related.

9. High frequency recombination cells arise when
   a. they have multiple F plasmids.
   b. F plasmid has been incorporated into the bacterial chromosome.
   c. conjugation and transformation happen at the same time.
   d. bacterial chromosomes do not break during conjugation.

10. Which of the following statements is *incorrect*?
    a. During conjugation of an F^+ and F^- *E. coli*, a copy of the F factor is almost always transferred to the recipient.
    b. Conjugation between Hfr strain and F^- strain leaves the recipient strain F^-.
    c. Conjugation between Hfr strain and F^- strains results in high frequency of recombination and therefore the recipient become Hfr or F^+.
    d. Conjugation between F^+ and F^- strains results in the recipient becoming F^+. [GATE]

11. During the transformation in *S. pneumoniae,* the following events happen
    P. binding of DNA fragment to cell surface receptor.
    Q. incorporation of DNA strand into bacterial chromosome.
    R. degradation of one strand.
    S. repair of hetroduplex.
    The *correct* order of these events are
    a. P, R, Q, S
    b. P, Q, R, S
    c. P, S, Q, R
    d. R, P, Q, S

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12. Bacterial conjugation can be used to map genes on a bacterial chromosome. Which of the following would be the best donor strain to use if we wish to map the bio gene?

a. Hfr: strS, bio$^-$, ade$^-$, pro$^-$
b. Hfr: strS, bio$^+$, ade$^+$, pro$^+$
c. Hfr: strR, bio$^-$, ade$^-$, pro$^-$
d. Hfr: strR, bio$^+$, ade$^+$, pro$^+$

[IISc]

13. Generalized transduction

a. is carried out by temperate phage only.
b. is an example of vertical gene transfer.
c. results in formation of merozygote.
d. transfers specific gene(s) from one bacteria to another.

14. Gal$^-$ mutants of *E. coli* cannot use galactose as a carbon source. When these mutants are infected with phage λ that has been grown on wild-type *E. coli*, however, the progeny bacteria can use galactose. This genome alteration is an example of bacterial

a. transformation
b. transduction
c. conjugation
d. transposition

15. Presence of maltose in the medium increases transduction frequency by lambda phage because maltose

a. provides the lambda phage with energy required for transduction.
b. induces the maltose receptor.
c. binds to the lambda phage surface protein, making it competent to infect.
d. keeps the bacterial cells dividing, which is required for infection by lambda phage. [IISc]

16. Transformation in *S. pneumoniae*

P. involves binding with dsDNA fragment.
Q. is a random process which is virus mediated.
R. is a natural process.
S. involves cleavage of one strand of DNA by exonuclease.

a. P and R
b. R and S
c. P, R and S
d. Q, R and S

17. Four Hfr strains of *E. coli* were generated from the same F$^+$ strain. The Hfr strains donated markers in the following order

Strain 1 : DQWMT
Strain 2 : AXPTM
Strain 3 : BNCAX
Strain 4 : BDQWM

The order of the markers in the original F$^+$ strain is

a. DQWMTPXACNB
b. AXPTMDQWBNC
c. BNCAXPTMDQW
d. BDQWMNCAXPT

[GATE]

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Toxins

1. Endotoxins are
   1. found in gram positive cells
   2. lipopolysaccharides
   3. found in gram negative cells
   4. lipoproteins
   5. found in mycoplasma

   a. 1, 2 and 5 b. 1 and 2
   c. 2 and 3 d. 3, 4 and 5

2. Which of the following is/are correct? Exotoxins are
   P. chemically proteins
   Q. highly toxic
   R. heat labile
   S. produced by only gram negative bacteria

   a. P and R b. Q and R
   c. P, Q and R d. P, Q, R and S

3. Match the correct combination of toxin and the mode of action

Toxin	*Mode of action*
P. Pertussis toxin	1. Prevents release of glycine by nerve end
Q. Diphtheria toxin	2. Blocks G-protein signal transduction
R. Botulinum toxin	3. Induces fluid loss from intestinal cells
S. Tetanus toxin	4. Inhibits protein synthesis in eukaryotes
	5. Causes haemolysis
	6. Blocks release of acetylcholine by nerve end

   a. P-3, Q-2, R-5, S-4 b. P-2, Q-4, R-6, S-1
   c. P-5, Q-6, R-3,S-2 d. P-1, Q-3, R-2, S-5

4. The pathogenesis associated with *Vibrio cholerae* infection depends on the colonization of the small intestine by the organism and secretion of an enterotoxin. Which of the following statement related to pathogenicity of cholera is *incorrect*?
   a. *Vibrio cholerae* secretes the enterotoxin choleragen.
   b. Choleragen consists of A (active) and B (binding) subunit.
   c. Choleragen toxin alone cannot reproduce the symptoms of cholera in the absence of *Vibrio cholerae*.
   d. CTX bacteriophage carries the genes of cholera toxin and other virulence factors. [JNU]

5. Match the following

Name of Bacteria	*Mode of action*
1. *Corynebacterium diphtheriae*	P. blocks release of acetylcholine
2. *Clostridium tetani*	Q. binds to class II MHC protein
3. *Clostridium botulinum*	R. inactivates EF-2 by ADP ribosylation
4. *Staphylococcus aureus*	S. blocks release of the inhibitory neurotransmitter glycine

   a. 1-R, 2-P, 3-Q, 4-S b. 1-P, 2-Q, 3-R, 4-S
   c. 1-S, 2-R, 3-P, 4-Q d. 1-R, 2-S, 3-P, 4-Q

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Each of the following statements concerning exotoxins is correct *except*

a. exotoxins are polypeptides.
b. exotoxins are more easily inactivated by heat than are endotoxins.
c. exotoxins are less toxic than the same amount of endotoxins.
d. exotoxins can be converted to toxoids by using formaldehyde.

7. Choose the *incorrect* combination

Toxin	*Gene location*	*Enzyme activity*
a. Botulinum toxin	prophage	cleavage of SNARE
b. Diphtheria toxin	prophage	ADP ribosylation of eEF_2
c. Tetanus toxin	plasmid	ADP ribosylation of G_s
d. Shiga toxin	chromosomal	inhibition of protein synthesis

8. Which of the following group of characteristics is *correct* with respect to *Shigella* species found as food pathogen?

a. Gram positive, motile by gliding, spore forming cocci and transmitted by contaminated food.
b. Gram negative, motile by flagella, spore forming bacilli and transmitted by contaminated water.
c. Gram positive, non-motile, non-spore forming cocci and transmitted by contaminated air and water both.
d. Gram negative, non-motile, non-spore forming and transmitted by fecal-oral route. **[GATE]**

9. Which of the statement about *Corynebacterium diphtheriae* is *incorrect*?

a. All strains of C. *diphtheria* are producers of diphtheria toxin.
b. Diphtheria toxin production can be minimized by high concentration of iron in the medium.
c. Diphtheria toxin inhibits protein synthesis.
d. Diphtheria toxin is an A-B toxin secreted as a polypeptide of 62 kDa. **[GATE]**

Virus

1. Viruses

a. are obligate intracellular parasites.
b. divide by binary fission.
c. have their own metabolism.
d. contain enzymes for formation of 16S rRNA.

2. Choose the *correct* statements. Viruses

P. are sensitive to antiviral agents.
Q. may produce cytopathic changes in cell culture.
R. can be inactivated by use of formaldehyde, UV light and phenol.
S. may contain double stranded DNA with single strand break.

a. P and S
b. P, Q and R
c. Q, R and S
d. P and Q

3. Viruses are obligate intracellular parasites. Each of the following statements concerning this fact is correct *except*

a. viruses cannot generate energy outside of cells.
b. viruses cannot synthesize proteins outside of cells.

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c. viruses must degrade host cell DNA in order to obtain nucleotides.
d. enveloped viruses require host cell membranes to obtain their envelops.

4. Which of the following statement is *not* true?
   a. Virus contains DNA or RNA.
   b. The nucleic acid of a virus is surrounded by a protein coat.
   c. Viruses multiply inside living cells using viral mRNA, tRNA and ribosomes.
   d. Viruses multiply inside living cells.

5. Some viruses are surrounded by a membranous envelope which originate from the
   a. plasma membrane of the host cell.
   b. synthesis of proteins and lipids inside the virus.
   c. synthesis of proteins in the virus and lipids in the host cell.
   d. synthesis of lipids in the virus and proteins in the host cell.

6. An RNA-dependent RNA polymerase is likely to be present in the virion of a
   a. DNA virus that multiplies in the cytoplasm.
   b. DNA virus that multiplies in the nucleus.
   c. minus-strand RNA virus.
   d. transforming virus.

7. An envelope is acquired during which of the following steps?
   a. Penetration
   b. Release
   c. Lysis
   d. Assembly

8. A virion is a
   a. naked, infectious piece of RNA.
   b. complete, infectious virus particle.
   c. nucleic acid without a capsid.
   d. a naked, infectious piece of DNA.

9. Which of the following statements about spikes present on virus is *true*?
   a. They contain lysozyme for rupturing the host cell.
   b. They function in recognition and attachment to host cells.
   c. They are glycoproteins.
   d. Both b and c are correct.

10. A virus with RNA-dependent RNA polymerase
    a. synthesizes DNA from an RNA template.
    b. synthesizes dsRNA from an RNA template.
    c. synthesizes dsRNA from a DNA template.
    d. transcribes mRNA from DNA.

11. Which of the following characteristics or processes is common to both bacteria and viruses?
    a. Nucleic acid as the genetic material
    b. Binary fission

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c. Ribosomes in the cytoplasm
d. Conjugation

12. RNA as a genetic material is found in which of the following organisms?
a. *Plasmodium*
b. *Staphylococcus aureus*
c. *Schizosaccharomyces cerevisiae*
d. Polio virus [IISc]

13. Which one of the following viruses replicate in the cytoplasm?
a. SV40
b. Adenovirus
c. Vaccinia virus
d. Herpex simplex virus [IISc]

14. The bacteriophage M13 contains as its genetic material
a. single stranded RNA
b. double stranded RNA
c. single stranded DNA
d. double stranded DNA [IISc]

15. Bacteriophage MS2
a. contains dsDNA as genetic material
b. contains ssRNA and infects only F^+ *E. coli*
c. contains minus(-) stranded RNA
d. does not require pili for infection to *E. coli*

16. The genome of cauliflower mosaic virus is
a. positive-stranded RNA
b. single-stranded DNA
c. double-stranded DNA
d. double-stranded RNA [GATE]

17. Match the following

Name of virus	*Type of genome*
1. Polyoma virus	P. plus ssRNA
2. Polio virus	Q. linear dsDNA
3. Cauliflower mosaic virus	R. circular dsDNA
4. Influenza virus	S. dsRNA
5. Wound tumor virus	T. minus ssRNA

a. 1-R, 2-P, 3-Q, 4-T, 5-S
b. 1-R, 2-T, 3-P, 4-S, 5-Q
c. 1-P, 2-R, 3-T, 4-S, 5-Q
d. 1-P, 2-S, 3-T, 4-S, 5-Q

18. Which of the following would be the first step in the biosynthesis of a virus with reverse transcriptase?
a. A complementary strand of RNA must be synthesized.
b. Double-stranded RNA must be synthesized.
c. A complementary strand of RNA must be synthesized from a RNA template.
d. A complementary strand of DNA must be synthesized from a RNA template.

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19. Which of the following statement is *false*?
    a. Most bacteria in nature are lysogens.
    b. Lysogenic phage contains dsDNA.
    c. Biosynthesis of DNA containing animal virus occur in the nucleus.
    d. TMV is minus ssRNA virus.

20. If one λ phage infects a bacterium lysogenic for λ which one of the following will probably happen?
    a. A normal phage cycle will commence producing 50-100 phage particles.
    b. The λ DNA will circularize but will not replicate.
    c. The bacterium will die.
    d. The λ DNA will not be injected.

21. Which of the following is *false* statement? [IISc]
    P. Retroviruses convert their RNA genomes into DNA that is longer than the RNA genome.
    Q. Double-stranded RNA viruses carry an RNA-dependent RNA polymerase known as a replicase.
    R. Replication of DNA of poxvirus occurs in the nucleus.
    S. Retrovirus genome contains *gag, pol* and *env* genes.
    a. P
    b. Q
    c. R
    d. S

22. Hydroxymethyl cytosine is found in the DNA of
    a. simian virus 40
    b. phage T4
    c. phage QB
    d. adenovirus

23. Retroviruses have RNA genome, however they replicate through double stranded DNA formation. This process involves [IISc]
    a. a polymerase coded by the virus itself
    b. a polymerase coded by the host
    c. host DNA polymerase
    d. unknown mechanism

24. Choose the *incorrect* statement. Retrovirus genome consists of [JNU]
    a. two molecules of ssRNA
    b. plus (sense) RNA having 5'-cap and 3'-poly (A) tail
    c. *gag, pol* and *env* genes
    d. dsRNA, non-covalently bonded to one another

25. Match the HIV region in Column B that specifies the protein in Column A. Choose the mismatch.

Column A	Column B
a. capsid	1. gag
b. protease	2. *rev*
c. reverse transcriptase	3. *pol*
d. envelope glycoprotein	4. *env*

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26. Which of the following statement about HIV is *incorrect*?
a. HIV is an enveloped retro virus.
b. HIV infects T-cells that carry the CD4 antigen on their surface.
c. HIV contains two copies of plus single stranded RNA genome.
d. HIV-1 contains only three genes namely *gag*, *pol* and *env*.

27. HIV, a retrovirus contains
a. two copies of negative sense ssRNA
b. one ssRNA
c. two copies of dsRNA
d. two copies of positive sense ssRNA

28. When an animal virus becomes integrated into the host's chromosome, it is then called
a. prophage
b. temperate virus
c. provirus
d. coli phage

29. Which of the following is *false* statement?
a. Retroviruses convert their RNA genomes into DNA that is longer than the RNA genome.
b. The gag region of retroviruses specifies only one polypeptide.
c. The human T-cell lymphotropic viruses (HTLV-I, -II) are in the subfamily of oncoviruses.
d. A number of DNA viruses (like SV40) carry oncogenes, but these oncogenes are not related to genes found in normal cells.

30. Which of the following is *not* related to HIV?
a. HIV belongs to lentiviridae subform.
b. gp 120 is the membrane protein that HIV uses to attach to host cells.
c. Long terminal repeat contains the promoter for HIV transcription.
d. HIV genome contains one copy of each gene.

31. The ability of a virus to produce a disease can result from a variety of mechanisms. Which of the following is *not* a likely option?
a. cytopathic effect in infected cells.
b. malignant transformation of infected cells.
c. secretion of exotoxin that activates adenylyl cyclase.
d. immune response to virus-induced antigens on the surface of infected cells. **[IISc]**

32. Match the viruses in group I will their host cell receptors in group II.

Group I	*Group II*
P. Hepatitis A virus	1. Heparan sulphate
Q. HIV	2. Acetylcholine receptor
R. Rabies virus	3. CD4 protein
S. Herpes simplex virus type I	4. α-2 macroglobulin

a. P-1, Q-3, R-2, S-4
b. P-3, Q-4, R-1, S-2
c. P-4, Q-3, R-2, S-1
d. P-2, Q-3, R-1 S-4 **[GATE]**

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Viroids and Prions

1. Infectious single stranded RNAs in plants that are not associated with any protein are called
   a. viruses
   b. viroids
   c. prions
   d. satellite viruses

2. Which of the following is *not* true of a viroid? [MKU]
   a. Viroids affect mostly plants.
   b. Its RNA is circular.
   c. Viroids possess about 300–400 nucleotides.
   d. Its capsid is quite small.

3. Virusoid is
   P. a satellite nucleic acid
   Q. a non-coding RNA
   R. a ribozyme
   S. a viroid with a capsid
   a. P and Q
   b. Q and R
   c. P, Q and R
   d. P, Q, R and S

4. Which of the following properties are *true* for prions?
   P. PrP^{Sc} is protease-resistant protein.
   Q. Gene for PrP^{C} normally express in the brain.
   R. PrP^{C} is a GPI-linked membrane protein.
   S. PrP^{Sc} form has a high content of beta sheets as compared to PrP^{C}.
   a. P and Q
   b. Q and R
   c. P, Q and R
   d. P, Q, R and S

5. Which of the following statement is *true*?
   a. A knocked out mouse for PrP gene becomes resistant to infection by PrP^{Sc}.
   b. A knocked out mouse for PrP gene becomes sensitive to infection by PrP^{Sc}.
   c. A mouse with no PrP^{C} develops neurodegenerative disorder.
   d. A PrP^{Sc} protein can infect an animal that has the different type of endogenous PrP^{C} protein.

Antibiotics

1. Which of the following statements are *true*?
   P. Chloramphenicol inhibits protein synthesis on ribosomes in mitochondria.
   Q. Puromycin is a structural analogue of a tRNA molecule linked to an amino acid.
   R. Erythromycin blocks the translocation reaction on ribosomes.
   S. Tetracycline blocks binding of aminoacyl-tRNA to A-site of ribosome.
   T. Anisomycin blocks the peptidyl transferase reaction on ribosomes.
   a. P, R and S
   b. P, Q, R and S
   c. P, R, S and T
   d. P, Q, R, S and T

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2. Sulpha drugs are effective antibiotics which inhibits the formation of folic acid by acting as a
   a. non-competitive inhibitor
   b. competitive inhibitor
   c. uncompetitive inhibitor
   d. irreversible inhibitor

3. Match the activity spectrum of the following antibiotics
   P. actinomycin D
   Q. daunorubicin
   R. rifamycin
   S. griseofulvin
   1. antifungal
   2. antituberculosis
   3. antitumor
   4. antiprotosoal
   a. P-3, Q-4, R-2, S-1
   b. P-3, Q-1, R-4, S-2
   c. P-3, Q-1, R-2, S-4
   d. P-2, Q-1, R-4, S-3 **[GATE]**

4. Penicillin-resistant bacteria produce-lactamase which inactivates penicillin by
   a. esterifying the carboxyl group
   b. oxidising S atom
   c. hydrolysing the side chain
   d. hydrolysing the C–N bonds **[IISc]**

5. Penicillin and lysozyme prevent synthesis and cause lysis, respectively, of cell walls of
   a. *Micrococcus lysodeikticus*
   b. *Escherichia coli*
   c. *Saccharomyces cerevisiae*
   d. *Methanobacterium barkeri* **[GATE]**

6. Mycoplasma are *not* inhibited by penicillin because they
   a. produce penicillinase
   b. are gram-positive
   c. are gram-negative
   d. do not have a cell wall **[IISc]**

7. Which of the following antibiotics are used to treat fugal infections?
   P. Aminoglycosides
   Q. Cephalosporins
   R. Griseofulvin
   S. Polyenes
   T. Bacitracin
   a. P, Q and R
   b. R and S
   c. R, S and T
   d. S and T **[MKU]**

Miscellaneous

1. Chose the *incorrect* statement.
   a. Rickettsia is a gram negative and obligate intracellular parasitic bacteria.
   b. Cyanobacteria is an example of photoorganoautotroph.
   c. Mutations are expressed more frequently in bacteria because they are monoploid.
   d. A micro organism which required a growth factor in order to grow is called an auxotroph.

2. Identify the *correct* statement.
   a. Mycoplasma does not have cell wall.
   b. A single stranded DNA with non-histone is called genophore.
   c. Pasteurized milk contains lactobacillus.
   d. Virus have DNA and cytoplasm enclosed in a capsule.

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3. Choose the *correct* statement(s).
   P. Genes for exotoxins are located on plasmid or temperate phage.
   Q. Surface appendages used by bacteria to attach to one another are called pili.
   R. Most bacteria grow best at pH 7.
   a. P and Q
   b. P and R only
   c. Q and R only
   d. P, Q and R

4. What is true about *E. coli* ?
   a. It is a parasite in the liver of man.
   b. It shows syngamy and meiosis.
   c. It exhibits alternation of generation.
   d. It lives symbiotically in the colon of humans. **[IISc]**

5. Which of the following statements is *false*?
   a. *E. coli* exhibits alteration of generation.
   b. Partial zygote is formed in bacteria as a result of genetic recombination.
   c. Broad spectrum antibiotics act against both gram positive and gram negative bacteria.
   d. Cell wall synthesis in gram positive bacteria is inhibited by penicillin.

6. Which of the following statement regarding bacteria is *incorrect*?
   a. Genome of *Haemophilus influenzae* is the first bacterial genome that was sequenced.
   b. DNA binding protein HU and H1 play structural role in chromosomal organization.
   c. Flagella in *Vibrio cholerae* remain surrounded with lipopolysaccharide sheath.
   d. Bacterial chromosome may be single stranded.

7. Which one of the following combinations of the name of the disease, its causative agent and the discoverer is *correct*?
   a. Diphtheria, *Corynebacterium diphtheriae, Koch.*
   b. Anthrax, *Bacillus anthracis, Koch.*
   c. Malaria, *Plasmodium species, Neisser.*
   d. Typhoid, *Salmonella typhi, Ogston.* **[IISc]**

8. Which of the following statement is *incorrect*?
   a. Antibiotic rifampicin inhibits the cross linking in peptidoglycan cell wall of bacteria.
   b. Rickettsia is a gram negative obligate intracellular parasite.
   c. Bacterial chromosome rarely contains introns.
   d. Endospore formation is common in genera *Bacillus* and *Clostridium.*

9. Severe acute respiratory syndrome (SARS)
   a. is caused by a variant of *Pneumococcus pneumonia.*
   b. is caused by a variant of the common cold virus (corona virus).
   c. is an acute form of asthma.
   d. affects non-vegetarians much faster than the vegetarians.

10. Pasteurization of milk is done by
    a. boiling the milk for 20 minutes.
    b. heating the milk at 72°C for 30 minutes.
    c. heating the milk at 72°C for 20 minutes.
    d. heating the milk at 62°C for 30 minutes.

 [GATE]

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Chapter 12

Immunology

Innate and Adaptive immune response

1. Rapid but non-antigen specific immune responses are produced by the
   a. adaptive immune system
   b. innate immune system
   c. leukocytes
   d. lymphatic system

2. Innate immune responses are most effective against
   a. antigens resembling self antigens
   b. common epitopes on bacteria
   c. genetically engineered antigens
   d. viruses that have previously caused infection

3. Which situation below describes an example of innate immunity?
   a. Antibody production by plasma cells
   b. Antigen removal by cilia in the respiratory tract
   c. Complement activation by antibody bound to the surface of a bacterium
   d. Memory response to influenza virus

4. Inflammation does not involve
   a. cytokine production by macrophages
   b. migration of leukocytes out of the circulation
   c. pain and swelling at the site of infection
   d. secretion of antibodies

5. The antigen specificity of an adaptive immune response is due to
   a. activation of antigen-specific lymphocytes.
   b. folding of antibody to fit the pathogen.
   c. lysis of only certain pathogens by neutrophils.
   d. phagocytosis of only certain pathogens by macrophages.

6. The primary purpose of the adaptive immune system is to
   a. block all pathogens from entering the body
   b. cure allergic reactions
   c. kill tumor cells
   d. protect from disease upon re-infection with a specific pathogen

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7. Viral replication within cells is inhibited by
   a. IL-4
   b. IL-1
   c. IFNα
   d. TNFα [GATE]

8. Active immunity means
   a. resistance developed after disease
   b. resistance developed before disease
   c. increasing rate of heart beat
   d. increasing quantity of blood

9. Interferons
   a. activate B-cells to make virus-specific antibodies
   b. are Th2 cytokines
   c. are virus proteins that interfere with activation of cytotoxic T-cells
   d. inhibit virus replication by infected cells [IISc]

10. Toll-like-receptors (TLRs) play an important role in immune defense by recognizing
    a. microbial components
    b. conformational differences in antigenic proteins
    c. MHC-peptide complexes
    d. anti-idiotypic immunoglobulins [IISc]

Adaptive immunity

1. The humoral immune response involves
   P. stimulation of B-cells by helper T-cells.
   Q. secretion of antibodies that specifically recognize foreign molecules.
   R. presentation of an antigen-antibody complex to cytotoxic T-cells.
   S. destruction of the antigen-antibody complex by macrophages.
   a. P and S
   b. P, R and S
   c. P, Q and S
   d. P, Q, R and S

2. The cell-mediated immune response
   P. is dependent upon the humoral response.
   Q. is usually used to respond to virus-infected cells.
   R. involves direct recognition of the antigen by killer T-cells.
   S. requires that the antigen be presented to killer T-cells by an MHC protein.
   a. P and S
   b. Q and S
   c. P, Q and S
   d. P, Q, R and S

3. Individuals exposed to smallpox virus are immune to same disease due to
   a. presence of larger quantities of antibodies
   b. presence of long lived memory cells
   c. healthy lifestyle
   d. generation of antigen specific macrophages [JNU]

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4. Which of the following is *not* a characteristic of a secondary immune response?
   a. IgG isotype
   b. Low affinity antibodies
   c. High affinity antibodies
   d. No lag phase [JNU]

5. Which of the following statements is *incorrect* concerning antigen-specific receptors on both B and T-cells?
   a. They are clonally distributed transmembrane molecules.
   b. They have extensive cytoplasmic domains that interact with intracellular molecules.
   c. They consist of polypeptides with variable and constant regions.
   d. They are associated with signal transduction molecules at the cell surface.

Cells and organs of the immune system

1. Central (primary) lymphoid organs
   a. are efficient in exposing T-cells to foreign antigen.
   b. are the primary site of antibody synthesis and maturation.
   c. filter blood and trap blood-borne antigens.
   d. provide the microenvironment for synthesis and maturation of T and B-cells.

2. Peripheral lymphoid organs
   a. are centrally located in the abdomen to protect their vital functions.
   b. are designed to maximize contact between antigen and lymphocytes.
   c. produce antigen-specific lymphocytes from stem cells in response to antigen.
   d. sequester antigen to minimize its damage to the body.

3. Lymphocytes continually recirculate through peripheral lymphoid tissue in order to
   a. be killed before they cause autoimmunity.
   b. efficiently encounter antigen.
   c. mature from stem cells into lymphocytes.
   d. phagocytose antigen and kill it.

4. Hematopoietic stem cells are multipotent, which means that they are
   a. antigen-specific cells.
   b. capable of developing into any blood cells.
   c. committed to produce cells of a single lineage.
   d. not self-renewing.

5. The antigen processing cell in higher organisms is
   a. T-cell
   b. macrophage
   c. megakaryocyte [IISc]
   d. eosinophil

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6. Choose the mismatch

Cell types	*Descriptions*
a. Eosinophils	phagocytic cell important in the body's defense against parasitic organisms
b. Kupffer	macrophages found in liver
c. Dendritic	an antigenic presenting cell that is not phagocytic
d. Mast	generally first cells to arrive at the site of inflammation

7. Which one group of the following cells is *not* phagocytic in nature?
   a. Neutrophil polymorphonuclear leukocytes
   b. B-lymphocytes
   c. Microglial cells
   d. Macrophages

8. Specialized group of leukocytes which have the ability to ingest and damage large intracellular parasites such as schistosomes are called
   a. dendritic cells
   b. eosinophils
   c. basophils
   d. neutrophils

9. The macrophage rich mass found at the site of injection of an adjuvant is called
   a. myeloma
   b. granuloma
   c. adjuvant activated lymphoma
   d. none of the above

10. Which one of the following matching of leukocyte types and its functions is *incorrect*?
    a. T-lymphocytes – secrete T-cell receptors
    b. Basophils – produce histamine
    c. Macrophages – engulf and digest bacteria
    d. Natural killer cells – infected or cancerous body cells, lyse virus infected cells

11. Which of the following cells play a major role in defense against parasitic microorganisms and constitute 1–3% of the white blood cell population?
    a. Macrophages
    b. Neutrophils
    c. Eosinophils
    d. Basophils

12. Mature dendritic cells are capable of which of the following?
    a. Activating antigen-specific T-cells
    b. Removing red blood cells
    c. Producing bradykinin
    d. Extracellular killing of target cells

13. Neutrophils provide protection from a variety of microbes and are arguably the most important WBC's eliminating
    a. non-viral infections
    b. all fungal infections
    c. viral infections
    d. parasitic infections

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14. Most antibodies are synthesized by the
a. central lymphoid organs
b. peripheral lymphoid organs
c. primary lymphoid organs
d. macrophages

[GATE]

15. The following two organs are examples of secondary lymphoid organs
a. spleen and thymus
b. lymph nodes and thymus
c. spleen and GALT
d. bone marrow and MALT

[IISc]

16. Removal of the bursa of Fabricius from a chicken results in
a. a markedly decreased number of circulating T-lymphocytes
b. anemia
c. a delayed rejection of skin grafts
d. low serum levels of antibodies

17. Lack of thymus leads to increased infection in humans such a condition is called
a. thymectomy
b. nude mice
c. Di George's syndrome
d. none of the above

18. Which of the following statements is *false* concerning the immune system?
a. Antigen presenting cells are found only in the primary immune organs.
b. B-lymphocytes are capable of processing and presenting antigens to T-lymphocytes.
c. T-lymphocytes are capable of helping to activate B-lymphocytes.
d. All nucleated cells have MHC class I antigens on their cell surface.

19. Which of the following statements is *correct*?
a. Helper T-cells express surface CD8 receptors.
b. Cytotoxic T-cells express surface CD4 receptors.
c. Helper T-cells express surface IgG molecules.
d. Cytotoxic T-cells express surface CD8 molecules.

20. The common myeloid progenitor cell gives rise to
a. neutrophils alone
b. neutrophils, basophils and mast cells
c. PMNs and mononuclear leukocytes
d. neutrophils, basophils and monocytes

21. Which of the following statement is *correct*?
a. T-lymphocytes are conditioned by the bone narrow.
b. B-lymphocytes are conditioned by thymus.
c. B-cells produce plasma and memory cells.
d. T-cells do not produce cytokines.

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22. Lymphocytes acquire their antigen specificity
    a. as they enter the tissues from the circulation.
    b. before they encounter antigen.
    c. depending on which antigens are present.
    d. from contact with self antigen.

23. Which of the following statements are *false*?
    1. Monocytes are circulating blood cells that differentiate into macrophages in the tissues.
    2. Eosinophils are phagocytic cells important in the body's defense against parasitic organisms.
    3. All vertebrates generate B-lymphocytes in bone marrow.
    4. Follicular dendritic cells can process and present antigen to T-lymphocytes.
    5. Spleen filters antigens out of the blood.
    6. Dendritic cell is an antigen-presenting cell derived from monocytes that is not phagocytic.
    7. The pluripotent stem cell is one of the most abundant cell types in the bone marrow.
    8. Basophils and mast cells are nonphagocytic cells.
    9. The primary lymphoid organs provide sites where lymphocytes mature and become antigenically committed.

    a. 1, 2 and 5
    b. 3, 4 and 7
    c. 7 and 9
    d. 2, 3 and 4

24. Which of the following is *not* true about macrophages?
    a. Macrophages have longer lifespan than neutrophils.
    b. Macrophages at inflammatory sites are responsible for phagocytosis of cellular debris and help in keeping the tissue clean.
    c. Macrophages contain many lysosomes and have cytoplasmic extensions called *pseudopodia*.
    d. Macrophages appear at the site of inflammation earlier than neutrophils. [JNU]

Antigens

1. Which one of the following does affect immunogenicity of a substance?
    a. Making a molecule more foreign.
    b. Making a molecule larger.
    c. Increasing chemical complexity by adding amino acids.
    d. All of the above.

2. Haptens
    a. require carrier molecules to be immunogenic.
    b. react with specific antibodies when homologous carriers are employed.
    c. interact with specific antibody, even if the haptens are monovalent.
    d. all of the above.

3. In addition to adjuvants, generation of anti-hapten antibodies will require
    a. injection of the hapten into mice.
    b. injection of a mixture of hapten and protein.
    c. injection of the hapten covalently coupled with protein.
    d. injection of a mixture of haptein and lipopolysaccharide. [IISc]

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4. The antibiotic penicillin is a small molecule that does not induce antibody formation. However, penicillin binds to serum proteins and forms a complex that in some people induces antibody formation resulting in an allergic reaction. Penicillin is therefore
   a. an antigen
   b. a hapten
   c. an immunogen
   d. both an antigen and a hapten

5. Epitope is the discrete region on the antigen that interacts with
   a. Fab
   b. antibody's light and heavy chains
   c. T-cell receptor
   d. all of the above

6. Agretope is the region of antigen that is interacting with
   a. TCR
   b. MHC
   c. both TCR and MHC
   d. antibody

7. Which one of the following statements regarding superantigens is *not* true?
   a. Staphylococcal toxic shock syndrome is an example of superantigen associated disease.
   b. Superantigens activate large numbers of B-cells.
   c. Superantigen associated diseases are characterized by markedly elevated circulating levels of pro-inflammatory cytokines.
   d. Superantigens are presented by antigen-presenting cells in an identical manner to conventional antigens.

MHC and Antigen presentation

1. Which of the following cell/cells express MHCII on the surface?
   a. B-lymphocytes
   b. Macrophages
   c. Dendritic cells
   d. All of the above

2. Major variability of MHC is found in
   a. α chains
   b. β_2 microglobulins
   c. amino terminus region
   d. domains

3. Class I MHC genes that encode for membrane bound glycoproteins are found in
   a. all nucleated cells
   b. B-cells, macrophages, dendritic cells only
   c. only B-cells and T-cells
   d. tumor cells

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4. Choose the correct combination of statements from the following for MHC II protein
   P. has α_1, α_2, β_1 and β_2 domains.
   Q. α_1 and β_1 are the polymorphic domains.
   R. involved in presenting antigen to cytotoxic T-cells.
   S. only α-chain is glycosylated.
   a. Q and R
   b. R and S
   c. P and S
   d. P and Q

5. Which of the following statements is *false?*
   a. MHC class II molecules are present on all nucleated cells.
   b. MHC class II molecules are made up of two polypeptides.
   c. MHC class I molecules are present on all nucleated cells.
   d. T-cell response is MHC-restricted. [GATE]

6. Which of the following is *not* true?
   a. Antigen-presenting cells express both class I and II MHC molecules on their membranes.
   b. Class III MHC genes encode membrane bound proteins.
   c. Class II MHC molecules typically bind to longer peptides than do class I molecules.
   d. Most nucleated cells express class I MHC.

7. Which of the following statements about the MHC is *incorrect*?
   a. It codes for complement components.
   b. It codes for both chains of the MHC class I molecule.
   c. It codes for both chains of the MHC class II molecule.
   d. It is associated with susceptibility and resistance to different diseases.

8. Which of the following *correctly* describes the structure of MHC class I gene products?
   a. A bimolecular structure comprising two glycosylated polypeptide chains encoded by MHC.
   b. The molecule has a polypeptide chain of molecular mass 45 kDa which is anchored through the cell membrane and is covalently bound to β_2 microglobulin.
   c. Polypeptide chain has 3 extracellular domains, one trans membrane domain and one cytoplasmic region.
   d. β_2 microglobulin is encoded by MHC and is membrane bound.

9. Which of the following statement is *correct*?
   a. Tc cells recognize MHC II molecule on the surface of B-cell and excretes cytokines.
   b. Tc cells recognize antigen presented by MHC II molecule on the surface of B-cell and attack cancer cells.
   c. Tc cells proliferate to form cytotoxic T-cells and interact with MHC II along with CD4 present on their surface.
   d. Tc cells recognize antigen presented by MHC I much on the surface of virally infected cells/cancer cells and lead to cell death/cytotoxicity.

10. Which is *not* true of Antigen Presenting Cells (APCs)?
    a. They express class II MHC molecules on their membranes.
    b. They can't deliver a co-stimulatory signal that is required for helper T-cell activation.
    c. These specialized cells include macrophages, B-lymphocytes and dendritic cells.
    d. Dendritic cells are widely distributed in the human body to trap lymph and blood borne antigens and activate helper T-cells which in turn induce B-cell growth and antibody.

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The role of the antigen-presenting cell in the immune response is all of the following *except*
a. the limited catabolism of polypeptide antigens.
b. to allow selective association of MHC gene products and peptides.
c. to supply second signals required to fully activated T-cells.
d. to present non-self peptides associated with MHC class II molecules to B-cells.

12. Antigenic peptides are presented to T-cells by
a. TCR/CD3
b. CD28
c. CTL A4
d. MHC [GATE]

13. Generation of antigenic peptide for MHC class-I occurs in
a. endoplasmic reticulum
b. cytosol
c. Golgi apparatus
d. endosome [JNU]

14. Choose the correct statements.
P. Tc cells recognize antigen with class I MHC molecules on target cells.
Q. Endogenous antigens are degraded into peptides within the cytosol by proteasomes.
R. Exogenous antigens are internalized and degraded within the acidic endocytic compartments.
S. Presentation of nonpeptide antigens derived from bacteria involves the class I-like CD1 molecules.
a. P and Q
b. P, R and S
c. P, Q and S
d. P, Q, R and S

Antibodies : Structure and Function

1. How many disulfide bonds are there in a typical immunoglobulin molecule?
a. 2-5
b. 6-9
c. 10-20
d. 28 [JNU]

2. Variable and hyper variable regions are found in
a. alpha chain of T-cell receptor
b. light chain of immunoglobulins
c. β-chain of T-cell receptor
d. all of the above [IISc]

3. An antibody molecule
P. is composed of two light and two heavy chains.
Q. is held together by disulfide bonds between the chains.
R. binds antigen through its constant region.
S. has certain effector functions as determined by its light chain.
a. P and Q
b. P, R and S
c. P, Q and S
d. P, Q, R and S

4. Antibody effector functions include all of the following *except*
a. activating complement on bacterial surfaces to promote phagocytosis by neutrophils.
b. binding extracellular viruses to block their entry into host cells.
c. binding intracellular viruses to initiate cytotoxicity.
d. blocking uptake of bacterial toxins by host cells.

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5. In an immunoglobulin molecule, the region determining complementarity is present in
   a. variable domain of heavy chain.
   b. variable domain of light chain.
   c. variable domains of heavy and light chain.
   d. variable and constant domains of the heavy chains. [IISc]

6. The major immunoglobulin family to which a particular immunoglobulin belongs can be determined by sequential analysis of the 100 amino acids beginning from the
   a. amino-terminal of the light chain.
   b. carboxy-terminal of the light chain.
   c. amino-terminal of the heavy chain.
   d. carboxy-terminal of the heavy chain.

7. In immunoglobulin molecule, the complementarity determining region is present in
   a. V_H domain only
   b. V_L domain only
   c. V_H and V_L domains
   d. V_H and C_H1 domains [IISc]

8. There are five classes of the antibodies (IgM, IgD, IgG, IgE, IgA). What determines the class to which an antibody belongs?
   a. structure of the light chain
   b. variable region of the antibody
   c. structure of the heavy chain constant region
   d. stage of the infection [GATE]

9. Antigen binding sites of an immunoglobulin are located in
   a. light chain alone
   b. heavy chain alone
   c. Fc region of the antibody
   d. Fab region of the antibody [GATE]

10. In the normal human being the concentration(s) of various antibodies in the serum in the order
   a. IgM > IgA > IgG > IgE
   b. IgG > IgA > IgM > IgE
   c. IgE > IgG > IgM > IgA
   d. IgA > IgM > IgE > IgG [GATE]

11. The class of immunoglobulin that can get transported across epithelial cells is
   a. IgG
   b. IgE
   c. IgA
   d. IgM [GATE]

12. IgM class of antibodies are more effective than IgG in bringing about complement mediated lysis because
   a. IgM has higher affinity for complement components.
   b. IgM has higher affinity for antigen binding.
   c. IgM is a pentamer.
   d. IgM is present in higher concentration. [GATE]

13. One molecule of secretory IgA can at most bind to
   a. 10 molecules of antigen
   b. 4 molecules of antigen
   c. 2 molecules of antigen
   d. 1 molecule of antigen [IISc]

14. Number of antigen binding sites in IgM
   a. 5
   b. 8
   c. 4
   d. 10

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15. J chain or joining chain is found in
a. IgM
b. IgA
c. IgM and IgA
d. IgE

16. The immunoglobulin that results in histamine release is
a. IgG
b. IgM
c. IgE
d. IgD
[IISc]

17. Surface receptor (IgA) on the target cell is the site of binding of
a. hepatitis B virus
b. HIV
c. rabies
d. influenza A, B viruses
[GATE]

18. The predominant antibody in saliva is
a. IgG
b. IgA
c. IgM
d. IgD
[GATE]

19. Pepsin digestion of immunoglobulin G yields
a. $F(ab)_2$ fragment and low molecular weight fragments.
b. 2 Fab fragments and Fc fragment.
c. one heavy chain and one light chain.
d. only light chains.

20. Choose the correct match

Group I	*Group II*
P. IgA	1. Basophils
Q. IgE	2. δ heavy chain
R. IgG	3. Secretory component
S. IgM	4. Pentamer
	5. Crosses placenta

a. P-3, Q-1, R-5, S-4
b. P-3, Q-5, R-2, S-1
c. P-2, Q-3, R-5, S-4
d. P-2, Q-1, R-3, S-5

21. Immunoglobulin in the serum of a newborn will be mostly
a. IgM of maternal origin
b. IgG of maternal origin
c. IgG of endogenous origin
d. IgM of endogenous origin

22. All humans start producing antibodies only after they are
a. infected with a pathogen
b. immunised with an antigen
c. exposed to an antigen
[JNU]
d. none of the above

23. Immunoglobulin G was fractionated by size exclusion chromatography and it was observed to have a molecular mass of 150 kDa. SDS-PAGE analysis under reducing conditions revealed two bands of size 50 kDa and 25 kDa. The oligomeric status of the protein consists of
a. 3 polypeptide chains of molecular mass 50 kDa.
b. 6 polypeptide chains of molecular mass 25 kDa.
c. one polypeptide chain of 50 kDa and 4 polypeptides of mass 25 kDa.
[IISc]
d. two polypeptides of 50 kDa and two of 25 kDa.

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24. The molecular mass of IgG is 150 kDa, and that of the antigen-antibody complex formed in the presence of large excess of antigen is 300 kDa. The molecular mass of the antigen is

a. 25 kDa
b. 50 kDa
c. 75 kDa
d. 150 kDa [IISc]

25. You have purified some Fab from IgG myeloma protein. Under appropriate conditions you could use this Fab to generate antibodies against

a. both κ and λ chain
b. γ chain hinge region
c. γ chain allotypic determinants
d. the idiotype of this myeloma [IISc]

26. Choose the mismatch

Isotypes	*Descriptions*
a. IgG	the most abundant type in serum
b. IgA	major antibody in secretions such as saliva, tears, and breast milk
c. IgD	plays an important role in protecting against pathogens that invade through the gut or respiratory mucosa
d. IgE	least abundant and latest isotype which plays an important role in hypersensitivity

27. Which one of the following statements is *incorrect*?

a. Plasma cells are terminally differentiated cells.
b. IgA is the predominant class of immunoglobulin in sero mucous secretions.
c. The secretory component is associated with IgM.
d. The pentamers of IgM are formed by disulphide linkages. [IISc]

28. The secretory IgA was electrophoresed on SDS-PAGE under reduced and denaturing condition. The number of polypeptide bands detected on the gel is/are

a. 2
b. 3
c. 4
d. 5 [GATE]

29. The relative level of specific IgM antibodies can be of diagnostic significance because

a. IgM is easier to detect than the other isotypes.
b. viral infection often results in very high IgM response.
c. IgM antibodies are more often protective against reinfections than are the other isotypes.
d. relative high levels of IgM often correlate with a first recent exposure to the inducing agent. [IISc]

30. IgG has four chains. Purified monoclonal IgG was subjected to electrophoresis. The number of bands visible by

P. reducing SDS-PAGE
Q. isoelectric focusing
R. native PAGE

a. P-2, Q-1, R-1
b. P-1, Q-1, R-2
c. P-2, Q-4, R-2
d. P-4, Q-2, R-4 [GATE]

31. In an antigen antibody interaction, in the zone of equivalence, the isolated antigen IgG complex was found to be in the molar ratio of antigen: IgG, 2 : 1. The number of epitope(s) present on the antigen is

a. 1
b. 2
c. 4
d. 10 [GATE]

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32. Class switching occurs in a B-cell to produce IgG from IgM. Which one of the following statements is *correct*?
a. Specificity of the IgG and IgM are different from each other.
b. Specificity of the IgG is the same as IgM.
c. The molecular weight of the new antibody is the same as the old one.
d. The valency of the new antibody is the same as the old one. **[GATE]**

33. Two types of IgMs are produced by the same B-cell, one that is secreted and the other that can bind to the membrane. Which one of the following statements is *correct*?
a. These two antibodies are coded by altogether two different genes.
b. The membrane anchor residues are added to the protein by post-translational modification.
c. The two different proteins associate, one of which provides the membrane anchor.
d. The proteins are produced by alternate splicing of its primary transcript. **[GATE]**

34. Monoclonal antibodies differ from polyclonal antibodies in their property of reacting with
a. specific epitope
b. specific antigen
c. specific clone of cells
d. all of the above **[IISc]**

35. Which of the following statement is *incorrect*?
a. The variable region domains of an immunoglobulin contain the antigen recognition site.
b. The class of an immunoglobulin is determined by its heavy chain.
c. An antigenic determinant is a paratope.
d. An IgG antibody is bivalent.

36. Which of the following statements about monoclonal antibodies is *incorrect*?
a. They can be produced in large amount by growing many mice, each of which is immunized with the same antigen.
b. They are useful in quantifying specific proteins in human blood.
c. They can be used to purify scarce proteins by affinity chromatography.
d. They can be used to prepare catalytic antibodies. **[GATE]**

37. Which of the following is a *correct* statement for a monoclonal antibody?
a. It is raised in a special breed of rabbit.
b. It is prepared by the biochemical purification from a pool of polyclonal antibodies.
c. It recognizes a single epitope on the antigen.
d. Its affinity towards the antigen is very high as compared to that of the polyclonal antibody. **[JNU]**

38. Monoclonal antibodies are produced in large scale outside the body by the method
a. Hybridoma technique which involves fusion of antigen exposed spleen cells with myeloma cells, and the hybrid cells keep producing antibodies specified to that antigen.
b. Culturing spleen cells previously exposed to a particular antigen and harvesting antibodies from the medium.
c. Antibodies multiply in culture medium if they are provided with nutrients and growth factors.
d. Antibodies can be collected from blood by centrifugation. **[JNU]**

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39. Aminopterin is used during the production of hybridoma cells because
    a. it blocks the *de novo* pathway
    b. it prevents the growth of B-cells
    c. it prevents the growth of myeloma cells
    d. it blocks the synthesis of Ig by B-cells [GATE]

40. Which of the following statements is most appropriate for recombinant antibody production in transgenic plants?
    a. A very high level expression is always obtained.
    b. Light promotes more antibody production.
    c. Such antibodies are free from other antigen of animal origin.
    d. Functional antibody cannot be produced in plants. [GATE]

41. Which of the following uniquely distinguishes the T-cell receptor (TCR) from an antibody?
    a. The TCR is composed of two different types of polypeptide chains.
    b. The TCR can function as a cell surface antigen receptor.
    c. The TCR can bind an antigen fragment only in a trimolecular complex with either the class I or class II surface proteins of the major histocompatibility complex.
    d. The TCR is capable of participating in a cytotoxic reaction.

42. All of the following are *true* with respect to IgE molecules, *except*
    a. they are the principal immunoglobulin class involved in allergic reactions.
    b. they will cross the placenta and fix complement.
    c. they are involved in mediating anti-parasitic immune responses.
    d. they can effect the release of histamine and other chemical mediators.

43. According to the Clonal Selection theory
    a. an antibody changes its shape according to the antigen it meets.
    b. an individual animal contains only one type of B-cell.
    c. the animal contains many types of B-cells, each producing one kind of antibody.
    d. each B-cell produces many types of antibodies.

44. HAT (Hypoxanthine, aminopterin and thymidine) is used for selecting the hybridomas based on the following
    P. Only hybridoma will grow since it inherited the HGPRT genes from B-cells and can synthesize DNA from hypoxanthine.
    Q. Myeloma cells will not grow in cultures since *de novo* synthesis is blocked by aminopterin and due to the lack of HGPRT enzyme.
    a. only P is true
    b. only Q is true
    c. both P and Q are true
    d. P is true and Q is false [GATE]

45. Considering only combinatorial joining of gene segments and association of light and heavy chains, how many different antibody molecules potentially could be generated from germ-line DNA containing 500 V_L and 4 J_L gene segments and 300 V_H, 15 D_H and 4 J_H gene segments?
    a. 2×10^3
    b. 1.8×10^4
    c. 3.6×10^7
    d. 3.24×10^8 [JNU]

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1. Genes for immunoglobulins (antibodies) are unlike other human genes in that
   a. antibody genes are composed of introns and exons.
   b. DNA for antibody molecules is inherited from only one parent.
   c. gene segments must be spliced together to make each unique antibody molecule.
   d. several exons encode each antibody molecule.

2. The potential repertoire of antibody diversity of any vertebrate organism is much higher than that can be calculated from the total number of V segments, J segments and D segments. This is due to
   a. gene rearrangements
   b. somatic hypermutation
   c. gene conversion
   d. genetic recombination

3. Antibody diversity is generated by [IISc]
   a. protein splicing
   b. somatic hypermutation
   c. allelic exclusion
   d. inter-chromosomal recombination

4. Class switching takes place in [GATE]
   a. Pro-B cell
   b. Pre-B cell
   c. immature B-cell
   d. activated B-cell

5. The immunoglobulin gene rearrangement that takes place in activated B-cells is [IISc]
   a. V-D-J recombination
   b. D-J recombination
   c. V-J recombination
   d. class switch recombination

6. The extraordinary diversity of antibodies results in part from [IISc]
   a. the action of monoclonal antibodies
   b. the splicing of protein molecules
   c. the action of cytotoxic T-cells
   d. the rearrangement of gene segments

7. Which of the following is *not* a feature of immunoglobulin gene rearrangements?
   a. V_K genes join J_K genes
   b. D_H genes join both V_H and J_H genes
   c. V_H genes are preceded by leader sequence [JNU]
   d. J_H genes join both V_H and D_H genes

8. Which of the following gene rearrangement may not occur?
   a. V_H with C_H
   b. V_L with C_H
   c. V kappa with C kappa
   d. V_L with C_L

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Antigen-antibody interactions : Principles and applications

1. Which of the following does *not* participate in the formation of antigen-antibody complexes?
   a. Hydrophobic bonds
   b. Covalent bonds
   c. Electrostatic interaction
   d. van der Walls forces [IISc]

2. Which of the following chemical bonds is *not* involved in antigen binding by antibody?
   a. Hydrogen bonds
   b. Hydrophobic forces
   c. Covalent bonds
   d. van der Waals forces

3. The protection against smallpox virus infection afforded by prior infection with cowpox virus represents
   a. antigenic specificity
   b. antigenic cross-reactivity
   c. enhanced viral uptake by macrophages
   d. innate immunity

4. Some antibodies like anti A, anti B antibodies in the serum act against some bacterial antigens, such cross interaction is due to
   a. shared epitopes
   b. cross-reactivity
   c. molecular mimicry
   d. all of the above

5. A crossed precipitin line following double immune diffusion is due to
   a. shared epitopes between antigens
   b. identity between antigens
   c. no common epitopes between antigens
   d. only few epitopes that are common between antigens

6. To prevent hemolytic disease of the newborn from occurring in a successive pregnancy where an Rh negative woman gives birth to an Rh positive child, the child will receive shortly after birth an injection of
   a. Rh antigens
   b. A and B factors
   c. Rh antibodies
   d. type O blood

7. Clumping of erythrocytes, caused by antibodies forms the basis of blood group typing is called
   a. haemagglutination
   b. molecular mimicry
   c. frustrated phagocytosis by antibody
   d. none of the above

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8. Antigen-antibody binding do not involve
   a. van der waals forces
   b. non covalent bonding
   c. hydrophobic interaction
   d. covalent bonding

9. In order to separate the antibodies in an antibody mixture, the laboratory technologist may use a procedure called
   a. transfusion
   b. complement fixation
   c. electrophoresis
   d. gene amplification

10. ELISA used to detect antigens or antibodies utilizes those enzymes that
   a. have a high turnover rate
   b. yield a stable colored product
   c. are stable on conjugation to proteins
   d. all of the above

11. Which of the following is *false*?
   a. Most antigens induce a polyclonal response.
   b. For precipitation to occur, both antigen and antibody must be multivalent.
   c. ELISA tests using chemiluminescence are the most sensitive immunoassays available.
   d. Western blotting is useful quantitative assay for measuring the levels of proteins in cells or tissues.

12. Indicate the most appropriate assay for detecting syphilis bacterium in the serum
   a. agglutination to detect titer of anti-syphilis antibodies
   b. rocket electrophoresis
   c. radial immunodiffusion
   d. SDS-PAGE and western blotting

13. Serological tests are used in diagnostic microbiology to detect
   a. the presence of antigens or antibodies in blood.
   b. whether bacterial flagella are present or absent.
   c. whether the serum of an individual contains red blood cells.
   d. whether protozoa and fungi have invaded the tissues of the body.

14. An example of an agglutination test used in serology is that in which
   a. the dissolved cells break open during lysis
   b. viruses multiply within bacteria
   c. a visible clump appears to the observer
   d. the body's B-cells diminish in number

15. The titer of an individual represents the
   a. amount of antigen present in the body tissues
   b. amount of antibody in the patients serum
   c. number of flagella present on a bacterial cell
   d. number of spores formed by a fungus

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16. In order to separate the antibodies in an antibody mixture, the laboratory technologist may use a procedure called
    a. transfusion
    b. complement fixation
    c. electrophoresis
    d. gene amplification

17. The immunofluorescence test can be used to identify
    a. protein molecules and polysaccharide molecules
    b. lipid molecules and nucleic acid molecules
    c. antibody molecules and antigen molecules
    d. cytoplasmic molecules and cell wall molecules

18. At the conclusion of the ELISA test,
    a. radioactivity is produced
    b. a clumping reaction is seen
    c. cells undergo lysis
    d. a color change takes place

19. Tests for typhoid fever and strep throat as well as blood typing utilize a serological test known as
    a. precipitation
    b. electrophoresis
    c. agglutination
    d. complement fixation

B-cell : Generation, activation, differentiation and Response

1. Humoral immunity can be acquired passively by
    a. catching a virus from a friend by shaking hands.
    b. receiving a vaccine of influenza virus grown in eggs.
    c. receiving serum from someone who has recovered from an infection.
    d. receiving leukocytes from an immune family member.

2. A secondary immune response is not
    a. faster than a primary response.
    b. long lasting than a primary response.
    c. more likely to result in increased adaptive immunity than a primary response.
    d. preceded by a longer lag period than a primary response.

3. When a B-cell undergoes immunoglobulin class switching
    a. the variable region of the light chain changes, but its constant region remains the same.
    b. the variable region of the light chain remains the same, but its constant region changes.
    c. the variable region of the heavy chain changes but its constant region remains the same.
    d. the variable region of the heavy chain remains the same but its constant region changes. [GATE]

4. Antigen binding to the B-cell receptor
    a. transduces a signal through the antigen-binding chains.
    b. invariably leads to B-cell activation.
    c. transduces a signal through the IgA and IgB molecules.
    d. results in macrophage activation.

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5. Which of the following statements concerning B-cell maturation are *true*?
P. The surrogate light chain is expressed by pre-B cells.
Q. Most of the B cells generated every day never leave the bone marrow as mature B-cells.
R. Heavy chain V_H-D_H-J_H rearrangement begins in the pre-B-cell stage.
S. Immature B cells express membrane IgM and IgD.
a. P and Q
b. Q and R
c. R and S
d. P and S

6. Naïve B cells require several signals in order to become activated. Which are these signals?
a. Signal 1 from CD4 positive T-cells and signal 2 from follicular dendritic cells.
b. Signal 1 from antigen through the B cell receptor and signal 2 from CD8 positive T-cells.
c. Signal 1 from antigen through the B cell receptor and signal 2 from CD4 positive T-cells.
d. Signal 1 from CD4 positive T-cells and signal 2 from cytokines produced by macrophages.

7. Which of the following comparison of primary and secondary antibody responses is *not* correct?

	Primary responses	*Secondary responses*
a. Responding B cell	Naive B-cell	Memory B-cell
b. Lag period	Generally 4–7 days	Generally 1–3 days
c. Isotype produced	IgM predominates	IgG predominates
d. Antibody affinity	Higher	Lower

8. Which of the following is/are the property of thymus-dependent antigens?
P. It is chemically soluble protein.
Q. Isotype switching occurs during humoral response.
R. Affinity maturation occurs during humoral response.
S. No immunologic memory develops.
a. P and Q
b. Q and R
c. P, Q and R
d. P, Q, R and S

T-cell : Maturation, activation, differentiation and Response

1. T-cells, like B-cells, express antigen specific receptors, T-cell receptor (TCR). Which of the following statement is *incorrect* about TCR?
a. It is monovalent.
b. Antigen recognized by the TCR is a combination of MHC molecule and a linear sequence of peptide.
c. It does not exist in a secretory form.
d. It is a four chain molecule with two antigen-binding sites.

2. The T-cell receptor can bind to antigenic peptides
a. only in the free form
b. only when loaded on to MHC molecules
c. only when bound to hapten **[IISc]**
d. only when bound by antibody

3. Which of the following is *not* true of T-cell receptor (TCR)?
a. The T-cell receptor is membrane bound.
b. TCR does not appear in a soluble form as B-cell receptor does.
c. Most TCRs are specific for antigen combined with molecules encoded by MHC.
d. TCRs are not antigen specific.

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4. CD3 molecules are present on the surface of both these types of cells
   a. B-cell and T_H cell
   b. T_H cell and NK cell
   c. T_C cell and T_H cell
   d. T_C cell and NK cell

5. The T-cell receptor
   P. is required for helper T-cells to stimulate B-cells in the humoral response.
   Q. is required for killer T-cells to recognize antigen.
   R. recognizes antigen fragments and MHC molecules independently.
   S. has a domain organization that is roughly similar to that of immunoglobulin molecules.
   a. P and S
   b. P, R and S
   c. P, Q and S
   d. P, Q, R and S

6. Development T-cell that reacts strongly with self-peptides bound to self-MHC molecules are
   a. eliminated in the thymus
   b. eliminated in the bone marrow
   c. allowed to function normally
   d. are suppressed in peripheral blood circulation [JNU]

7. A T-cell is most likely to be eliminated by the process of negative selection in the thymus if it expresses a TCR which can bind strongly to
   a. foreign MHC and self antigen
   b. self MHC and foreign antigen
   c. self MHC and self antigen
   d. foreign MHC and foreign antigen

8. A fundamental difference between the antigen receptors on B-cells (BCR) and on T-cells (TCR) is their
   a. different requirements for antigen presentation
   b. function following antigen binding
   c. heterogeneity from one lymphocyte to the next
   d. heterogeneity on each lymphocyte

9. Natural killer cells
   P. mediate lysis of tumor cells and virus-infected cells.
   Q. cause perforin-induced pore formation.
   R. mediated killing of normal cells are protected due to the expression of high levels of class I MHC molecules on normal cells.
   S. have only activation receptors.
   a. P and Q
   b. Q and S
   c. P, Q and R
   d. P, Q, R and S

10. Which one of the following characters is *not* found in natural killer cells?
    a. Ability to lyse a variety of tumor cells and virally infected cells without prior sensitization.
    b. Inability to rearrange TCR (Antigen specific T-cell Receptor) genes.
    c. Presence of two cell-surface molecules, CD16 and CD56.
    d. Much smaller than T and B-lymphocytes. [JNU]

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11. Which of the following statements is/are *correct*?

P. T-helper lymphocytes secrete a factor, which stimulates formation of other types of lymphocytes.

Q. T-suppressor lymphocytes cause lowering of the intensity of the immune response.

R. T-cytotoxic lymphocytes recognize nonself antigens on cells and lyse them.

S. B-lymphocytes develop into cells that secrete antibodies.

a. P, Q
b. P, R
c. R, S
d. P, Q, R, S

12. Tolerance refers to the finding that **[GATE]**

a. the T-cell repertoire for recognition of foreign Ag is very diverse.
b. CD4 T-cells do not respond to foreign Ag associated with MHC class I molecules and $CD8^+$ T-cells do not responds to foreign Ag associated with MHC class II molecules.
c. in the absence of foreign antigen, mature T-cells are non-responsive to the host's tissues and cells.
d. mature T-cells responds to foreign MHC molecules at a high frequency.

Cytokines and Complement system

1. Cytokines are *not*

a. able to induce increased blood vessel permeability.
b. antigen-specific.
c. made in response to bacterial antigens.
d. signals from one cell that affects the behavior of another cell.

2. Effector functions of complement include all of the following *except*

a. attracting phagocytes to the site of infection.
b. facilitating phagocytosis of complement-coated bacteria.
c. lysing bacterial cells.
d. presenting antigen to B-cells.

3. Which of the following statements about cytokines are *correct*?

P. Cytokines are low-molecular-weight proteins.

Q. There are over 100 different cytokines.

R. Cytokines act by binding to cytokine receptors.

S. A cytokine can only act on a cell that expresses a receptor for it.

a. P and Q
b. Q and S
c. P, R and S
d. P, Q, R and S

4. Which of the following statement(s) about cytokines functions are *correct*?

P. Cytokines can lead to the generation of subpopulations of helper T-cells known as T_H1 and T_H2.

Q. Cytokines of T_H1 cells induce CTLs and NK cells to eliminate intracellular pathogens.

R. Cytokines produced by T_H2 cells support synthesis of specific Ig isotypes and IgE-mediated responses.

S. Cytokines activate a Ras-MAP Kinase signal-transduction pathway.

a. P and Q
b. Q and S
c. P, Q and R
d. P, Q, R and S

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5. Interferon
   a. is species specific.
   b. reacts directly with virus particles to inactivate them.
   c. reacts with cells, and the affected cells then become resistant to a number of different viruses.
   d. is constitutively produced at high levels in cells but requires an inducer for activity.

6. Each of the following statements concerning interferon is correct *except*
   a. it inhibits the growth of both DNA and RNA virus.
   b. its synthesis may induced by dsRNA.
   c. interferons produced by cells of one species acts more effectively in the cells of that species than in the cells of other species.
   d. it acts by preventing viruses from entering the cell.

7. Complement
   a. is a group of active proteolytic enzymes found in serum.
   b. is secreted by macrophages and hepatocytes in response to antigen binding.
   c. participates in both innate and adaptive immune responses.
   d. all of the above statements are true.

8. Which of the following statement(s) about complements are *correct*?
   P. Classical pathway is initiated by IgM and certain IgG subclasses of antibodies.
   Q. Alternative and lectin pathways are antibody independent.
   R. The complement system mediates opsonization of bacteria.
   S. Nucleated cells tend to be more resistant to complement mediated lysis than RBCs.
   a. P and Q
   b. Q and S
   c. P, Q and R
   d. P, Q, R and S

Hypersensitivity and Autoimmunity

1. Type I hypersensitivity is known as
   a. anaphylaxis
   b. agglutination
   c. the transfusion reaction
   d. contact hypersensitivity

2. The antibody IgE and mast cells are associated with
   a. contact hypersensitivity
   b. immunofluorescence tests
   c. electrophoresis
   d. immediate hypersensitivity

3. Allergy symptoms are produced when antigen binds to IgE on FcR on
   a. T_H1 cells
   b. macrophages
   c. mast cells
   d. neutrophils [IISc]

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4. The most common manifestation of penicillin drugs in hypersensitive patients is
   a. pernicious anemia
   b. hemolytic anemia
   c. autocatalytic degradation of penicillin
   d. penicillities

5. Gell and Coombs classified hypersensitivity into 4 types. IgE mediated hypersensitivity belongs to
   a. Type I
   b. Type II
   c. Type III
   d. Type IV [IISc]

6. Cells that release histamine and other vasoactive substances in response to allergens are
   a. neutrophils
   b. macrophages
   c. NK cells
   d. mast cells [IISc]

7. Which of the following disease is *not* an autoimmune disease?
   a. Rheumatoid arthritis
   b. Lupus erythematosus
   c. Bovine spongiform encephalitis [IISc]
   d. Grave's disease

8. Which of the following is *least* likely to lead to autoimmunity?
   a. Loss of suppressor T-cells
   b. Release of sequestered self antigen
   c. Genetic predisposition
   d. Increased clearance of immune complexes

9. The form of diabetes curable by insulin injection is
   a. insulin dependent diabetes mellitus
   b. diabetes rugosa
   c. insulin independent diabetes mellitus
   d. none of the above

10. In myasthenia gravis blocking antibodies bind to
   a. acetylcholine
   b. dopamine
   c. acetylcholine receptors
   d. axons

11. An autoimmune disease of humans usually involving antinuclear antibodies
   a. sclerosis
   b. SLE
   c. rheumatic fever
   d. myasthenia gravis

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Vaccine

1. Live vaccine is
   a. low dose of the infectious bacteria administered as prophylactic.
   b. a dose of bacterial strain in a modified form which retains immunogenicity but is not pathogenic.
   c. a low dose of toxin that is produced by the bacterium.
   d. a sample of cells from a patient who recently recovered from the disease. [JNU]

2. The tuberculin skin test is an example of
   a. type IV delayed hypersensitivity
   b. allergy reaction
   c. serum sickness
   d. precipitation reaction

3. Immunosuppressive measures are most effective when administered
   a. just prior to antigen exposure
   b. one week before antigen exposure
   c. at the time of antigen exposure
   d. following antigen exposure [GATE]

4. Mycobacterial cell wall is the immune system booster present in
   a. alum
   b. Freund's incomplete adjuvant
   c. Freund's complete adjuvant
   d. all of the above

5. Which of the following statement(s) are *correct*?
   P. Attenuated vaccines are more likely to induce cell-mediated immunity than killed vaccines.
   Q. Multivalent subunit vaccines generally induce a greater response than synthetic peptide vaccines.
   R. A DNA vaccine only induces a response to a single epitope.
   S. Transplacental transfer of maternal IgG antibodies against measles confers short-term immunity on the fetus.
   a. P and Q
   b. Q and S
   c. P, Q and S
   d. P, Q, R and S

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Chapter 13

Plant Physiology

Plant water relationship/Transport of minerals and water

1. Which of the following statements is *correct* about diffusion?
   a. It is very rapid over long distances.
   b. It requires an expenditure of energy by the cell.
   c. It is a passive process.
   d. It occurs when molecules move from a region of lower concentration to one of higher concentration.

2. Osmosis is a form of diffusion in which
   a. the solute moves freely from a region of higher concentration to lower concentration through a semipermeable membrane.
   b. the solvent moves through a semipermeable membrane from region, where a solute is in higher concentration to region of lower concentration.
   c. the solvent moves through a semipermeable membrane from higher solvent concentration to lower solvent concentration.
   d. solute as well as solvent moves freely from a region of higher concentration to lower concentration through a semipermeable membrane.

3. I place a cell in a solution. Over a period of time, I notice that the cell shrinks, as if it is losing water. Which of the following seems likely?
   a. The solution is a strong buffer.
   b. The solution is an acid.
   c. The solution has more dissolved solutes than the cell does.
   d. The solution has fewer dissolved solutes than the cell does.

4. Water potential
   a. of a solution is always greater than for pure water.
   b. is the potential energy of water in a system.
   c. is a measure of the level of the active movement of water through a system.
   d. is never zero.

5. If a plant cell is placed in deionised water the water potential of that cell becomes
   a. more positive because the pressure potential becomes more positive.
   b. more positive because the osmotic potential becomes more negative.
   c. more negative because the pressure potential becomes more negative.
   d. less negative because the pressure potential becomes more positive.

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6. The sap of a plant cell has an osmotic potential of −10 bars and there is a wall pressure of 2 bars. When this cell is placed in a solution present in an open beaker having an osmotic potential of −3 bars the force causing water to enter the cell is

a. −8 bar
b. −7 bar
c. −5 bar
d. −3 bar

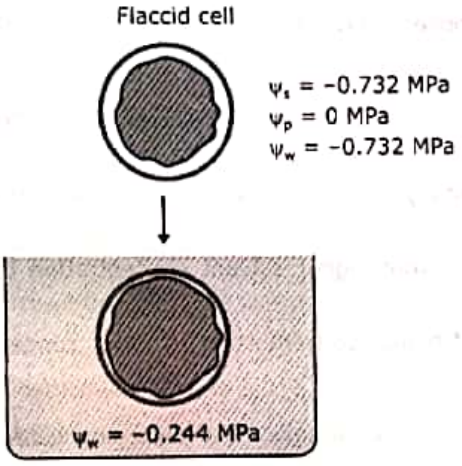
7. A root cortex cell has a solute potential of −0.5 MPa. The water potential of the soil is −0.3 MPa. At what turgor pressure would the root cortex cell no longer take up water?

a. 0 MPa
b. −0.15 MPa
c. +0.15 MPa
d. +0.2 MPa

8. A limp lettuce leaf has an osmotic potential of −4 MPa. The lettuce leaf is placed into a beaker of pure water. When the cells of the lettuce leaf are fully turgid the water potential of the leaf cells will be

a. +4 MPa
b. +8 MPa
c. −4 MPa
d. 0.0 MPa

9. A flaccid cell (in air) is dropped in the 0.1 M sucrose solution. Because the starting water potential of the cell is less than the water potential of the solution, the cell takes up water. After equilibration, what will be the value of positive turgor pressure?



a. 0 Mpa
b. 0.244 Mpa
c. 0.732 Mpa
d. 0.488 Mpa

10. Which of the following statements are *correct*?

P. Water potential is a measure of the free energy of water per unit volume.
Q. Water potential is influenced by concentration, pressure and gravity.
R. An open beaker full of pure water at 20°C has 0 MPa water potential value.
S. Water in plants moves from higher to lower water potentials.

a. P and Q
b. P and S
c. Q, R and S
d. P, Q, R and S

11. A cell at incipient plasmolysis, with a solute potential of −2000kPa, is placed in a solution of water potential −1200kPa. The direction of flow of water will be

a. from cell to solution
b. from solution to cell
c. data incomplete
d. no flow of water

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12. If both leaf water potential and leaf osmotic potential in mid afternoon on a hot summer day equals −15 bars
    a. the leaf is probably wilted.
    b. the whole plant is at the permanent wilting point.
    c. leaf cells have highest wall pressure.
    d. the soil is at the plant's permanent wilting point.

13. Which of the following plays no role in the movement of water through the xylem of plants?
    a. Capillarity
    b. Root pressure
    c. H^+/ATPase pump at the xylem element membrane
    d. Transpirational pull

14. Water cohesion
    a. creates the pulling force which pulls water upward in the xylem.
    b. causes water to move from the xylem into the phloem in response to differences in water concentration.
    c. is an energy requiring process.
    d. is responsible for making the column of water.

15. The class of water in the soil, that provides most of the water to plants and is thus the most important, is
    a. gravitational
    b. field capacity
    c. capillary
    d. hygroscopic

16. Halophytes such as mangroves meet high osmotic pressures in the soil. They overcome the problem of water uptake by
    a. increase in the root to shoot ratio.
    b. reduction in the number of stomata to reduce transpiration.
    c. accumulation of electrolytes in the vacuoles.
    d. growth at relatively high humidity to reduce transpiration.

17. Root pressure is a/an
    a. non-osmotic phenomenon
    b. osmotic phenomenon
    c. positive hydrostatic pressure
    d. more than one statement is correct

18. Which of the following statement is *correct*?
    a. Members of gymnosperm show high root pressure.
    b. Actively transpiring plants show high root pressure.
    c. Root pressure is mainly responsible for ascent of sap.
    d. None of the above.

19. Transpiration pull depends on
    a. adhesion of water molecules to the walls of phloem cells
    b. capillarity
    c. the very negative water potential of the atmosphere
    d. cohesion of water molecules to each other

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20. The source of energy driving the transport of water through the xylem is
    a. ATP produced by photosynthesis
    b. ATP generated by respiration
    c. the sun
    d. transpiration

21. The water within xylem vessels moves toward the top of a tree (long distances) as a result of
    a. atmospheric pressure on roots
    b. active transport of ions into the vascular bundle
    c. evaporation of water through stoma
    d. the force of root pressure

22. Which combination of characteristic of a vessel element is most important for water movement in the xylem?
    a. Rigid cell wall, cell death at maturity, end walls absent
    b. Rigid cell wall, reduction in size of plastids and mitochondria, end walls present
    c. Rigid cell wall, living cell membrane, and walls absent
    d. Flexible cell wall, nucleus anchored to the cell membrane, end walls present [MKU]

23. Which of the following properties of water is most directly related to its ability to rise in the capillary spaces of plants?
    a. Neutral pH
    b. High density
    c. Low compressibility
    d. High surface tension [MKU]

24. Which of the following observations shows that ion uptake in plants is energy dependent?
    a. Ion uptake shows saturation kinetics.
    b. Ion uptake rates are different for different ions.
    c. Some ions accumulate against a concentration gradient.
    d. Some ions enter the symplast before reaching the endodermis.

25. How does a plant accumulate ions from the soil, which has a lower concentration than that in the plant?
    a. Applies pressure to push ions into the cell.
    b. Applies tension to pull the ions into the cell.
    c. Expends energy to transport ions against the concentration gradient.
    d. Roots grow around soil particles trapping the ions in a cell.

26. Which statement about the function of the casparian strip is *correct*?
    a. It prevents excess transpiration from leaves.
    b. It regulates ions movement into the root vascular cylinder.
    c. It prevents disease causing organisms from invading the plant.
    d. It is the pathway for nutrient transfer from xylem to phloem.

27. Mark the *correct* statements.
    P. Resistance to water flow in root cortex will be higher for symplastic pathway.
    Q. Major pathway followed by ions from the epidermis to tracheary element of root is symplastic.
    R. The Casparian strip prohibits H_2O from passing through the apoplast.

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S. Absorption of ions in plants is controlled by carriers and pumps.

a. P and Q
b. P, Q and R
c. P, Q, R and S
d. none of the above

28. Ion absorption by plant roots

a. can be explained by the nutrient-carrier hypothesis.
b. is basically ions passively soaking in root cells along with soil water.
c. explains the movement of ions from a dilute soil solution into a more concentrated solution.
d. more than one statement is correct.

29. When a cell membrane moves substances from a region of lower concentration to a region of higher concentration, and expends energy in the process, this type of movement is called

a. osmosis
b. diffusion
c. active transport
d. facilitated diffusion

30. Ion absorption by symplast occurs

a. passively by facilitated diffusion
b. actively by primary and secondary active transport
c. passively by free diffusion
d. both a and b

31. In plant roots, the casparian strip is *correctly* described by which of the following?

a. It is located in the walls between endodermal cells and cortex cells.
b. It provides energy for the active transport of minerals into the vascular bundle from the cortex.
c. It ensures that all minerals are absorbed from the soil in equal amounts.
d. It ensures that all water and dissolved substances must pass through a cell before entering the vascular bundle.

32. Ion transport in root occurs

P. passively through channels
Q. actively through channels
R. actively through carriers
S. through both symplast and apoplast

a. P and R
b. Q, R and S
c. P, R and S
d. R and S

33. Ion channels

P. always require ATP in order to function
Q. are always open
R. are integral membrane proteins
S. are responsible for the selective permeability of a membrane

a. P and Q
b. R and S
c. P, R and S
d. P, Q, R and S

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34. The Casparian strip

P. limits the pathway available to water and solutes, forcing them to enter the symplast

Q. surrounds the root vascular tissue

R. allows water to move down a water potential gradient

S. is made of suberin

a. P and Q
b. R and S
c. P, R and S
d. P, Q, R and S

Transpiration

1. Which of the following statement(s) is/are *correct*?

a. Water lost by transpiration is pure water.
b. The cause of guttation is mainly root pressure.
c. Rate of transpiration is dependent upon vapour pressure deficit.
d. All the above statements are correct.

2. Which statement concerning stomatal function is *false*?

a. To close stoma, potassium ions diffuse passively out of the guard cells.
b. When stomata are about to open, potassium ions are actively transported into the guard cells from the surrounding cells of the epidermis.
c. Stomata are open when the turgor pressure of the guard cells decreases.
d. Low levels of carbon dioxide stimulate the opening of stomata.

3. Below are given some statements related to transpiration in plants. Which of the following statement is *correct*?

P. The major function of stomata in plants is transpiration.

Q. The movement of water during transpiration is due to adhesion of water to the xylem walls.

R. Transpiration through stomata is a regulated process.

S. Maximum stomatal transpiration occurs in green light.

a. P, Q and R
b. Q and R
c. P, Q and S
d. Q, R and S

4. Stomatal pores open when

a. H^+ ions are pumped into the guard cells.
b. K^+ ions are pumped out of the guard cells.
c. water moves into the guard cells in response to a decrease in water potential in the guard cells.
d. water moves out of the guard cells in response to a decrease in water potential of the guard cells.

5. Which of the following statements is *incorrect*?

a. A root concentrates minerals by active transport.
b. To develop root pressure, energy is used to actively transport minerals into root cells.
c. Excretion of minerals is not a function of transpiration.
d. High carbon dioxide concentrations cause stomata to open.

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6. Match the different types of instruments mentioned in column I with their uses in column II

Column-I Instrument	Column-II Use
P. Auxanometer	1. Measurement of size of stomatal pore
Q. Potometer	2. Measurement of blood pressure
R. Porometer	3. Measurement of transpiration rate
S. Sphygmomanometer	4. Measurement of growth rate in plants

a. P-4, Q-1, R-3, S-2
b. P-4, Q-3, R-2, S-1
c. P-1, Q-3, R-2, S-4
d. P-4, Q-3, R-1, S-2

7. Most of the water taken up by the plant is
a. split during photosynthesis as a source of electrons and hydrogen.
b. lost by transpiration through stomata.
c. absorbed by cells during their elongation.
d. incorporated directly into organic material.

8. Which of the following statement(s) is/are *incorrect*?
a. Classical theory of opening and closing of stomata by Sayre and Scarth is based on change in pH.
b. Pigments which absorbs blue light in guard cell is zeaxanthin.
c. Radial micellation on guard cell is made up of cellulose microfibril.
d. None of the above.

9. The *correct* match is
1. instrument used to measure transpiration pull
2. instrument used to measure the size of stomata
3. instrument used to measure the atmospheric pressure
4. amount of osmosis per unit time

a. 1. Potometer	2. Manometer	3. Porometer	4. Osmometer
b. 1. Porometer	2. Manometer	3. Potometer	4. Osmometer
c. 1. Manometer	2. Potometer	3. Porometer	4. Osmometer
d. 1. Potometer	2. Porometer	3. Manometer	4. Osmometer

10. If you put a cell into pure water from a wilted plant, which of the following will happen?
a. More water molecules will move out of the cell than moves in.
b. There will be no change in the concentration of water inside or outside of the cells.
c. More water will move into the cell than moves out.
d. The cell will plasmolyze.

Plant nutrition

1. Essential elements are
a. not required for normal reproduction
b. not replaceable by other elements
c. not directly involved with metabolism
d. required only in those metabolic processes leading to increased growth

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2. Choose the mismatch

Mineral nutrient	Function
a. B	cell elongation
b. Mo	N_2 fixation
c. Na	C_3 pathway
d. Cl	photolysis of water

3. Choose the *incorrect* pair

Mineral nutrient	Deficiency symptom
a. Nitrogen	chlorosis
b. Zinc	reduction in internodal growth
c. Nickel	accumulation of urea
d. Sodium	whiptail disease

4. Which of the following is an *incorrect* match of essential element and function?
   a. Manganese–structural component of chlorophyll
   b. Calcium–component of the middle lamella
   c. Zinc–enzyme activator
   d. Iron–component of ferredoxin

5. Legume root nodules contain leghemogobin. Its function is to regulate
   a. expression of *nif* genes
   b. dinitrogenase activity
   c. oxygen supply
   d. nodule growth [JNU]

6. How many ATP are required for the conversion of one N_2 to $2NH_4^+$ during biological nitrogen fixation?
   a. 8 ATP
   b. 10 ATP
   c. 12 ATP
   d. 16 ATP

7. Conversion of NO_3^- to NO_2^- occurs in plant cell
   P. occur in the cytosol
   Q. catalyzed by nitrate reductase
   R. requires Mo and FAD as cofactor
   S. reduced ferredoxin acts as electron donor
   Which of the following is *correct*?
   a. P and S
   b. P, Q and R
   c. P, Q, R and S
   d. only Q

8. Carnivorous adaptations of plants mainly compensate for soil that has a relatively low content of
   a. potassium
   b. nitrogen
   c. phosphate
   d. calcium

9. A mineral deficiency is likely to affect older leaves more than younger leaves if
   a. the mineral is a micronutrient.
   b. the mineral is very mobile within the plant.
   c. the mineral is required for chlorophyll synthesis.
   d. the deficiency persists for a long time.

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10. Which of the following would be most likely to develop deficiency symptoms?
a. A plant growing in a tropical rainforest with associated mycorrhizae.
b. Cacti growing in the desert.
c. A crop plant growing in a field that has been cultivated for many years.
d. A plant growing naturally on calcareous soil.

11. Which of the following is correct about *nitrate reductase*?
a. FAD, Mo and heme act as prosthetic groups.
b. It catalyzes conversion of NO_3^- to NH_4^+.
c. $FADH_2$ acts as electron donor.
d. It is very sensitive to O_2.

12. Which of the following is *not* correct about nitrogen fixation?
a. It requires metabolic energy.
b. It is carried out only by few prokaryotic organisms termed diazatrophs.
c. Enzyme that catalyzes the reduction of N_2 is called nitrogenase complex.
d. None of the above.

13. Which of the following is *false* statement?
a. All NH_4^+ seems to be converted first into the amide group of glutamine.
b. Conversion of organic nitrogen to NH_4^+ by soil microbes is called ammonification.
c. Cl^- is required for photochemical reaction involved in photolysis of water.
d. Plant roots absorb ions as well as the salts.

Plant hormones

1. Which of the following is *incorrect* about plant hormones?
a. Influence plant growth and development at low concentrations.
b. Abscisic acids are synthesized from carotenoids.
c. Are organic compounds.
d. Include compounds such as florigen, which has been isolated from the phloem and induces a plant to flower.

2. Which one of the following is *not* a characteristic of plant hormone?
a. They are active in small concentration.
b. They are pleiotropic in their effects.
c. They are synthesized in specific glands and tissues. **[JNU]**
d. Plant hormone responses are tissue and time specific.

3. Charles Darwin and his son Francis experimented with phototropism of grass seedlings by placing a metal for blindfold over different parts of the seedlings coleoptiles. A simplified version of their results is shown below. Which of the following statements best explains their results?
a. The light signal is perceived below the tip, and these cells cause the coleoptiles to grow toward the light.
b. Both the seedling root and coleoptiles perceive and respond to light in the same manner.
c. A chemical messenger must travel from the base of the coleoptile to the tip.
d. The light signal is perceived at the tip of the coleoptile, but the growth response occurs a few millimeters below the tip.

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4. Which of the following statement is *false* about auxin?
   a. Accumulates in root tips and branch root initials.
   b. Transport away from leaf primordia causes cells to differentiate into vascular tissue.
   c. In high concentrations in tissue culture media will promote root formation.
   d. Synthesis by shoot apex results in the phenomenon of apical dominance.

5. The geotropic response in roots
   a. involves the redistribution of calcium ions.
   b. primarily involves the basipetal transport of ABA in the root cortex.
   c. can be abolished by the removal of the root cap.
   d. involves the increased transport of auxin into the lower side of the root elongation zone.

6. Bioassay for gibberellins is
   a. *Avena* coleoptile test
   b. α-amylase induction in cereal endosperms
   c. callus formation in tobacco from pith tissue
   d. formation of abscission layer in cotton [JNU]

7. Which of the following plant hormones is *incorrectly* paired with its function?
   a. Auxins — responsible for apical dominance
   b. Abscisic acid — regulates the rate of transpiration
   c. Cytokinins — delays senescence
   d. Gibberellins — promotes bud and seed dormancy

8. In 1925, the Japanese biologist Kurosawa showed that the *bakanae* (foolish seedling) disease of rice was caused by a substance produced by the fungus *Gibberella fujikuroi*. The substance was named gibberellin and was shown to cause elongation of stem tissues. Which of the following statements about gibberellin is false?
   a. Pea plants carrying the dwarf mutation would be expected to have higher levels of gibberellin in their stems than normal plants.
   b. If gibberellin is applied to the stems of dwarf pea plants, the stems elongate so that plant reach normal height.
   c. Dwarf pea plants have a mutation in the gibberellin biosynthetic pathway.
   d. The gibberellin produced by the fungus caused normal healthy rice plants to become unusually tall.

9. Lateral stem development is controlled by the relative levels of
   a. cytokinins and auxins
   b. abscisic acid and auxins
   c. auxins and gibberellins
   d. auxins and ethylene

10. Which of the following hormone, in combination with auxin, stimulates cell division in plants and determines the course of differentiation?
   a. Ethylene
   b. Indoleacetic acid
   c. Gibberellins
   d. Cytokinin

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11. Gravisensing (the mechanism by which plant tissues sense gravity)
P. is thought to involve *statoliths*, starch grains inside of cells which fall to the lower side of the cell.
Q. is thought to involve pressure-sensitive channels at the top and bottom of cells.
R. leads to a redistribution of auxin in stems and roots, causing differential growth.
S. in roots, relies on the presence of a root cap.
a. P and Q are correct
b. P, Q and R are correct
c. only P is correct
d. P, Q, R and S are correct

12. In vascular plants, most cytokinins are produced in the
a. roots
b. shoots
c. leaves
d. lateral branches

13. Although abscisic acid is not involved in the formation of the abscission layer, it does have many roles in the life of plants. Which of the following is a function of abscisic acid?
P. In times of water stress, it causes the stomata to close.
Q. It promotes tolerance to stress.
R. It induces leaf and flower senescence.
S. It induces and maintains the dormancy of seeds.
a. P and Q
b. R and S
c. P, R and S
d. P, Q, R and S

14. If an inhibitor of metabolism were applied to a tissue transporting auxin, which of the following would happen?
a. It would have no effect
b. The direction of transport would reverse
c. Transport would be inhibited
d. The rate of transport would increase

15. Which of the following is *incorrect*?
a. Root apical meristem is major site of synthesis of the free ctytokinin.
b. Amino acid methionine is the precursor of ethylene.
c. Kinetin is the most abundant naturally occurring cytokinin in higher plants.
d. IAA is synthesized from the amino acid tryptophan.

16. Which of the following is not part of the triple response of etiolated pea seedlings to ethylene?
a. Stunted growth
b. Increased radial expansion of cells of the stem
c. Loss of gravity perception
d. Unfolding of the apical hook of the seedling

17. Which of the following statement is *correct*?
a. Blue light causes phototropism.
b. Auxin movement is non-polar.
c. TIBA is not an anti auxin substance.
d. ABA is transported only through phloem.

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18. Ethylene receptor
    a. is a membrane-bound dimer of two component system.
    b. is a soluble protein present in the cytoplasm.
    c. has a chromophore similar to phytochrome.
    d. has pterin and flavin as prosthetic groups. [JNU]

19. Environmental signals influence the distribution of an auxin in a plant by
    a. decreasing the cell's sensitivity to the auxin.
    b. causing auxin to migrate to the lighted portion.
    c. destroying the auxin.
    d. causing auxin to migrate into the shaded portion.

20. What does leaf abscission have in common with fruit ripening?
    a. Both involve cell swelling.
    b. Both are induced by ethylene.
    c. Both involve the dissolution of the cell wall by cellulase.
    d. Both are inhibited by auxin.

21. The accumulation of one of the following causes seed dormancy
    a. cytokinin
    b. auxin
    c. abscisic acid
    d. gibberellin [JNU]

22. Which of the following is *incorrect* about ABA?
    a. Inhibits germination of many mature seeds and vivipary.
    b. Causes seed and bud dormancy in some species.
    c. Causes leaves senescence independently of ethylene.
    d. Synthesized only in roots and transported to shoots.

23. Which of the following statement(s) is/are *correct*?
    a. Acid growth hypothesis states that auxin causes receptive cells to pump protons into their cell walls.
    b. Unidirectional blue light causes a redistribution of auxin in the stem apex.
    c. ABA synthesis occurs in chloroplast by the mevalonic acid pathway.
    d. All the above statements are correct.

24. Leaf abscission
    a. is followed by leaf senescence.
    b. is regulated by the hormone abscisic acid.
    c. is a passive process, involving nothing more than death of petiole cells.
    d. is associated with an increase in ethylene production by petiole cells.

25. Which of the following is *incorrect* about ethylene?
    a. Promotes elongation of stems and root especially in dicot.
    b. Biosynthesis involves the enzyme ACC synthase.
    c. Will promote flowering in Bromeliads (the pineapple family).
    d. It enhances lateral or radial expansion of cell.

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26. The production of α-amylase in cereal grains
a. results in the degradation of sugar into starch.
b. is induced by gibberellin from the germinating embryo.
c. occurs in the starchy endosperm cells.
d. occurs throughout seed development.

27. Which of the following is *not* related to cytokinin?
a. Promotes apical dominance
b. Promotes chloroplast maturation
c. Cause Richmond-Lang effect
d. Promotes nutrient mobilization

28. Which one of the following phytohormones is produced under water-deficient condition and plays an important role in the tolerance response of plants to drought?
a. Abscisic acid
b. Cytokinin
c. Ethylene
d. Gibberellin [IISc]

29. Which of the following phytohormones play a role in seed dormancy?
a. Gibberellins
b. Abscissic acid
c. Cytokinins
d. Auxins [IISc]

30. Abscisic acid controls which of the following?
a. Abscission
b. Fruit climacteric
c. Stomatal closure
d. Cell elongation [MKU]

31. Abscisic acid
P. is synthesized from mevalonic acid
Q. is synthesized in mature leaves (major site)
R. is transported from both xylem and phloem
S. causes differentiation of vascular tissues
a. P and Q
b. Q and R
c. P, Q and R
d. P, Q, R and S

32. Ethylene
P. is synthesized from methionine
Q. shows polar transport
R. causes leaf and fruit abscission
S. is synthesized primarily in root tips
a. P and Q
b. Q and R
c. P and R
d. P, Q, R and S

33. Which one of the following statements is correct about indole acetic acid (IAA) polar transport?
a. Cytosolic IAA is largely uncharged and can flow out passively.
b. Cytosolic IAA is largely charged and cannot flow out passively.
c. Apoplastic IAA is largely uncharged and thus requires carrier proteins.
d. Apoplastic IAA is largely charged and enters the cell passively.

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Photomorphogenesis and Vernalization

1. Light stimulates seed germination, promotes chlorophyll synthesis and the development of chloroplasts and promotion of leaf expansion. These responses induced by light are examples of
   a. photoperiodism
   b. photomorphogenesis
   c. phototropism
   d. photosynthesis

2. Photoperiodism
   a. is a plant's response to the direction of light.
   b. is a plant's response to the absence of light.
   c. is a plant's response to a change in the amount of light in a 24 hour period.
   d. is a plant's response to presence of light.

3. Which of the following is *false* about phytochrome?
   a. Phytochrome pigment present in two photo reversible forms Pfr and Pr.
   b. Phytochrome in all plants is synthesized entirely as Pr.
   c. Chromophore present in phytochrome is an open chain tetrapyrole ring.
   d. Phytochrome is involved in light energy transfer.

4. The pigment that plays a key role in photomorphogenesis is
   a. chlorophyll
   b. phytochrome
   c. cytochrome
   d. anthocyanin [IISc]

5. Photoreversebility defines involvement of which plant pigment/photoreceptor in plant growth and development
   a. chlorophyll
   b. cryptochrome
   c. phototropin
   d. phytochrome [MKU]

6. There are three modes for phytochrome responses. These responses can be distinguished by the amount of light required. Choose the *correct* statements about the different phytochrome responses.
   P. VLFRs are activated by extremely low light intensities.
   Q. HIR does not show red and far-red photoreversibility.
   R. VLFR does not obey the law of reciprocity.
   S. Classical red and far-red photoreversible responses of phytochromes are example of LFRs.
   a. P, R and S
   b. Q, R and S
   c. P, Q and S
   d. P, Q, R and S

7. Type I phytochrome
   a. is destroyed in the light
   b. is synthesized in the far-red-absorbing form (Pfr)
   c. reverts to type II phytochrome in the dark
   d. senses red light, while type II senses far-red light

8. Which of the following function is *not* associated with phytochrome?
   a. Nyctinastic movement in *Mimosa*
   b. Seed germination in lettuce

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c. Stomatal opening
d. Circadian rhythms

9. Which statement about photomorphogenesis is *false*?
a. Involves the protein pigment phytochrome.
b. Is only important at the seedling stage of plant development.
c. Is affected by red, far-red, and blue light.
d. Can be defined as light dependent changes in plant growth, development and metabolism.

10. Which of the following statement is *incorrect*?
a. Photoperiodic stimulus is perceived by leaves.
b. Climacteric is the rise in respiration in the ripening fruit.
c. Photoperiodism is non-directional developmental response to a non-directional, periodic light stimulus.
d. None of the above.

11. The light which stimulates flowering in an inducible plant
a. is perceived by the apical meristem
b. is perceived throughout the plant
c. is perceived by the leaves
d. none of the above are correct

12. Which of the following is *true* for photomorphogenesis?
a. It is non-directional developmental process
b. It involves phytochrome pigment
c. It does not depend on direction of light stimulus
d. All of the above

13. Phototropins
P. are blue-light receptors
Q. include phototropism
R. are autophosphorylating protein kinases
S. absorbs red light and undergo phosphorylation
a. P, Q and R
b. P, R and S
c. P, Q and S
d. Q, R and S

14. Vernalization
a. prepares plant for flowering
b. initiates flowering
c. is low temperature treatment for seed germination
d. all of the statements are correct

15. Which of the following is *false* about vernalization?
a. Site of perception of vernalization stimulus is stem apex.
b. Vernalization does not induce flowering but prepares the plants for flowering.
c. Low temperature effect on germination.
d. Genetic makeup is not changed by the cold temperature treatment.

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Seed physiology

1. To break dormancy in a seed with a rudimentary embryo
   a. one should soak the seed to remove inhibitors
   b. scarify, to break the seed coat
   c. germinate under red light, to promote germination
   d. none of these will work

2. Seed coat imposed dormancy (in which a seed will not germinate under the usual conditions) is due to
   P. prevention of water takes up as in many legumes
   Q. interference with gas exchange as in cocklebur
   R. mechanical constraints as in lettuce seed
   S. retention of inhibitor i.e. prevents escape of inhibitors in xanthium
   a. P and Q are correct
   b. P, Q and R are correct
   c. only P is correct
   d. all P, Q, R and S are correct

3. Which statement about seed germination is *false*?
   a. The radicle elongates first and absorbs water needed for germination.
   b. The shoot apical meristem is formed after germination.
   c. The plumule expands and functions in photosynthesis.
   d. The hypocotyl elongates and carries the plumule into the light. [MKU]

4. Which of the following is *false* about glyoxylate cycle?
   a. It is a variant of citric acid cycle.
   b. It occurs in glyoxisome.
   c. It results in the formation of succinate from acetyl-coA.
   d. It occurs in all eukaryotic cells.

5. Which of the following statements are *correct*?
   P. Gibberellin and cytokinin promotes seed germinations.
   Q. *Latuca sativa* have positive photoblastic seeds.
   R. Orthodox seeds can withstand extremely low temperature and reduced moisture.
   S. Primary effect of light on seed germination is mediated by phytochrome.
   a. P and Q
   b. Q and R
   c. P, Q and R
   d. P, Q, R and S

Plant movements

1. Phototropism
   P. involves chromoprotein phototropin
   Q. is induced by blue light
   R. is an auxin mediated phenomenon
   S. is a phytochrome mediated response
   a. P and S
   b. P, R and S
   c. P, Q and R
   d. Q, R and S

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Phototropism refers to how plants respond to sunlight by bending towards the light. This is due to
a. cell division on the side of the stem closer to the light source.
b. cell division on the side of the stem away from the light source.
c. cell differentiation on the side of the stem away from the light source.
d. cell elongation on the side of the stem away from the light source.

Unlike tropisms, nastic movements are in response to [MKU]
a. paratonic stimuli
b. gravity
c. non-directional stimuli
d. directional stimuli

Which of the following movement of plants are reversible and allow the plant to advantageously orient leaves?
a. Thigmotropisms
b. Turgor movements
c. Phototropisms
d. Gravitropisms

Choose the *correct* statement?
a. Tropic movement is autonomic movement
b. Nastic movement depends on direction of stimulus
c. Growth of pollen tube is chemotactic movement
d. None of the above

Nyctinasty
P. occurs due to differential growth rate
Q. occurs due to change in pressure potential
R. involves motor cells
S. mediated by blue light
a. P and Q
b. Q, R and S
c. P, R and S
d. P, Q, R and S

Tactic movement
P. is a movement of locomotion
Q. is dependent on direction of stimulus
R. occurs due to differential growth
S. is a paratonic movement
a. P and Q
b. Q and R
c. P, Q and S
d. P, Q, R and S

Which of the following is/are *incorrect* about nastic movement?
P. It is a paratonic movement.
Q. It occurs due to change in turgor pressure as well as differential growth.
R. Nyctinasty is an example of nastic movement.
S. It is induced by endogenous stimulus.
a. only P
b. only S
c. P and Q
d. P and R

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Chapter 14

Human Physiology

Nervous system

1. Which is *true* for the autonomic nervous system?
   a. The sympathetic nervous system always stimulates the organ system.
   b. The sympathetic nervous system always inhibits the organ system.
   c. The parasympathetic nervous system always stimulates the organ system.
   d. It depends on the organ system whether the division stimulates or inhibits it.

2. Which of the following statement is *true*?
   a. Peripheral nerve is similar to smooth muscle in terms of the connective tissue investment.
   b. Most nerves contain afferent and efferent fibers and thus carry both motor and sensory signals.
   c. Nodes of Ranvier are most easily seen in cross-section of peripheral nerve.
   d. None of the above

3. Which of the following cells is responsible for myelin formation in the peripheral nervous system?
   a. Astrocyte
   b. Oligodendrocyte
   c. Schwann cell
   d. Microglial cell

4. Which of the following is *not* done by glial cells?
   a. Receiving and conducing electrochemical signals.
   b. Giving metabolic support to neurons.
   c. Producing insulating sheaths around axons.
   d. Removing debris after the death of a neuron. [JNU]

5. The perineurium is the connective tissue layer
   a. surrounding an entire nerve.
   b. surrounding individual axons in the CNS.
   c. surrounding individual axons in the PNS.
   d. surrounding fascicles of axons in the PNS.

6. The system that controls smooth muscle, cardiac muscle and gland activity is the
   a. somatic nervous system
   b. autonomic nervous system
   c. skeletal division
   d. sensory nervous system

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Neurotransmitters are removed from the synaptic cleft by
1. axonal transport
2. diffusion
3. neurosecretory cells
4. enzymatic breakdown
5. cellular uptake

a. 1, 2, 3 and 4
b. 2, 4 and 5
c. 2, 3 and 4
d. 1, 4 and 5

8. Which of the following is *true* of the sympathetic nervous system?
a. It's voluntarily controlled via the forebrain.
b. It's voluntarily controlled via the reticular formation.
c. It uses different neurotransmitters at the ganglion and at the synaptic cleft.
d. It's a subdivision of the somatic nervous system.

9. The effect of parasympathetic nervous stimulation on the heart is
a. slowing of the heart
b. increased activity of the SA node
c. increased activity of the AV node
d. increased force of contraction

10. The action potential of a neuron
a. is initiated by efflux of Na^+
b. is terminated by efflux of K^+
c. declines in amplitude as it moves along the axon
d. results in a transient reversal of the concentration gradient

11. Action potentials are conducted more rapidly in
a. small diameter axons than large diameter axons.
b. large diameter axons than small diameter axons.
c. unmyelinated axons than myelinated axons.
d. axons that lack a wrapping of Schwann cells.

12. An inhibitory neuron could affect the neuron with which it synapses by
a. producing an IPSP within the neuron
b. hyperpolarizing the neuron
c. increasing K^+ efflux from the neuron
d. all of the above

13. Which of the following comparisons are *true*?
1. Nerve impulses produce their effects quickly, whereas hormonal responses generally are slower.
2. Nervous system effects are brief, whereas endocrine system effects are longer lasting.
3. The nervous system controls homeostasis through nerve impulses; the endocrine system works through hormones.
4. The nervous system can stimulate or inhibit the release of hormones by the endocrine system.
5. The nervous system includes organs scattered throughout the body; the endocrine system consists of only the pituitary gland, thyroid gland, adrenal gland, and pancreas.

a. 1, 2, 3, 4 and 5
b. 1, 2, 3 and 4
c. 2, 3, 4 and 5
d. 2, 4 and 5

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14. Long term reflex actions such as cycling and swimming are controlled by
    a. cerebellum
    b. spinal cord
    c. hypothalamus
    d. cerebrum

[IISc]

15. Which of the following sequences describes the passage of an action potential in the neuron?
    a. Axon, cell body, dendrite, synaptic cleft
    b. Synaptic cleft, axon, dendrite, cell body
    c. Dendrite, synaptic cleft, cell body, axon
    d. Dendrite, cell body, axon, synaptic cleft

[MKU]

16. Which of the following statements are *true*?
    1. The frequency of impulses and the number of activated sensory neurons encode differences in stimuli intensity.
    2. Larger-diameter axons conduct nerve impulses faster than smaller-diameter ones.
    3. Continuous conduction is faster than salutatory conduction.
    4. The diameter of an axon and the presence or absence of a myelin sheath are the most important factors that determine the speed of nerve impulse propagation.
    5. Action potentials are localized, whereas graded potentials are propagated.

    a. 1, 3 and 5
    b. 3 and 4
    c. 2, 4 and 5
    d. 1, 2 and 4

17. Which of the following statements are *true*?
    1. Two or more neurotransmitters may be present in many neurons.
    2. Some amino acids, including glutamate and aspartate, are neurotransmitters.
    3. The catecholamine neurotransmitter acetylcholine is synthesized from the amino acid tyrosine.
    4. The simple gas nitric oxide functions as a neurotransmitter.
    5. An excitatory neurotransmitter can never be inhibitory regardless of the neuron that produces it.

    a. 1, 2 and 4
    b. 2, 3 and 5
    c. 1, 3 and 5
    d. 1, 3 and 4

18. Match the following

Column A	*Column B*
1. Graded potential	a. A sequence of rapidly occurring events that decreases and eventually reverses the membrane potential and then restores it to the resting state; a nerve impulse.
2. Action potential	b. A small deviation from the resting membrane potential that makes the membrane either more or less negative.
3. Excitatory postsynaptic potential	c. A neurotransmitter-caused hyperpolarization of the postsynaptic membrane.
4. Inhibitory postsynaptic potential	d. Polarization that is less negative than the resting level.
5. Depolarizing graded potential	e. A neurotransmitter-caused depolarization of the postsynaptic membrane.

    a. 1-a, 2-b, 3-e, 4-c, 5-d
    b. 1-b, 2-a, 3-e, 4-c, 5-d
    c. 1-d, 2-a, 3-c, 4-e, 5-b
    d. 1-d, 2-e, 3-a, 4-c, 5-b

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19. Match the following

Column: A	*Column: B*
1. Myelin sheath	a. The part of the neuron that contains the nucleus
2. Neurolemma	b. Rough endoplasmic reticulum in neurons
3. Nodes of Ranvier	c. A multilayered lipid and protein covering for axons produced by neuroglia
4. Cell body	d. The outer nucleated cytoplasmic layer of a Schwann cell
5. Nissl bodies	e. Form the cytoskeleton of a neuron
6. Neurofibrils	f. Gaps in the myelin sheath of an axon
7. Axon hillock	g. Area where the axon joins the cell body

a. 1-d, 2-c, 3-f, 4-a, 5-b, 6-e, 7-g
b. 1-c, 2-d, 3-f, 4-a, 5-b, 6-e, 7-g
c. 1-f, 2-d, 3-c, 4-b, 5-a, 6-e, 7-g
d. 1-c, 2-d, 3-a, 4-f, 5-b, 6-g, 7-e

Sense organs

1. The most sensitive vertebrate chemoreceptors known are the
   a. rod and cone cells of mammals
   b. taste receptors of fishes
   c. organs of Corti of humans
   d. olfactory receptors of mammals

2. Which of the following statement is *correct*?
   a. In order to produce a clear image, the aqueous humor is involved in the process of accommodation.
   b. The space anterior to the lens is filled with the vitreous humor.
   c. The function of the crystalline lens is to bend light rays and focus them on the optic nerve.
   d. Near sightedness is a condition resulting from loss of lens elasticity.

3. The fovea of the eye
   a. has the lowest light threshold
   b. is the region of highest visual acuity
   c. contains only red and green cones
   d. contains only rods

4. The following events that occur in rods in response to light are listed in random sequence:
   1. activation of transducin
   2. decreased release of synaptic transmitter
   3. structural changes in rhodopsin
   4. closure of Na^+ channels
   5. decrease in intracellular cGMP

   What is the sequence in which they normally occur?
   a. 2, 1, 3, 5, 4
   b. 1, 2, 3, 5, 4
   c. 5, 3, 1, 4, 2
   d. 3, 1, 5, 4, 2

5. The fluid that fills the posterior chamber of the eye is the
   a. lacrimal fluid
   b. aqueous humor
   c. choroid humor
   d. vitreous humor

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6. Some diseases damage the hair cells in the ear. When the damage to the outer hair cells is greater than the damage to the inner hair cells
   a. the perception of vertical acceleration is disrupted.
   b. the K^+ concentrations in endolymph is decreased.
   c. the K^+ concentrations in perilymph is decreased.
   d. the affected hair cells fail to shorten when exposed to sound.

7. The basilar membrane of the cochlea
   a. is unaffected by movement of fluid in the scala vestibule.
   b. covers the oval window and the round window.
   c. vibrates in a pattern determined by the form of the traveling wave in the fluids in the cochlea.
   d. vibrates when the body is subjected to linear acceleration.

Blood vascular system

1. Most of the CO_2 transported in the blood
   a. dissolved in plasma
   b. in carbamino compounds formed from plasma proteins
   c. bound to Cl^-
   d. in the form of HCO_3^-

2. Which of the following has the greatest effect on the ability of blood to transport oxygen?
   a. Capacity of the blood to dissolve oxygen
   b. Amount of hemoglobin in the blood
   c. pH of plasma
   d. Temperature of the blood

3. Variations in which of the following components of blood or cerebrospinal fluid do not affect respiration?
   a. arterial HCO_3^- concentration
   b. arterial H^+ concentration
   c. arterial Na^+ concentration
   d. cerebrospinal fluid CO_2 concentration

4. The most important factor in determining the percent oxygen saturation of hemoglobin is
   a. the partial pressure of oxygen
   b. acidity
   c. the partial pressure of carbon dioxide
   d. temperature

5. The most abundant protein in human blood is
   a. transferrin
   b. albumin
   c. γ-globulin
   d. hemoglobin [IISc]

6. Leukopenia is a term used to describe
   a. high WBC count
   b. low WBC count
   c. high RBC count
   d. low RBC count [MKU]

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7. Which of the following contains oxygenated blood in an adult human?
   a. Right atrium
   b. Pulmonary artery
   c. Pulmonary vein
   d. All of the above

8. Which of the following is the most muscular chamber in a bird's heart or a mammal's heart?
   a. The right atrium
   b. The left atrium
   c. The left ventricle
   d. The right ventricle

9. Which of the following statement about circulatory systems is *true*?
   a. Hormones are transported in the blood.
   b. All invertebrates have an open circulatory system.
   c. Capillaries have thicker walls than veins.
   d. The systemic circulation carries blood to and from the lungs.

10. Which one of the following series represents the *correct* path of blood circulation?
   a. Left atrium, left ventricle, lungs, right atrium, right ventricle, body
   b. Right atrium, right ventricle, lungs, left atrium, left ventricle, body
   c. Left atrium, left ventricle, right atrium, right ventricle, lungs, body
   d. Right atrium, lungs, right ventricle, left atrium, body, left ventricle

11. Which of the following statement about the heart is *false*?
   a. Contraction is initiated by a nerve impulse.
   b. Impulses are conducted from one muscle cell to the next.
   c. The heart contains a number of cells with an unstable membrane potential.
   d. The heart contains a number of cells with a stable membrane potential.

12. The *lub* of the *lub-dub* sound the heart makes is caused by the
   a. closing of the mitral and tricuspid valves.
   b. closing of the pulmonary and aortic valves.
   c. sound of blood rushing into the atria.
   d. sound of blood rushing into the ventricles.

13. If communication between the SA node and the AV node became blocked, which will most likely occur?
   a. The rate of ventricular contraction will decrease.
   b. Afterload will increase.
   c. The rate of atrial contraction will decrease.
   d. Stroke volume will increase to 5L/beat.

14. When compared to arteries, veins generally
   a. are thinner walled
   b. have more muscle in the tunica media
   c. carry faster moving blood
   d. have thicker endothelium

15. Which of the following is the primary factor regulating normal coronary blood flow?
   a. Aortic diastolic pressure
   b. Coronary perfusion pressure
   c. Systolic wall tension
   d. Myocardial oxygen consumption

[MKU]

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16. Administration of a local anaesthetic with epinephrine will most likely produce which of the following cardiovascular effects?

a. Increased heart rate
b. Decreased hear rate
c. Increased diastolic blood pressure
d. Decreased systolic blood pressure [MKU]

17. The lymphatic system

a. is an open circulatory system
b. contains one-way valves
c. returns fluids to the bloodstream
d. all of the above

18. Fluid is driven through the lymphatic system by

a. contraction of the walls of the lymphatic vessels
b. pressure created by the pumping of the heart
c. contractions of the lymph nodes
d. squeezing of the lymphatic vessels by the body's muscles

Respiratory system

1. Which of the following statements are *correct*?

1. Normal exhalation during quiet breathing is an active process involving intensive muscle contraction.
2. Passive exhalation results from elastic recoil of the chest wall and lungs.
3. Air flow during breathing is due to a pressure gradient between the lungs and the atmospheric air.
4. During normal breathing, the pressure between the two pleural layers is always subatmospheric.
5. Surface tension of alveolar fluid facilitates inhalation.

a. 1, 2 and 3
b. 2, 3 and 4
c. 3, 4 and 5
d. 1, 3 and 5

2. Which of the following factors affect the rate of external respiration?

1. Partial pressure differences of the gases
2. Surface area for gas exchange
3. Diffusion distance
4. Solubility and molecular weight of the gases
5. Presence of Bis-phosphoglycerate (BPG)

a. 1, 2 and 3
b. 2, 4 and 5
c. 1, 2, 4 and 5
d. 1, 2, 3 and 4

3. Which gas law that states that each gas in a mixture exerts its own pressure as if all the other gases were *not* present?

a. Boyle's law
b. Henry's law
c. Dalton's law
d. Haldane's law

4. Which of the following statements are *true*?

1. It is impossible for people to kill themselves by holding their breath.
2. The cerebral cortex provides voluntary control over breathing.
3. Emotional stimuli can alter respiration.
4. Certain chemical stimulation alter the rate and depth of breathing.

a. 1, 2 and 4
b. 2, 3 and 4
c. 1, 3 and 4
d. 1, 2, 3 and 4

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5. Match the following

Column A	*Column B*
1. eupnea	a. a deficiency of oxygen at the tissue level
2. apnea	b. above-normal partial pressure of carbon dioxide
3. hyperventilation	c. normal quiet breathing
4. diaphragmatic breathing	d. deep, abdominal breathing
5. hypoxia	e. absence of breathing
6. hypercapnia	f. rapid and deep breathing

a. a-5, b-6, c-1, d-4, e-2, f-3
b. a-6, b-5, c-4, d-1, e-2, f-3
c. a-5, b-6, c-4, d-1, e-3, f-2
d. a-4, b-6, c-2, d-5, e-1, f-3

6. Match the following

Column A	*Column B*
1. tidal volume	a. tidal volume and inspiratory reserve volume and expiratory reserve volume
2. residual volume	b. additional amount of air inhaled beyond tidal volume when taking a very deep breath
3. expiratory reserve volume	c. amount of air remaining in lungs after expiratory reserve volume is expelled
4. inspiratory reserve volume	d. tidal volume and inspiratory reserve volume
5. inspiratory capacity	e. volume of air in one breath
6. vital capacity	f. amount of air exhaled in forced exhalation

a. 1-c, 2-e, 3-b, 4-f, 5-d, 6-a
b. 1-e, 2-f, 3-c, 4-b, 5-a, 6-d
c. 1-e, 2-c, 3-f, 4-b, 5-d, 6-a
d. 1-e, 2-c, 3-b, 4-f, 5-a, 6-d

7. On the summit of Mt. Everest, where the barometric pressure is about 250 mm Hg, the partial pressure of O_2 is about

a. 0.1 mm Hg
b. 0.5 mm Hg
c. 5 mm Hg
d. 50 mm Hg

8. The tidal volume in a normal man at rest is about

a. 0.5 L
b. 1.2 L
c. 2.5 L
d. 4.9 L

9. Which of the following is responsible for the movement of O_2 from the alveoli into the blood in the pulmonary capillaries?

a. active transport
b. filtration
c. secondary active transport
d. passive diffusion

Excretory system

1. Concerning the functional histology of the kidney

a. the superficial nephrons have short loops of Henle; thus, they have a low capacity to reabsorb salt.
b. the deep nephrons have long loops of Henle; thus have a high capacity to reabsorb salt and water.
c. in dehydration, the blood flow to deep nephrons tends to increase.
d. all are correct.

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2. Which of the following is a function of the kidneys?
   a. Release of hormones
   b. Maintenance of plasma pH
   c. Maintenance of plasma (electrolyte)
   d. All of the above

3. In the distal convoluted tubule of the nephrons
   a. sodium reabsorption requires energy
   b. secretion of potassium does not require energy
   c. water reabsorption requires energy
   d. ammonia is secreted

4. Reabsorption of chloride ions from the glomerular filtrate in the kidney tubule is carried out by
   a. osmosis
   b. diffusion
   c. active transport
   d. brownian movement

5. Drinking which of the following would lead to the highest rate of ADH secretion and release?
   a. Two liters of distilled water
   b. Two liters of sea water (mainly hyperosmotic saline)
   c. Two liters of iso-osmotic (isotonic) saline
   d. Two liters of human blood plasma

6. An increase secretion of renin would be expected to have what effect on sodium excretion and potassium excretion in urine?
   a. Increase in Na^+ excretion and increase K^+ excretion.
   b. Increase in Na^+ excretion and decrease K^+ excretion.
   c. Decrease in Na^+ excretion and increase K^+ excretion.
   d. Decrease in Na^+ excretion and decrease K^+ excretion.

7. Which of the following is *not* a function of atrial natriuretic peptide?
   a. It acts to decrease aldosterone release from the adrenal cortex
   b. It acts to increase urine output
   c. It acts to increase blood pressure
   d. It acts to decrease ADH release

8. Which of the following is *incorrect*?
   a. Aldosterone stimulates the reabsorption of Na^+.
   b. Aldosterone stimulates the secretion of K^+.
   c. Aldosterone affects water reabsorption.
   d. Aldosterone is made in the hypothalamus and released from the anterior pituitary.

9. The majority of reabsorption occurs in the
   a. renal capsule
   b. proximal convoluted tubule
   c. collecting duct
   d. ascending limb of the loop of Henle

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10. Concerning water reabsorption by the proximal tubule
    a. main driving forces for water reabsorption in the proximal tubule are solute uptake and oncotic pressure in peritubular capillaries.
    b. a significant amount of water uptake in the proximal tubule is dependent on sodium uptake by the Na^+/H^+ antiports present in their luminal membrane.
    c. aquaporin-I are abundantly present in the cellular membranes of proximal tubule cells.
    d. all are correct.

11. Concerning arterial blood pressure regulation
    a. prostaglandins, dopamine and bradykinin are vasodilators.
    b. ADH, angiotensin II and epinephrine are vasoconstrictors.
    c. the vasodilator/vasoconstrictor ratio that regulates total peripheral resistance plus cardiac stroke volume and heart rate are determinants of blood pressure.
    d. none are correct.

12. Which of the following is/are *correct* for the countercurrent multiplier system?
    a. the ascending limb of the Loop of Henle transports NaCl by active transport.
    b. the descending limb is quite permeable to water.
    c. the fluid of the ascending limb becomes relatively dilute (low osmolarity).
    d. all of the above statements are correct.

Digestive system

1. The site of production of cholecystokinin and secretin is the
    a. stomach
    b. pancreas
    c. small intestine
    d. large intestine

2. Which of the following is *not* a function of the liver?
    a. Production of bile
    b. Detoxification of drugs
    c. Storage of glucose
    d. Storage of vitamin C

3. The gallbladder
    a. produces bile
    b. is attached to the pancreas
    c. stores and concentrates bile
    d. produces cholecystokinin

4. Which structure thickens in certain regions of the alimentary canal in order to act as a sphincter?
    a. Circular layer of the muscularis mucosae
    b. Longitudinal layer of the muscularis mucosae
    c. Circular layer of the muscularis externa
    d. Longitudinal layer of the muscularis externa

5. The 3 pairs of extrinsic salivary glands are the
    a. parotid, sublingual, and ethmoidal
    b. parotid, buccal, and submaxillary
    c. parotid, submandibular, and buccal
    d. parotid, submandibular, and sublingual

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6. The gastric gland cell whose absence could lead to pernicious anemia is the
   a. chief cell
   b. goblet cell
   c. mucous neck cell
   d. parietal cell

7. A major function of the large intestine is to
   a. secrete digestive enzymes
   b. remove waste materials
   c. regulate the release of bile
   d. secrete water in order to regulate blood volume

8. Which of the following statements are *correct*?
   P. Mechanical digestion occurs in the stomach.
   Q. Cholecystokinin, gastrin and secretin are produced by an enteroendocrine cells.
   R. Pancreas, small intestine and salivary glands produce digestive enzymes.
   S. Mucous neck cells, chief cells, parietal cells present in the stomach.
   a. P and Q
   b. Q and R
   c. P, Q and R
   d. P, Q, R and S

9. Each of the following statements about Brunner's glands is correct *except*
   a. they are a characteristic component of the duodenal wall.
   b. they produce a serous secretion rich in digestive enzymes.
   c. they lie in the submucosal layer.
   d. they empty their secretions into the crypts of Lieberkuhn.

10. Which of the following statement(s) is/are *correct*?
    a. Pepsinogen is synthesized and released by chief cells.
    b. Hormones mainly involved in controlling pancreatic exocrine secretions are cholecystokinin and secretin.
    c. Chief cells in children secrete an enzyme Rennin.
    d. All of the above statements are correct.

11. The function of the hepatic portal circulation is to
    a. carry toxins to the venous system for disposal through the urinary tract.
    b. collect absorbed nutrients for metabolic processing or storage.
    c. hormone distribution.
    d. transfer bile to the liver from the pancreas.

12. Which of the following is *false* about the small intestine?
    a. Site of carbohydrate, protein and fat digestion.
    b. Site of the majority of water absorption in the GI tract.
    c. First site of protein hydrolysis.
    d. Most rapid absorption of galactose.

13. Digestive processes in the large intestine include
    a. mass peristalsis
    b. absorption of some vitamins and electrolytes
    c. elimination of cellulose-based material
    d. all of the above

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1. A Graafian follicle is
   a. an immature developing follicle
   b. a mature follicle ready to ovulate
   c. a follicle undergoing apoptosis
   d. ovulated follicle

[IISc]

2. In species with two sexes, *males* are defined by which of the following characters?
   a. Males have brighter colouration than females.
   b. Males have smaller gametes compared to females.
   c. Males are larger than females.
   d. Males defend territories while females brood the eggs.

[IISc]

3. Which of the following are functions of Sertoli cells?
   1. Protection of developing spermatogenic cells.
   2. Nourishment of spermatocytes, spermatids, and sperm.
   3. Phagocytosis of excess sperm cytoplasm as development proceeds.
   4. Mediation of the effects of testosterone and FSH.
   5. Control of movements of spermatogenic cells and release of sperm into the lumen of seminiferous tubules.

   a. 1, 2, 4 and 5
   b. 1, 2, 3 and 5
   c. 2, 3, 4 and 5
   d. 1, 2, 3, 4 and 5

4. Which of the following statements are *correct*?
   1. A sperm head contains DNA and an acrosome.
   2. An acrosome is a specialized lysosome that contains enzymes that enable sperm to produce the ATP needed to propel themselves out of the male reproductive tract.
   3. Mitochondria in the midpiece of a sperm produce ATP for sperm motility.
   4. A sperm's tail, a flagellum, propels it along its way.
   5. Once ejaculated, sperms are viable and normally are able to fertilize a secondary oocyte for 5 days.

   a. 1, 2, 3 and 4
   b. 2, 3, 4 and 5
   c. 1, 3 and 4
   d. 2, 4 and 5

5. Which of the following statements are *correct*?
   1. Spermatogonia are stem cells because when they undergo mitosis, some of the daughter cells remain to serve as a reservoir of cells for future mitosis.
   2. Meiosis I is a division of pairs of chromosomes resulting in daughter cells with only one member of each chromosome pair.
   3. Meiosis II separates the chromatids of each chromosome.
   4. Spermiogenesis involves the maturation of spermatids into sperm.
   5. The process by which the seminiferous tubules produce haploid sperm is called spermatogenesis.

   a. 1, 2, 3 and 5
   b. 1, 2, 3, 4 and 5
   c. 1, 3, 4 and 5
   d. 1, 2, 3 and 4

6. In the 1st phase of the menstrual cycle
   a. oogonia differentiate into primary oocytes.
   b. the thickness of the stratum basalis decreases dramatically.
   c. the Graafian follicle ruptures.
   d. the dominant follicle is opsonized.

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7. During ovulation, all of the following occur *except*
   a. rupture of the Graafian follicle.
   b. estrogen production reaches its lowest point.
   c. FSH and LH plasma levels surge.
   d. corpus luteum is formed.

8. Human chorionic gonadotropin
   a. begins to be produced 4 months after fertilization
   b. is produced by the prostate gland
   c. is produced by the corpus luteum
   d. is present only if a successful fertilization has occurred

9. Which of the following statements are *correct*?
   1. Cells from the endoderm of the yolk sac give rise to oogonia.
   2. Ova arise from the germinal epithelium of the ovary.
   3. Primary oocytes enter prophase of meiosis I during fetal development but do not complete it until after puberty.
   4. Once a secondary oocyte is formed, it proceeds to metaphase of meiosis II and stops at this stage.
   5. The secondary oocyte resumes meiosis II and forms the ovum and a polar body only if fertilization occurs.
   6. A primary oocyte gives rise to an ovum and four polar bodies.

   a. 1, 3, 4 and 5
   b. 1, 3, 4 and 6
   c. 1, 2, 4 and 6
   d. 1, 2, 4 and 5

Endocrine system

1. Hormones
   1. generally utilize negative feedback mechanisms to regulate their secretion.
   2. will only affect target cells far removed from the hormone-producing secretory cells.
   3. must bind to transport proteins in order to circulate in the blood.
   4. will only cause an effect on cells with receptors for the hormone.
   5. can regulate the responsiveness of the target tissue by controlling the number of receptors for the hormone.

   a. 1, 2 and 3
   b. 1, 2, 4 and 5
   c. 2, 3 and 4
   d. 1, 4 and 5

2. The pituitary gland's posterior lobe produces two hormones, i.e.
   a. vasopressin and oxytocin
   b. cortisone and corticosterone
   c. progesterone and estradiol
   d. testosterone and aldosterone [JNU]

3. In which of the following combinations is the name of the hormone, its chemical type and its tissue of origin correctly matched?
   a. Aldosterone → peptide → pancreas
   b. Glucagons → peptide → adrenal cortex
   c. ACTH → polypeptide → adrenal cortex
   d. Vasopressin → peptide → posterior pituitary [MKU]

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4. Which of the following hormones does *not* act by a second messenger system?
   a. Glucagon
   b. Epinephrine
   c. Follicle stimulating hormone
   d. Testosterone [IISc]

5. Which hormone binds to intracellular receptors?
   a. Insulin
   b. Growth hormone
   c. Triiodothyronine
   d. Thyroid stimulating hormone [IISc]

6. Which of the following hormones does not act by a second messenger system?
   a. Glucagon
   b. Epinephrine
   c. Luteinizing hormone
   d. Aldosterone [IISc]

7. Which one of the following statements with respect to testosterone is *not* true?
   a. Testosterone receptor mutant is embryonic lethal.
   b. Testosterone receptor is essential for male reproduction.
   c. Testosterone is produced in female rats.
   d. Testosterone is not essential for fetal growth. [IISc]

8. Administration of estrogen to adult male rats results in
   a. decreased testosterone production
   b. decreased luteinizing hormone and testosterone production
   c. increased estrogen secretion in the testis
   d. none of the above [IISc]

9. Radioactive iodine can be incorporated into
   a. serine
   b. threonine
   c. tyrosine
   d. leucine [IISc]

10. The receptor for which of the following hormones is a transcription factor?
    a. Insulin
    b. Glucagon
    c. Estradiol
    d. Adrenaline [IISc]

11. The development of adult characteristics in a molting insect is promoted by
    a. thyroxine
    b. ecdysone
    c. juvenile hormone
    d. a pheromone

12. The fight-or-flight response is developed by hormones of the
    a. hypothalamus
    b. adrenal medulla
    c. adrenal cortex
    d. adrenal-pancreatic complex

13. Epinephrine and norepinephrine function as both hormones and
    a. fuel for cellular respiration
    b. neurotransmitters
    c. ions to promote action potentials
    d. solutes to promote osmotic flow

14. All the hormones of the adrenal cortex are synthesized from
    a. tyrosine
    b. glycoproteins
    c. cholesterol
    d. fats

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15. Thyroxine and triiodothyronine, produced by the thyroid gland, are synthesized from iodine and
    a. phenylalanine
    b. cholesterol
    c. glycoproteins
    d. tyrosine

16. The parathyroid glands are located adjacent to the
    a. adenoids
    b. thyroid gland
    c. adrenal glands
    d. pancreas

17. A person with diabetes mellitus does not secrete enough
    a. sugar
    b. insulin
    c. glucagons
    d. epinephrine

18. The neurons of a person with diabetes mellitus do not produce sufficient
    a. ATP
    b. fatty acids
    c. enzymes
    d. steroids

19. Which of the following statements are *true* concerning androgens?
    1. They stimulate the male pattern of development.
    2. They stimulate the male and female patterns of development.
    3. They contribute to sex drive in males and females.
    4. They contribute to male secondary sex characteristics.
    5. They stimulate protein synthesis.
    a. 1, 2 and 3
    b. 1, 3, 4 and 5
    c. 1, 2, 4 and 5
    d. 1, 2, 3 and 4

20. Which of the following statements are *true* concerning estrogens?
    1. They promote development and maintenance of female reproductive structures and secondary sex characteristics.
    2. They help control fluid and electrolyte balance.
    3. They increase protein catabolism.
    4. They lower blood cholesterol.
    5. In moderate levels, they inhibit the release of GnRH and the secretion of LH and FSH.
    a. 1, 4 and 5
    b. 1, 3, 4 and 5
    c. 1, 2, 3 and 5
    d. 1, 2, 3 and 4

21. A peptide hormone typically alters the activity of its target cells by
    a. digesting holes in the cell's plasma membrane
    b. entering the cell and altering gene expression
    c. passing its message to an intracellular messenger
    d. digesting holes in the cell's lysosomes

22. Human chorionic gonadotropin responsible for maintenance of pregnancy in women originates from
    a. ovary
    b. pituitary
    c. uterus
    d. placental trophoblast [IISc]

23. A steroid hormone typically alters the activity of its target cells by
    a. digesting holes in the cell's plasma membrane.
    b. entering the cell and altering gene expression.
    c. passing its message to an intracellular messenger.
    d. digesting holes in the cell's lysosomes.

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24. Which of the following hormones is a modified amino acid?
a. prostaglandin
b. estrogen
c. epinephrine
d. progesterone

25. Thyroxine is important in the control of
a. cellular metabolic rates
b. diabetes mellitus
c. mitochondrial respiration
d. calcium uptake [IISc]

26. The primary target organ of aldosterone action is
a. liver
b. pancreas
c. kidney
d. heart [IISc]

27. Which of the following factors affect the responsiveness of a target cell to a hormone?
1. Hormone concentration
2. Target cell type
3. Abundance of hormone receptors
4. Influences of other hormones
5. Age of the target cell
a. 1, 2, 3 and 4
b. 2, 3, 4 and 5
c. 1, 2 and 3
d. 1, 3 and 4

28. The class of adrenal gland hormones that provide resistance to stress, produce anti-inflammatory effects, and promote normal metabolism to ensure adequate quantities of ATP is
a. glucocorticoids
b. mineralocorticoids
c. androgens
d. catecholamines

29. Place in correct order the action of a water-soluble hormone on its target cell
1. adenylate cyclase activated, catalyzing the conversion of ATP to cAMP.
2. enzymes catalyze reactions that produce physiological response attributed to the hormone.
3. hormone binds to plasma membrane receptor.
4. activated protein kinases phosphorylate cellular proteins.
5. hormone-receptor complex activates G proteins.
6. cyclic AMP activates protein kinases.
a. 3, 5, 1, 6, 4, 2
b. 3, 1, 5, 6, 4, 2
c. 5, 1, 4, 2, 3
d. 3, 4, 5, 1, 6, 2

30. Match the following hormone-secreting cells to the hormone(s) they release

Column: A	*Column: B*
1. beta cells of pancreatic islets	a. ACTH
2. alpha cells of pancreatic islets	b. TSH
3. follicular cells of thyroid gland	c. glucagon
4. parafollicular cells of thyroid gland	d. calcitonin
5. testes	e. insulin
6. ovary	f. progesterone
7. somatotrophs	g. FSH and LH
8. thyrotrophs	h. hGH
9. gonadotrophs	i. testosterone
10. corticotrophs	j. thyroxine and triiodothyronine

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a. 1-e, 2-c, 3-j, 4-d, 5-i, 6-f, 7-h, 8-b, 9-g, 10-a
b. 1-c, 2-e, 3-j, 4-d, 5-i, 6-f, 7-h, 8-b, 9-g, 10-a
c. 1-e, 2-c, 3-j, 4-d, 5-f, 6-i, 7-b, 8-h, 9-g, 10-a
d. 1-c, 2-e, 3-j, 4-d, 5-i, 6-f, 7-h, 8-b, 9-a, 10-g

31. Match the following

Column: A	*Column: B*
1. inhibin	a. relaxes the uterus by inhibiting myometrial contractions during monthly cycles; increases flexibility of the pubic symphysis during childbirth
2. LH	b. stimulates Leydig cells to secrete testosterone
3. FSH	c. inhibits production of FSH by the anterior pituitary
4. testosterone	d. stimulates male pattern of development; stimulates protein synthesis; contributes to sex drive
5. estrogens	e. maintains the corpus luteum during the first trimester of pregnancy
6. progesterone	f. promote development of female reproductive structures; lower blood cholesterol
7. relaxin	g. stimulates the initial secretion of estrogens by growing follicles; promotes follicle growth
8. human chorionic gonadotropin	h. is secreted by the corpus luteum to maintain the uterine lining during the first trimester of pregnancy

a. 1-c, 2-b, 3-g, 4-d, 5-h, 6-f, 7-e, 8-a
b. 1-c, 2-b, 3-g, 4-f, 5-d, 6-a, 7-h, 8-e
c. 1-c, 2-b, 3-g, 4-d, 5-f, 6-h, 7-a, 8-e
d. 1-b, 2-c, 3-d, 4-g, 5-f, 6-h, 7-a, 8-e

32. Match the endocrine disorder to the problem that produced the disorder

Column:A	*Column:B*
1. gigantism	a. hypersecretion of glucocorticoids
2. acromegaly	b. hypersecretion of hGH before closure of epiphyseal plates
3. pituitary dwarfism	c hyposecretion of hGH before closure of epiphyseal plates
4. diabetes insipidus	d. hypersecretion of hGH after closure of epiphyseal plates
5. myxedema	e. hyposecretion of glucocorticoids and aldosterone
6. graves' disease	f. hyposecretion of ADH
7. cushing's syndrome	g. hyposecretion of thyroid hormone in adults
8. Addison's disease	h. hyperthyroidism, an mellitus autoimmune disease

a. 1-d, 2-b, 3-f, 4-c, 5-g, 6-h, 7-a, 8-e
b. 1-b, 2-d, 3-c, 4-f, 5-h, 6-g, 7-e, 8-a
c. 1-b, 2-d, 3-c, 4-f, 5-g, 6-h, 7-a, 8-e
d. 1-f, 2-c, 3-d, 4-b, 5-g, 6-h, 7-a, 8-e

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Chapter 15

Diversity of Life

Taxonomy

1. Prokaryotic organisms includes
   a. archaebacteria, eubacteria and protists
   b. archaebacteria and protists
   c. protists and eubacteria
   d. eubacteria and archaebacteria

2. Unicellular eukaryotes are grouped into
   a. fungi and plantae
   b. fungi
   c. protists
   d. eubacteria

3. The largest unit within which gene flow can readily occur is a
   a. population
   b. species
   c. genus
   d. phylum

4. If humans and pandas belong to the same class, then they must also belong to the same
   a. order
   b. genus
   c. phylum
   d. species

5. Eukaryotes differ from prokaryotes in that
   a. eukaryotes do not have organelles
   b. eukaryotes have a single chromosome
   c. eukaryotes have a nucleus and organelles
   d. eukaryotes do not divide by mitosis

6. The table shows the distribution of traits (A-E) in six species (1-6). A **0** indicates the ancestral condition; a **1** is the derived condition. With the help of following data, find out which pair of species diverged most recently?

	Trait				
Species	A	B	C	D	E
1	1	0	0	0	0
2	1	1	1	1	1
3	1	1	1	1	1
4	1	1	1	1	0
5	1	1	1	0	0
6	1	1	0	0	0

   a. 1 and 4
   b. 2 and 3
   c. 2 and 4
   d. 6 and 4

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Monera

1. Bacterial cells are prokaryotic in comparison to a typical eukaryotic cell they would
   a. be smaller
   b. have a smaller nucleus
   c. lack a plasma membrane
   d. have a greater variety of organelles

2. Which of the following information would tell you whether a cell is prokaryotic or eukaryotic?
   a. The presence or absence of a rigid cell wall.
   b. Whether or not the cell is partitioned by internal membranes.
   c. The presence or absence of ribosomes.
   d. Whether or not the cell carries out cellular metabolism.

3. Among the common features shared by prokaryotes and eukaryotes are
   a. same shapes and sizes.
   b. same types of movement.
   c. common organic substances such as proteins and carbohydrates.
   d. ribosomes of the same sedimentation coefficient.

4. The cell walls of gram-positive and gram-negative bacteria
   a. are the sites of spore formation.
   b. both contain peptidoglycan.
   c. are connected to the cell nucleus by the endoplasmic reticulum.
   d. are composed exclusively of proteins.

Protists

1. Which of the following are the features of protozoa?
   P. Unicellular (single celled)
   Q. Aquatic, free-living or parasitic organisms
   R. Locomotion with the help of pseudopodia, cilia or flagella
   S. Nutrition is heterotrophic
   a. P and Q
   b. Q and S
   c. P, Q and R
   d. P, Q, R and S

2. In ciliates, the process that produces genetic variation through the exchange of nuclei is
   a. mixotrophy
   b. endosymbiosis
   c. meiosis
   d. conjugation

3. Which algal group is *mismatched* with its descriptions?
   a. Dinoflagellates—glassy, two-part shells
   b. Green algae—closest relatives of land plants
   c. Red algae—no flagellated stages in life cycle
   d. Brown algae—include the largest seaweeds

4. *Dinoflagellates* are characterized by which of the following features?
   P. Unicellular, flagellated
   Q. Mainly marine
   R. Contains chlorophylls a and c
   S. Asexual reproduction by binary fission
   a. P and Q
   b. Q and S
   c. P, Q and R
   d. P, Q, R and S

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5. Which of the following feature is *not* present in slime mold?
   a. A fungus-like organisms
   b. Exhibit characteristics of both fungi and animals
   c. Use spores to reproduce
   d. Composed of networks of long hollow tubes called hyphae

6. Which of the following is a mismatch?

Disease	*Causative agent*
a. Amoebiasis	*Entamoeba histolytica*
b. *Giardiasis*	*T. cruzi*
c. African sleeping sickness	*Trypanosoma brucei*
d. Leishmaniasis	*Leishmania donovani*

7. Which of the following statements is *true* of amoebas?
   a. They form pseudopodia
   b. None has an outer shell
   c. They have a flagellated phase in their life cycle
   d. They do not have food vacuoles

8. The protozoans that cause malaria in humans are
   a. radiolarians
   b. trichosomes
   c. dinoflagellates
   d. sporozoans

9. Ciliates differ from all other protozoans in
   a. having two types of nuclei
   b. capturing prey with pseudopodia
   c. having contractile vacuoles
   d. not using flagella for locomotion

10. Slime molds in the division Myxomycota (true slime molds) have
    a. feeding stages consisting of solitary, individual cells.
    b. spores that develop into flagellated gametes.
    c. pseudoplasmodia.
    d. spores that develop into free-living amoeboid cells.

11. Which species of protists are known as the *whirling whips* because of the spin produced by two flagella beating in opposing grooves along their hard-surfaced bodies?
    a. Diatoms
    b. Chrysophytes
    c. Dinoflagellates
    d. Golden-brown algae

12. Which of the following statements is *true* of euglena?
    a. They are negatively phototactic.
    b. They move by means of flagella.
    c. They have a rigid cell wall.
    d. They do not have chloroplasts.

Fungi

1. Which of the following is *not* a characteristic of the fungi?
   a. They are all absorptive heterotrophs.
   b. They have cell walls made of chitin.
   c. They are eukaryotic heterotrophs.
   d. They always have motile stages.

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2. Which of the following cells or structures are associated with asexual reproduction in fungi?
   a. Ascospores
   b. Basidiospores
   c. Conidiophores
   d. Zygosporangia

3. Sporangia on upright hyphae that produce asexual spores are characteristic of
   a. ascomycetes
   b. basidiomycetes
   c. club fungi
   d. zygomycetes

4. The photosynthetic symbiont of a lichen is often a(n)
   a. moss
   b. green alga
   c. brown alga
   d. ascomycete

5. An ascomycete can be distinguished from other fungi
   a. because ascomycetes are mainly diploid.
   b. because ascomycetes lack a dikaryotic phase.
   c. by the presence of sexual spores in an ascus.
   d. by the presence of gills on the mycelium.

6. Which of the following is *not* part of asexual reproduction in fungi?
   a. Formation of asexual spores
   b. New individuals growing from hyphal fragments
   c. Budding
   d. Development of dikaryons

7. The oomycota, such as water molds, are so different from other fungi that some biologists think they evolved separately. Which of the following is *not* characteristic of oomycota?
   a. flagellated zoospores
   b. lack of chitin in cell walls
   c. oogamous reproduction
   d. haploid during most of their life cycle

Animals

1. Which feature is *not* characteristic of the animals?
   a. Multicellular heterotrophs
   b. Sexual reproduction
   c. Embryonic development
   d. Presence of cell walls

2. A key evolutionary development seen for the first time in the sponges is
   a. a complete digestive system
   b. tissue organization
   c. body symmetry
   d. multicellularity

3. Which of the following is an example of an organism with the medusa body form?
   a. A hydra
   b. A coral
   c. An anemone
   d. A jellyfish

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4. Members of which group are *not* deuterostomes?
   a. Chordates
   b. Echinoderms
   c. Arthropods
   d. None of these, all are deuterostomes

5. Which of the following is *not* a characteristic of most members of the phylum annelida?
   a. Hydrostatic skeleton
   b. Segmentation
   c. Metanephridia
   d. Pseudocoelom

6. Which phylum is characterized by animals that have a segmented body?
   a. Cnidaria
   b. Platyhelminthes
   c. Porifera
   d. Arthropoda

7. Which of the following characteristics is probably most responsible for the incredible diversification of insects on land?
   a. Segmentation
   b. Antennae
   c. Tracheal system
   d. Flight

8. The water vascular system of echinoderms
   a. functions as a circulatory system that distributes nutrients to body cells.
   b. functions in locomotion, feeding, and gas exchange.
   c. is bilateral in organization, even through the adult animal has radial anatomy.
   d. moves water through the animals body during suspension feeding.

9. Which of the following combinations of phylum and description is *incorrect*?
   a. Echinodermata—bilateral and radial symmetry, coelom from archenteron
   b. Nematode—roundworms, pseudocoelomate
   c. Cnidaria—radial symmetry, polyp and medusa body forms
   d. Porifera—gastrovascular cavity, coelomate

10. The type of body cavity seen in the roundworms is called a(n)
    a. coelom
    b. acoelom
    c. pseudocoelom
    d. gastrovascular cavity

11. Segmentation was first apparent in the
    a. flatworms
    b. annelids
    c. mollusks
    d. arthropods

12. The segments of annelids are
    a. apparent in the embryo but not in the adult.
    b. specialized for different functions.
    c. most obvious in the mesoderm (muscles) but not in the ectoderm.
    d. repetitive, each able to develop a complete set of adult organs.

13. The phylum that shows the greatest diversity, or the greatest number of species, is
    a. Arthropoda
    b. Brachiopoda
    c. Echinodermata
    d. Mollusca

14. Which animal group has radial symmetry, a water-vascular system, moves with tube feet and has an endoskeleton?
    a. Arachnids
    b. Crustaceans
    c. Echinoderms
    d. Cnidarians

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15. The very first vertebrates were
    a. cartilaginous fish
    b. fishes with jaws
    c. amphibians
    d. jawless fish

16. In order for amphibians to be successful on land, they had to develop which of the following?
    a. A more efficient swim bladder
    b. Cutaneous respiration and lungs
    c. More efficient gills
    d. Shelled eggs

17. Vertebrates and tunicates share
    a. jaws adapted for feeding
    b. a high degree of cephalization
    c. the formation of structures from the neural crest
    d. a notochord and a dorsal, hollow nerve cord

18. Mammals and living birds share all of the following characteristics *except*
    a. endothermy
    b. descent from a common amniotic ancestor
    c. a dorsal, hollow nerve cord
    d. a hole behind the eye socket on each side of the skull

19. Which one of the following statements is *not* correct?
    a. All sponges are exclusively marine.
    b. Amphiblastula is a landform of certain sponges.
    c. Calcareous and siliceous spicules are found in many sponges.
    d. All sponges are sessile.

20. Which one of the following is the *correct* sequence of larval stage in the development of *Fasciola*?
    a. Miracidium, redia, sporocyst metacercaria, cercaria
    b. Cercaria, miracidium, sporocyst, metacercaria, redia
    c. Miracidium, sporocyst, redia, cercaria, metacercaria
    d. Redia, cercaria, metacercaria, sporocyst, miracidium

21. Which one of the following animal groups belongs to same class?
    a. *Pheretima*, *Lumbricus*, *Hirudinaria*
    b. Spider, louse, millipede
    c. Cuttlefish, devilfish, squids
    d. Silverfish, crayfish, razor fish

22. Which one of the following pairs is *not* correctly matched?
    a. *Aedes aegypti* : yellow fever
    b. *Glossina palpalis* : sleeping sickness
    c. *Xenopsylla cheopis* : plague
    d. *Phlebotomus papatasi* : hay-fever

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23. Which of the following is *not* an arthropod characteristic?
a. Jointed appendages
b. Nonsegmented body
c. Periodic molting
d. Articulated exoskeleton

24. Which of the following is *not* a characteristic of insects?
a. There main body segments: head, thorax, abdomen
b. Three pairs of legs
c. Two pairs of antennae
d. Excretion by malpighian tubules

25. Which of the following traits is *not* characteristic of echinoderms?
a. Water-vascular system
b. Trochophore larva
c. Tube feet
d. Interior skeleton with projection spines

Plants

1. All plants exhibit alternation of generations. This means their life cycle
a. includes both haploid and diploid gametes.
b. shows only asexual reproduction.
c. has both a multicellular haploid stage and a multicellular diploid stage.
d. does not include meiosis.

2. The plant life cycle has both a sporophyte and a gametophyte generation. In the sporophyte stage
a. gametes are produced
b. meiosis occurs
c. only mitosis takes place
d. gametophytes form

3. Which of the following is *not* common to all phyla of vascular plants?
a. The development of seeds
b. Alternation of generations
c. Dominance of the diploid generation
d. Xylem and phloem

4. A heterosporous plant species is one that
a. produces a gametophyte that bears both antheridia and archegonia.
b. produces microspores and megaspores, which give rise to male and female gametophytes.
c. produces microspores and megaspores, which give rise to male and female gametophytes.
d. produces two kinds of spores, one asexually by mitosis and the other sexually by meiosis.

5. Which of the following is *not* a characteristic that distinguishes gymnosperms and angiosperms from other plants?
a. Alternation of generations
b. Ovules
c. Integuments
d. Pollen

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6. Which of the following is *true* of the bryophytes?
   a. It is the only group that shows an alternation of generations.
   b. Bryophytes exhibit extensive vascular tissue.
   c. The sporophyte (multicellular diploid) is the dominant stage.
   d. The gametophyte (multicellular haploid) is the dominant stage.

7. Which of the following best describes a fern gametophyte?
   a. Its cells are haploid.
   b. It lacks chlorophyll.
   c. It is tough and woody.
   d. It is larger than the sporophyte.

8. Which of the following characteristics is common to all vascular plants that exhibit an alternation of generations in their life cycle?
   a. Large, independent gametophytes
   b. Multicellular sporophytes
   c. Fertilization in water
   d. Diploid spores

9. Which of the following characters is seen in the gymnosperms, but is not seen in other seeded vascular plants?
   a. Occurrence of alternation of generations
   b. Naked seeds
   c. Dominant sporophyte stage
   d. Formation of pollen grains

10. In the angiosperms, the
   a. gametophyte is prominent, and the sporophyte is dependent upon the gametophyte.
   b. sporophyte is prominent, with the sporophyte and gametophyte living independently.
   c. sporophyte is prominent, and the gametophyte is dependent upon the sporophyte.
   d. gametophyte is prominent, and the sporophyte stage has disappeared.

11. Which of the following is *not* characteristic of a monocot?
   a. Leaves with parallel veins
   b. Flower parts usually in threes or multiples of three
   c. Lateral meristems occurring rarely
   d. Seed with two cotyledons

12. The fruit produced by angiosperms is an evolutionary adaptation that most often
   a. nourishes the seeds within the fruit on ripening
   b. aids in seed dispersal
   c. attracts pollinators
   d. inhibits seed germination until favorable environmental conditions occur

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Anatomy of angiospermic plants

1. Which one of the following can be considered to be dead mechanical tissue?
   a. Aerenchyma
   b. Collenchyma
   c. Parenchyma
   d. Sclerenchyma

2. Cork cells are impervious to water because of the presence of
   a. lignin
   b. cutin
   c. suberin
   d. pectin

3. Match list-1 with list-2 and select the *correct* answer using the codes given below the lists

List-1	*List*-2
P. apical meristem	1. cambium
Q. lateral meristem	2. internode
R. intercalary meristerm	3. root apex
S. secondary meristem	4 cork cambium

   a. P-3, Q-1, R-2, S-4
   b. P-1, Q-2, R-4, S-3
   c. P-3, Q-2, R-4, S-1
   d. P-4, Q-3, R-1, S-2

4. Vessels and companion cells are the characteristic features of
   a. gymnosperms
   b. angiosperms
   c. pteridophytes
   d. fungi

5. According to histogen theory, the meristematic zone responsible for the development of pericycle is
   a. periblem
   b. plerome
   c. dermatogen
   d. calyptrogen

6. Consider the following statement:
   Lateral roots originate
   P. endogenously
   Q. from pericycle cells
   R. exogenously
   S. from endodermal cells
   Which of these statement are *correct*?
   a. P and Q
   b. R and S
   c. P and S
   d. Q and R

7. Match list-1 (type of tissues/structures) with list-2 (characteristic features of the tissue) and select the *correct* answer using codes given below the lists:

   List - 1
   P. hydathodes
   Q. collenchyma
   R. periblem
   S. tyloses

   List - 2
   1. balloon-like protrusions into tracheary elements, developed from xylem parenchyma.
   2. mechanical tissue that consists of living cells, walls rich in pectin and hemicellulose.
   3. occur on leaf tips responsible for guttation.
   4. meristematic tissue responsible for the development of cortex.

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a. P-2, Q-3, R-4, S-1
b. P-3, Q-2, R-1, S-4
c. P-2, Q-3, R-1, S-4
d. P-3, Q-2, R-4, S-1

8. Wood with vessels of equal diameters uniformly distributed through a growth ring is referred to as
a. diffuse porous wood
b. ring porous wood
c. heart wood
d. sap wood

9. Wood with vessels of unequal diameter and with the largest vessels localized in the early wood is called
a. diffuse porous wood
b. ring porous wood
c. stratified wood
d. non-stratified wood

10. Periderm includes
a. phelloderm, collenchyma and cortex
b. phellem, cambium and cortex
c. all the tissues between epidermis and pith
d. phellogen, phellem and phelloderm

11. Vertically elongated cells in the lateral meristem producing axial system in the secondary xylem and phloem are known as
a. fusiform initials
b. procambial cells
c. phellogen cells
d. ray initials

12. In the roots apex, the meristem from which the root cap develops independently of all other initials of the apical meristem, is known as
a. calyptrogen
b. casparain strip
c. columella
d. corpus

Reproduction in angiosperms

1. With reference to sexual incompatibility in angiosperms, consider the following statements
P. gemetophytic self-incompatibility is found in all species of brassicaceae
Q. sporophytic self-incompatibility is found in all species
Which of the statement given above is/are *correct*?
a. P only
b. Q only
c. both P and Q
d. neither P nor Q

2. Which one of the following statement is *not correct*?
a. Pollen tube grows by cell elongation.
b. Endosperm is always triploid.
c. Pseudoembryo sac is formed by the breakdown of nucellar tissue.
d. Development of egg without fertilization is termed as parthenogenesis.

3. In maize (*zea mays*) fertilization takes place between a female parent having colorless endosperm and the male parent having yellow endosperm. After fertilization, the endosperm of the new seed becomes yellow. This phenomenon is known as:
a. dominance
b. xenia
c. metaxenia
d. paternal influence

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4. A minimum number of meiotic divisions required to produce 50 seeds of an angiosperm is:
   a. 13 b. 26
   c. 50 d. 63

5. Match list-1 (ovular structures) with list-2 (post-fertilization structures) and select the *correct* answer using the codes given below the lists:

List-1	*List-2*
P. ovule	1. endosperm
Q. funiculus	2. aril
R. nucellus	3. seed
S. polar nuclei	4. perisperm

   a. P-2, Q-3, R-4, S-1 b. P-2, Q-3, R-1, S-4
   c. P-3, Q-2, R-4, S-1 d. P-3, Q-2, R-1, S-4

6. What would be the number of chromosomes in the aleurone cells of a plant with 42 chromosomes in the root tip cells?
   a. 21 b. 42
   c. 63 d. 84

7. Which one of the following pairs is *not* correctly matched?
   a. inbreeding depression : homozygosity
   b. hybrid vigour : heterozygosity
   c. apogamy : fertilization
   d. male sterility : cross-pollination

8. How many meiotic divisions are required to produce 12 pollen grains in *Cyperaceae*?
   a. 3 b. 4
   c. 6 d. 12

9. If more than one pollen tubes discharge their contents inside an embryo sac and the sperm fusion with the egg is derived from one tube and the one fusion with the secondary nucleus from anther tube, it is called:
   a. heterofertilization b. polyspermy
   c. heterospermy d. polygamy

10. In a cross between a tetraploid female plant and a diploid male plant of mulberry, the endosperm would be
   a. hexaploid b. triploid
   c. tetraploid d. pentaploid

11. The exine of mature pollen grain is composed chiefly of
   a. lipids b. cellulose
   c. pectocellulose d. sporopollenin

12. During fertilization, the pollen tube discharges its contents in the
   a. central cell b. egg
   c. micropyle d. synergid

13. Dormancy caused by seed coat impermeability to water and gases can be broken by
   a. stratification b. scarification
   c. low temperature d. abscisic acid

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14. Which one of the following chemical ions plays an important role in the growth of pollen tube?
    a. Calcium
    b. Chlorine
    c. Magnesium
    d. Sulphate

15. Perisperm in the seed is a
    a. persistent endosperm
    b. persistent nucellus
    c. persistent hypostase
    d. persistent epistase

16. Which one of the following shows the last diploid stage in the life cycle of angiosperms?
    a. Microspore mother cell
    b. Zygote
    c. Nucellus
    d. Pollen grain

17. Apospory is the development of an offspring from the
    a. cell of nucellus
    b. synergids or antipodals
    c. haploid female gamete
    d. haploid microspore

18. Development of embryo from gametophyte without the intervention of the gamete is known as
    a. apospory
    b. apogamy
    c. apomixis
    d. aposporogamy

19. The maturation of anther and stigmas at different times in the same flower is known as
    a. herkogamy
    b. cleistogamy
    c. chasmogamy
    d. dichogamy

20. One of the most notable differences between gamete formation in animals and gamete formation in plants is that
    a. plants produce gametes in somatic tissue, while animals produce gametes in germ tissue.
    b. plants produce gametes by mitosis, while animals produce gametes by meiosis.
    c. plants produce only one of each gamete, while animals produce many gametes.
    d. plants produce gametes that are diploid, while animals produce gametes that are haploid.

21. In angiosperms, normally after fertilization
    a. the zygote divides before the division of the primary endosperm.
    b. the primary endosperm nucleus divides before the division of the zygote.
    c. both the zygote and the primary endosperm nucleus divide simultaneously.
    d. both the zygote and the primary endosperm nucleus undergo a resting period and then divide simultaneosly.

22. Plant development process differs from animal development in which of the following parameters?
    P. Plants do not gastrulate.
    Q. Plants have sporic meiosis rather than gametic meiosis.
    R. Plants have tremendous plasticity in development.
    S. Maternal RNA plays only a minor role in plant embryo development.
    a. P and Q
    b. R and S
    c. P, Q and S
    d. P, Q, R and S

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Chapter 16

Ecology and Evolution

Ecosystem

1. Which of the following statement is not *correct*?
   a. Ecosystem is an open system.
   b. Ecosystem is self-sustaining and dynamic structure.
   c. Sun is the ultimate source of energy for any ecosystem.
   d. In an artificial ecosystem flow of energy is not unidirectional. [JNU

2. The length of the food chain in an ecosystem is generally limited to 3-4 trophic levels because
   a. quality of energy increases from lower to higher trophic levels.
   b. quantity of energy decreases from lower to higher trophic levels.
   c. quantity of energy increases from lower to higher trophic levels.
   d. flow of energy is unidirectional. [IISc]

3. Which of the following statements is *not* correct?
   a. Nutrient cycling time tends to increase with increasing latitude.
   b. In general, the transfer of energy from one trophic level to the next is in the order of 10%.
   c. Open ocean and deserts are less productive than ocean-land interface region.
   d. None of the above statements are incorrect.

4. Which of the following statements about food chains and energy flow through ecosystem is *false*?
   a. A single organism can feed at several trophic levels.
   b. Detritivores feed at all trophic levels except the producer level.
   c. The lower the trophic level at which an organism feeds, the more energy available.
   d. Those who are not producers are consumers.

5. The concept of trophic structure of a community emphasizes the
   a. prevalent form of vegetation
   b. keystone predator
   c. feeding relationships within a community
   d. effects of coevolution

6. Rate of energy storage at consumer level is known as
   a. net community productivity
   b. net primary productivity
   c. secondary productivity
   d. gross primary productivity

7. Ecological pyramid
   a. graphically represents the trophic structure and function
   b. based on rate of energy flow is always upright
   c. producer always occupies the base of pyramid
   d. all statements are true

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8. The greatest amount of free energy is available at which of the following levels?
   a. Tertiary consumers
   b. Secondary consumers
   c. Decomposers
   d. Producers [IISc]

9. Ecologists generally separate organisms in an ecosystem into three categories (producer, consumer, decomposer) based on
   a. how they obtain their energy
   b. how rapid their metabolism is
   c. the rate of succession in the ecosystem
   d. the size and complexity of the ecosystem

10. Energy pyramids of an ecosystem tend to diminish at higher trophic levels. This is because at each successive level
   a. the organisms become smaller.
   b. energy is stored and therefore less is passed on to the next level.
   c. energy is being concentrated in fewer and larger organisms.
   d. energy is being lost from one level to the next.

11. One of the lessons from pyramid of productivity is that
   a. only one half of the energy in one trophic level is passed onto the next trophic level.
   b. most of the energy from one trophic level is incorporated into the biomass of the next level.
   c. ecological efficiency is highest for primary consumers.
   d. eating grain fed beef is an inefficient means of obtaining the energy trapped by photosynthesis.

12. Which of the following is *not* recycled in an ecosystem?
   a. Water
   b. Carbon
   c. Energy
   d. Nitrogen

13. What percentage of the incident solar energy do plants typically harvest during photosynthesis?
   a. 1–2%
   b. 5–10%
   c. 10–20%
   d. 20–50% [GATE]

14. The transfer of energy through a terrestrial ecosystem is often depicted by energy pyramids. Which one of the following statement is *true*?
   a. Ecological efficiency is highest for top consumers.
   b. About 10% of the energy from one trophic level is incorporated into the biomass of the next level.
   c. The energy lost as the heat in respiration is 10% of the available energy of each trophic level.
   d. Only 25% of energy in one trophic level is passed on to the next level.

15. In a stable terrestrial (land base) food chain, which of the following statements is *correct*?
   a. The total mass of the primary producers will be less than the mass of any other trophic level.
   b. Tertiary consumers consume secondary consumers, which in turn consume primary consumers.
   c. At any given time, the biomass of decomposers is directly proportional to the biomass of primary producers.
   d. All tertiary consumers are mammals.

16. Net primary productivity is
   a. the difference between primary productivity and community respiration.
   b. the rate production of biomass by heterotrophs.
   c. the mass of living plants per unit area in a community.
   d. the total rate of fixation of energy by photosynthesis.

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17. In relation to the sulfur biogeochemical cycle, which of the following is *true*?
   a. The sulfur cycle has an atmospheric phase and a lithospheric phase of similar magnitude.
   b. It has an atmospheric phase of overwhelming importance.
   c. It derives mainly from the weathering of rock.
   d. None of the above.

18. The role of decomposers in nitrogen cycle is to
   a. fix N_2 into ammonia.
   b. release ammonia from organic compounds, thus returning it to the soil.
   c. denitrify ammonia thus returning N_2 into the soil.
   d. incorporate nitrogen into amino acids and organic compounds.

19. Which of the following statement is *correct*?
   a. The niche is the habitat where organisms live.
   b. The realized niche is always larger than fundamental niche.
   c. The niche includes the behavioural characteristics of organisms.
   d. The niche is defined only in terms of the dominant environmental factors.

20. Which of the following biomes is correctly paired with the description of its climate?
   a. Savanna–cool temperature, precipitation uniform during the year.
   b. Tundra–long summers, mild winters.
   c. Temperate deciduous forest–relatively short growing season, mild winters.
   d. Tropical forests–nearly constant photoperiod and temperature.

21. Which of the following is an *incorrect* comparison between tropical rainforests and temperate grasslands?
   a. They are both highly productive biomes.
   b. They both have rich, deep topsoil.
   c. They are both characterized by rapid decomposition of dead plant material.
   d. Tropical rain forests, but not grasslands, have a high degree of vertical stratification in their vegetation.

22. Quantities of mineral nutrient in soils of tropical rainforests are relatively low because
   a. the standing crop is small.
   b. microorganisms that recycle chemicals are not very abundant.
   c. decomposition of organic matter and reassimilation of chemicals by plants occur rapidly.
   d. nutrient cycles occur at a relatively slow rate.

23. Which of these ecosystem has the lowest primary productivity per square meter?
   a. A salt marsh
   b. A grassland
   c. An open ocean
   d. A tropical rainforest

24. Choose the mismatch

Column A	*Column B*
a. neuston	resting or swimming on the surface
b. benthos	rests on the bottom or in the bottom sediment
c. periphyton	attached to the rooted plants or other surfaces
d. plankton	actively floating

25. Which of the following is *correctly* paired with its description?
   a. Benthic zone–surface water of shallow sea
   b. Neritic zone–shallow area over continental shelf

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c. Pelagic zone–seafloor
d. Aphotic zone–zone in which light penetrates

26. Choose the mismatch

a. limnology	–	study of fresh water ecosystem
b. phenology	–	study of periodic changes in plants and animals in relation to season
c. autoecology	–	study of interaction of individual organism or species with environment
d. standing crop	–	amount of inorganic matters at any given time in an ecosystem

Biotic community and Succession

1. Commensalism is a symbiotic relationship between two organisms where
   a. both partners benefit
   b. one partner benefits and causes harm to the other
   c. one partner benefits but the other receives no harm
   d. both partners are disadvantaged **[IISc]**

2. Which of the following is *not* a character of r-selected species?
   a. Small adult size
   b. Long generation time
   c. A high energy allocation to reproduction
   d. Maximum population growth

3. Population of individuals of a species having genetically fixed variations but are interfertile and placed under same taxonomic species is known as
   a. ecads
   b. ecotype
   c. endemic species
   d. keystone species

4. Parthenium is
   a. a K-selected species
   b. an r-selected species
   c. none of the above
   d. both of the above **[GATE]**

5. In leguminous plants there exists a symbiotic association between the plant and the bacterium. Choose the *correct* statement from below.
   a. The bacteria derive reduced N_2 from plant.
   b. The bacteria supply reduced N_2 to the plant and receive sugars from plant.
   c. The nodule is produced by the plant independent of bacteria.
   d. The plant growth is inhibited by the factors produced by bacteria. **[IISc]**

6. Which biological characteristic is *not* commonly found in invasive species?
   a. Optimal local adaptation
   b. High reproductive capacity
   c. Broad ecological tolerance
   d. Well-developed dispersal mechanism

7. Which of the following best describes an ecological race?
   a. A locally adapted population with distinct features.
   b. A population that is fertile with other population.
   c. A species with more than one polymorphism.
   d. A morphological polymorphism maintained by ecological factors.

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8. According to the concept of competitive exclusion
   a. Two species cannot coexist in the same habitat.
   b. Extinction or emigration are the only possible results of competitive interaction.
   c. Intraspecific competition results in the success of the best adapted individuals.
   d. Two species cannot share the same niche in a habitat.

9. The Competitive Exclusion Principle states that
   a. if two competitive species coexist in a stable environment, they do so as a result of niche differentiation.
   b. if two competitive species coexist in an unstable environment, they do so as a result of differentiation.
   c. niche differentiation proves that there are coexisting competitors.
   d. competitors can coexist because of environmental heterogeneity.

10. The effect of keystone predator within a community may be to
   a. competitively exclude other predators from the community.
   b. maintain species diversity by preying on the prey species that is the dominant competitor.
   c. increase the relative abundance of the most competitive prey species.
   d. encourage the coevolution of predator and prey adaptations.

11. Which one of the following is *not* generally a feature of shifting cultivation?
   a. Relatively small field
   b. Slash and burn forest clearing
   c. Regular pesticide application
   d. Local consumption of crops

12. Biomagnification refers to
   a. absorption of a chemical from the media to concentrations in the organisms tissue that are greater than in surrounding environment.
   b. tendency to some chemicals to become increasingly concentrated at successively higher trophic levels in a food chain.
   c. the tendency for a compound to accumulate in an organisms tissue.
   d. all the statements are incorrect.

13. Of the following ecological relationships, which one is the most different from the other three?
   a. Algae embedded in coral tissues
   b. Salmonella in human gastric tract
   c. Cellulolytic bacteria in a termite gut
   d. Pollen-collecting bees visiting flowers

[IISc]

14. Choose the *incorrect* statement.
   a. Coral reefs constitute the most diverse and productive biomes on earth.
   b. In pond ecosystems a narrow vertical zone of rapid temperature change is called thermocline.
   c. An ecosystem's energy budget depends on primary productivity.
   d. Red data book compiles data on extinct species.

15. Choose the *incorrect* statement.
   a. Secondary succession are more rapid as compared to primary succession.
   b. Autogenic succession begins in a predominantly inorganic environment.
   c. Successful establishment of the species as a result of a adjustment with the condition prevailing there called ecesis.
   d. The ratio of primary productivity and community respiration in heterotrophic succession is greater than one.

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16. Which of the following is least likely to be *true* of an ecological succession?
    a. The species composition of the community changes continuously during the succession.
    b. The total number of species rises initially, then stabilizes.
    c. The total amount of non living organic matters in the ecosystem increases.
    d. The total biomass in the ecosystem declines after the initial stages.

17. When a forest that is destroyed by fire and again reoccupied by herbs in the initial stage is an example of
    a. primary succession
    b. autogenic succession
    c. secondary succession
    d. climax stage

18. Which of the following is *not* true about succession?
    a. It is a natural process of change in species structure and community with time.
    b. It is a directional and predictable changes.
    c. Progressive increase in the amount of living biomass as succession proceeds.
    d. Food chain relationship becomes less complex as succession proceeds.

19. Which of the following statement is *correct*?
    a. According to competitive exclusion principle, two species cannot coexist in the same community if they have the same niche.
    b. Niche describes the physical space occupied by an organism not its functional role in community.
    c. Ecospecies is an unit of classification which contain one or more ecads.
    d. All statements are correct.

20. The traditional concept of succession includes the idea of an equilibrium state called a climax community. Ecologists now think there may be no such thing as a climax community because
    a. disturbance is ongoing in ecosystem
    b. all organisms eventually die
    c. species diversity generally increases
    d. extinction increases in stable environments

21. The climax plants in a desert would be expected to have the following adaptations
    a. broad leaves and shallow root
    b. dull leaves and a thin epidermis
    c. large number of stomata on the upper surface of the leaves
    d. reduced leaf surface and deep root system

[IISc]

Population ecology

1. A population that has a relatively low *r* value will most likely
    a. have large clutch sizes with relatively small offspring.
    b. be found in environments that are highly variable.
    c. have an early age of first reproduction and a short generation time.
    d. produce fewer offspring with more competitive capabilities.

2. A population's carrying capacity is
    a. the number of individuals in that population.
    b. reached when the number of deaths exceeds the number of births.
    c. inversely related to r_{max}.
    d. the population size that can be supported by available resources for that species within the habitat.

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3. All these terms are characteristics of the human populations in industrial countries except
   a. relatively small clutch size.
   b. several potential reproductions per lifetime.
   c. r-selected life history.
   d. type I survivorship curve.

4. Expanding population is characterized by
   P. a pyramid-shaped age structure.
   Q. an urn-shaped age structure.
   R. pre-reproductive and reproductive age groups become more or less equal in size.
   S. rapidly growing population with high birthrate.
   a. P and R
   b. P and S
   c. Q and R
   d. R and S

5. What is the defining feature of exponential growth?
   a. It lasts indefinitely.
   b. The growth rate is constant.
   c. The growth rate increases rapidly over time.
   d. The growth rate is very high.

6. Which of the following is *not* a reason why population growth declines as population size approaches the carrying capacity?
   a. Climate becomes unfavorable
   b. Competition for resources increases
   c. Predation rates increase
   d. Disease rates increase

7. Why have population biologists become particularly interested in the dynamics of metapopulations?
   a. Because humans exist as a metapopulation.
   b. Because whooping cranes exist as a metapopulation.
   c. Because many populations are becoming restricted to small islands of habitat.
   d. Because metapopulations explain why populations occupying large, contiguous areas are vulnerable to extinction.

8. Assume that you have been studying a population of cattails at the edge of a pond. After 10 years of observations, you notice that the population has remained steady. What is the *most likely* explanation?
   a. The birth rate and death rate are both increasing at the same rate.
   b. The pond is drying up.
   c. The carrying capacity of the pond has been reached.
   d. Nutrient levels in the pond are fluctuating.

9. The life history table below contains information on a single cohort of individuals born in the year 2000. What is the net reproductive rate (R_0) of this population?

Year	Number of individuals	Number of offspring produced per female	Age specific survivorship	$l_x m_x$
2000	500	0		
2001	250	0.5		
2002	200	0.4		
2003	100	0.2		
2004	50	0.1		
2005	0	0		

   a. 0.46
   b. 1
   c. 0.76
   d. More than 1

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Biodiversity

1. Which of the following is *not* true about biodiversity?
   a. Maximum biodiversity occurs in trophic.
   b. Introduction of exotic species may cause the loss of biodiversity.
   c. Diversity of the habitats over the total landscape or geographical area is called gamma diversity.
   d. Hotspot concept developed by Norman Mayers to designate priority areas for ex-situ conservation of biodiversity.

2. Which of the following character make a species susceptible to extinction?
   a. Large population size
   b. Feeding at high trophic level in food chain
   c. Wide range of distributions
   d. Small body size

3. A species facing an extremely high risk of extinction in the immediate future is called
   a. vulnerable
   b. rare
   c. endangered
   d. link species

4. Choose the *incorrect* statement.
   a. Nilgiri biosphere reserve is the 1st biosphere reserve in India.
   b. all land with tree cover of canopy density of between 10 to 40% is known as open forest.
   c. Western ghats and Eastern Himalayas are two hotspots in India.
   d. the Earth summit held in 1992 at Rio de Janerio to prevent ozone depletion at stratosphere.

5. Which of the following statements is/are *false?*
   P. Genetic diversity helps prevent a species from becoming extinct.
   Q. The phenomenon in which animals with favorable adaptation reproduce more rapidly is called differential reproduction.
   R. Geographic isolation is a common mechanism contributing to speciation.
   S. Biologists estimate that 99% of all the species that have ever existed are now extinct.
   a. P and R
   b. P and S
   c. Q and R
   d. P, Q and R

Evolution

1. Darwin synthesized information from several sources in developing his theory of evolution by natural selection. Which of the following did *not* influence his thinking?
   a. Linneaus hierarchical classification of species, which could imply evolutionary relationships.
   b. Lyell's *Principles of Geology*, which described the gradualness and uniformity of geologic changes over long periods of time.
   c. Mendel's paper describing the basic principles of inheritance.
   d. The biogeographic distribution of species that he observed on the Galapagos Islands and during his journey around South America.

2. Within a population and due to competition between individuals, those individuals with an advantage will survive to pass their genes on to the next generation. This idea is called
   a. adaptation
   b. Darwinian theory
   c. speciation
   d. natural selection

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3. The order of appearance of the main groups of organisms during evolution can best be described as follows
   a. autotrophic, anaerobic heterotrophic, aerobic heterotrophic.
   b. aerobic heterotrophic, anaerobic heterotrophic, autotrophic.
   c. anaerobic heterotrophic, aerobic heterotrophic, water oxidizing autotrophic.
   d. anaerobic heterotrophic, water oxidizing autotrophic, aerobic heterotrophic.

4. The *most* likely sequence for the biological evolution of life is
   a. aerobic prokaryotes—photosynthetic prokaryotes—anaerobic prokaryotes—eukaryotes.
   b. photosynthetic prokaryotes—anaerobic prokaryotes—aerobic prokaryotes—eukaryotes.
   c. anaerobic prokaryotes—photosynthetic prokaryotes—aerobic prokaryotes—eukaryotes.
   d. aerobic prokaryotes—eukaryotes—anaerobic prokaryotes —photosynthetic prokaryotes.

5. Which of the following is a post-zygotic isolating mechanism in speciation?
   a. Behavioural isolation
   b. Seasonal isolation
   c. Fertilization failure
   d. Hybrid sterility [IISc]

6. The phenomenon of genetic drift is most likely to occur in populations that are
   a. small and inbred
   b. undergoing gene flow
   c. allopatric
   d. large and panmictic [IISc]

7. In vertebrate evolution, reptiles were able to move completely onto land due to
   a. their scales
   b. the amniotic egg
   c. their increase in size
   d. the availability of food [IISc]

8. In zoos, lions and tigers mate and produce viable hybrid offspring called tigons or ligers. Hybrids were never recorded in Asia, however, where their ranges once overlapped. Hybrids were probably not produced because of
   a. hybrid infertility
   b. gametic mortality
   c. behavioral isolation
   d. mechanical isolation [IISc]

9. Assume that an animal generates heat at a rate proportional to its volume and can radiate heat at a rate proportional to its body surface area. Which of these animals would be best at maintaining its body temperature in a cold climate?
   a. Mouse
   b. Rabbit
   c. Fox
   d. Bear [IISc]

10. A species of flowering plant has a wide altitudinal distribution. Which of the following methods is most appropriate to determine whether phenotypic variation in these plants is due to genetic variation?
   a. Find chromosomal differences between populations of plants at the extremes of the altitudinal range.
   b. Examine amount of phenotypic variation within populations along entire altitudinal range.
   c. Determine if phenotypic differences are retained when plants from different altitudes are grown under the same environmental conditions.
   d. Determine whether hybrids between phenotypically different populations grow at altitudes intermediate between parent populations. [IISc]

11. Well over 90% of all species that have ever existed are extinct today. This shows that
   a. evolution is inefficient.
   b. evolution does not take long-term advantage into account.
   c. extinction is inevitable.
   d. a very long time has passed since life began on earth. [IISc]

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12. In summer, the oxygen consumptions of a tree squirrel and of a lobster were first measured after 1 hour at 20°C and then after 1 hour at 10°C. Which of the following best describes the respective oxygen consumptions as the temperature changed from 20°C to 10°C?

	Tree squirrel	*Lobster*
a.	increased	remained constant
b.	decreased	remained constant
c.	decreased	increased
d.	increased	decreased

13. In allopatric speciation
    a. populations occupy the same habitat and there is a reduction in gene flow.
    b. populations occupy the same habitat and there is normal gene flow.
    c. populations occupy a different habitat and there is a reduction in gene flow.
    d. populations occupy a different habitat and there is normal gene flow. **[MKU]**

14. Fossils of marsupial lions were discovered recently in Australia. The fact that such lions existed points to
    a. convergent evolution
    b. divergent evolution
    c. neutral evolution
    d. reverse evolution **[IISc]**

15. The main feature of the biological species concept is its emphasis on the
    a. large morphological differences between different species.
    b. genetic variation within populations.
    c. role of sexual reproduction in maintaining diversity within a species.
    d. absence of gene flow between different species.

16. Why do scientists think RNA, rather than DNA, may have been the original genetic material?
    a. Most organisms on Earth use RNA as their genetic material.
    b. The simplest life forms, viruses, use RNA.
    c. RNA is more stable than DNA.
    d. RNA has the ability to catalyze a few simple, chemical reactions.

17. Which of the following is *true* of homologous structures?
    a. They are characteristics present in two species that were not present in their common ancestor.
    b. They are characteristics present in two species that were present in a common ancestor.
    c. They generally result from convergent evolution.
    d. b and c are true.

18. Choose the *correct* sequence of evolutionary events in one form of allopatric speciation, using the codes given below
    1. geographical isolation
    2. ecological isolation
    3. increased pre-mating reproductive isolation
    4. increased genetic divergence
    5. selection completed

    a. $3 \rightarrow 4 \rightarrow 2 \rightarrow 5$
    b. $1 \rightarrow 4 \rightarrow 3 \rightarrow 5$
    c. $3 \rightarrow 2 \rightarrow 4 \rightarrow 5$
    d. $1 \rightarrow 2 \rightarrow 3 \rightarrow 5$

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19. A small, isolated population is more likely to undergo speciation than a large population because a small population
    a. is more affected by genetic drift and natural selection
    b. contains, relatively, more genetic diversity
    c. is more susceptible to gene flow
    d. has a higher mutation rate

20. Which of the following evolutionary processes are random?
    1. Gene flow
    2. Mutation
    3. Natural selection
    4. Speciation
    5. Genetic drift
    a. 1 and 2
    b. 1, 2 and 4
    c. 2, 4 and 5
    d. 2 and 5

21. What is the difference between genetic drift and change due to natural selection?
    a. Genetic drift does not require the presence of variation.
    b. Genetic drift does not involve competition between members of a species.
    c. Genetic drift never occurs in nature, natural selection does.
    d. There is no difference.

22. Mitochondrial DNA is advantageous for evolutionary studies because
    a. it is inherited only through the female parent and thus evolves in a way that allows trees of relationship to be easily constructed.
    b. it is inserted into the X-chromosome.
    c. it first appeared in humans and is not found in other animals.
    d. it evolves more slowly than the genes in the nucleus.

23. The allele associated with sickle cell anemia apparently reached a high frequency in some human populations due to
    a. random mating.
    b. superior fitness of heterozygotes in areas where malaria was present.
    c. migration of individuals with the allele into other populations.
    d. a high mutation rate at that specific gene.

24. An increase in the inbreeding coefficient, F, is likely to result in
    a. reduced likelihood of heterozygotes being present in a population.
    b. higher proportion of genes that show linkage.
    c. higher proportion of genes with introns.
    d. lower level of difference between proteins in two daughter cells.

25. Which of the following is required to see a *founder effect*?
    a. High rate of new mutation.
    b. Population *bottleneck* in which the population is reduced to a relatively small number of breeding individuals.
    c. Lethality of homozygotes with a mutant allele.
    d. Selection in favor of heterozygotes.

26. The movement of tectonic plates influences evolution by
    a. increasing atmospheric carbon, changing the location of different types of ecosystems.
    b. changing the locations of continents, causing some species to be geographically isolated from others.
    c. increasing soil richness and thus species diversity.
    d. causing mass extinctions that open up opportunities of new species.

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27. In the context of the evolution of man, who of the following left evidence of artistic ability in the form of drawings and paintings on the walls of caves?
    a. Cro-magnon man
    b. Java-ape man
    c. Neanderthal man
    d. Peking man

28. In which one of the following geological periods did the first bird appear?
    a. Permian
    b. Triassic
    c. Jurassic
    d. Cretaceous

29. Match list I with List II and select the correct answer using the codes given below the lists.

List I	*List II*
P. Allopatric species	1. species inhabiting the same geographical area
Q. Sibling species	2. species inhabiting different geographical areas
R. Sympatric species	3. species which are morphologically identical but reproductively isolated
S. Parapatric species	4. species occupying separate areas that share a common boundary

    a. P-2, Q-1, R-3, S-4
    b. P-2, Q-3, R-1, S-4
    c. P-1, Q-2, R-3, S-4
    d. P-1, Q-3, R-2, S-4

30. The term *living fossil* refers to a
    a. life like fossil well preserved in amber
    b. fossil formed from a living animal
    c. living animal which is about to be fossilized
    d. living organisms showing some very primitive characters

31. A.I. Oparin postulated that colloidal bodies arose in the seas during the primordial stage of the earth. He referred to them as
    a. protobionts
    b. coacervate droplets
    c. microspheres
    d. spherules

32. Which one among the following is the most important factor in speciation?
    a. Geographic isolation
    b. Reproductive isolation
    c. Ethological isolation
    d. Ecological isolation

33. Which one of the following is an evidence to show that the birds evolved from reptiles?
    a. Scales of birds and reptiles are shed and replaced.
    b. Bones of some reptiles are pneumatic like birds.
    c. Presence of pygostyle in reptiles.
    d. Presence of synapsid skull and the oviparous nature.

34. Consider the following statements.
    P. Wings of insects and birds are homologous organs.
    Q. Wings of insects and bats are analogous organs.
    R. Wings of insects and birds are analogous organs.
    S. Wings of birds and bats are homologous organs.
    a. P and Q
    b. Q, R and S
    c. P and S
    d. P, Q and S

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35. The first vertebrates appeared in which one of the following periods of the Palaeozoic era?
a. Ordovician
b. Silurian
c. Devonian
d. Mississippian

36. The shape of the beaks of Darwin's finches, industrial melanism, and the changes in horse teeth are all examples of
a. artificial selection
b. natural selection
c. convergent evolution
d. homologous structures

37. All of the following are believed to contribute to genomic diversity among various species, *except*
a. gene duplication
b. gene transcription
c. lateral gene transfer
d. chromosomal rearrangements

38. Stabilizing selection favours
a. both extreme forms of a trait
b. intermediate forms of a trait
c. environmental differences
d. one extreme form over the other extreme form and over intermediate forms of a trait

39. The genus *Rhagoletis* is a group of small, brightly colored flies. Each species in the genus feeds, during its larval stage, on the fruit of just one plant family, and when the larvae hatch into adults, they court and mate on the same fruit. The species that feed on fruits from different plant families are reproductively isolated by
a. ecological isolation
b. temporal isolation
d. behavioral isolation
d. mechanical isolation

40. When two related populations that have been geographically isolated come back into contact, retain their reproductive isolation, and compete for resources, there may be a divergence in traits known as
a. character convergence
b. sibling speciation
c. phyletic evolution
d. character displacement

Population genetics

1. The Hardy-Weinberg law describes
a. genotype frequencies of a population when evolutionary forces are not acting.
b. how sexual reproduction would change the relative gene frequencies in a population.
c. how mutations occur and balance each other.
d. genotype frequencies of a population when evolutionary forces are acting. **[IISc]**

2. What are the assumptions of Hardy-Weinberg equilibrium?
a. Small population size, random mating, no selection, no migration, no mutation. **[IISc]**
b. Large population size, random mating, no selection, no migration, no mutation.
c. Large population size, random mating, heterozygotes survive the best, no migration, no mutation.
d. Large population size, random mating, no selection, migrants enter from other populations, no mutation.

3. Genetic diversity is required for natural selection to act, but natural selection can reduce or eliminate diversity. What process can restore genetic diversity to a population?
a. Genetic drift
b. Mutation
c. Sexual selection
d. Stabilizing selection

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4. Deviation from the Hardy-Weinberg assumption of infinitely large population size results in
   a. genetic lethal
   b. heterozygote advantage
   c. consanguinity
   d. genetic drift

5. In a plant species, a segregating line (one that contains both homozygotes and heterozygotes at a locus) can be made homozygous by repeated selfing for several generations. What is the level of remaining heterozygosity after three generations of selfing, if the level of heterozygosity in generation '0' is denoted as 1?
   a. 0.125
   b. 0.25
   c. 0.50
   d. 0.0625

6. In a population that is in equilibrium, the proportion of individuals showing the dominant trait at a given locus having two alleles is 84%. The frequency of the recessive allele in the population is
   a. 0.40
   b. 0.30
   c. 0.20
   d. 0.16 [IISc]

7. How many *A* and *a* alleles are present in a sample of organisms consisting of 10 AA, 15 Aa and 4 aa individuals? What are the allele frequencies in this sample?
   a. A = 0.60; a = 0.40
   b. A = 0.40; a = 0.60
   c. A = 0.60; a = 0.60
   d. A = 0.40; a = 0.40 [IISc]

8. In a diploid organism, what is the maximum number of alleles that can exist in a population for any given gene?
   a. 1
   b. 2
   c. 4
   d. unlimited [IISc]

9. Cystic fibrosis is an autosomal recessive inherited disease. Its frequency in newborns is 1 in 1700. One of the following indicates the frequency of the disease carriers
   a. 0.024
   b. 0.047
   c. 0.976
   d. 0.148

10. Which statement about allele frequencies is *not* true?
    a. The sum of any set of allele frequency is always 1.
    b. If there are two alleles at a locus and we know the frequency of one of them, we can obtain the frequency of the other by subtraction.
    c. If an allele is missing from a population, its frequency is 0.
    d. If two populations have the same gene pool for a locus, they will have the same proportion of homozygotes at that locus.

11. If an X-linked recessive disorder is in Hardy-Weinberg equilibrium and the incidence in males is 1 in 100, then the expected incidence of affected homozygous females would be
    a. 1 in 1000
    b. 1 in 4000
    c. 1 in 10000
    d. 1 in 40000 [JNU]

12. The colour of flowers of an annual species of plants is controlled by a single locus with two alleles R and r, and the genotypes RR, Rr and rr and red, pink and white respectively. A large number of seeds from individuals with pink flowers were collected and planted on an island, where, because of absence of pollinators, only self-pollination is possible. What will be the most likely outcome after 25 years?
    a. About 50% plants with red flowers, 50% with white flowers.
    b. Almost 100% plants with pink flowers.
    c. Red, pink and white flowered plants in a ratio of 1:2:1.
    d. Red, pink and white flowered plants in equal proportion. [IISc]

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Model Paper – I

1. The clonal selection theory best accounts for
   a the production of memory cells
   b. the mechanism whereby antigen and antibody unite
   c. the maturation of B-cells into plasma cells
   d. the specificity of the immune response

2. What happens to the Cdk-cyclin A complex at metaphase?
   a. Only Cdk is degraded
   b. Only cyclin A is degraded
   c. Both Cdk and cyclin A are degraded
   d. None of the above

3. Which of the following statement is *incorrect*? A monoclonal antibody
   a. could theoretically be isolated from serum.
   b. is active against multiple epitopes on one protein.
   c. is produced by cells that must be grown in HAT medium.
   d. is produced by immortal cells.

4. A measure of the repetitiveness of a DNA sequences based on its reassociation rate is the
   a. C-value paradox
   b. hypochromicity
   c. buoyant density
   d. $Cot_{1/2}$ value

5. Transport vesicles
   a. can have a clathrin coat
   b. can be coated with COPs
   c. contain assembly proteins
   d. all of the above

6. What type of DNA-binding domains do homeodomains have?
   a. Helix-turn-helix
   b. Helix-zinc finger
   c. Helix-leucine zippers
   d. β-ribbon

7. In human males, testosterone is produced mainly by the
   a. leydig cells
   b. sertoli cells
   c. seminiferous tubules
   d. epididymis

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8. The minimum components of an artificial yeast chromosome include all of the following *except*
   a. all the histones except H1
   b. an autonomously replicating sequence (ARS)
   c. a centromere (CEN) sequence
   d. a telomere (TEL) sequence

9. The DNA polymerase responsible for replication of mitochondrial DNA is
   a. α
   b. β
   c. γ
   d. ε

10. Which of the following protein's mRNA lacks a poly-A tail?
   a. Immunoglobulin
   b. Albumin
   c. Keratin
   d. Histone

11. Sequences at the ends of viral retrotransposons that also present in retroviruses are
   a. direct repeat
   b. palindromic sequence
   c. long terminal repeat
   d. LINE

12. In bacteria, which of the following is *not* an inducer of an operon?
   a. tryptophan
   b. allolactose
   c. IPTG
   d. L-arabinose

13. The first ribozyme was found in
   a. a nuclear gene for a DNA replicating enzyme
   b. a mitochondrial gene for a respiratory enzyme
   c. an mRNA for a mitochondrial enzyme
   d. an intron within a pre-rRNA molecule

14. Which of the following is a fact that argues against a symbiotic origin of eukaryotic flagella from bacterial flagella?
   a. Bacterial flagella typically contain flagellin; eukaryotic flagella contain tubulin.
   b. Bacterial flagella rotate; eukaryotic flagella do not rotate.
   c. Bacterial flagella are directly energized by electrons and protons from the respiratory system of the plasma membrane; eukaryotes are not.
   d. More than one of the above argue against the symbiotic theory.

15. MHC class I molecules are important for which of the following?
   a. Binding to CD8 molecules on T-cells.
   b. Presenting exogenous antigen (e.g. bacterial protein) to B-cells.
   c. Presenting intact viral proteins to T-cells.
   d. Binding to CD4 molecules on T-cells.

16. Which of the following regulators are said to act in "cis"?
   a. The *lac* repressor and the *lac* operator.
   b. The lac operator and mammalian transcription factors.
   c. The *lac* operator and mammalian enhancers.
   d. Mammalian transcription factors and enhancers.

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17. Which part of coenzyme A is found attached to a serine residue in the acyl carrier protein of the fatty acid synthase?

a. Phosphopantetheine
b. Phosphopantothenate
c. Pantothenate
d. β-alanine

18. Which dietary fat is typically nonesterified?

a. Cholesterol
b. Glycerophospholipids
c. Triacylglycerols
d. Fatty acids

19. Which of the following proteins in the photosynthetic electron-transport chain is *not* a transmembrane protein?

a. ATP synthase
b. LHC
c. PSII
d. Ferrodoxin

20. Which of the following is *incorrect*?

a. Root apical meristem is major site of synthesis of the free cytokinin.
b. Amino acid methionine is the precursor of ethylene.
c. Kinetin is the most abundant naturally occurring cytokinin from higher plants.
d. IAA is synthesized from the amino acid tryptophan.

21. A retrotransposon would be best described as

a. a mobile DNA element that can also reproduce as a virus.
b. a transposable element that passes through an RNA stage.
c. a transposable element that cuts and moves the DNA from one site to another.
d. all of the above.

22. In the pentose phosphate pathway

a. only the C-1 carbon of glucose is oxidized to CO_2.
b. all the carbons of glucose is oxidized to CO_2.
c. no decarboxylation occurs.
d. C-4 and C-5 of glucose is oxidized to CO_2.

23. Which of the following correctly explains how a favorable genetic trait can increase in frequency in a population?

a. Lamarck's principle
b. Natural selection
c. Adaptive radiation
d. Genetic recombination

24. Mammalian chromosomes have specialized structures with highly repetitive DNA at their ends (telomeres). Which of the following aspects of telomeric DNA replication is different from that of other chromosomal regions?

a. The DNA polymerase uses an RNA primer but does not degrade it.
b. The DNA polymerase contains an RNA molecule that serves as template for DNA synthesis.
c. The DNA polymerase must cross-link the 5′ and 3′ termini.
d. The DNA polymerase has a subunit that facilitates binding to repetitive DNA.

25. Parvovirus DNA is sensitive to digestion with S1 nuclease. This is evidence that

a. the DNA is linear
b. the DNA is single-stranded
c. the DNA has hairpin ends
d. the DNA is double-stranded

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26. Most of the reducing equivalents utilized for synthesis of fatty acids can be generated from
    a. the pentose phosphate pathway
    b. glycolysis
    c. the citric acid cycle
    d. mitochondrial malate dehydrogenase

27. Most of the CO_2 that is transported in blood
    a. is dissolved in the plasma
    b. is bound to hemoglobin
    c. is in carbonic acid
    d. is in bicarbonate ion

28. The predominant secondary structural motif in an antibody molecule is
    a. antiparallel beta-sheet
    b. a 4-helix bundle
    c. alpha-helical coiled coil
    d. parallel beta-sheet

29. Which of the following properties is not exhibited by T_H cells?
    a. stimulate division of B-cells
    b. stimulate division of cytotoxic T-cells (CTLs)
    c. are cytotoxic for other cells
    d. stimulate migration of macrophages

30. Biochemical mechanisms that regulate progression through the cell cycle include
    a. phosphorylation
    b. dephosphorylation
    c. protein degradation
    d. all of the above

31. Silent Mutations in DNA are not expressed due to
    a. universality of genetic code
    b. non-ambiguous nature of genetic code
    c. degeneracy of genetic code
    d. none of the above

32. Signal hypothesis refers to
    a. the hormonal control of eukaryotic proteins synthesis.
    b. the synthesis of secreted proteins by membrane bound ribosomes.
    c. characteristics of 70S ribosomes involved in the synthesis of proteins for export by the cell.
    d. sequence of amino acids at the carbonyl end of a secretory proteins which is cleaved off after the secretion.

33. When DNA is denatured by heating and then allowed to renature, there will be
    a. an increase in absorbance at 595 nm.
    b. a decrease in absorbance at 260 nm.
    c. an increase in absorbance at 260 nm.
    d. no change in absorbance at any wavelength.

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34. Which one of the following statement is irrelevant to proteins glycosylation?
a. Protein glycosylation is one of the major biosynthetic functions of SER.
b. It is a process of covalent addition of sugars to secretory proteins.
c. During process an oligosaccharide containing 14 sugars is transferred to the NH_2 group of asparagines residue of a protein.
d. Asn-X-Ser and Asn-X-Thr sequences function as signals for N-linked glycosylation.

35. The rate-controlling step in the synthesis of fatty acids is catalyzed by
a. acetyl-CoA carboxylase
b. fatty acid synthase
c. HMG CoA synthase
d. thiolase

36. Choose the mismatch

Organelle	*Marker molecule*
a. lysosomes	acid phosphatase
b. peroxisomes	catalase
c. mitochondria	cytochrome oxidase
d. smooth endoplasmic reticulum	amino acid permease

37. What is the smallest number of molecules of ATP and GTP consumed in the synthesis of a 200-residue protein, starting from amino acids?
a. 400 ATP + 400 GTP
b. 100 ATP + 600 GTP
c. 100 ATP + 598 GTP
d. 200 ATP + 200 GTP

38. What is the role of the cI gene of phage lambda?
a. It encodes a protein that enables the phage to enter lysis.
b. Its gene product helps the prophage excise from the host DNA.
c. It encodes the lambda repressor which represses the lytic cycle.
d. Its gene product helps the prophage integrate into the host DNA.

39. What clones should a eukaryotic cDNA library contain?
a. Clones that represent every fragment of DNA in approximately equal frequencies.
b. Clones that represent one copy of every coding region.
c. Clones representing transcribed DNA in approximately equal frequencies.
d. Clones representing transcribed DNA in frequencies that reflect their level of expression.

40. What is the role of the bicoid gene in *Drosophila* development?
a. To determine the anterior end of a fly embryo.
b. To determine the thoracic region.
c. To determine even numbered segments.
d. To determine odd numbered segments.

41. Which of the following cofactors is required for conversion of proline residues into hydroxyproline in procollagen?
a. TPP
b. Pantothenate
c. Niacin
d. Ascorbic acid

42. Which of the following type of cross-link present in β-keratin?
a. Cysteine cross-link
b. Cross-link via isodesmosine
c. Cross-link via two lysine residues
d. None of the above

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43. Which of the following statement is *incorrectly* paired?
   a. Gluconeogenesis : cortisol
   b. Free fatty acid mobilization : dehydroepiandrosterone
   c. Muscle glycogenolysis : epinephrine
   d. Kaliuresis : aldosterone

44. Which of the following is the principal buffer in interstitial fluid?
   a. Hemoglobin
   b. Albumin
   c. Carbonic acid
   d. H_2PO_4

45. Choose the mismatch

Inhibitor	Function
a. Malonate	Prevents oxidation of succinate
b. Cyanide	Inhibits cytochrome oxidase
c. Oligomycin	Inhibits ATP synthase
d. Rotenone	Blocks $CoQH_2$-cyt C oxidoreductase

46. Which of the following statement is *false* about somaclonal variation?
   a. Genetic variability is present among cultured cells.
   b. Genetic variability may arise due to change in chromosome number/structure and gene mutation.
   c. Only undergoes epigenetic changes.
   d. None of the above.

47. A molecular biologist analyzed a DNA sample and found that it contained the nucleotide bases in the following proportion : A : T : G : C :: 1 : 1.33 : 0.98 : 0.65. This data indicates that
   a. DNA sample is highly thermolabile
   b. DNA sample is from a eukaryotic cell
   c. DNA in the sample is single stranded
   d. more data required for any deduction

48. Virus-free plants have been propagated commercially through
   a. cell culture
   b. pollen culture
   c. apical meristem culture
   d. embryo culture

49. Although the Seal and the Penguin both have streamlined, fish-like bodies with a layer of insulating fat, they are not closely related. This similarity results from
   a. homologous evolution
   b. convergent evolution
   c. adaptive radiation
   d. coevolution

50. In a population of diploid individuals that is in Hardy-Weinberg equilibrium, the frequency of a dominant allele for a certain hereditary trait is 0.3. What percentage of individuals in the next generation would be expected to be homozygous for the dominant trait?
   a. 9%
   b. 14%
   c. 42%
   d. 49%

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Model Paper – II

1. Which of the following statements about enhancers is *incorrect*?
   a. They function as promoters.
   b. They function when in either orientation in the DNA.
   c. They function even when located many base pairs away from the promoter.
   d. They function only in specific types of cells.

2. Which of the following statements about the σ subunit of RNA polymerase is *incorrect*?
   a. It enables the enzyme to transcribe asymmetrically.
   b. It confers on the core enzyme the ability to initiate transcription at promoters.
   c. It decreases the affinity of RNA polymerase for regions of DNA that lack promoter sequences.
   d. It facilitates the termination of transcription by recognizing hairpins in the transcript.

3. Which of the following statements about topoisomerases is *incorrect*?
   a. They alter the linking numbers of topoisomers.
   b. They break and reseal phosphodiester bonds.
   c. They require NAD^+ as a cofactor to supply the energy to drive the conversion of a supercoiled molecule to its relaxed form.
   d. They form covalent intermediates with their DNA substrates.

4. Activation of phospholipase C causes all of the following *except*
   a. cleavage of membrane-bound phosphatidylinositol 1,4,5-trisphosphate.
   b. production of diacylglycerol.
   c. release of Ca^{2+} from intracellular stores.
   d. dephosphorylation of proteins.

5. In the absence of the bile salts, glycocholic and taurocholic acid, the intestinal absorption of all of the following would be impeded except
   a. riboflavin
   b. oleic acid
   c. cholesterol
   d. vitamin A

6. The synthesis of glucose from pyruvate by gluconeogenesis
   a. occurs exclusively in the cytosol.
   b. is inhibited by an elevated level of glucagon.
   c. requires the participation of biotin.
   d. involves lactate as an intermediate.

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7. Which one of the following compounds cannot give rise to the net synthesis of glucose?
   a. Lactate
   b. Glycerol
   c. Oxaloacetate
   d. Acetyl-CoA

8. Which one of the following is *not* a characteristic of the hexose monophosphate pathway?
   a. It produces CO_2
   b. It is controlled by inhibition of glucose 6-phosphate dehydrogenase by NADPH
   c. It requires ATP for phosphorylation
   d. It produces ribose 5-phosphate

9. Reduced glutathione functions in red blood cells to
   a. produce NADPH
   b. reduce methemoglobin to hemoglobin
   c. reduce pyruvate to lactate
   d. reduce oxidizing agents such as H_2O_2

10. Which of the following statements are features of the 26S proteosome?
   a. Consists of a 20S catalytic subunit and a 19S regulatory subunit.
   b. Release only free amino acids as products.
   c. Digests ubiquitin along with the protein to which it is attached.
   d. All of the above.

11. Which of the following answers complete the sentence correctly? During the reduction of FAD,
   a. a flavin group is transferred.
   b. an equivalent of a hydride ion is transferred.
   c. the isoalloxazine ring becomes charged.
   d. two hydrogen atoms are added to the isoalloxazing ring.

12. The enzyme alkaline phosphatase
   a. joins double-stranded DNA fragments.
   b. adds a phosphate group to the 5'-end of DNA.
   c. cuts double-stranded DNA.
   d. removes a phosphate group from the 5'-end of DNA.

13. If a chiasma forms between the loci of genes A and B in 20% of the tetrads of an individual of genotype *AB/ab*, the percentage of gametes expected to be *Ab* is
   a. 40
   b. 20
   c. 10
   d. 5

14. The chemical that causes an A-T base pair to replace a G-C base pair is
   a. proflavin
   b. acridine dye
   c. ethylmethane sulfonate
   d. ethidium bromide

15. According to classical genetics, which of the following statement is *true*?
   a. Recessive alleles are detected by the phenotype of the F1 generation.
   b. The closer two genes are, the more frequently they recombine.
   c. Genes on different autosomes segregate independently.
   d. Genes on sex chromosomes segregate with the same pattern as autosomal genes.

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16. Which one of the following cofactors must be utilized during the conversion of acetyl-CoA to malonyl-CoA?
a. thiamine pyrophosphate
b. acyl carrier protein (ACP)
c. NAD
d. biotin

17. Fatty acids are transported into the mitochondria bound to
a. thiokinase
b. coenzyme A (CoA)
c. acetyl-CoA
d. carnitine

18. If the 16-carbon saturated fatty acid palmitate is oxidized completely to carbon dioxide and water (via the β-oxidation pathway and the citric acid cycle), and all of the energy conserving products are used as ATP to drive ATP synthesis in the mitochondria, the net yield of ATP per molecule of palmitate is, in round numbers
a. 3
b. 10
c. 25
d. 129

19. Bacteria that are resistant to penicillin and related antibiotics produce an enzyme that breaks the ..... in these antibiotics.
a. side groups (R groups)
b. benzene ring
c. beta-lactam ring
d. disulfide bonds

20. Increase number of chromosomes occurs in
a. Turner's syndrome
b. fragile-X syndrome
c. Klinefelter's syndrome
d. retinoblastoma

21. Loss of function mutations (involving a single allele) that affect the production on an enzyme are usually autosomal ....., while loss of function mutations that affect structural proteins are usually autosomal.....
a. recessive, recessive
b. recessive, dominant
c. dominant, dominant
d. dominant, recessive

22. Gal⁻ mutants of *E. coli* cannot use galactose as a carbon source. When these mutants are infected with phage λ that has been grown on wild-type *E. coli*, however, the progeny bacteria can use galactose. This genome alteration is an example of bacterial
a. transformation
b. transduction
c. conjugation
d. transposition

23. How do eukaryotic ribosomes identify the start sites of proteins?
a. Via their Shine-Delgarno sequences.
b. By scanning from the 5' end of the mRNA for the first AUG.
c. The *cap* structure of a eukaryotic mRNA overlaps with the first AUG.
d. The transcriptional start of a eukaryotic mRNA overlaps with the translational start.

24. *Bacillus anthracis* capsules consist of which one of the following?
a. γ-glutamyl polypeptide
b. Hyaluronic acid
c. Repeating polysaccharide capsule of glucose and glucuronic acid
d. Sialic acid polymers

25. Which of the following cofactors participates in acyl group transfer reactions?
a. Thiamine pyrophosphate
b. S-adenosylmethionine
c. Coenzyme A
d. NADH

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26. Among the following components of chloroplast membrane which one is the strongest reducing agent?
    a. Reduced cytochrome b_6
    b. Ubiquinone
    c. P_{680}
    d. Ferredoxin

27. When a plant of chromosomal type aa pollinates a plant of type AA, what chromosome constitution of embryo and endosperm is expected in the resulting seeds?
    a. Diploid zygote of type Aa and triploid endosperm of type AAa.
    b. Diploid zygote of type aa and triploid endosperm of type Aaa.
    c. Diploid zygote of type AA and triploid endosperm of type AAa.
    d. Diploid zygote of type Aa and triploid endosperm of type aaa.

28. The growth kinetics that result from metabolizing one sugar before another is referred to as
    a. exponential growth
    b. diphasic growth
    c. diauxic growth
    d. chemotaxis

29. The mode of action of a steriod hormone involves
    a. binding to a cell membrane receptor
    b. activation of protein kinases
    c. covalent modification of enzyme activity
    d. modifying gene transcription

30. Which of the following statements correctly describes p53?
    a. It is a membrane receptor that binds to a cell growth hormone.
    b. Cell division stops until p53 binds to DNA and repairs the damage.
    c. p53 is an example of an oncogene, and its expression causes uncontrolled cell division.
    d. Activation of p53 leads to inhibition of the S-phase cyclin-CDK complex.

31. Membrane potential
    a. occurs only in nerve and muscle cells
    b. is the same in all cells
    c. is oriented so that the cell's interior is positive with respect to the extracellular fluid
    b. none of the above is correct

32. Which of the following is *not* a mechanism for avoiding or correcting mutations?
    a. Recombination
    b. Proofreading by DNA polymerases
    c. Okazaki fragment synthesis
    d. Discrimination between nucleotides during polymerization by DNA polymerases

33. What is the force that is primarily responsible for stabilizing the tertiary structure of globular proteins?
    a. Disulfide bonding
    b. The hydrophobic effect
    c. Hydrogen bonding
    d. Ionic interactions

34. Which one of the following antibiotics attaches to 50S ribosome and inhibits peptidyl transferase?
    a. Amdinocillin
    b. Amphotericin
    c. Chloramphenicol
    d. Penicillin

35. Interferon is produced by cells in tissue culture when the cells are stimulated with which of the following?
    a. Botulinum toxin
    b. Chlamydiae
    c. Gram-positive bacteria
    d. Viruses

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36. With regard to histones which is *incorrect*?
   a. They are basic proteins.
   b. The amino acid composition of histones show great variability.
   c. Mitochondria do not contain histones.
   d. In the chromosomes, the weight of the histones equal to that of the DNA.

37. Dynein and kinesin
   a. are the two subunits of microtubules.
   b. are motor proteins that generate movements associated with microtubules.
   c. are motor proteins that generate sliding of the sarcomere.
   d. create the crawling motion typical of amoeba.

38. In *Drosophila* development, what sequence is followed in the hierarchy of expression of the gene types listed below?
   1. Homeotic genes
   2. Maternal effect genes
   3. Gap genes
   4. Pair rule genes
   5. Segment polarity genes

   a. 1, 2, 3, 4, 5
   b. 2, 3, 4, 5, 1
   c. 2, 3, 4, 1, 5
   d. 1, 2, 4, 3, 5

39. After interacting with a cell-surface receptor that signals activation to a trimeric G-protein, the first event that happens is
   a. GTP is hydrolyzed to produce GDP.
   b. tyrosines on the G-protein become phosphorylated.
   c. the G-protein dissociates into a β- and αγ-subunits.
   d. GTP replaces GDP in the α-subunit.

40. The Lineweaver-Burk plot is used to graphically determine K_m and V_{max} for an enzyme that obeys classic Michaelis-Menten kinetics. When V is the reaction velocity at substrate concentration S, the Y axis experimental data in the Lineweaver-Burk plot are expressed as
   a. V
   b. 1/V
   c. S
   d. 1/S

41. On average, how many fragments would a restriction enzyme which recognizes a specific 4 base sequence in DNA be expected to cleave a double-stranded bacteriophage with a genome size of 5,000 bp into?
   a. About 2
   b. About 4
   c. About 20
   d. About 50

42. Which of the following enzyme catalyzes the principal regulatory step in cholesterol synthesis?
   a. HMG-CoA reductase
   b. HMG-CoA synthase
   c. Thiolase
   d. Xanthine oxidase

43. The class of adrenal gland hormones that provide resistance to stress, produce anti-inflammatory effects, and promote normal metabolism to ensure adequate quantities of ATP is
   a. glucocorticoids
   b. mineralocorticoids
   c. androgens
   d. catecholamines

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44. The antigen specificity of a particular B-cell
    a. is induced by interaction with antigen.
    b. is determined only by the L-chain sequence.
    c. is determined by H + L-chain variable region sequences.
    d. changes after isotype switching.

45. Which of the following is *incorrect* concerning MHC class II molecules?
    a. B-cells may express different allelic forms of MHC class molecules on their surface.
    b. MHC class II molecules are synthesized in the endoplasmic reticulum of many cell types.
    c. Genetically different individuals express different MHC class II alleles.
    d. MHC class II molecules are associated with β_2-microglobulin on the cell surface.

46. Which of the following about ecosystems is *false*?
    a. Only some specialized organisms are capable of obtaining energy directly from sunlight.
    b. Energy flows through the system, matter cycles.
    c. Radiant energy from the sun is the ultimate source of energy for all of the earth's ecosystems.
    d. None of the above.

47. If two populations of bacteria cultured in a U-shaped tube are separated by a membrane filter (which does not allow phage particles to pass), but recombination takes place anyway, the mechanism of genetic exchange is
    a. specialized transduction
    b. site-specific recombination
    c. conjugation
    d. transformation

48. During the processing of introns, a single snRNP complex catalyzes both the cleavage of the RNA and the joining of the cut ends. What would be the consequence if these two processes were catalyzed by separate enzymes not associated in a single complex?
    a. The rate of RNA processing would be much faster.
    b. The cell would be unable to identify the correct cleavage sites.
    c. The exons might not be joined in the correct sequence.
    d. Exons instead of introns would be cleaved from the RNA molecule.

49. Malonyl ACP, a three-carbon unit, is the starting material for fatty-acid biosynthesis. However, the product fatty acid, at least in mammals, contains an even number of carbons. How is this accomplished?
    a. Malonyl ACP is decarboxylated during the condensation step.
    b. Malonyl ACP is used as a temporary catalytic carrier of two carbon units (acetyl ACP).
    c. Only even numbers of malonyl groups are used in fatty-acid biosynthesis.
    d. Malonyl ACP transfers only one carbon to the nascent fatty acid during the condensation step.

50. The allele associated with sickle cell anemia apparently reached a high frequency in some human populations due to
    a. random mating.
    b. superior fitness of heterozygotes in areas where malaria was present.
    c. migration of individuals with the allele into other populations.
    d. a high mutation rate at that specific gene.

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Answers

CHAPTER 1. AMINO ACIDS AND PROTEINS

Amino acids and peptides

01. c	02. a	03. b	04. c	05. d	06. c	07. a
08. d	09. c	10. d	11. c	12. a	13. b	14. d
15. a	16. b	17. b	18. a	19. d	20. b	21. a
22. b	23. d	24. a	25. c	26. b	27. a	28. b
29. b	30. d	31. d	32. b	33. d	34. c	35. b
36. c	37. a	38. c	39. d	40. a	41. c	42. d
43. c	44. d	45. c	46. a	47. b	48. a	49. a

Protein structure

01. b	02. b	03. b	04. d	05. a	06. b	07. b
08. d	09. c	10. c	11. c	12. c	13. b	14. d
15. b	16. b	17. a	18. b	19. a	20. b	21. a
22. d	23. d	24. a	25. d	26. b	27. a	28. d
29. c	30. c	31. b				

Globular and Fibrous protein

01. d	02. d	03. c	04. b	05. b	06. c	07. d
08. c	09. c	10. b	11. c	12. b	13. d	14. a
15. c	16. a	17. b	18. c	19. d	20. d	21. c
22. c	23. d	24. c	25. a	26. d	27. d	28. b
29. d	30. b	31. d				

Protein techniques

01. b	02. b	03. d	04. d	05. b	06. b	07. c
08. a	09. d	10. c	11. d	12. a	13. c	14. b
15. b	16. d	17. a	18. a	19. d	20. d	21. a
22. d	23. c	24. b	25. a	26. b	27. c	28. a
29. c	30. d	31. c	32. c	33. c		

Protein sequencing

01. a	02. c	03. b	04. b	05. a	06. a	07. c
08. b	09. a	10. b	11. b	12. d		

Amino acid metabolism

01. a	02. d	03. d	04. c	05. a	06. d	07. b
08. d	09. d	10. a	11. c	12. d	13. a	

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CHAPTER 2. NUCLEIC ACIDS

Nucleosides and Nucleotides

01. d	02. b	03. a	04. b	05. b	06. b	07. a
08. b	09. d					

Nucleic acids

01. c	02. c	03. c	04. d	05. a	06. d	07. c
08. d	09. b	10. d	11. c	12. a	13. c	14. c
15. d	16. d	17. b	18. a	19. a	20. c	21. b
22. b	23. b	24. b	25. d	26. b	27. b	28. b
29. a	30. a	31. d	32. a			

DNA

01. d	02. d	03. c	04. d	05. c	06. c	07. c
08. c	09. c	10. d	11. a	12. b	13. c	14. d
15. a	16. b	17. d	18. d	19. c	20. d	21. a
22. a						

RNA

01. b	02. b	03. b	04. c	05. c	06. a	07. b
08. c	09. d	10. d	11. a			

Electrophoresis and Sequencing

01. d	02. b	03. c	04. c	05. d	06. c	07. b

Nucleotide metabolism

01. c	02. b	03. c	04. c	05. b	06. d	07. c
08. c	09. a	10. a				

CHAPTER 3. CARBOHYDRATES AND LIPIDS

Monosaccharides and Disaccharides

01. d	02. c	03. d	04. d	05. b	06. d	07. a
08. d	09. d	10. d	11. a	12. b	13. d	14. a
15. d	16. c	17. d	18. b	19. d	20. c	

Polysaccharides

01. a	02. a	03. d	04. d	05. c	06. c	07. c
08. c	09. d	10. d	11. d	12. d		

Glycogenesis, Glycogenolysis and Gluconeogenesis

01. b	02. b	03. c	04. c	05. b	06. d	07. b
08. b	09. d	10. a	11. d			

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Fatty acids

01. d 02. b 03. d 04. d 05. d 06. a 07. c
08. d 09. d 10. d 11. c

Triacylglycerol, Phospholipid and Glycolipid

01. a 02. b 03. c 04. d 05. d 06. c 07. a
08. b 09. d 10. a 11. a 12. d 13. c 14. d
15. a 16. d 17. d 18. d 19. d 20. d 21. a
22. d

Cholesterol and Lipoproteins

01. d 02. c 03. d 04. a 05. a 06. b 07. a
08. d 09. a 10. d

Fatty acid metabolism

01. d 02. c 03. d 04. a 05. b 06. d 07. c
08. b 09. d 10. d 11. d 12. b 13. c

Cholesterol metabolism

01. b 02. b 03. d 04. b 05. b

CHAPTER 4. ENZYME AND VITAMINS

Enzyme : General features

01. a 02. d 03. b 04. d 05. c 06. c 07. c
08. d 09. c 10. a 11. d 12. a 13. c 14. d
15. c 16. a 17. c 18. d 19. c 20. b 21. b
22. d 23. c 24. c

Enzyme kinetics

01. a 02. b 03. c 04. d 05. d 06. d 07. a
08. c 09. c 10. b 11. c 12. b 13. d 14. c
15. a 16. a 17. b 18. d

Enzymatic inhibition

01. a 02. a 03. d 04. b 05. c 06. c 07. d
08. a 09. c

Regulatory enzymes

01. d 02. b 03. c 04. b 05. d 06. b

Enzymatic reactions

01. c 02. b 03. d 04. c 05. a 06. b 07. b
08. c

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Vitamins

01. c	02. a	03. b	04. a	05. d	06. c	07. d
08. b	09. a	10. d	11. c	12. d	13. d	14. c
15. b	16. a	17. d	18. d			

CHAPTER 5. CELL BIOLOGY

Eukaryotic cell : Structure and Function

01. b	02. d	03. a	04. c	05. b

Plasma membrane : Structure and Transport

01. c	02. c	03. b	04. c	05. d	06. d	07. d
08. b	09. a	10. d	11. c	12. c	13. d	14. d
15. c	16. c	17. b	18. d	19. a	20. d	21. c
22. b	23. c	24. d	25. b			

Membrane transport

01. d	02. b	03. d	04. c	05. d	06. d	07. c
08. d	09. b	10. b	11. c	12. c	13. b	14. a
15. a	16. c	17. d	18. c	19. a	20. b	21. c
22. c	23. a	24. c	25. b	26. a	27. d	

Membrane potential

01. a	02. a	03. c	04. a	05. b	06. b	07. d
08. d	09. d	10. a	11. d			

Endoplasmic reticulum

01. c	02. c	03. b	04. c	05. a	06. d	07. b
08. d	09. d	10. a	11. c	12. c	13. a	14. a

Golgi complex

01. b	02. d	03. b	04. a	05. a	06. d	07. c
08. c	09. c	10. c	11. b			

Lysosome

01. d	02. b	03. c	04. b	05. a	06. d	07. a
08. d	09. c	10. c	11. c	12. c	13. b	

Intracellular trafficking

01. a	02. d	03. d	04. d	05. d	06. b	07. c
08. a	09. d	10. d	11. c	12. d	13. a	14. b
15. d	16. c	17. a	18. d	19. c	20. b	21. a
22. c	23. b					

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Cytoskeleton and Motility

01. b	02. d	03. b	04. c	05. d	06. d	07. b
08. a	09. b	10. a	11. d	12. c	13. b	14. b
15. d	16. b	17. d	18. a	19. c	20. b	21. d
22. d	23. c	24. d	25. a	26. b	27. d	28. d
29. b	30. a	31. d	32. c	33. d	34. d	35. a
36. d	37. d					

Extracellular matrix and Cell junctions

01. b	02. a	03. a	04. a	05. d	06. a	07. c
08. c	09. a	10. a	11. b			

Mitochondria and Chloroplast

01. d	02. b	03. a	04. b	05. b	06. c	07. b
08. d	09. c	10. b	11. c	12. b	13. a	14. a
15. c	16. b	17. a	18. a	19. d	20. d	

Peroxisomes

01. c	02. a	03. d	04. c	05. d	06. b	07. c
08. d	09. d					

Nucleus

01. d	02. b	03. b	04. d	05. d	06. c	07. d
08. c	09. c	10. d	11. d	12. b	13. a	14. b
15. d	16. b	17. d	18. b	19. b	20. c	21. d
22. d	23. b	24. c	25. a	26. b	27. c	28. c

Cell signaling

01. d	02. d	03. a	04. c	05. b	06. b	07. d
08. d	09. c	10. b	11. a	12. d	13. b	14. a
15. c	16. d	17. b	18. a	19. b	20. b	21. b
22. c	23. a	24. c	25. d	26. d	27. d	28. d
29. b	30. c	31. c	32. b	33. d	34. b	35. a
36. c	37. c	38. d	39. c			

Cell cycle and Cell division

01. b	02. d	03. d	04. c	05. b	06. b	07. d
08. a	09. b	10. a	11. b	12. b	13. c	14. d
15. d	16. a	17. b	18. d	19. c	20. b	21. d
22. c	23. d	24. b	25. c	26. b	27. b	28. a
29. c	30. a	31. d	32. b	33. c	34. a	35. b
36. b	37. b	38. a	39. c	40. b	41. d	42. a
43. b	44. d	45. d	46. a	47. c	48. d	49. b
50. c						

Cancer

01. d	02. d	03. c	04. d	05. d	06. a	07. d
08. c	09. d	10. d	11. c	12. c	13. b	14. b
15. c	16. c	17. b	18. c	19. a	20. c	21. b
22. a						

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Answers of fill in the blanks

Eukaryotic cell : Structure and function

01. lipid bilayer
02. phospholipids
03. amphipathic
04. polar, hydrophobic hydrocarbon (fatty acid)
05. micelles, bilayers
06. liposomes, black. (The black membranes are so called destructive interference between the reflected light from the two surfaces makes them appear black).
07. phospholipids translocators
08. phase transition
09. cholesterol
10. glycolipids
11. gangliosides, GM1
12. transmembrane
13. glycosylphosphatidylinositol (GPI) anchor
14. peripheral membrane, integral membrane
15. single-pass transmembrane, multipass transmembrane
16. porins
17. detergents
18. vectorial labeling
19. spectrin
20. band 3 protein
21. freeze-fracture
22. bacteriorhodopsin
23. rotational, lateral, flip-flop
24. heterocaryons
25. florescence recovery after photobleaching (FRAP)
26. cell coat, glycocalyx
27. lectins
28. selectins
29. membrane transport
30. carrier, channel
31. passive (facilitated diffusion), active
32. electrochemical
33. ionophores, mobile-ion, channel
34. uniporters
35. symport
36. anion, antiport
37. Na^+-K^+ pump (Na^+-K^+ ATPase)
38. sarcoplasmic reticulum
39. secondary, primary
40. transcellular transport
41. microvilli
42. ABC transporter superfamily
43. multidrug resistance (MDR) protein

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44. channel, ion channels
45. voltage changes, mechanical stimulation, ligand binding
46. membrane potential, K^+ leak
47. resting membrane potential, Nernst
48. cell body, axon, dendrites
49. action potential
50. voltage-gated Na^+ channels, Na^+
51. voltage-gated K^+ channels, K^+
52. myelin sheath
53. patch-clamp
54. synapse, neurotransmitter, transmitter-gated
55. excitatory neurotransmitters, inhibitory neurotransmitters
56. neuromuscular junction
57. spatial, temporal, grand
58. axon hillock
59. delayed K^+, early K^+
60. Ca^{2+}-activated K^+, adaptation
61. long-term potentiation
62. cytoskeleton
63. centrosome
64. desmin
65. plus end, minus end
66. aster
67. centrosome
68. pericentriolar material, centrosome matrix
69. microtubule-associated proteins (MAPs)
70. cytoplasmic dyneins
71. kinesins

Cell Signaling

01. receptor
02. local mediators, paracrine signaling
03. neurotransmitter, synaptic signaling
04. endocrine, hormones
05. autocrine signaling
06. eicosanoids
07. gap junctions
08. nitric oxide (NO)
09. intracellular receptor superfamily
10. steroid
11. thyroid
12. retinoids
13. ion-channel-linked
14. G-protein-linked
15. enzyme-linked
16. trimeric GTP-binding proteins (G proteins)

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17. cyclic AMP, adenylyl cyclase, cyclic AMP phosphodiesterase
18. stimulatory G protein (G_s)
19. β-adrenergic receptors
20. cholera toxin
21. inhibitory G protein (G_i)
22. pertussin toxin
23. cyclic-AMP-dependent protein kinase (A-kinase)
24. serine/threonine phosphoprotein phosphatases
25. phosphatidylinositol (PI)
26. protein kinase C (C-kinase, or PKC)
27. calmodulin
28. olfactory receptors
29. cyclic GMP
30. rod photoreceptors (rods), rhodopsin
31. cyclic GMP phosphodiesterase
32. receptor guanylyl cyclases
33. receptor tyrosine kinases, protein kinases
34. SH2 domains
35. Ras superfamily, monomeric GTPases
36. Ras proteins
37. mitogen-activated protein (MAP) kinases
38. Src family
39. receptor serine/threonine kinases

Cell Cycle and Cell Division

01. cyclin-dependent protein kinase (Cdk)
02. mitotic cyclins, G_1 cyclins
03. maturation-promoting factor (MPF)
04. cyclin
05. *cdc* (cell-division cycle)
06. *cdc*2
07. Start
08. p53
09. oncogene, proto-oncogene
10. tumor-suppressor genes
11. retinoblastoma (Rb)

CHAPTER 6. RESPIRATION

Glycolysis and Fermentation

01. a	02. b	03. b	04. c	05. d	06. c	07. b
08. b	09. d	10. d	11. b	12. b	13. c	14. c
15. d	16. c	17. a	18. c	19. b	20. d	

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Krebs cycle

01. d	02. b	03. b	04. c	05. b	06. a	07. a
08. a	09. c	10. d	11. d	12. b		

Oxidative phosphorylation

01. d	02. d	03. b	04. b	05. b	06. c	07. c
08. c	09. d	10. a	11. d	12. d	13. c	14. d
15. b	16. b	17. c	18. b	19. c	20. d	21. c
22. d	23. b	24. c	25. c	26. c	27. b	28. b
29. d	30. c	31. a	32. b	33. d		

Pentose phosphate pathway

01. a	02. d	03. b	04. a	05. d	06. b	07. d

CHAPTER 7. PHOTOSYNTHESIS

Photosynthesis : General features

01. b	02. c	03. c	04. d	05. d	06. a	07. b
08. a	09. a	10. d	11. c	12. c	13. a	14. c
15. d	16. b	17. d	18. d	19. d	20. a	21. a
22. d	23. c	24. b	25. b	26. c	27. d	28. c
29. b	30. c					

Light reactions

01. c	02. c	03. c	04. b	05. d	06. d	07. d
08. d	09. a	10. d	11. c	12. a	13. d	14. b
15. c	16. c	17. c	18. b	19. a	20. c	21. b
22. d	23. a	24. d	25. c	26. b	27. d	28. a
29. d	30. d	31. b	32. a	33. a	34. c	35. b
36. d	37. d	38. d				

Calvin cycle

01. d	02. c	03. d	04. c	05. c	06. b	07. a
08. d	09. d	10. c	11. b	12. d	13. c	14. c
15. c	16. d	17. a	18. c	19. b.	20. d	

Photorespiration, C_4 and CAM pathway

01. b	02. a	03. b	04. a	05. d	06. b	07. b
08. c	09. c	10. b	11. d	12. c	13. b	14. c
15. b	16. d	17. b	18. b	19. c	20. b	21. a
22. d	23. c	24. a	25. a	26. d	27. b	28. a
29. a	30. d					

Transport of photoassimilate

01. a	02. b	03. c	04. d	05. b	06. a	07. d
08. b	09. b	10. a	11. d	12. b	13. c	

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CHAPTER 8. MOLECULAR GENETICS

DNA replication

01. c	02. c	03. a	04. b	05. c	06. d	07. b
08. d	09. b	10. d	11. d	12. d	13. d	14. b
15. c	16. c	17. c	18. b	19. c	20. d	21. c
22. c	23. b	24. d	25. b	26. d	27. d	28. b
29. d	30. c	31. c	32. c	33. d	34. a	35. b
36. d	37. d	38. d	39. c	40. a	41. d	42. c
43. a	44. c	45. b	46. c	47. d	48. d	49. d
50. c	51. b	52. a	53. a	54. d	55. d	56. b
57. b	58. a	59. d	60. c	61. c	62. b	63. c

Genome complexity

01. c	02. a	03. d	04. b	05. d	06. a

Transposable elements

01. c	02. d	03. b	04. d	05. d	06. c	07. c
08. d	09. d					

Satellite DNA

01. c	02. b	03. a	04. c	05. a	06. d	07. d
08. b	09. d	10. d				

Gene families

01. c	02. b	03. d	04. b	05. a	06. b	07. d
08. a	09. c					

Transcription

01. d	02. c	03. a	04. d	05. c	06. c	07. a
08. d	09. b	10. b	11. c	12. c	13. a	14. b
15. d	16. a	17. a	18. a	19. b	20. b	21. b
22. c	23. b	24. b	25. b	26. c	27. b	28. d
29. b	30. c	31. d	32. d	33. c	34. c	35. b
36. b	37. c	38. a	39. a	40. a	41. d	42. b
43. c	44. d	45. c	46. c	47. d	48. b	

RNA processing

01. a	02. b	03. d	04. b	05. d	06. c	07. c
08. c	09. b	10. b	11. d	12. d	13. b	14. b
15. a	16. c	17. c	18. d	19. d	20. c	21. a
22. a	23. b	24. d	25. b	26. d	27. d	28. c
29. c	30. c	31. d				

Prokaryotic gene regulation

Lac operon

01. c	02. d	03. b	04. c	05. d	06. c	07. b
08. d	09. b	10. b	11. c	12. c	13. b	14. b
15. c	16. c	17. c	18. b	19. b	20. a	21. d
22. d	23. a	24. a	25. b	26. c	27. a	28. c
29. c	30. d					

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Trp operon

01. c 02. d 03. c 04. a 05. d 06. b 07. c
08. b

Eukaryotic gene regulation

01. a 02. c 03. b 04. c 05. c 06. b 07. b
08. c 09. c 10. b 11. c 12. c 13. b 14. d
15. b 16. c

Genetic switch

01. b 02. a 03. d 04. c 05. c 06. d 07. a

DNA binding motifs

01. a 02. b 03. a 04. a 05. b 06. d 07. c
08. b 09. a

Genetic code

01. b 02. c 03. d 04. d 05. d 06. b 07. a
08. c 09. a 10. c 11. b 12. c 13. c 14. a
15. d 16. c 17. a

Ribosomes and tRNAs

01. a 02. d 03. c 04. c 05. a 06. c 07. a
08. b 09. a

Protein synthesis

01. d 02. d 03. d 04. c 05. d 06. c 07. a
08. b 09. a 10. c 11. b 12. d 13. a 14. b
15. a 16. d 17. d 18. d 19. b 20. d 21. b
22. d 23. c 24. b 25. a 26. d 27. d 28. d
29. b 30. c 31. d 32. d 33. b 34. d 35. b
36. a 37. d 38. d 39. d 40. c 41. b 42. a
43. b 44. d 45. b 46. c 47. a 48. c 49. d
50. c 51. a 52. d 53. d 54. c

DNA recombination

01. a 02. b 03. b 04. c 05. b 06. c 07. b
08. a 09. c 10. d

DNA repair

01. b 02. d 03. a 04. d 05. d 06. c 07. a
08. d 09. d 10. d

Gene mutation

01. d 02. d 03. b 04. b 05. c 06. b 07. d
08. c 09. c 10. d 11. a 12. a 13. d 14. b
15. c 16. c 17. b 18. c 19. a 20. b 21. d
22. d 23. b 24. d 25. b 26. d 27. b 28. d
29. c 30. c 31. c 32. b 33. b 34. b 35. a
36. d 37. c 38. b 39. c 40. a 41. c 42. b
43. d 44. d 45. c 46. d 47. b 48. c 49. d
50. d

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Answers of fill in the blanks

01. aminoacyl-tRNA synthetases, aminoacyl-tRNA
02. ribosome, peptidyl-tRNA-binding site, aminoacyl-tRNA-binding site
03. peptidyl transferase, rRNA
04. stop
05. reading frames
06. initiation factors (IFs)
07. initiator tRNA, start, methionine
08. DNA repair, mutation
09. fibrinopeptides
10. depurination, deamination
11. DNA repair nucleases, DNA polymerase, DNA ligase
12. base excision repair, DNA glycosylases
13. nucleotide excision repair
14. SOS response
15. DNA polymerase
16. DNA replication fork
17. DNA ligase
18. leading strand, lagging strand
19. proofreading exonuclease
20. RNA primers, DNA primase
21. DNA helicase
22. single-stranded DNA-binding (SSB) proteins
23. primosome
24. mismatch proofreading (mismatch repair)
25. replication origins
26. DNA topoisomerases
27. general recombination
28. heteroduplex joint (staggered joint)
29. DNA renaturation (hybridization), helix nucleation
30. RecA protein
31. branch migration
32. cross-strand exchange, Holliday junction
33. gene conversion
34. site-specific

CHAPTER 9. RECOMBINANT DNA TECHNOLOGY

Enzymes

01. c	02. b	03. d	04. b	05. a	06. d	07. b
08. b	09. b	10. b	11. c	12. a	13. a	14. b
15. b	16. d	17. b	18. d	19. c	20. c	21. a
22. d	23. c	24. d	25. c	26. a	27. d	28. c
29. b	30. d	31. c				

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Vector

01. d	02. c	03. d	04. b	05. c	06. b	07. b
08. d	09. d	10. c	11. d	12. c	13. b	14. c
15. a	16. c	17. a	18. c			

DNA cloning

01. a	02. c	03. a	04. b	05. d	06. c	07. c
08. d	09. a					

PCR

1. a.	2. b	3. a	4. d	5. d	6. d	7. b
8. d	9. b	10. b.	11. c			

Protein expression

01. d	02. b	03. c	04. b	05. c	06. b	07. d
08. b						

DNA library

01. b	02. a	03. c	04. b	05. d

Engineering plants and animals

01. c	02. d	03. d	04. b	05. a	06. d	07. b
08. d	09. a	10. c	11. b	12. c	13. c	14. a
15. c	16. b					

Sequencing/Blotting/Electrophoresis/Labelling

01. a	02. c	03. d	04. a	05. d	06. a	07. a
08. d	09. d	10. b	11. b	12. b	13. d	14. c
15. c						

Applications of recombinant DNA technology

01. a	02. c	03. c	04. d	05. a	06. d	07. b
08. b	09. a	10. d				

Answers of fill in the blanks

01. DNA cloning
02. plasmid vectors
03. genomic DNA clone, genomic DNA library
04. cDNA clone, cDNA library
05. substrate hybridization
06. chromosome walking
07. polymerase chain reaction (PCR)
08. genetics
09. reverse genetics
10. fusion proteins
11. epitope tagging
12. dominant-negative
13. antisense RNA
14. transgenic, transgenes

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CHAPTER 10. CLASSICAL GENETICS

Mendel's principle

01. b	02. b	03. a	04. a	05. a	06. b	07. a
08. b	09. c	10. c	11. a	12. c	13. b	14. d
15. a	16. c	17. d	18. a	19. c	20. a	21. a
22. a	23. c	24. c	25. a	26. c	27. d	28. c
29. a	30. d					

Linkage and Mapping

01. b	02. d	03. c	04. c	05. c

Sex determination and Sex-linked inheritance

01. a	02. b	03. d	04. a	05. b	06. d	07. d
08. d	09. b	10. b	11. c	12. d	13. d	14. c
15. d						

Quantitative inheritance

01. c	02. a	03. d	04. c	05. a	06. c

Cytogenetics

01. c	02. b	03. c	04. c	05. b	06. a	07. a
08. c	09. d	10. c	11. d	12. a	13. b	14. d
15. b	16. d	17. b				

CHAPTER 11. PROKARYOTES AND VIRUS

Bacterial cell structure

01. c	02. b	03. a	04. d	05. d	06. c	07. a
08. c	09. b	10. b	11. d	12. a	13. b	14. d
15. a						

Bacterial growth and Cultivation

01. a	02. d	03. b	04. d	05. b	06. b	07. a
08. c	09. d	10. d	11. b	12. a	13. a	14. d
15. a	16. b	17. d	18. b	19. d		

Bacterial groups

01. c	02. c	03. d	04. b	05. a	06. d	07. d
08. d	09. c	10. c	11. a	12. b	13. d	14. d
15. d						

Archaebacteria

01. a	02. c	03. d	04. d	05. a	06. d

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Bacterial genome

01. d 02. c 03. a 04. a 05. a 06. b 07. d

Gene transfer and recombination

01. d 02. d 03. a 04. c 05. c 06. d 07. d
08. b 09. b 10. c 11. a 12. b 13. c 14. b
15. b 16. c 17. a

Toxins

01. c 02. c 03. b 04. c 05. d 06. c 07. c
08. d 09. a

Virus

01. a 02. c 03. c 04. c 05. a 06. c 07. b
08. b 09. d 10. b 11. a 12. d 13. c 14. c
15. b 16. c 17. a 18. d 19. d 20. b 21. c
22. b 23. a 24. d 25. b 26. d 27. d 28. c
29. b 30. d 31. c 32. c

Viroids and Prions

01. b 02. d 03. c 04. d 05. a

Antibiotics

01. b 02. b 03. a 04. d 05. a 06. d 07. b

Miscellaneous

01. b 02. a 03. d 04. d 05. a 06. d 07. b
08. a 09. b 10. d

CHAPTER 12. IMMUNOLOGY

Innate and Adaptive immune response

01. b 02. b 03. b 04. d 05. a 06. d 07. c
08. a 09. d 10. a

Adaptive immunity

01. c 02. b 03. b 04. b 05. b

Cells and organs of the immune system

01. b 02. b 03. b 04. b 05. b 06. d 07. b
08. b 09. b 10. b 11. c 12. a 13. a 14. b
15. c 16. d 17. c 18. a 19. d 20. d 21. c
22. b 23. b 24. d

Antigens

01. d 02. d 03. c 04. b 05. d 06. b 07. d

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MHC and Antigen presentation

01. d	02. c	03. a	04. d	05. a	06. b	07. b
08. c	09. d	10. d	11. d	12. d	13. b	14. d

Antibodies : Structure and Function

01. c	02. d	03. a	04. c	05. c	06. d	07. c
08. c	09. d	10. b	11. c	12. c	13. b	14. d
15. c	16. c	17. d	18. b	19. a	20. a	21. b
22. d	23. d	24. c	25. d	26. c	27. c	28. c
29. d	30. a	31. a	32. b	33. d	34. a	35. c
36. a	37. c	38. a	39. a	40. c	41. c	42. b
43. c	44. c	45. c				

Organization and Expression of Ig genes

01. c	02. b	03. b	04. d	05. d	06. d	07. d
08. b						

Antigen-antibody interactions : Principles and applications

01. b	02. c	03. b	04. d	05. c	06. c	07. a
08. d	09. c	10. d	11. d	12. a	13. a	14. c
15. b	16. c	17. c	18. d	19. c		

B-cell : Generation, activation, differentiation and Response

01. c	02. d	03. d	04. c	05. a	06. c	07. d
08. c						

T-cell : Maturation, activation, differentiation and Response

01. d	02. b	03. d	04. c	05. c	06. a	07. c
08. a	09. c	10. d	11. d	12. c		

Cytokines and Complement system

01. b	02. d	03. d	04. c	05. a	06. d	07. d
08. d						

Hypersensitivity and Autoimmunity

01. a	02. d	03. c	04. b	05. a	06. d	07. c
08. d	09. a	10. c	11. b			

Vaccine

01. b	02. a	03. c	04. c	05. c		

CHAPTER 13. PLANT PHYSIOLOGY

Plant water relationship/Transport of minerals and water

01. c	02. c	03. c	04. b	05. d	06. c	07. d
08. d	09. d	10. d	11. b	12. a	13. c	14. d
15. c	16. c	17. d	18. d	19. c	20. c	21. c
22. a	23. d	24. c	25. c	26. b	27. c	28. d
29. c	30. d	31. d	32. c	33. b	34. d	

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Transpiration

01. d	02. c	03. a	04. c	05. d	06. d	07. b
08. d	09. d	10. c				

Plant nutrition

01. b	02. c	03. d	04. a	05. c	06. d	07. b
08. b	09. b	10. c	11. a	12. d	13. d	

Plant hormones

01. d	02. c	03. d	04. a	05. c	06. b	07. d
08. a	09. a	10. d	11. d	12. a	13. d	14. c
15. c	16. c	17. a	18. a	19. d	20. b	21. c
22. d	23. d	24. d	25. a	26. b	27. a	28. a
29. b	30. c	31. c	32. c	33. a		

Photomorphogenesis and Vernalization

01. b	02. c	03. d	04. b	05. d	06. c	07. a
08. c	09. b	10. d	11. c	12. d	13. a	14. a
15. c						

Seed physiology

01. d	02. d	03. d	04. d	05. d

Plant movements

01. c	02. d	03. c	04. b	05. d	06. b	07. c
08. b						

CHAPTER 14. HUMAN PHYSIOLOGY

Nervous system

01. d	02. b	03. c	04. a	05. d	06. b	07. b
08. c	09. a	10. b	11. b	12. d	13. b	14. b
15. d	16. d	17. a	18. b	19. b		

Sense organs

01. b	02. d	03. b	04. d	05. d	06. d	07. c

Blood vascular system

01. d	02. b	03. c	04. a	05. d	06. b	07. c
08. c	09. a	10. b	11. a	12. a	13. a	14. a
15. a	16. a	17. d	18. d			

Respiratory system

01. b	02. d	03. c	04. d	05. a	06. c	07. d
08. a	09. d					

Excretory system

01. d	02. d	03. a	04. b	05. b	06. c	07. c
08. d	09. b	10. d	11. d	12. d		

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Digestive system

01. c	02. d	03. c	04. c	05. d	06. d	07. b
08. d	09. b	10. d	11. b	12. c	13. d	

Reproductive system

01. b	02. b	03. d	04. c	05. b	06. d	07. b
08. d	09. a					

Endocrine system

01. d	02. a	03. d	04. d	05. c	06. d	07. a
08. d	09. c	10. c	11. b	12. b	13. b	14. c
15. d	16. b	17. b	18. a	19. b	20. a	21. c
22. d	23. b	24. c	25. a	26. c	27. d	28. a
29. a	30. a	31. c	32. c			

CHAPTER 15. DIVERSITY OF LIFE

Taxonomy

01. d	02. c	03. b	04. c	05. c	06. b

Monera

01. a	02. b	03. c	04. b

Protists

01. d	02. d	03. a	04. d	05. d	06. b	07. a
08. d	09. a	10. b	11. c	12. b		

Fungi

01. d	02. c	03. d	04. b	05. c	06. d	07. d

Animals

01. d	02. d	03. d	04. c	05. d	06. d	07. d
08. b	09. d	10. c	11. b	12. d	13. a	14. c
15. d	16. b	17. d	18. d	19. a	20. c	21. c
22. d	23. b	24. c	25. b			

Plants

01. c	02. b	03. a	04. b	05. a	06. d	07. a
08. b	09. b	10. c	11. d	12. b		

Anatomy of angiospermic plants

01. d	02. c	03. a	04. b	05. b	06. a	07. d
08. a	09. b	10. d	11. d	12. a		

Reproduction in angiosperms

01. c	02. b	03. b	04. d	05. c	06. c	07. c
08. d	09. a	10. d	11. d	12. d	13. b	14. a
15. b	16. a	17. a	18. b	19. d	20. b	21. b
22. d						

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CHAPTER 16. ECOLOGY AND EVOLUTION

Ecosystem

01. d	02. b	03. d	04. b	05. c	06. c	07. d
08. d	09. a	10. d	11. d	12. c	13. a	14. b
15. b	16. a	17. a	18. b	19. c	20. d	21. b
22. c	23. c	24. d	25. b	26. d		

Biotic community and Succession

01. c	02. b	03. b	04. b	05. b	06. a	07. a
08. d	09. a	10. b	11. c	12. b	13. b	14. d
15. d	16. d	17. c	18. d	19. a	20. a	21. d

Population ecology

01. d	02. d	03. c	04. b	05. b	06. a	07. c
08. c	09. a					

Biodiversity

01. d	02. b	03. c	04. d	05. d

Evolution

01. c	02. d	03. d	04. c	05. d	06. a	07. b
08. c	09. d	10. c	11. d	12. d	13. c	14. b
15. d	16. d	17. b	18. d	19. a	20. d	21. b
22. a	23. b	24. a	25. b	26. b	27. a	28. c
29. b	30. d	31. b	32. b	33. b	34. b	35. a
36. b	37. b	38. b	39. a	40. d		

Population genetics

01. a	02. b	03. b	04. d	05. a	06. a	07. a
08. d	09. b	10. d	11. c	12. a		

MODEL PAPERS

Model Paper – I

01. d	02. b	03. b	04. d	05. d	06. a	07. a	08. a
09. c	10. d	11. c	12. a	13. d	14. d	15. a	16. c
17. a	18. a	19. d	20. c	21. b	22. a	23. b	24. b
25. b	26. a	27. d	28. a	29. c	30. d	31. c	32. b
33. b	34. a	35. a	36. d	37. a	38. c	39. d	40. a
41. d	42. d	43. b	44. c	45. d	46. c	47. c	48. c
49. b	50. a						

Model Paper – II

01. a	02. d	03. c	04. d	05. a	06. c	07. d	08. c
09. d	10. a	11. d	12. d	13. d	14. c	15. c	16. d
17. d	18. d	19. c	20. c	21. b	22. b	23. b	24. a
25. c	26. d	27. a	28. c	29. d	30. d	31. d	32. c
33. b	34. c	35. d	36. b	37. b	38. b	39. d	40. b
41. c	42. a	43. a	44. c	45. d	46. d	47. d	48. c
49. a	50. b						

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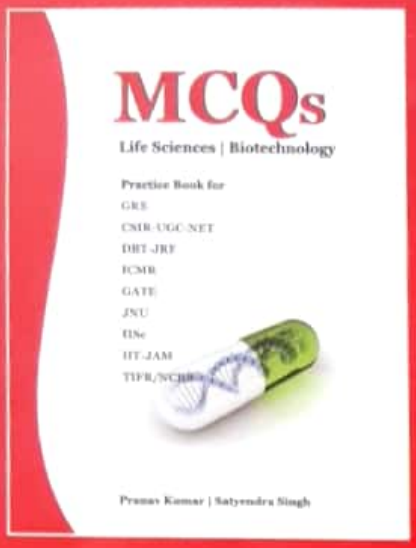

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